Reporte: Core vs Divergent
Metabolismo + Clínica + Supervivencia

Pacientes Core: 808 | Divergentes: 255

Variables metabólicas analizadas: 248 | Significativas (p_adj<0.05): 7

Soluciones FBA: FBA, pFBA, L1w, L2, L2w

Estructura: Clínica −> Metabolismo −> Integración −> Supervivencia

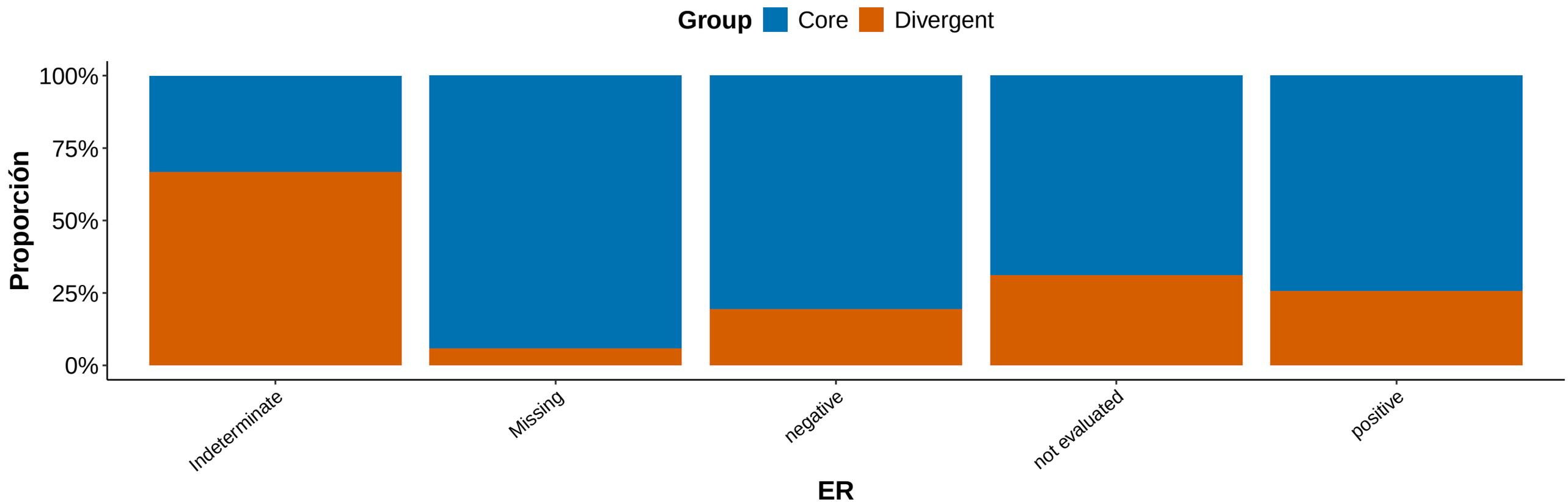
SECCIÓN I
Diferencias Clínicas: Core vs Divergent

Macro−categorías: Biología Tumoral | Estadio | Historia Terapéutica | Factores del Paciente

Clínica Categórica − . A. Biología Tumoral

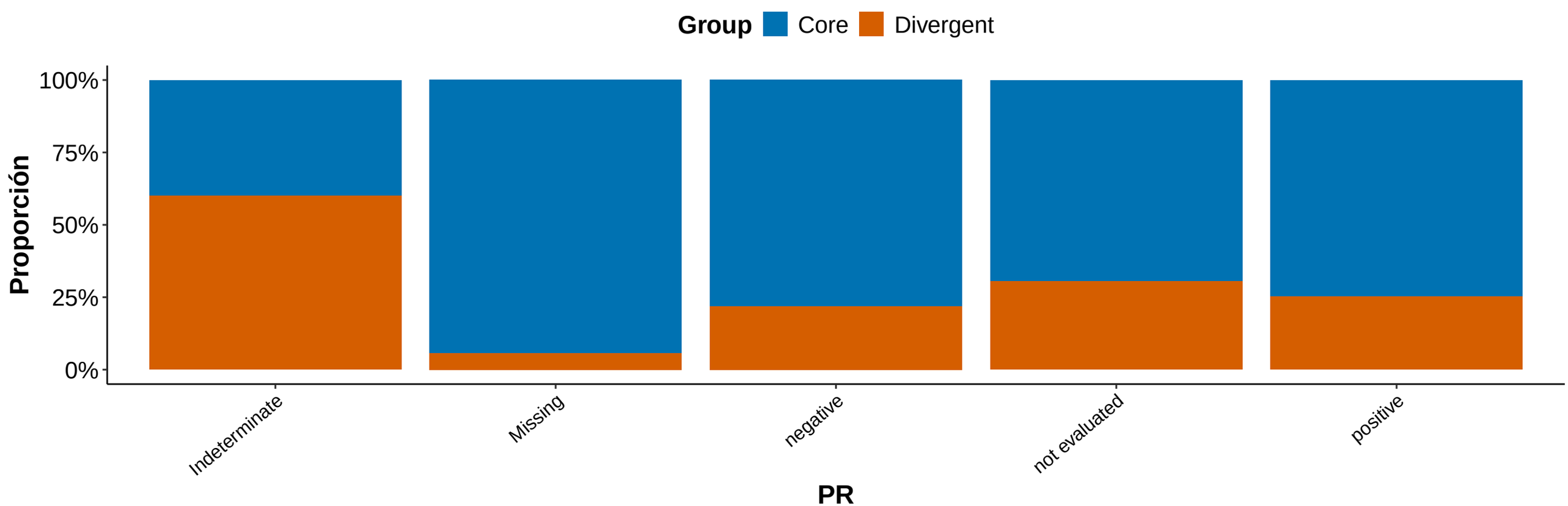
ER

Chi² p: 0.00636



PR

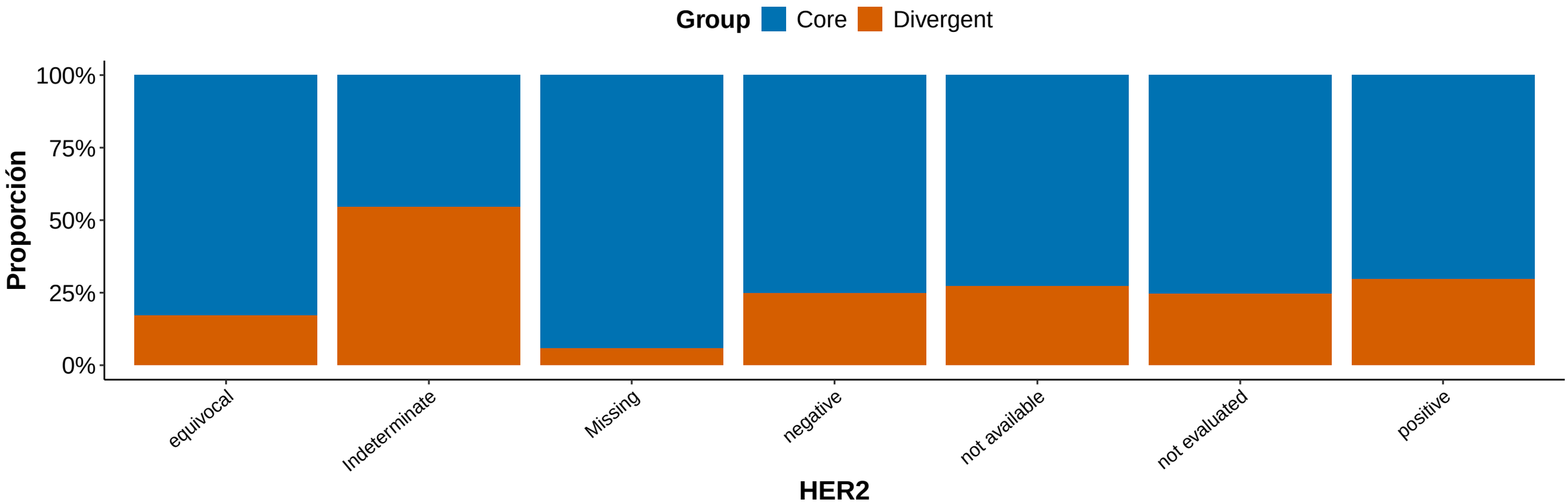
Chi² p: 0.015



Clínica Categórica − . A. Biología Tumoral

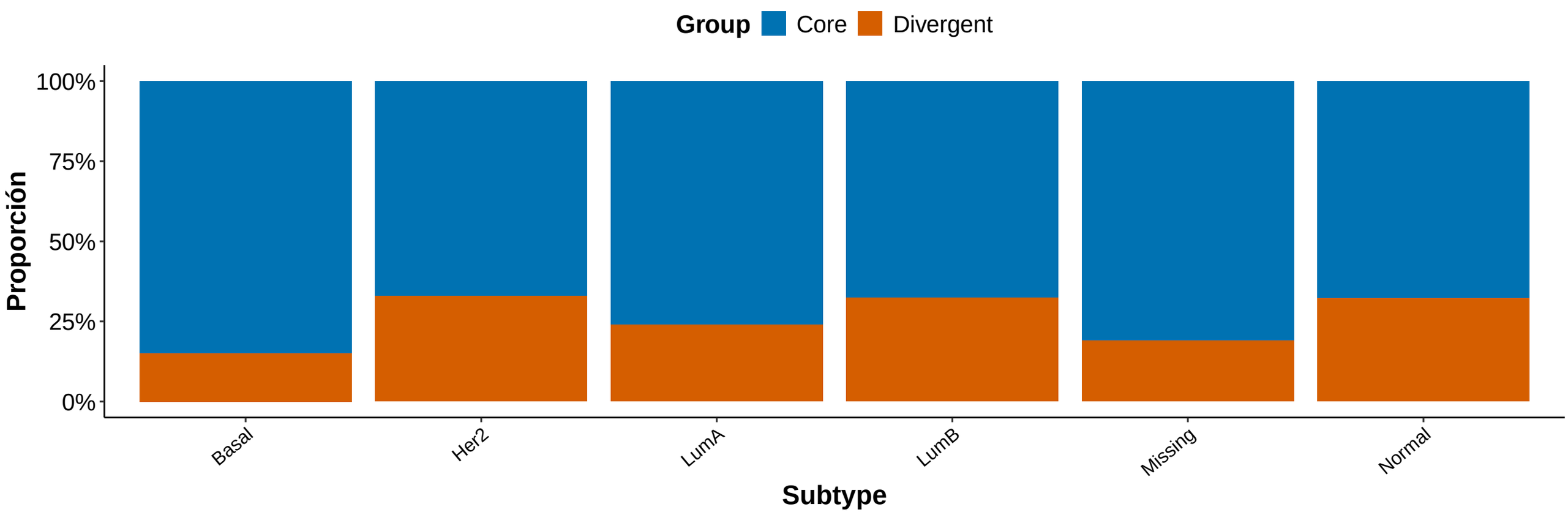
HER2

Chi² p: 0.00371



Subtype

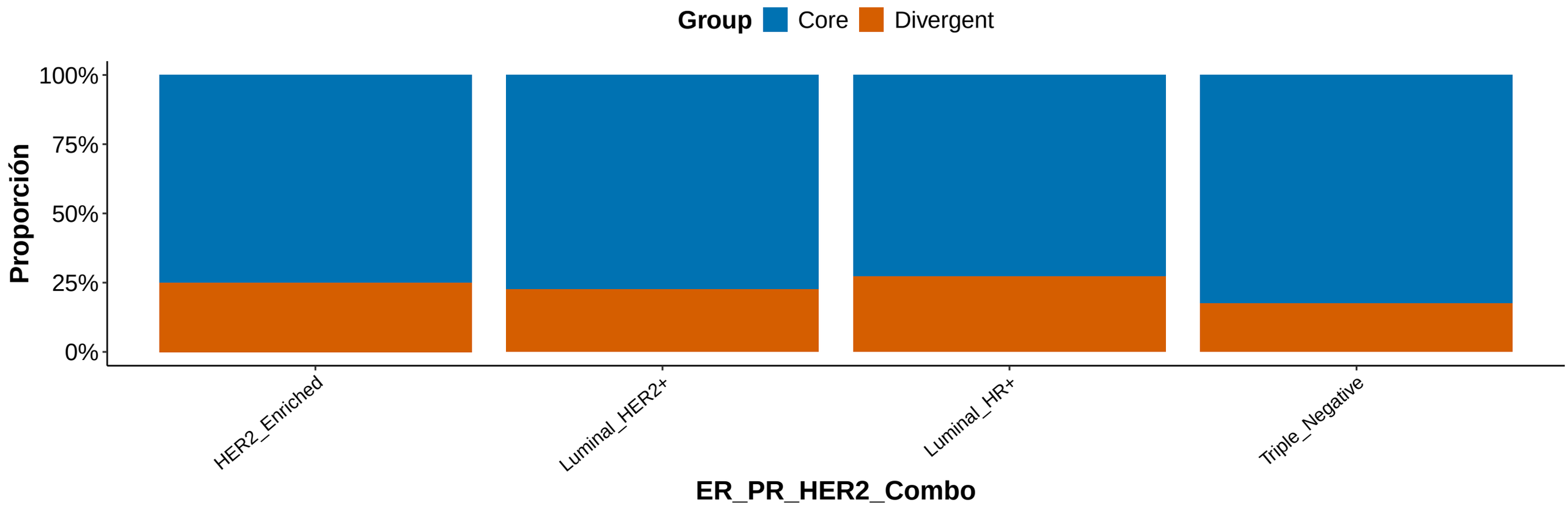
Chi² p: 0.00126



Clínica Categórica − . A. Biología Tumoral

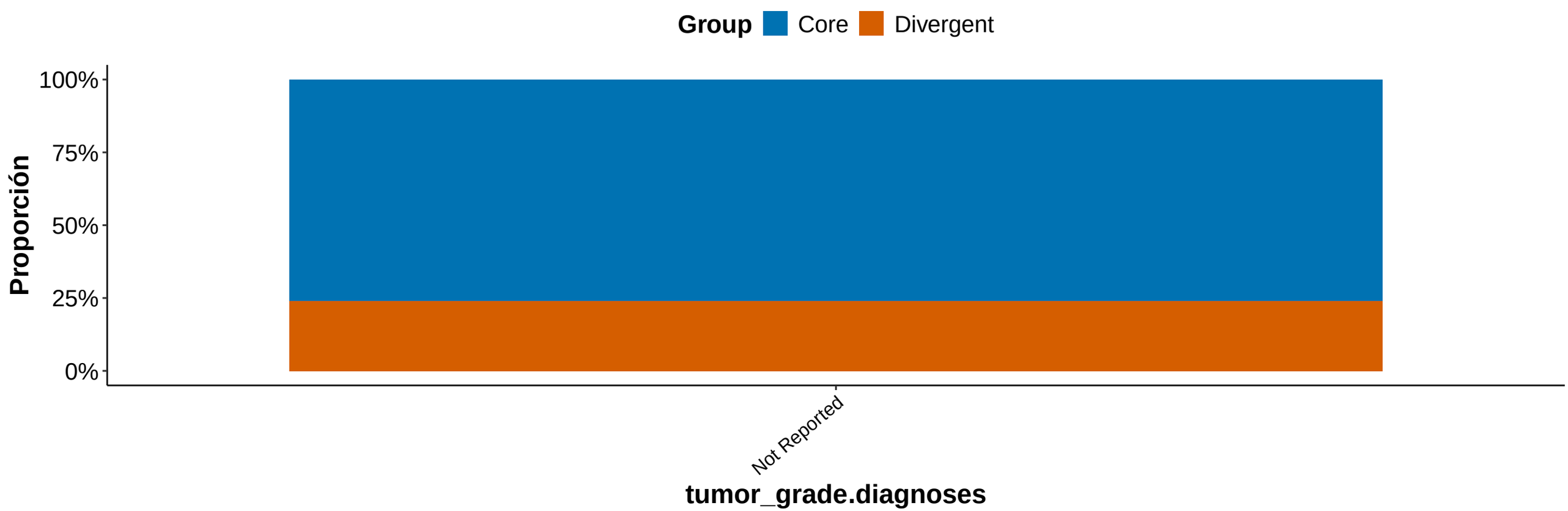
ER_PR_HER2_Combo

Chi² p: 0.0415



tumor_grade.diagnoses

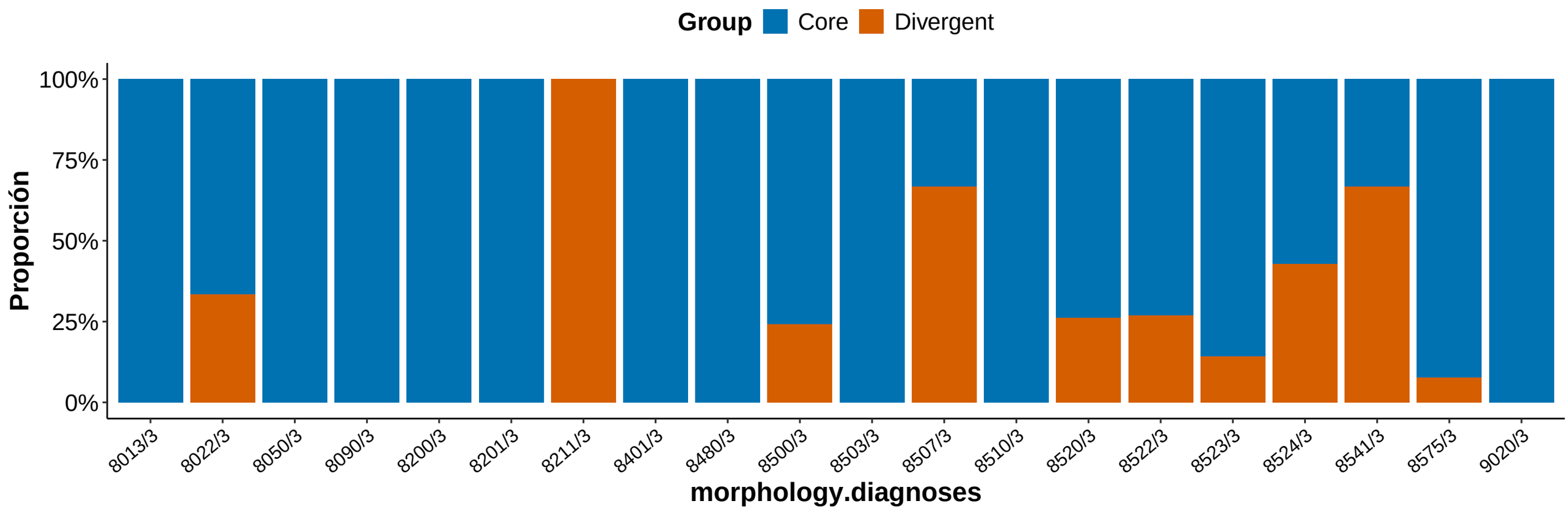
Chi² p: 1.59e−64



Clínica Categórica − . A. Biología Tumoral

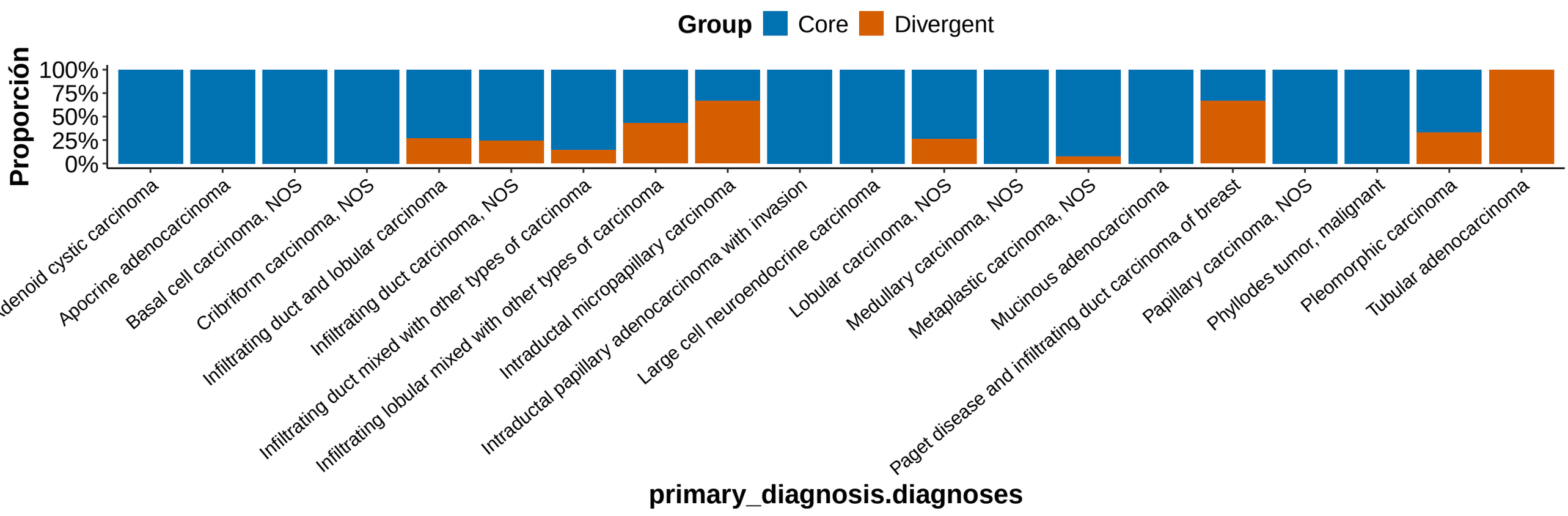
morphology.diagnoses

Chi² p: 0.172

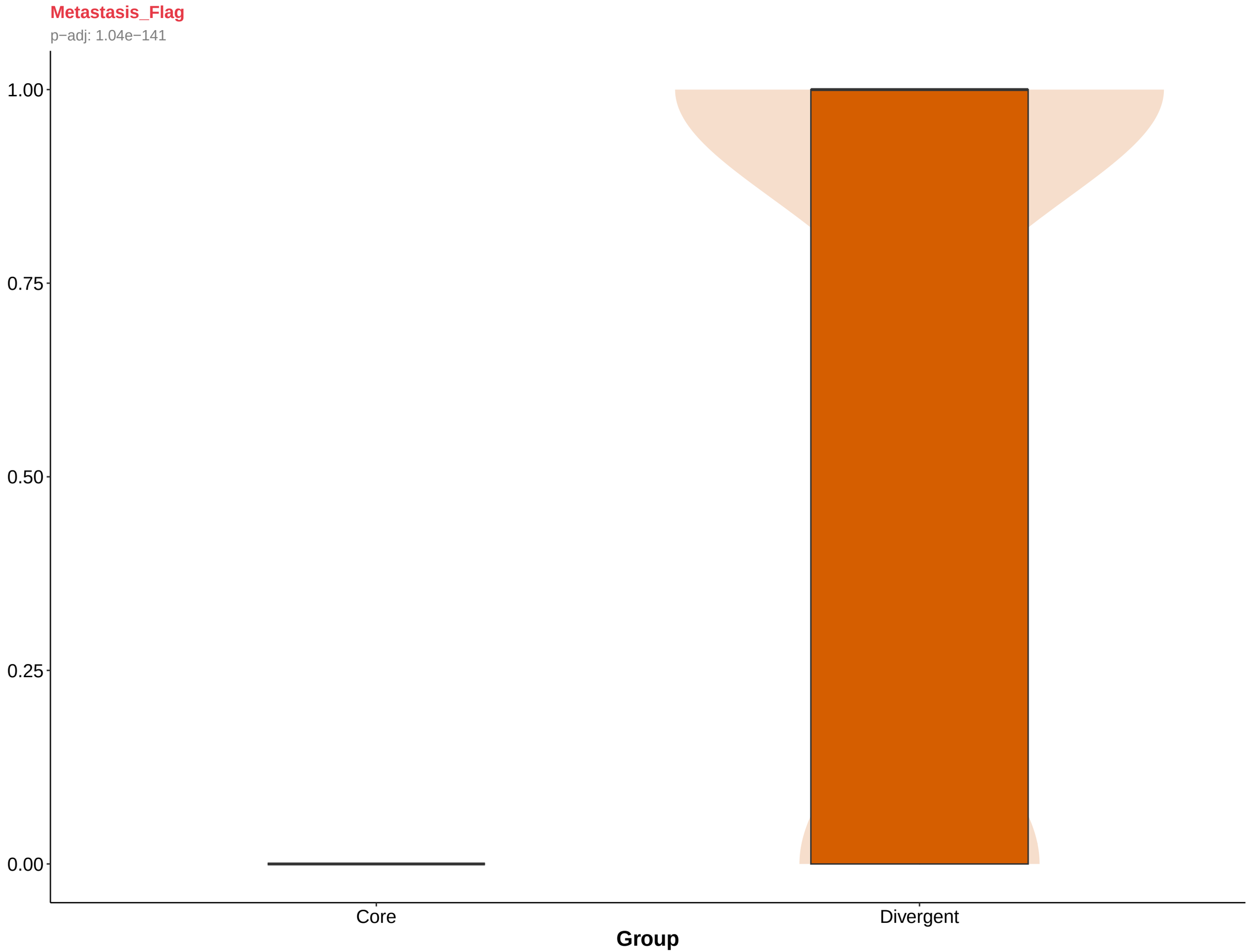


primary_diagnosis.diagnoses

Chi² p: 0.172



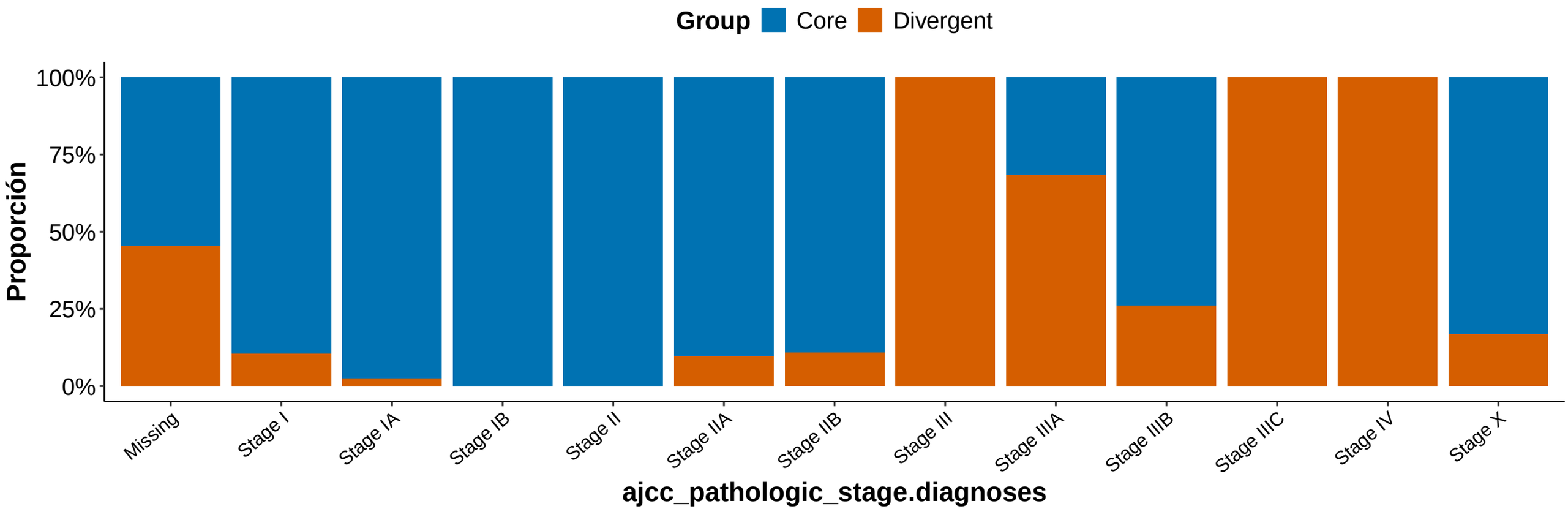
Clínica Numérica − . B. Estadio y Carga Tumoral



Clínica Categórica − . B. Estadio y Carga Tumoral

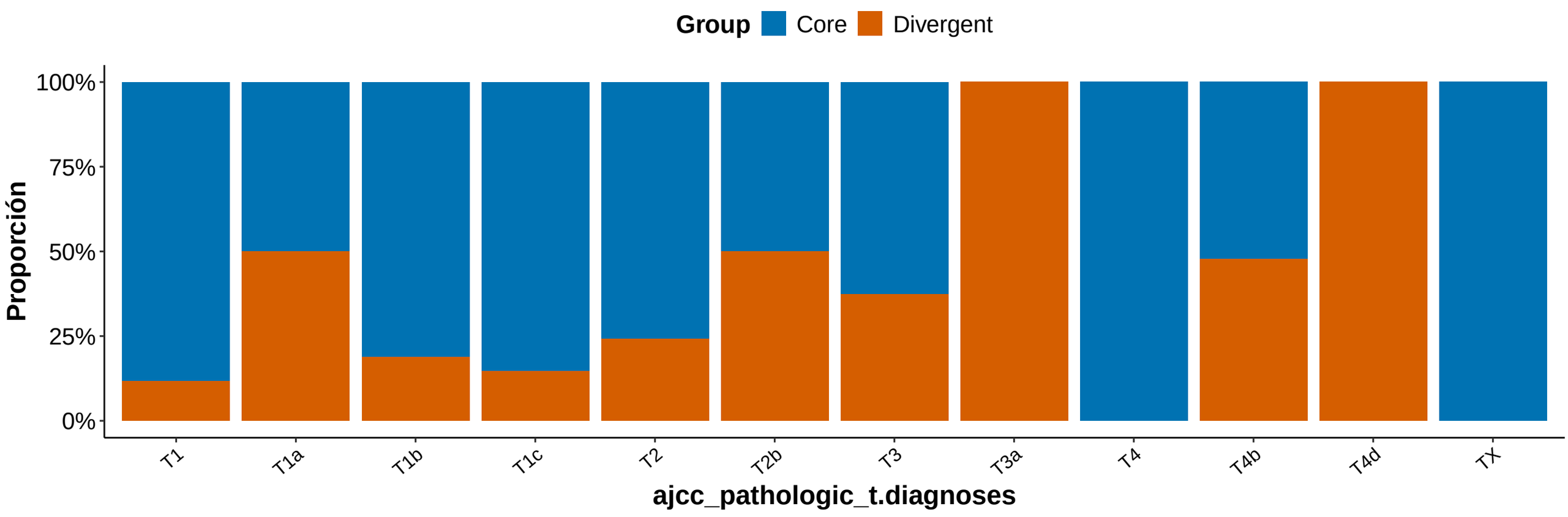
ajcc_pathologic_stage.diagnoses

Chi² p: 3.63e−94



ajcc_pathologic_t.diagnoses

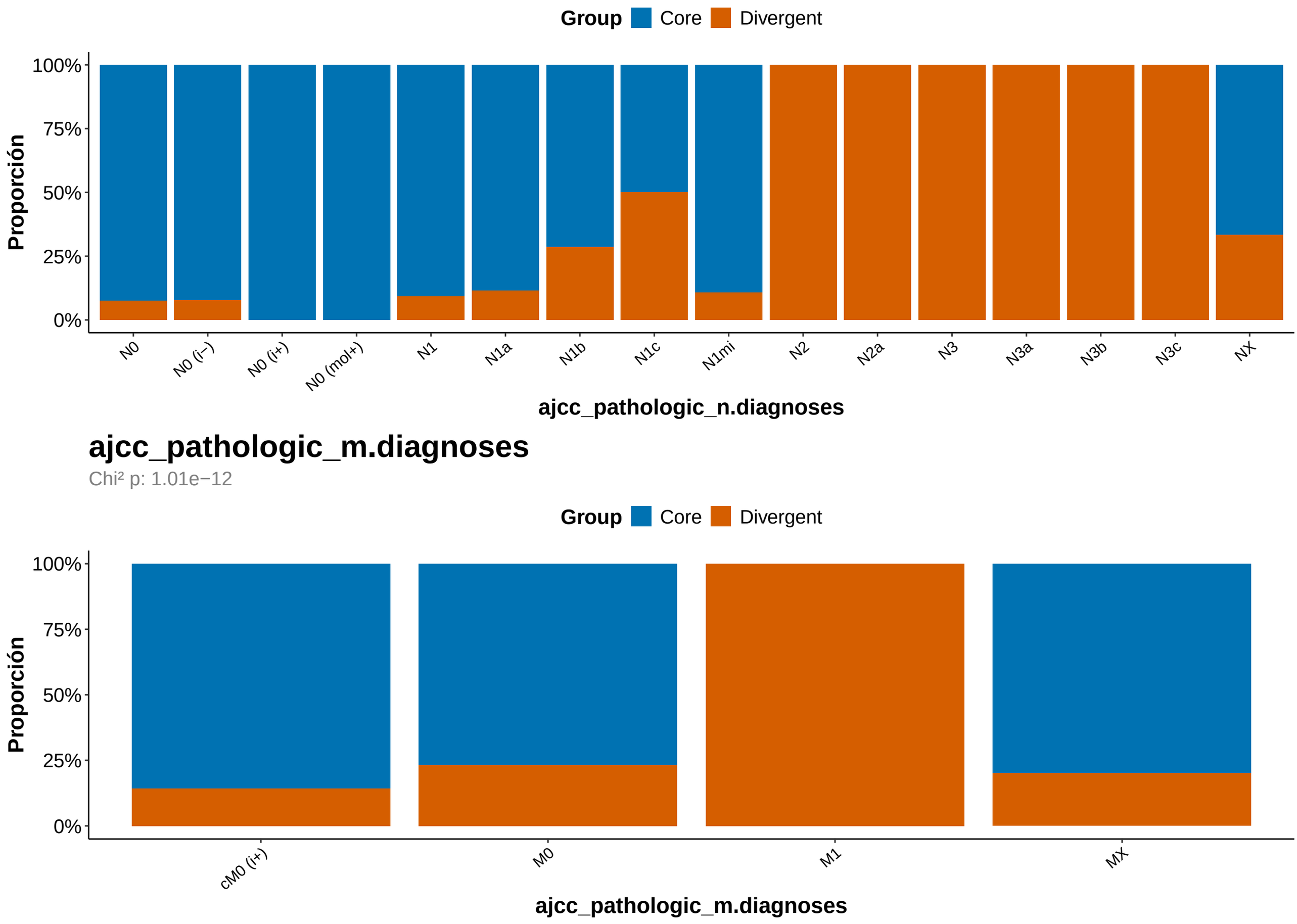
Chi² p: 4.13e−07



Clínica Categórica − . B. Estadio y Carga Tumoral

ajcc_pathologic_n.diagnoses

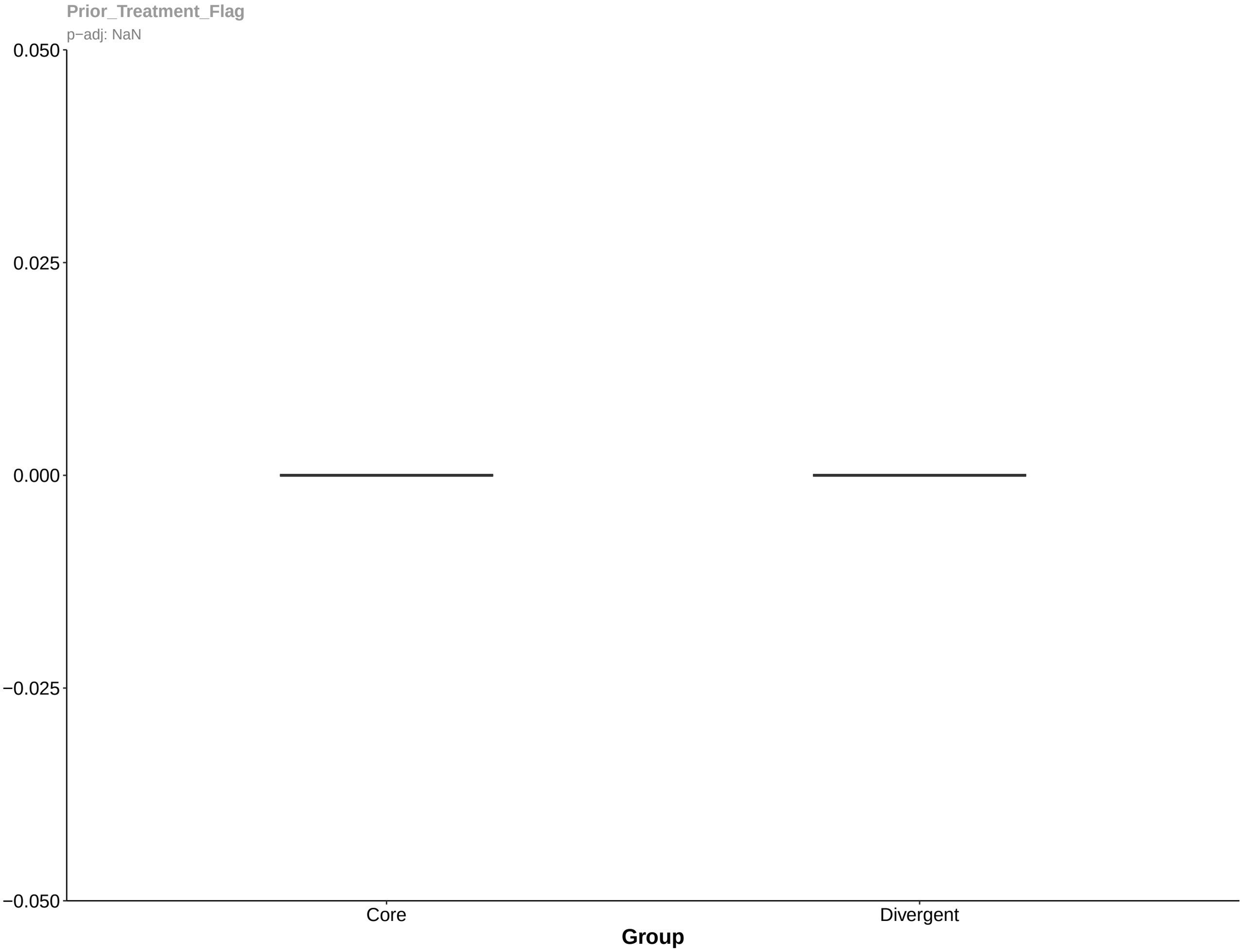
Chi² p: 1.56e−125



ajcc_pathologic_m.diagnoses

Chi² p: 1.01e−12

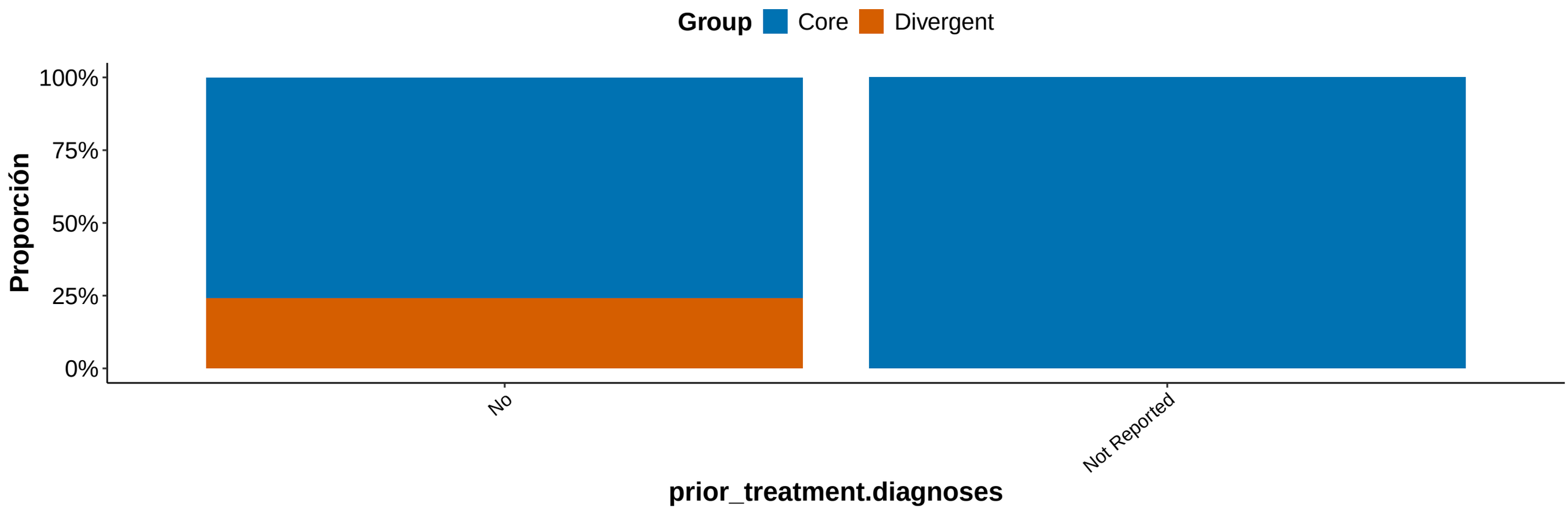
Clínica Numérica − . C. Historia Terapéutica



Clínica Categórica − . C. Historia Terapéutica

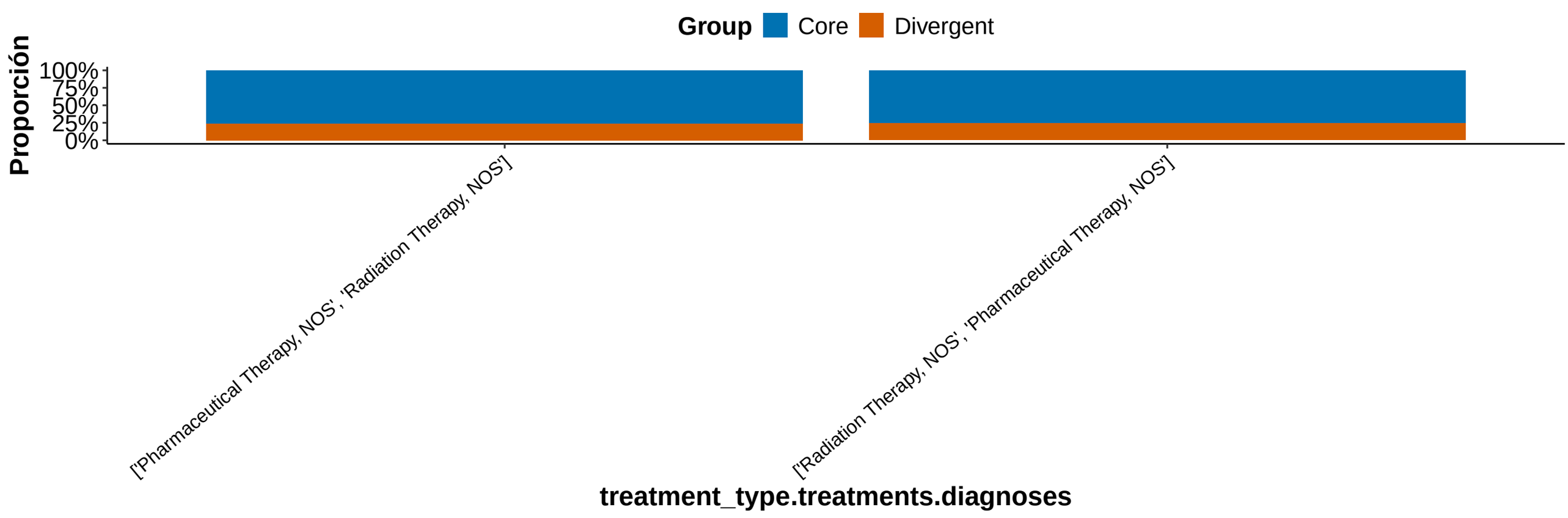
prior_treatment.diagnoses

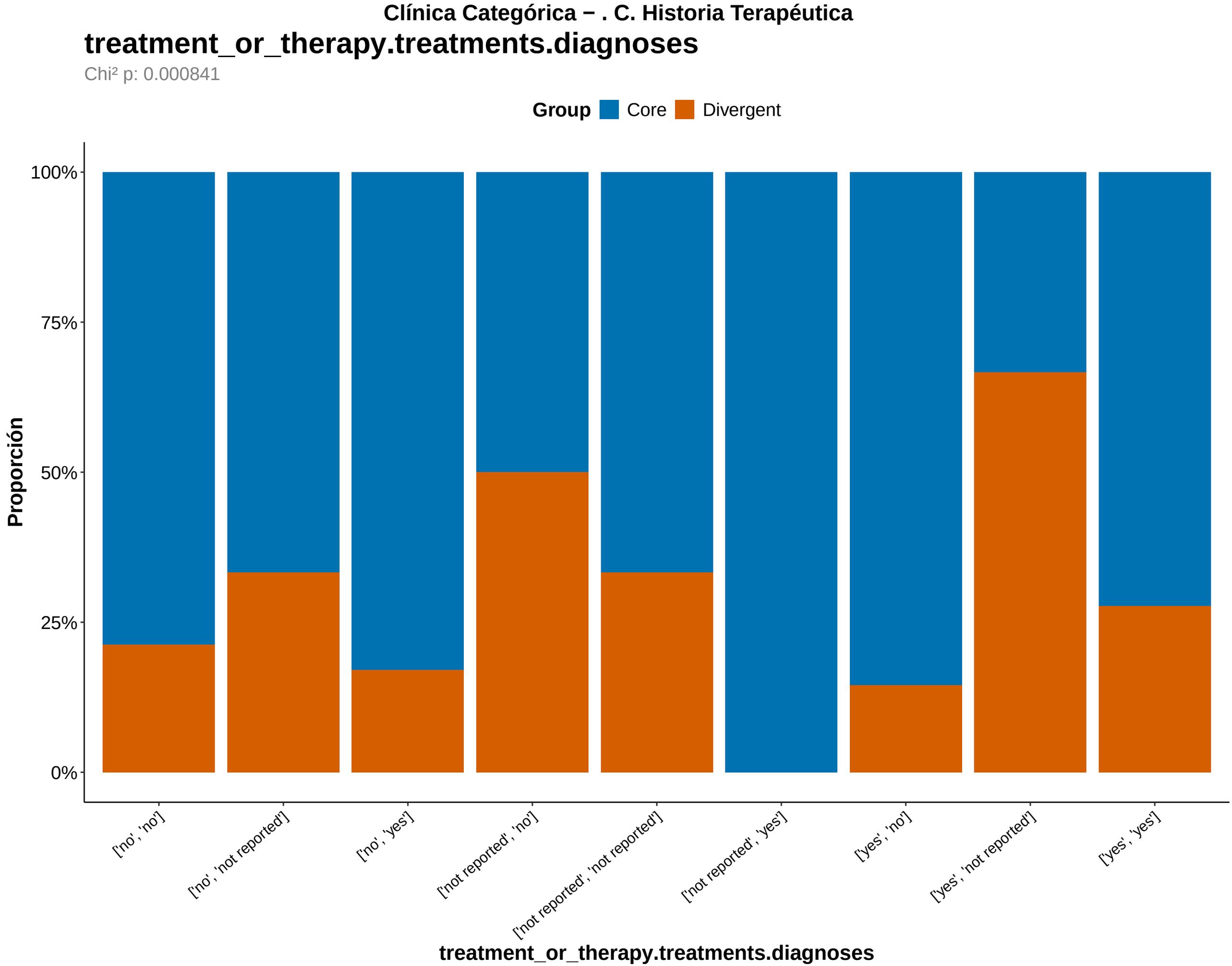
Chi² p: 1



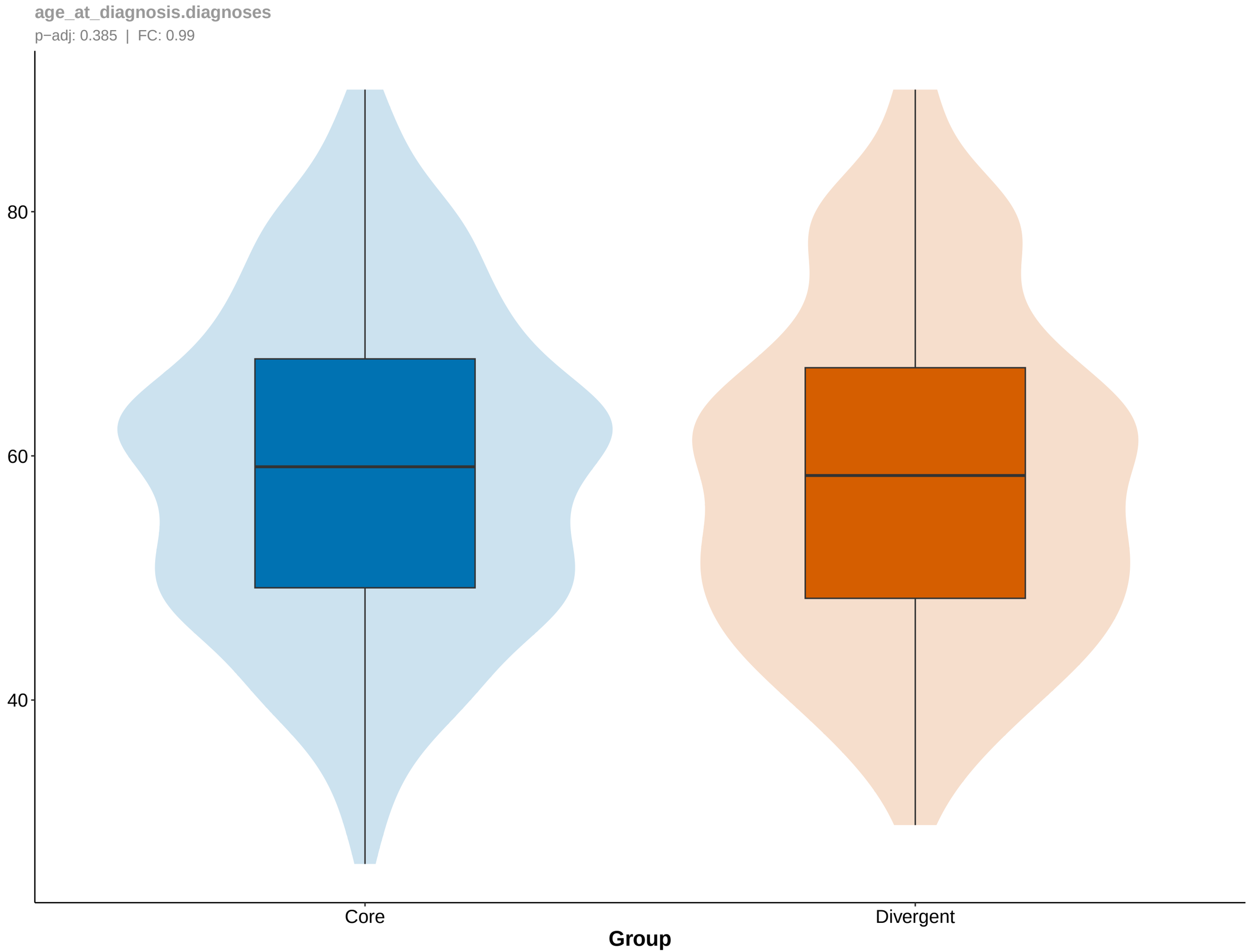
treatment_type.treatments.diagnoses

Chi² p: 0.95




Clínica Categórica − . C. Historia Terapéutica
treatment_or_therapy.treatments.diagnoses
Chi² p: 0.000841
Group
Core
Divergent
100%
75%
50%
25%
0%
Proporción
['no', 'no']
['no', 'not reported']
['no', 'yes']
['not reported', 'no']
['not reported', 'not reported']
['not reported', 'yes']
['yes', 'no']
['yes', 'not reported']
['yes', 'yes']
treatment_or_therapy.treatments.diagnoses

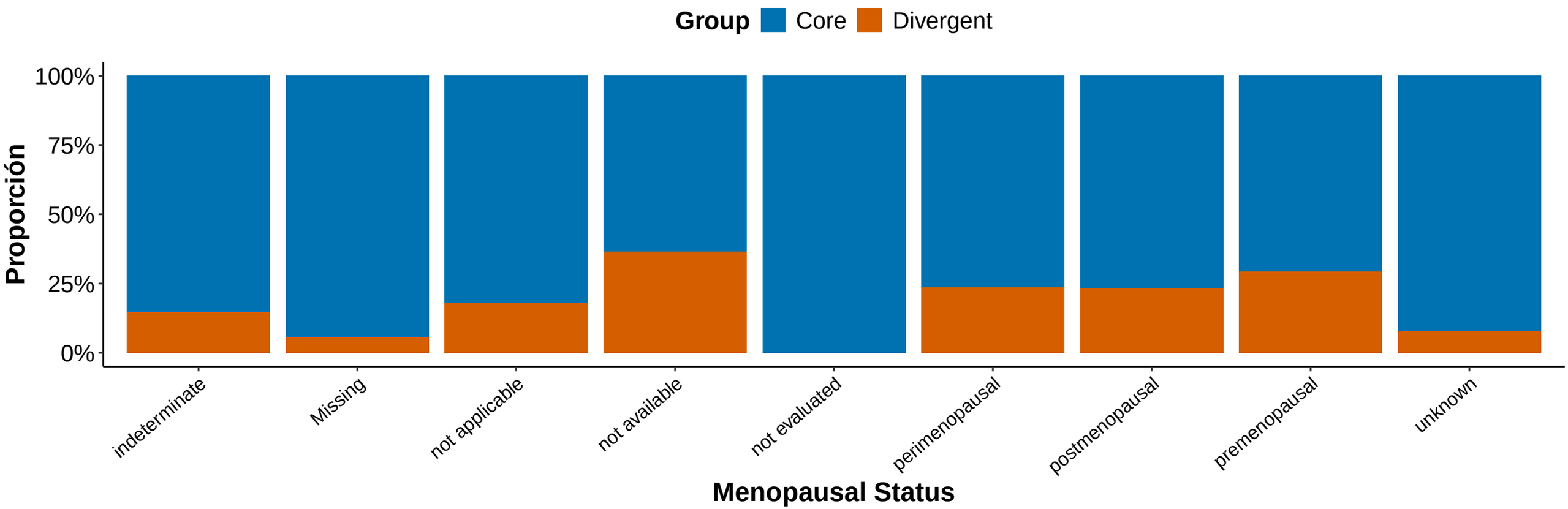
Clínica Numérica − . D. Factores del Paciente



Clínica Categórica − . D. Factores del Paciente

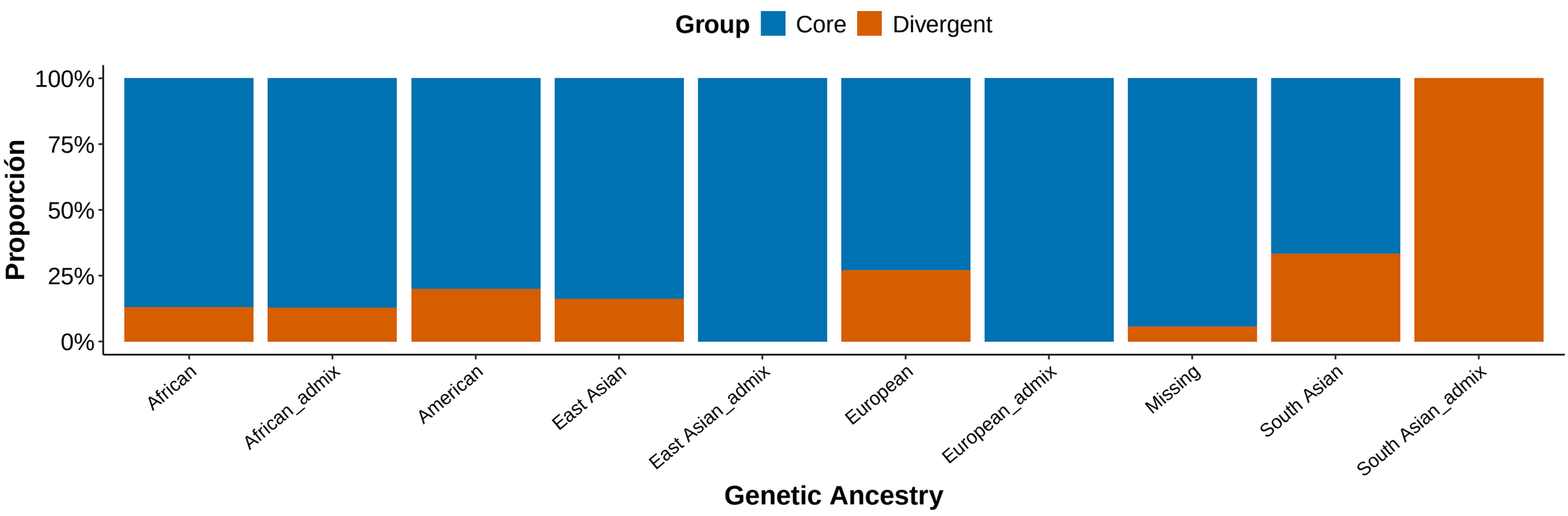
Menopausal Status

Chi² p: 0.00877



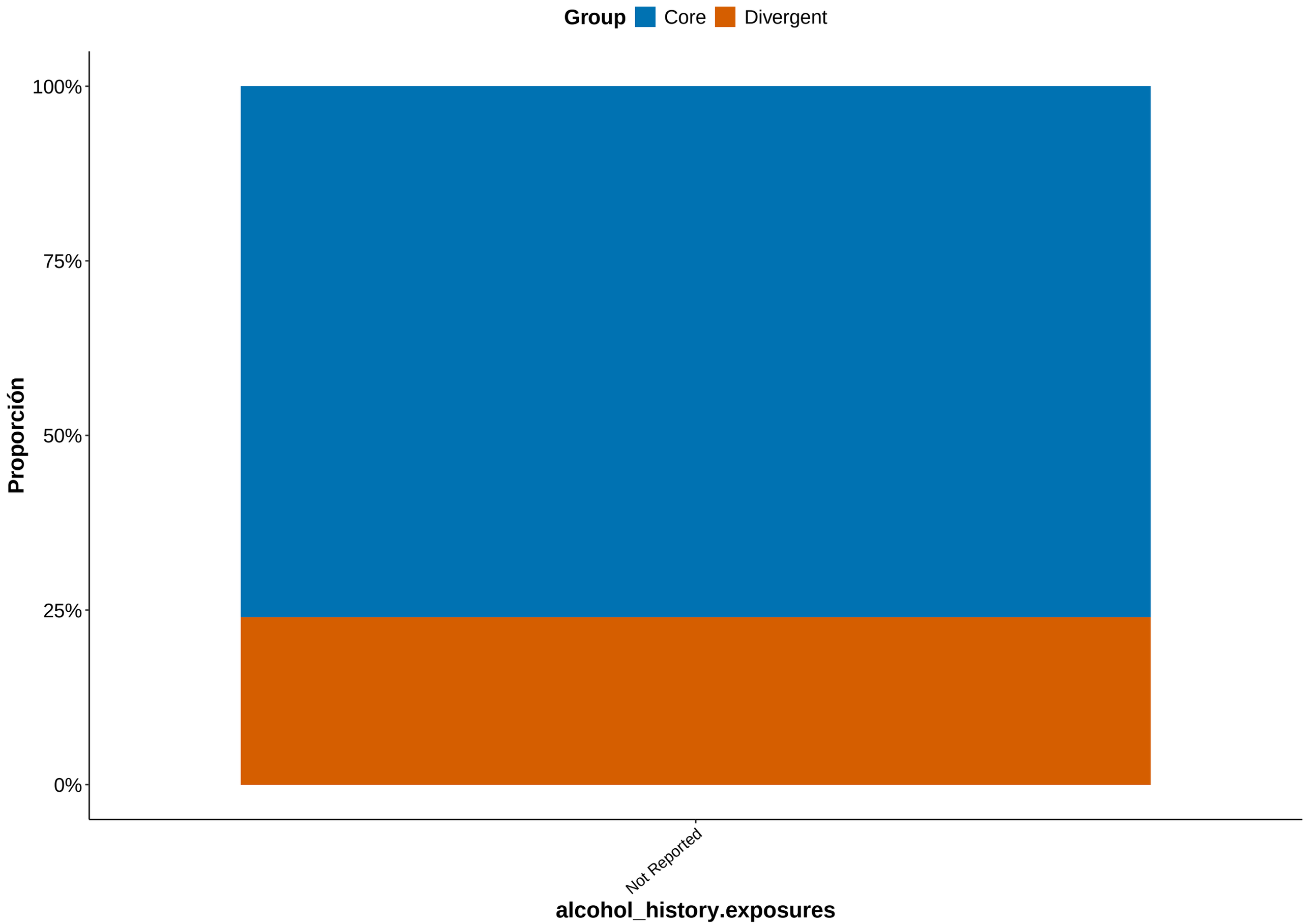
Genetic Ancestry

Chi² p: 0.00135



alcohol_history.exposures

Chi² p: 1.59e−64

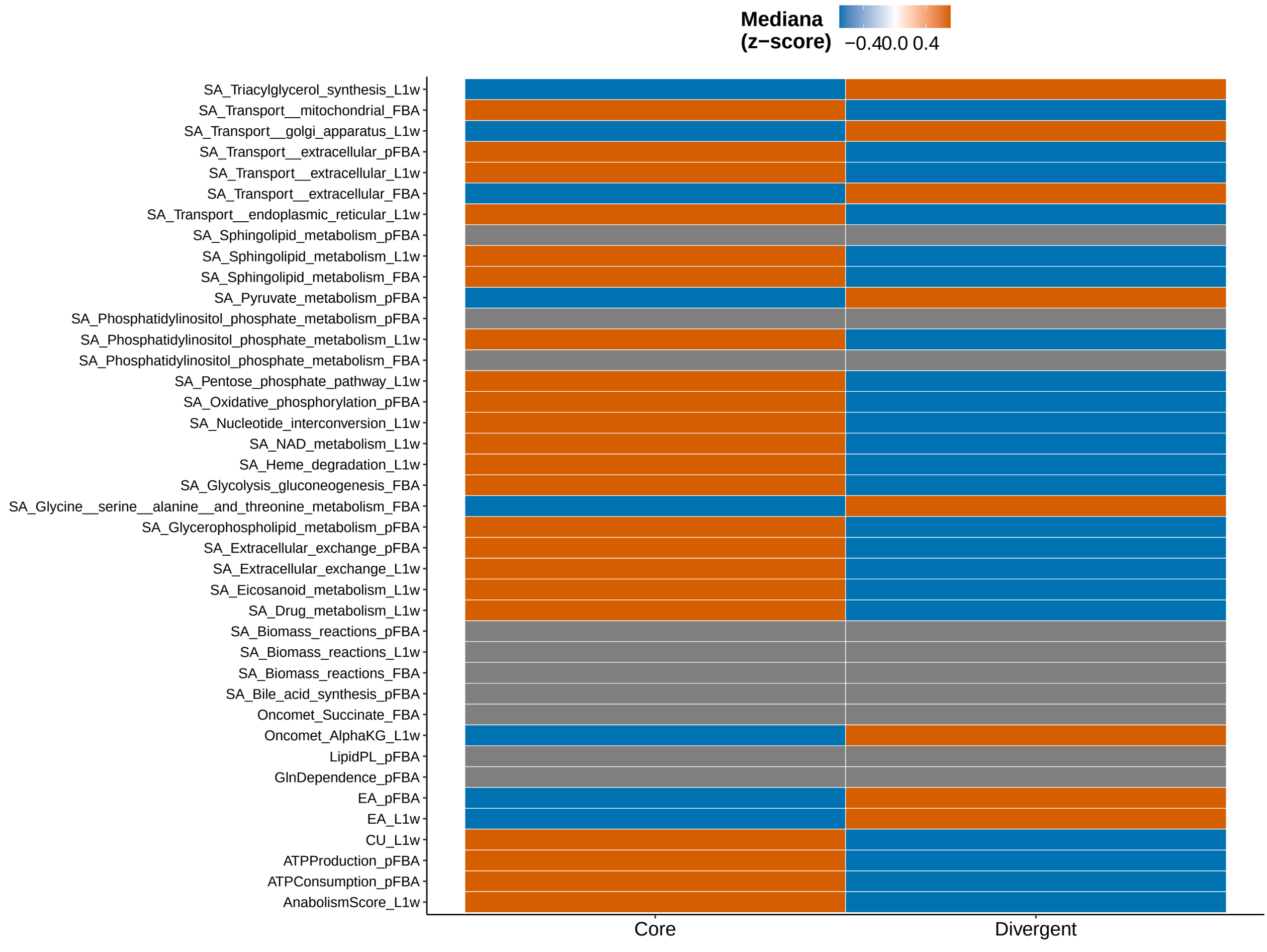


SECCIÓN II

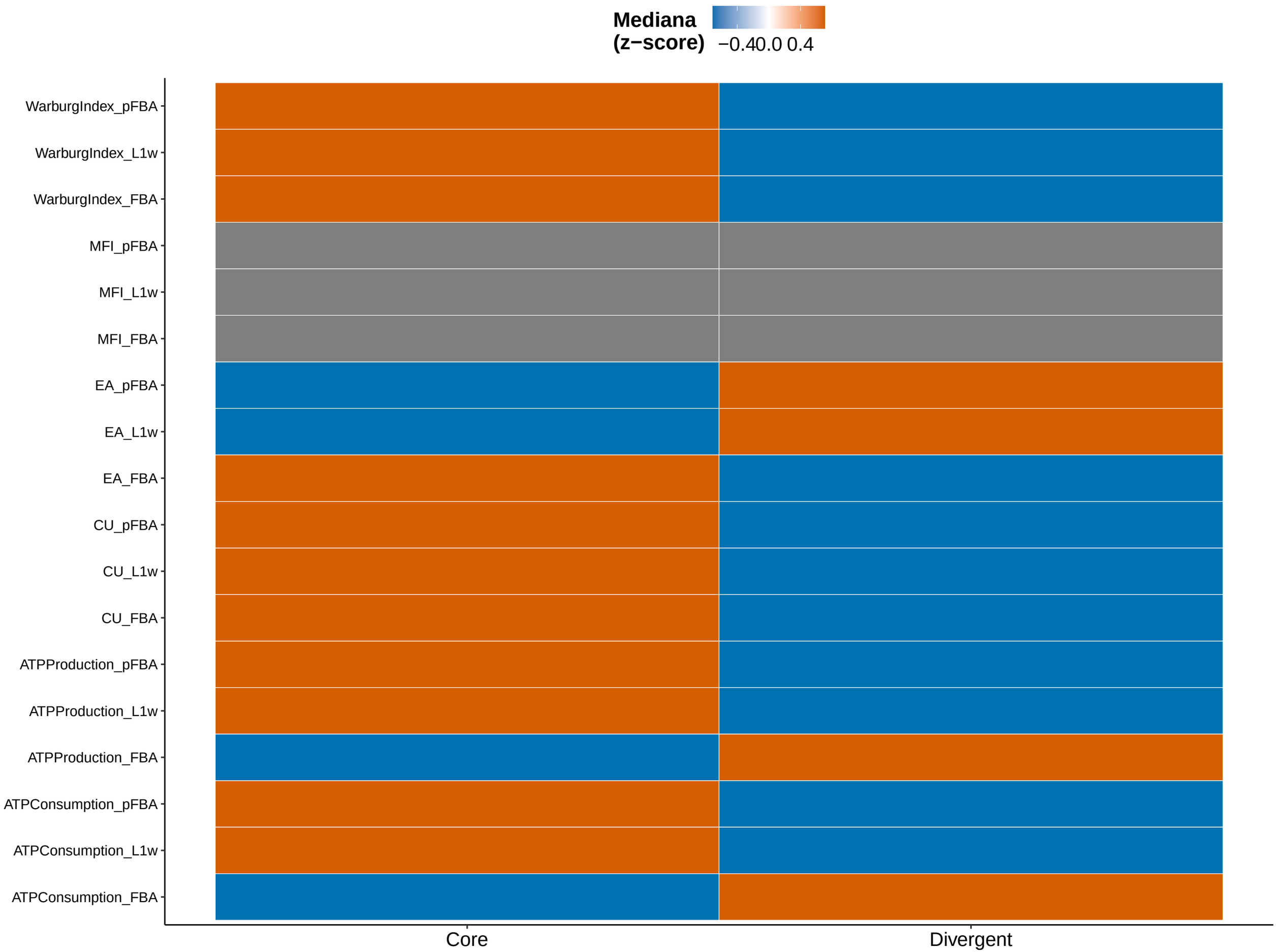
Diferencias Metabólicas: Core vs Divergent

5 bloques funcionales: Energía | Redox | Lípidos | Dependencias | Reprogramación Sistémica

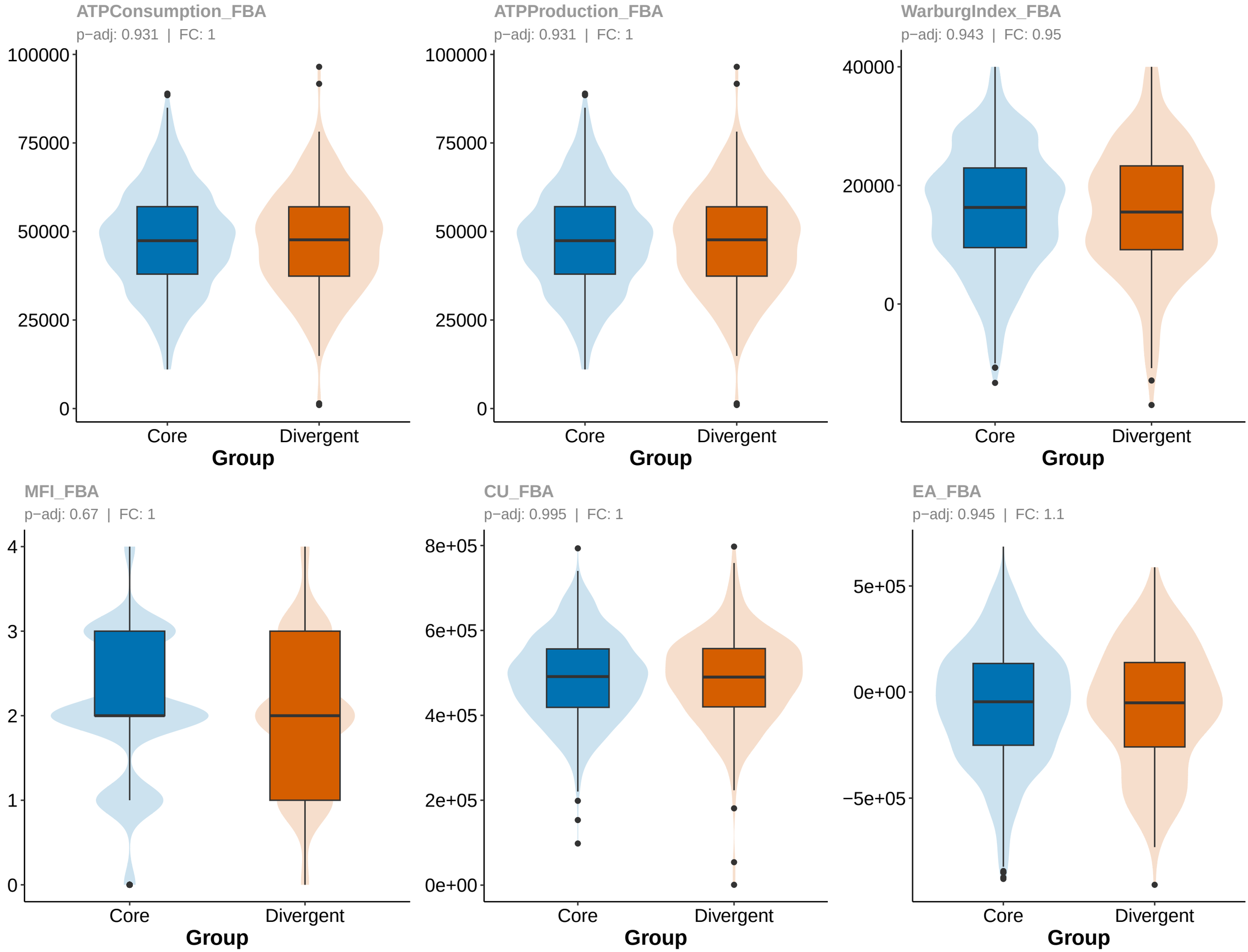
Heatmap: Top 40 variables metabólicas (por p_adj)



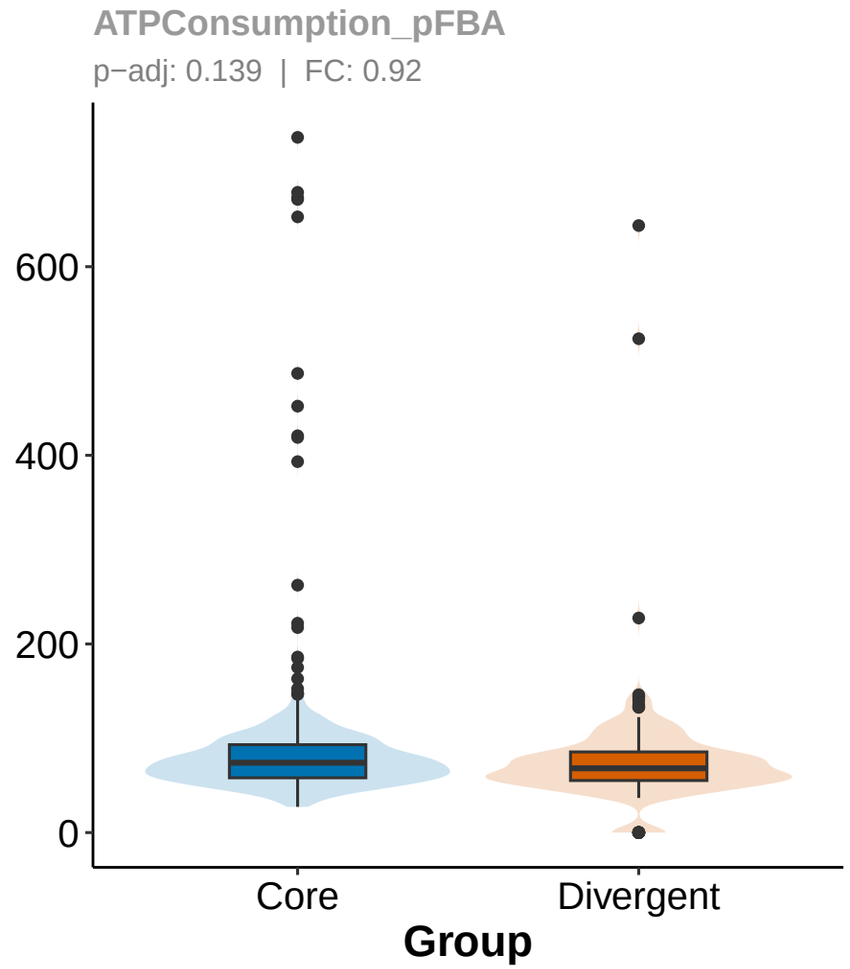
Heatmap: . 1. Energía y Bioenergética

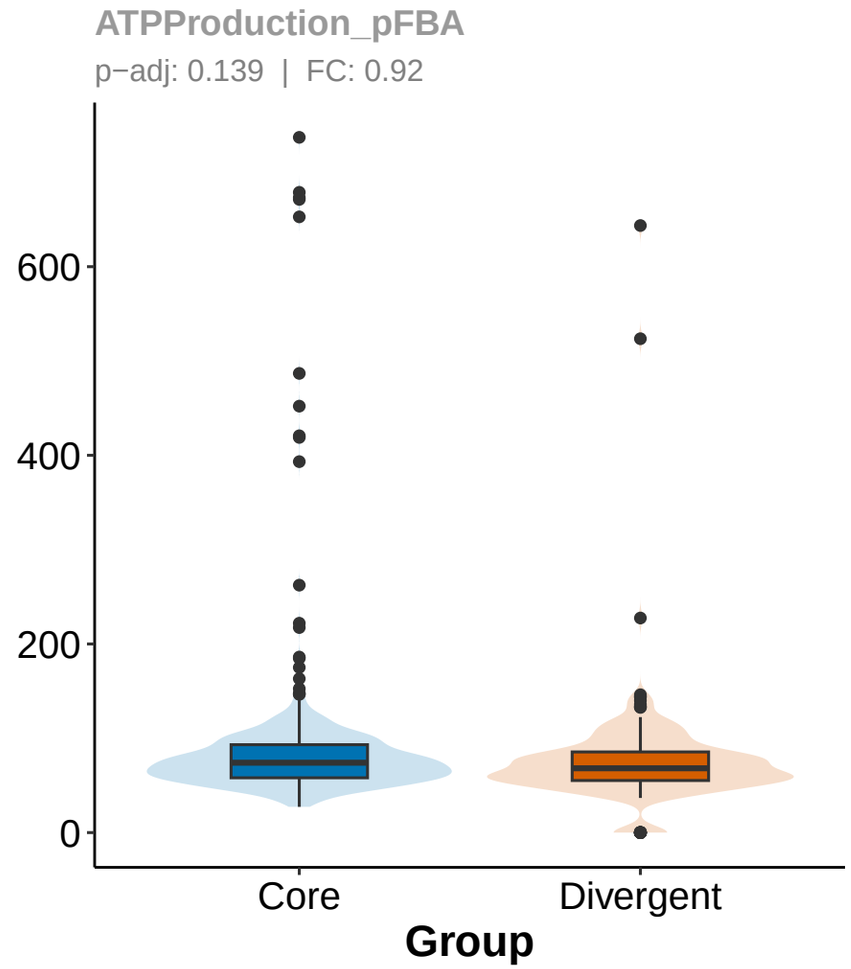


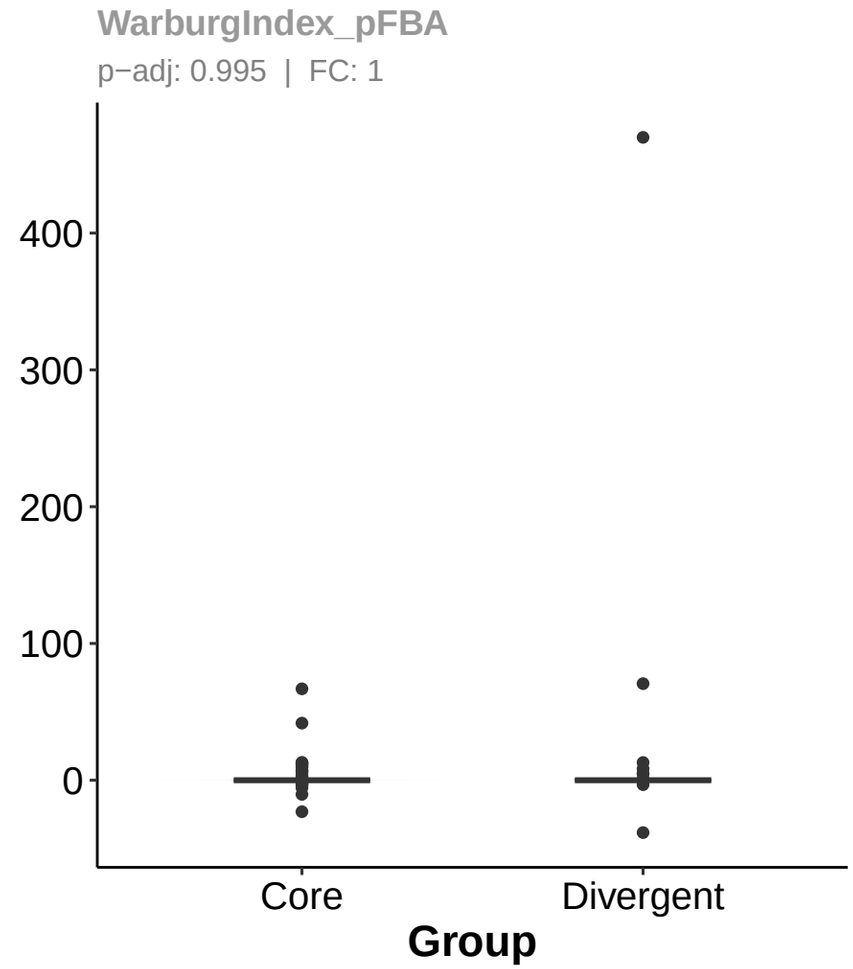
Metabolismo − . 1. Energía y Bioenergética

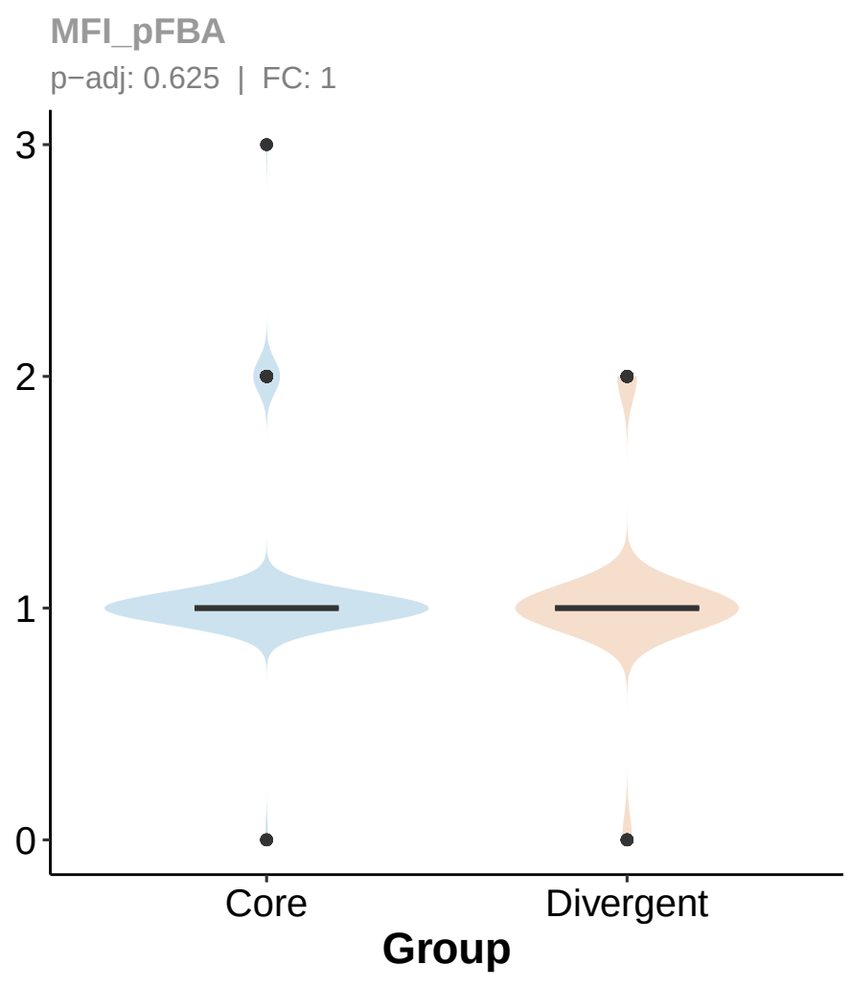


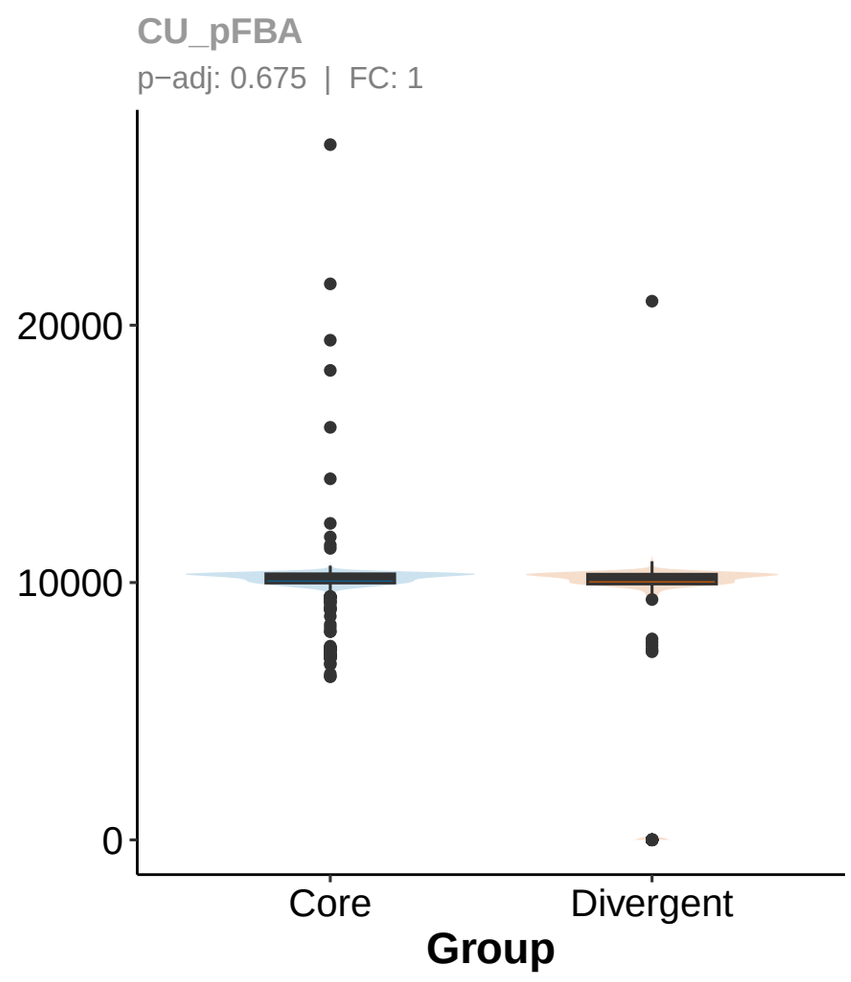
Metabolismo − . 1. Energía y Bioenergética

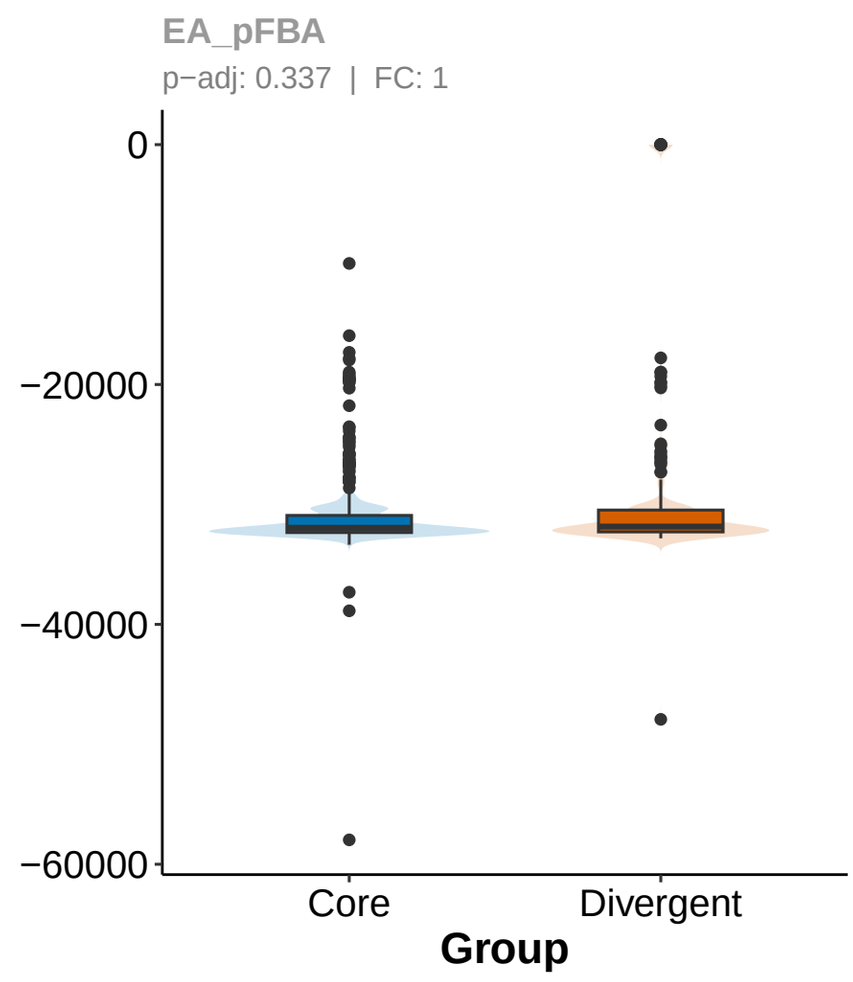




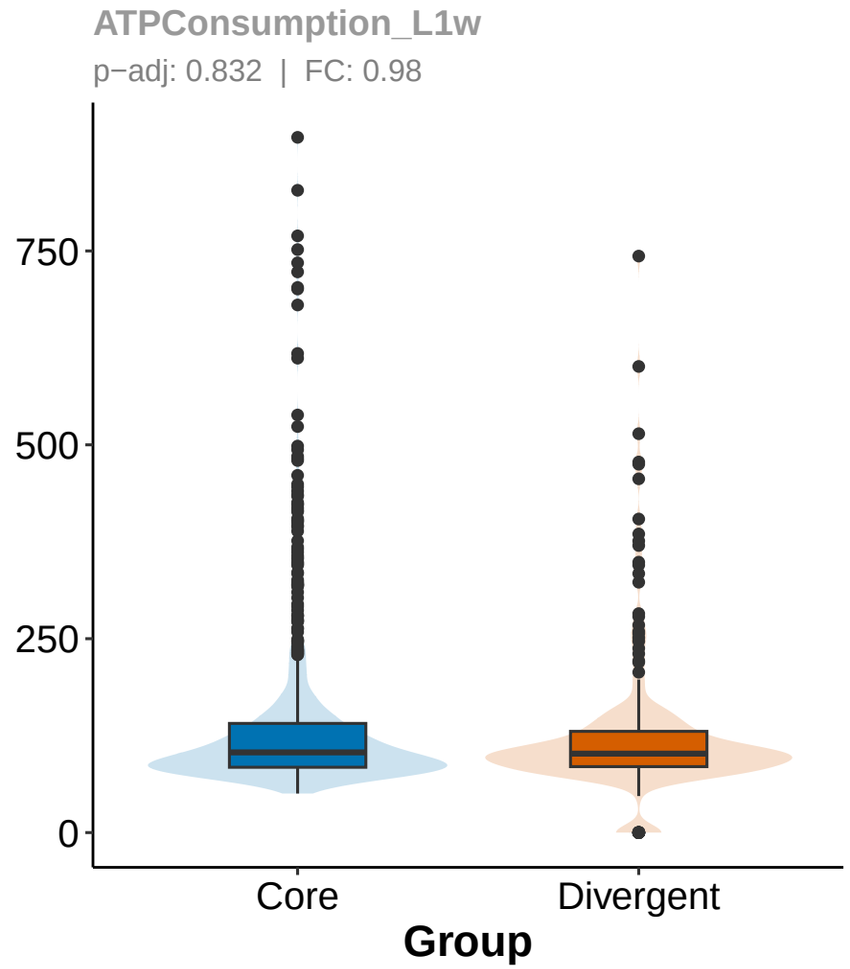


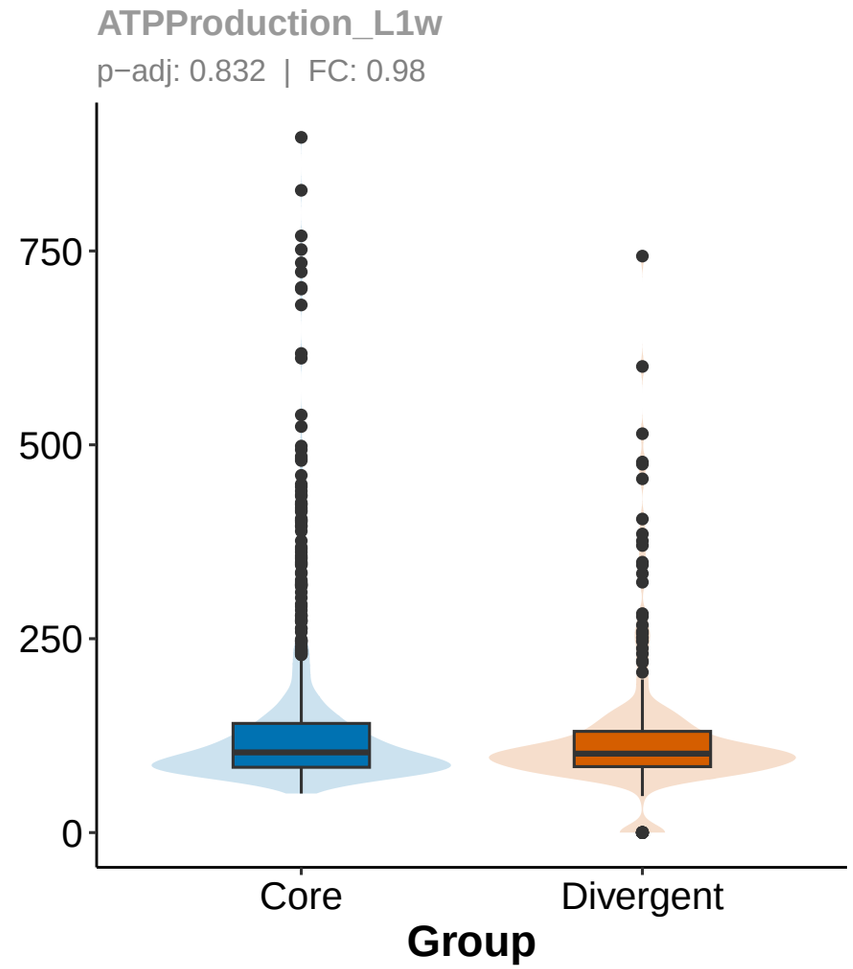


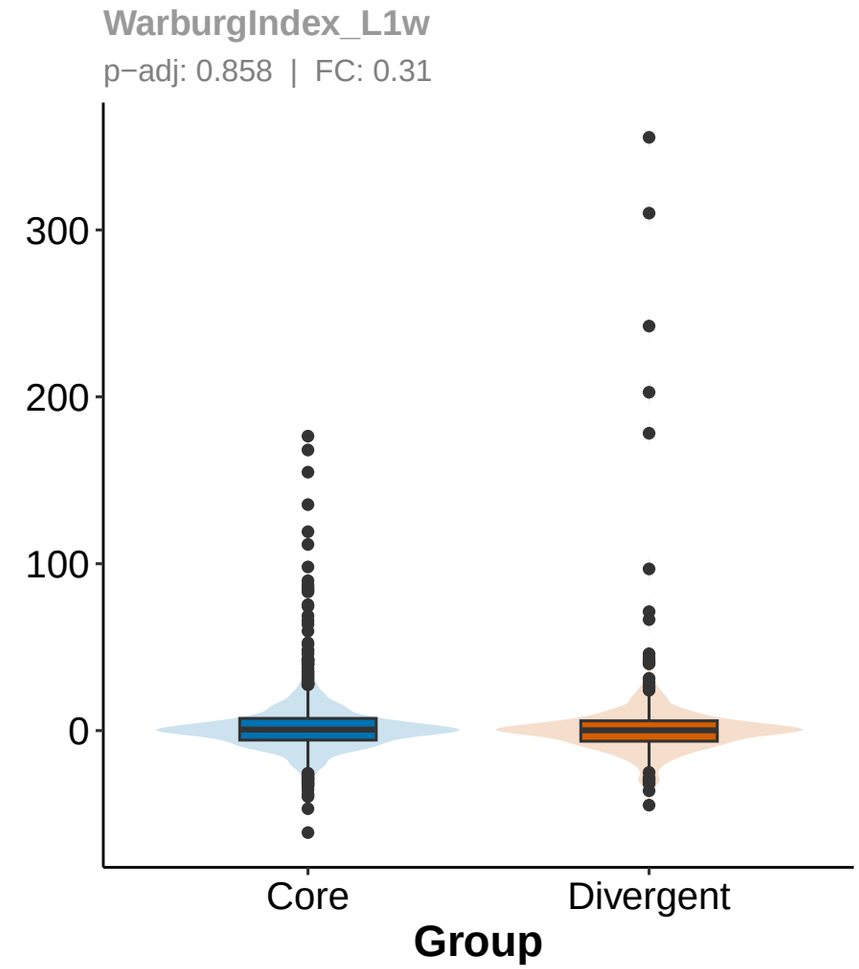


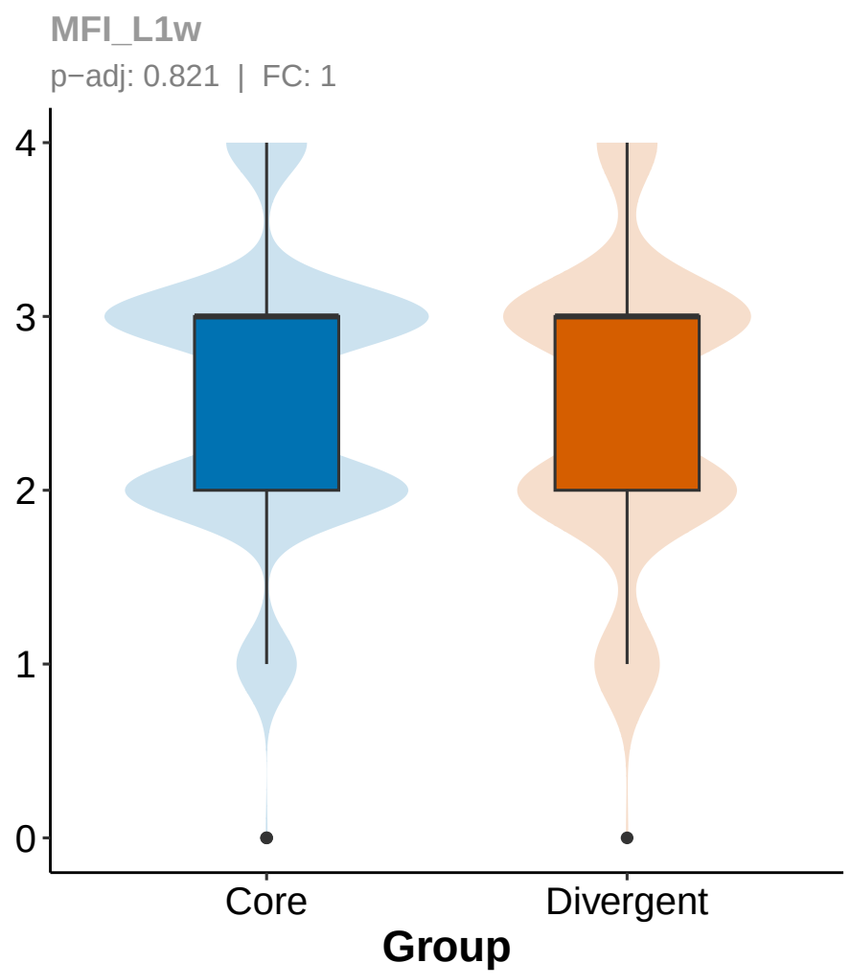


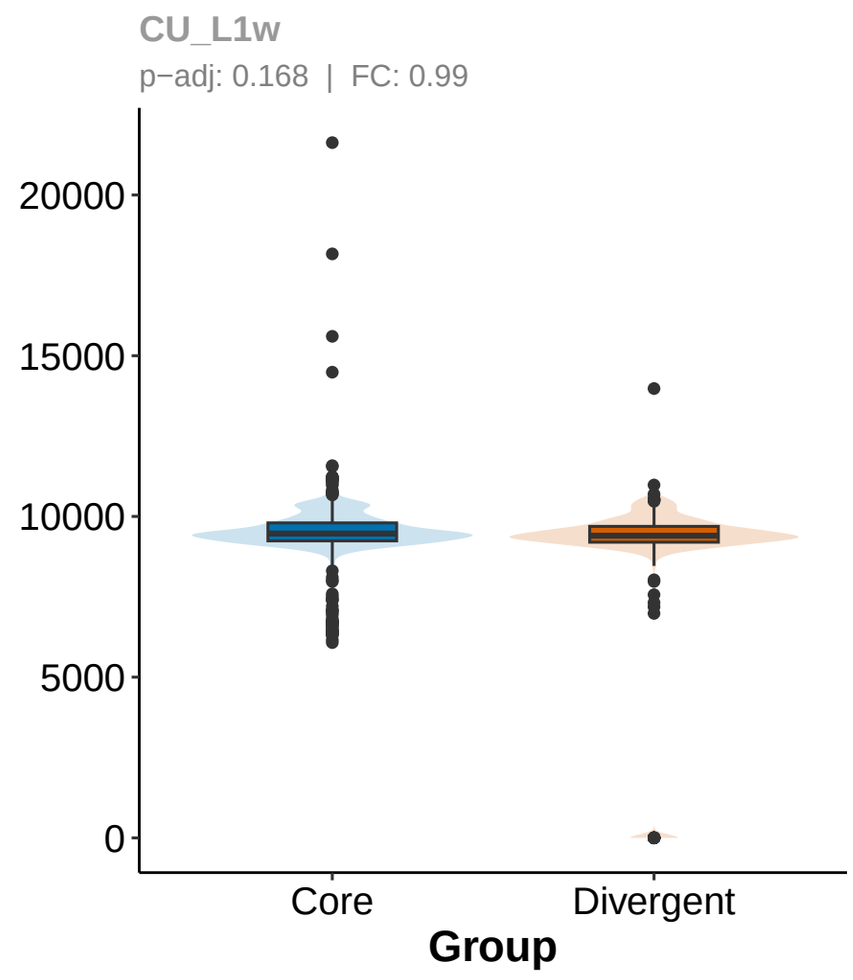
Metabolismo − . 1. Energía y Bioenergética

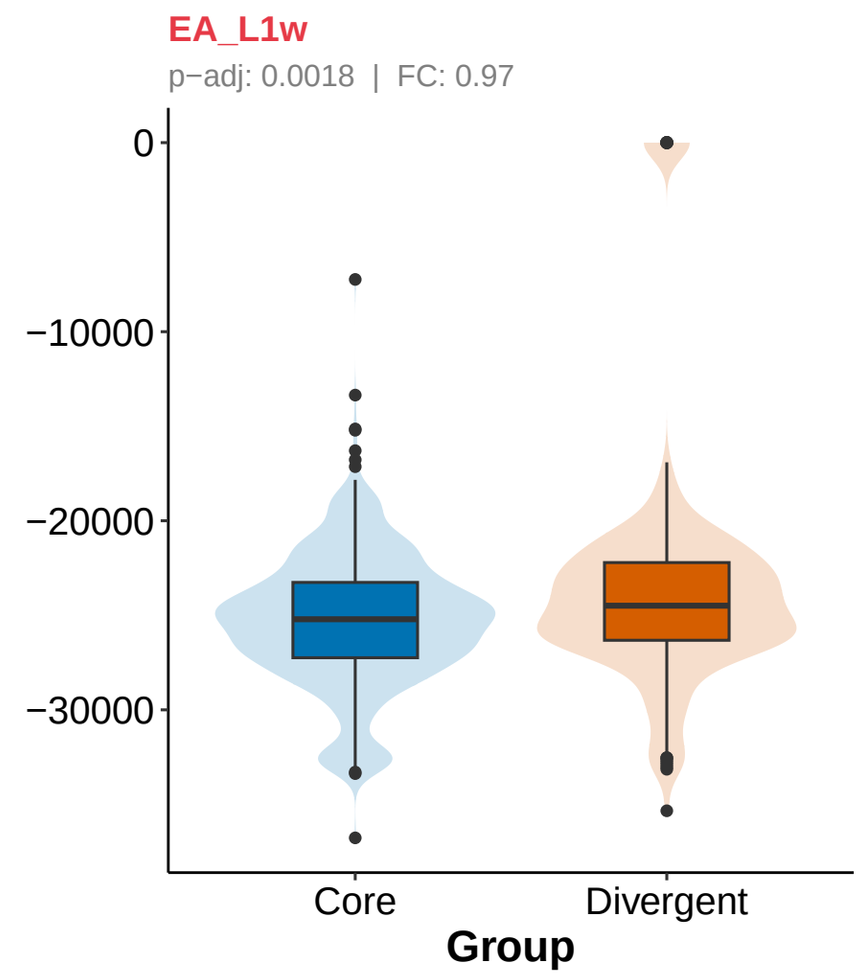




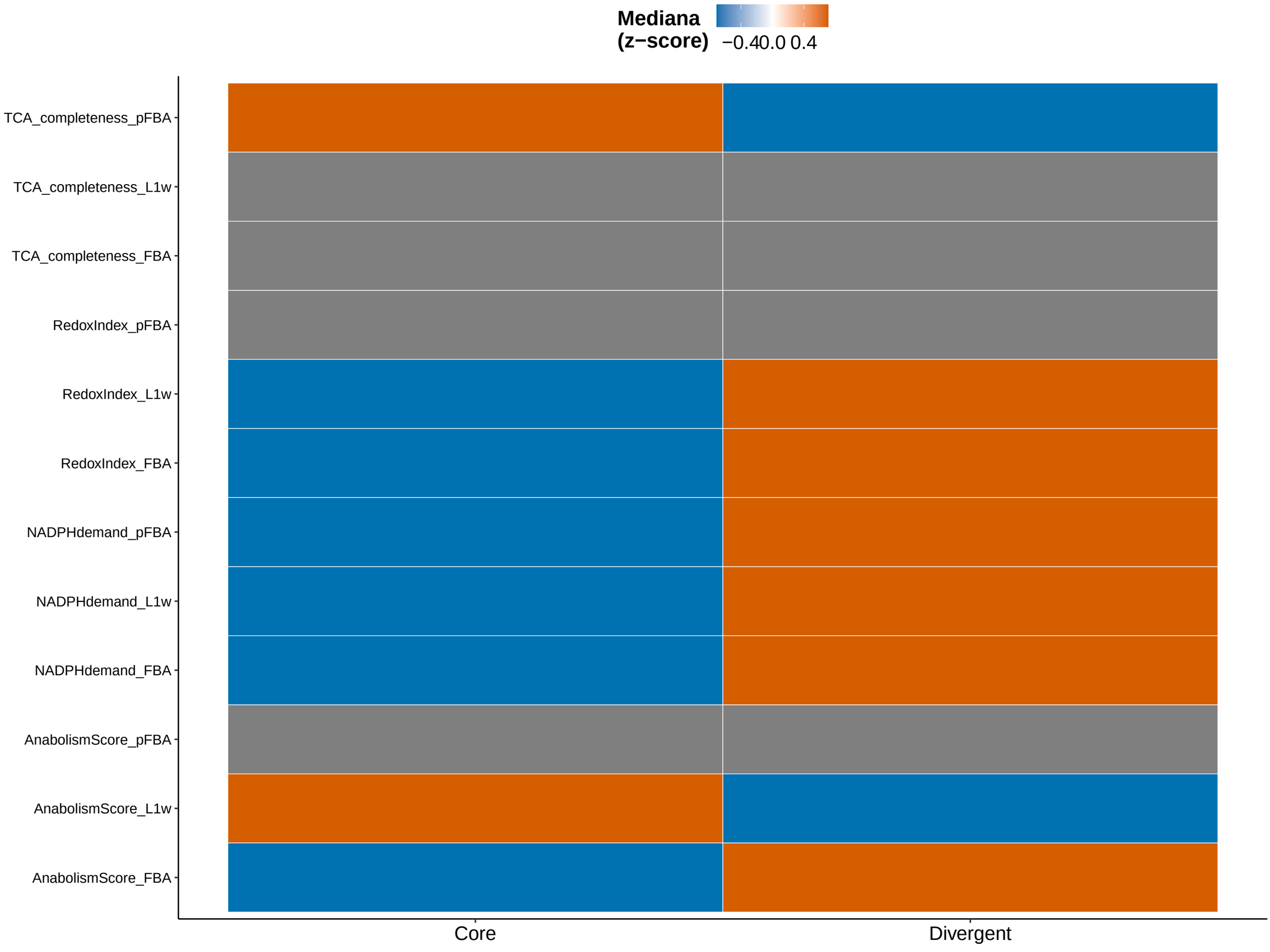




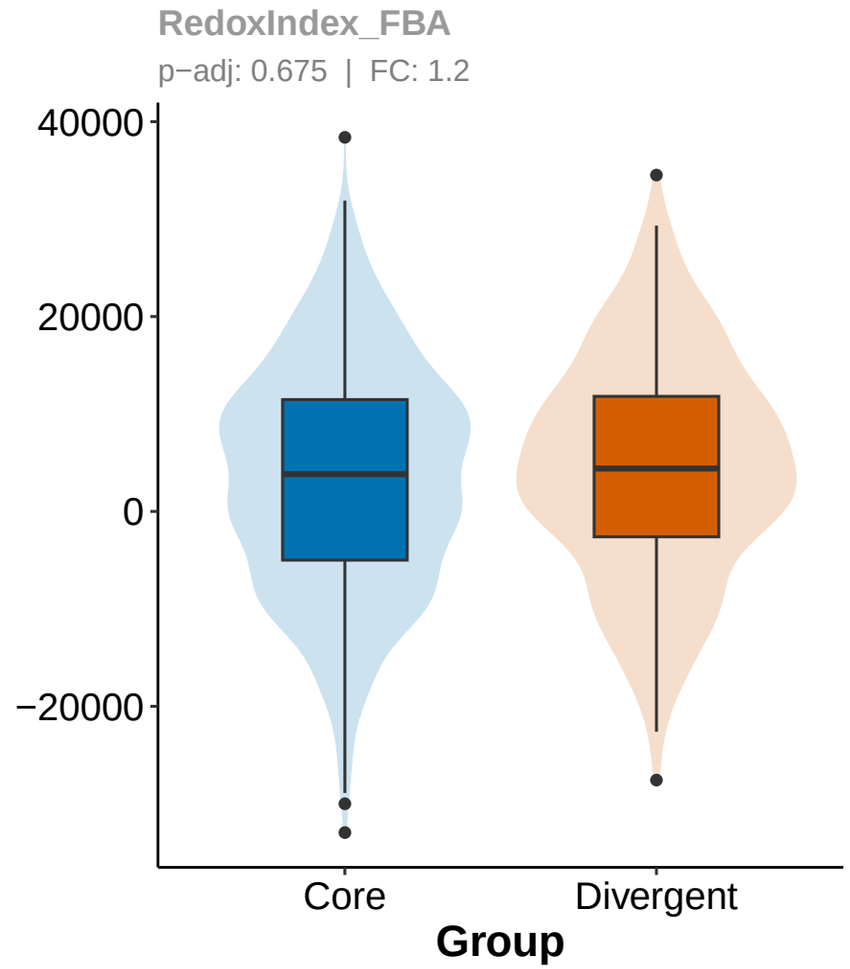


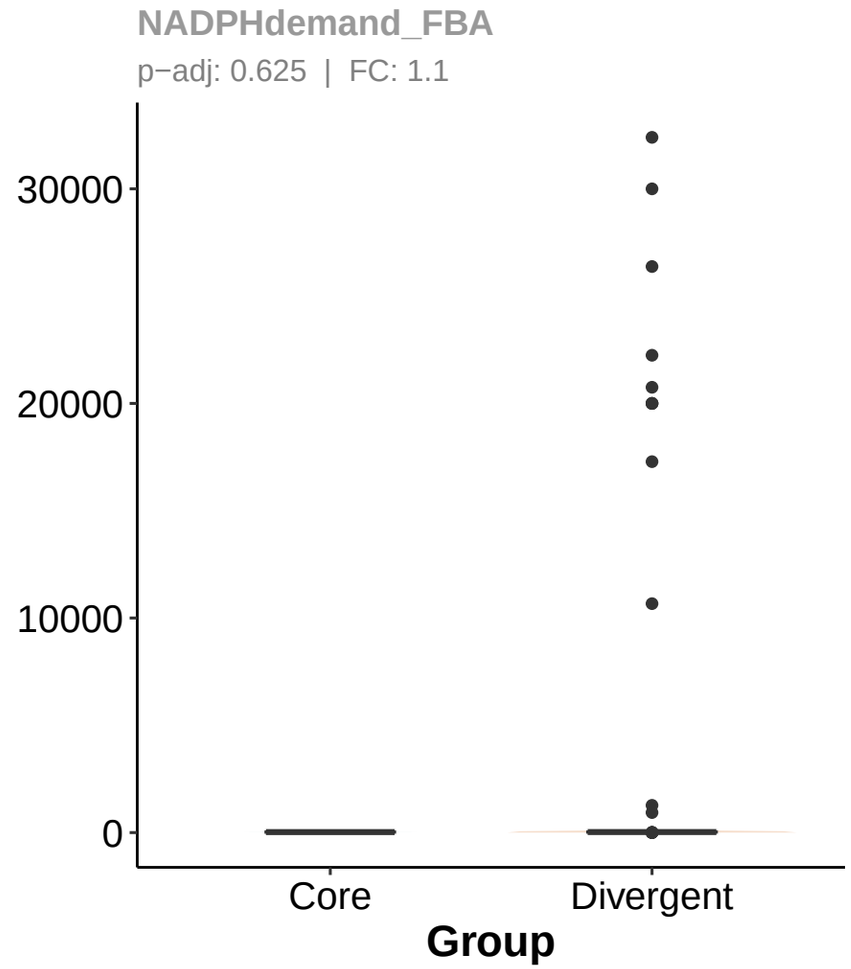


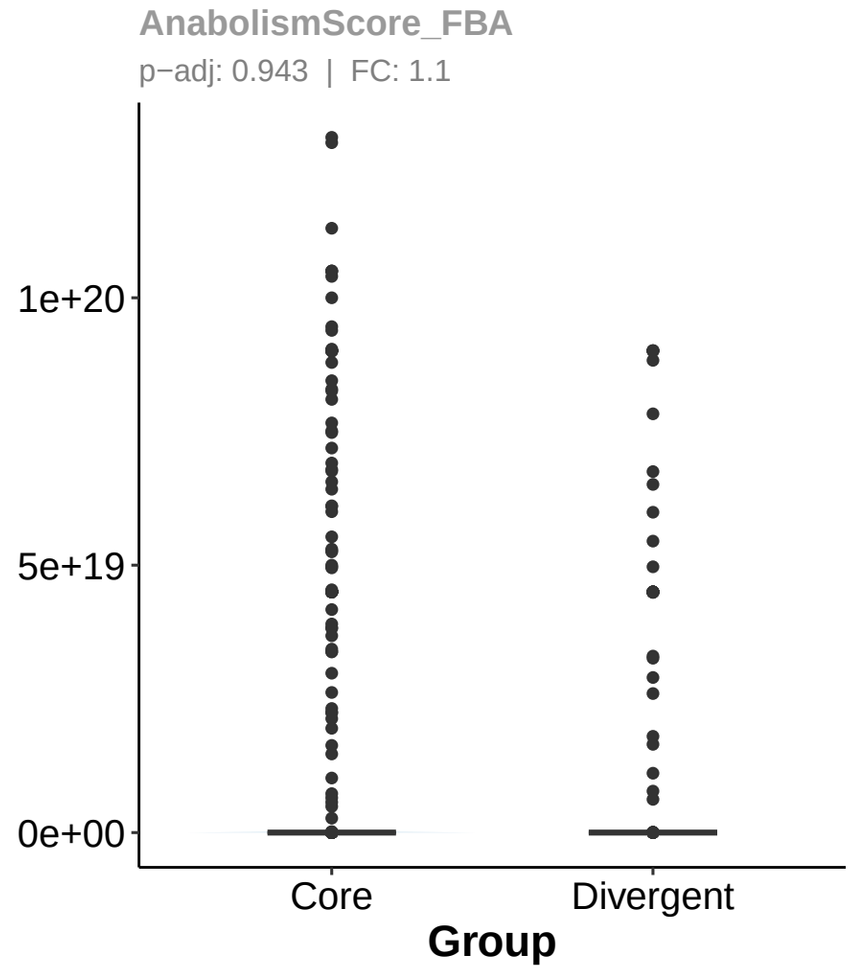
Heatmap: . 2. Estado Redox y Anabolismo

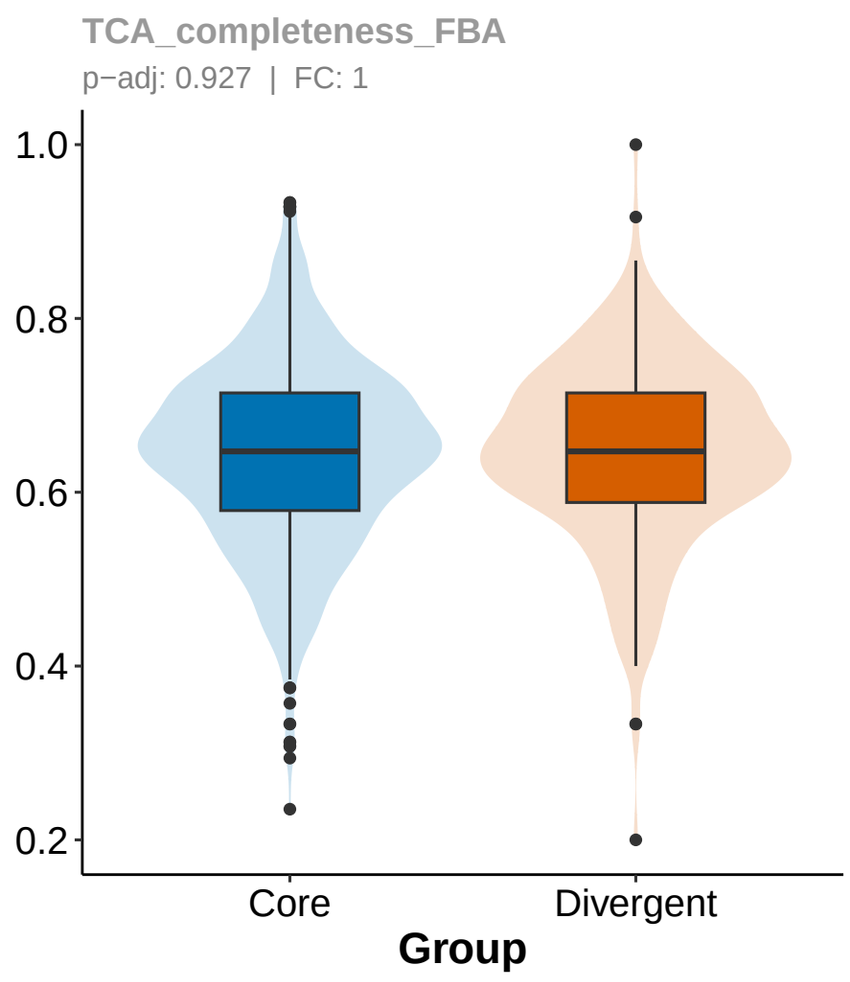


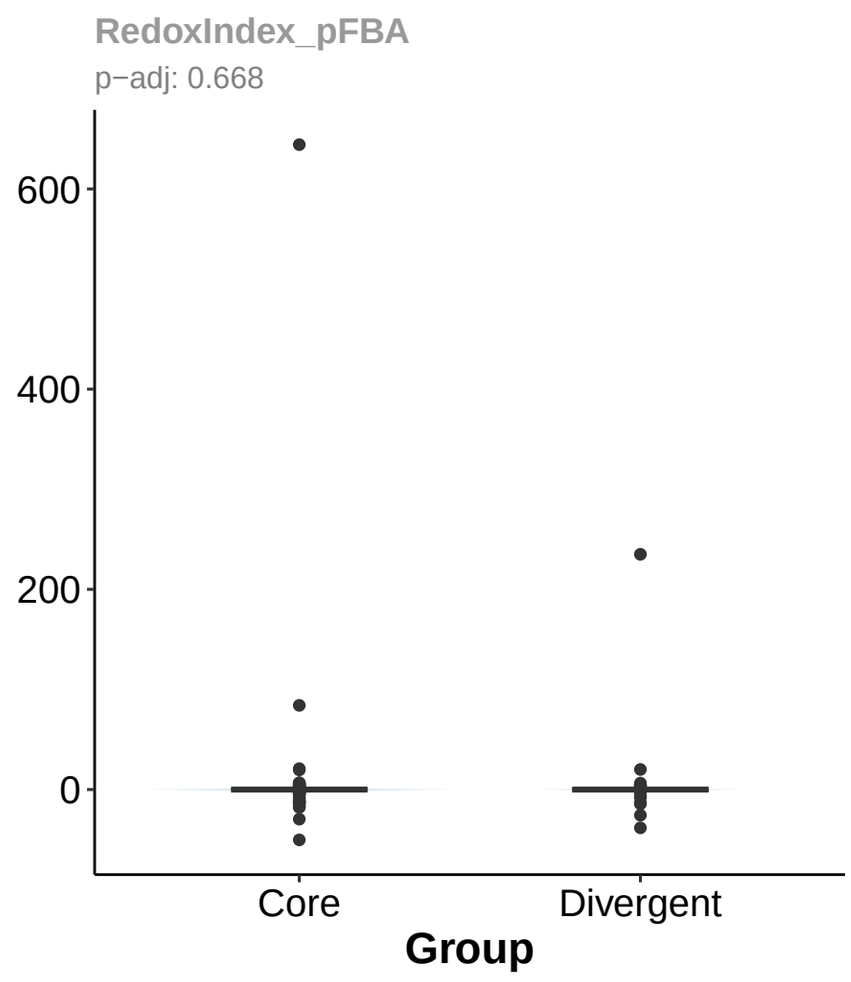
Metabolismo − . 2. Estado Redox y Anabolismo

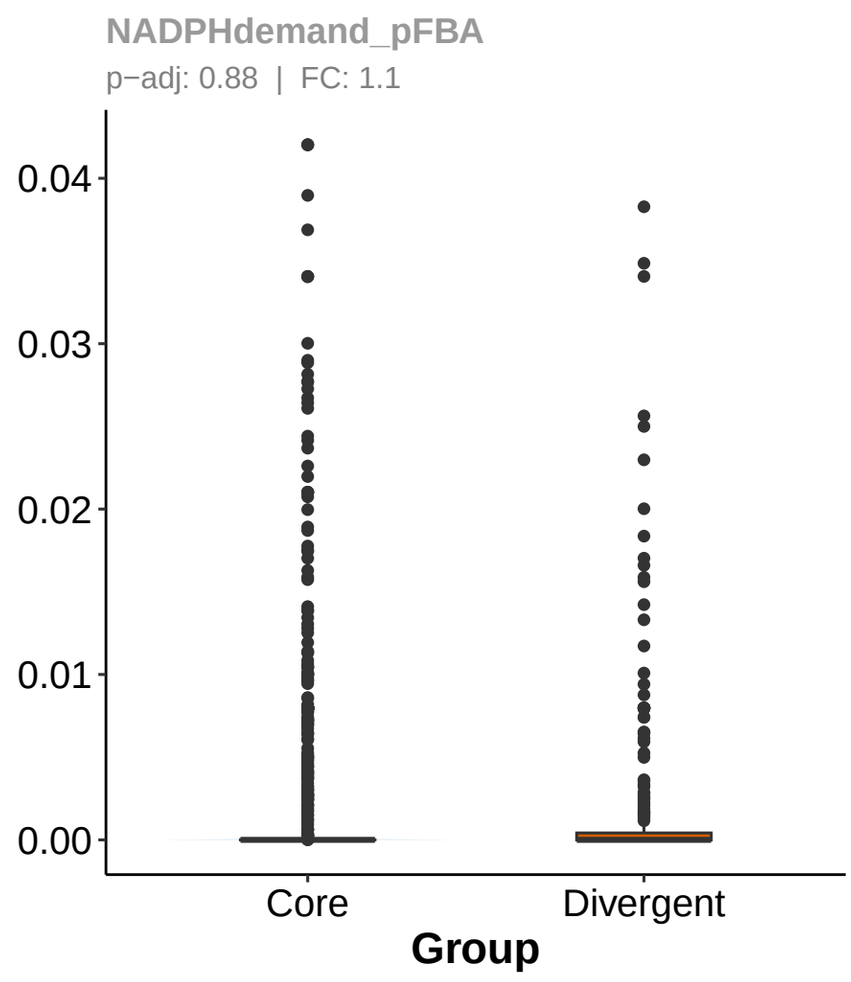




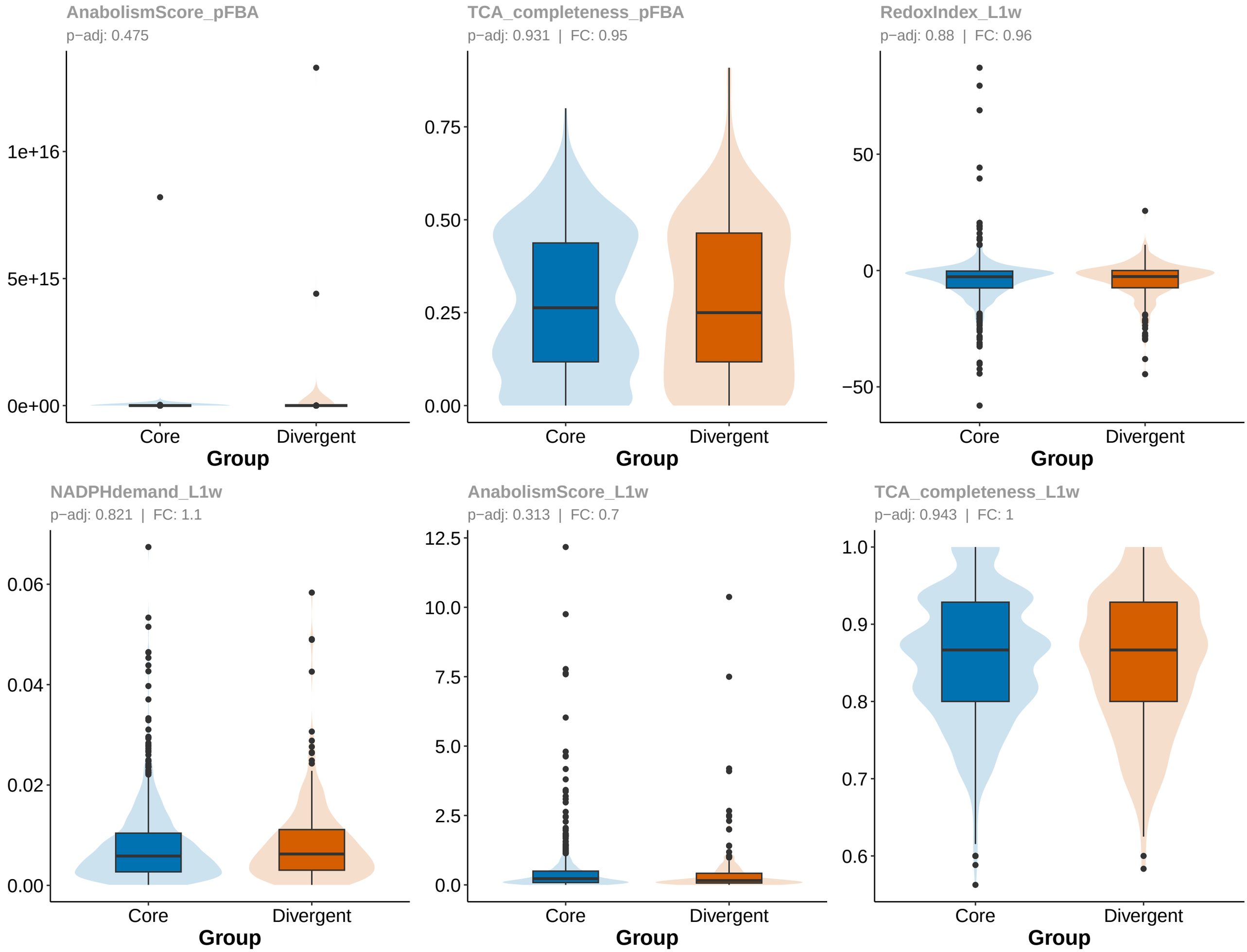




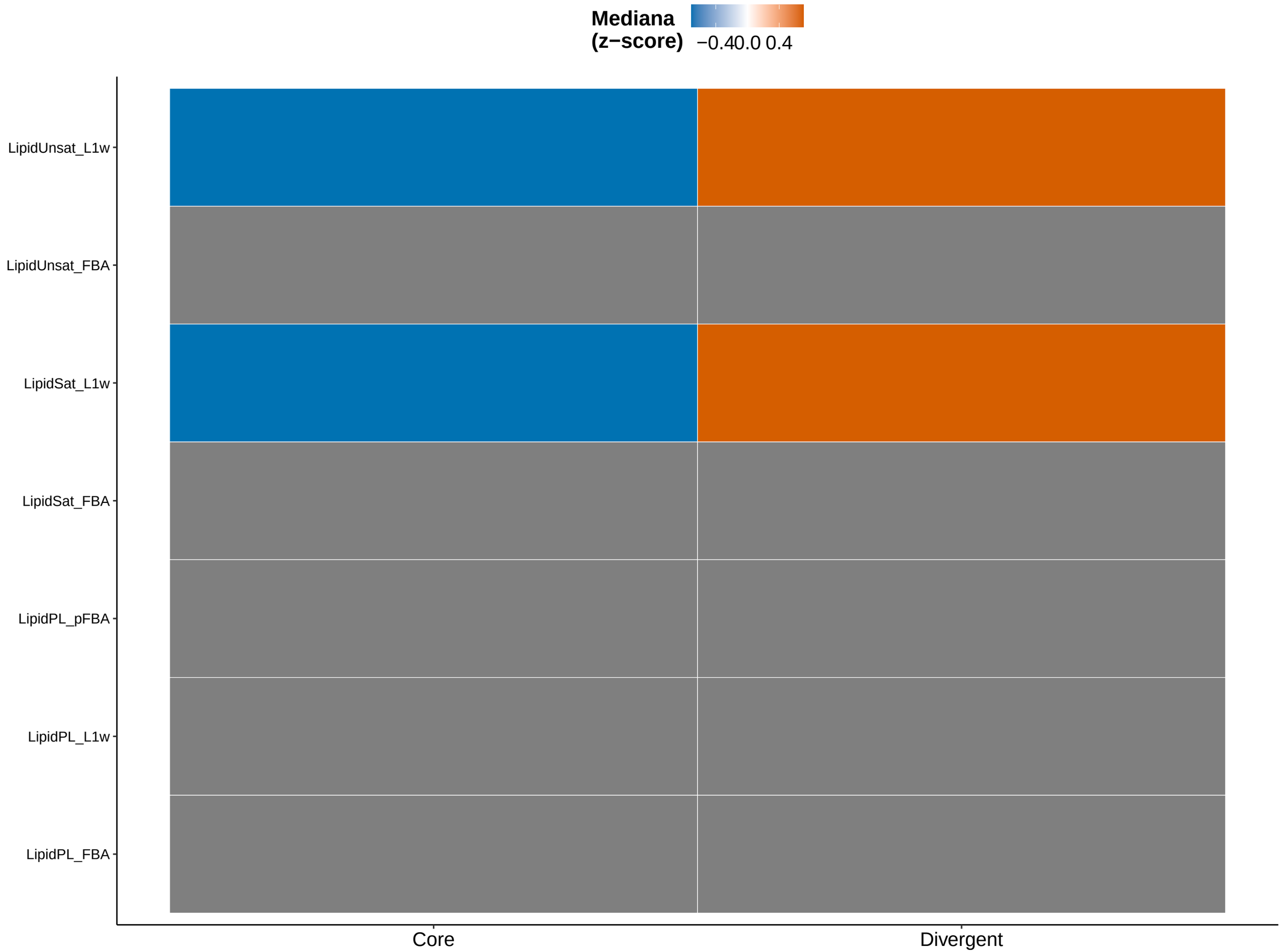




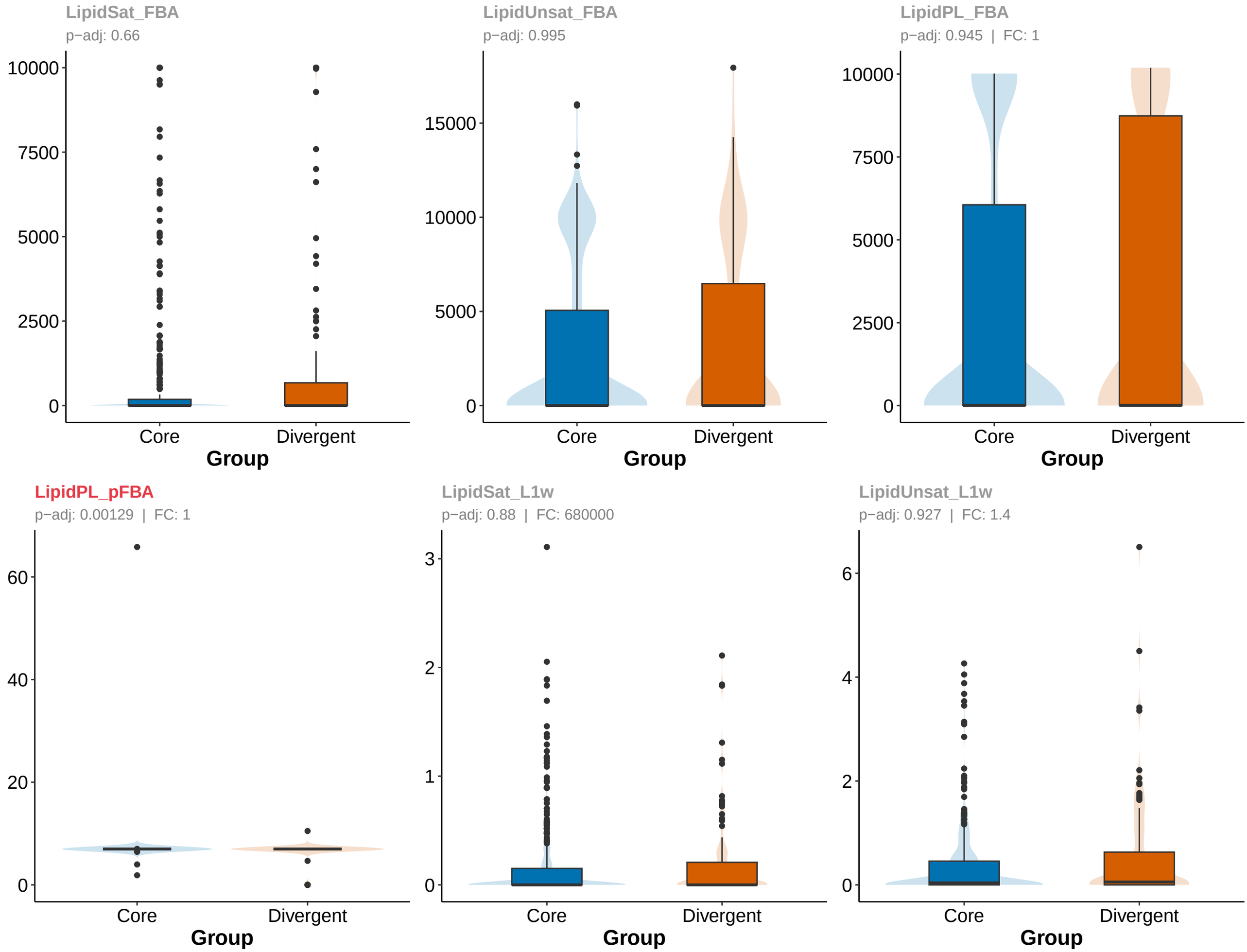
Metabolismo − . 2. Estado Redox y Anabolismo



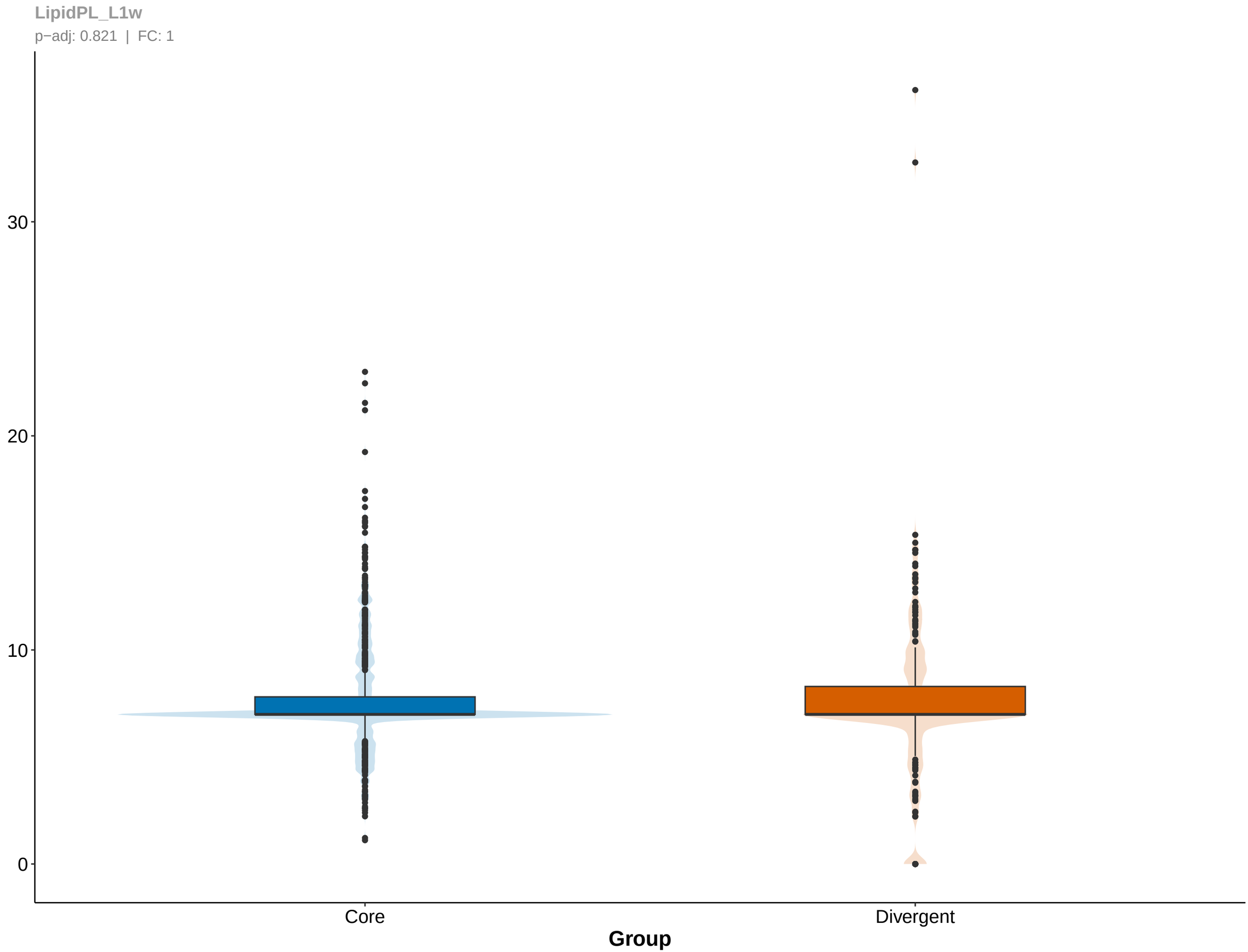
Heatmap: . 3. Metabolismo Lipídico



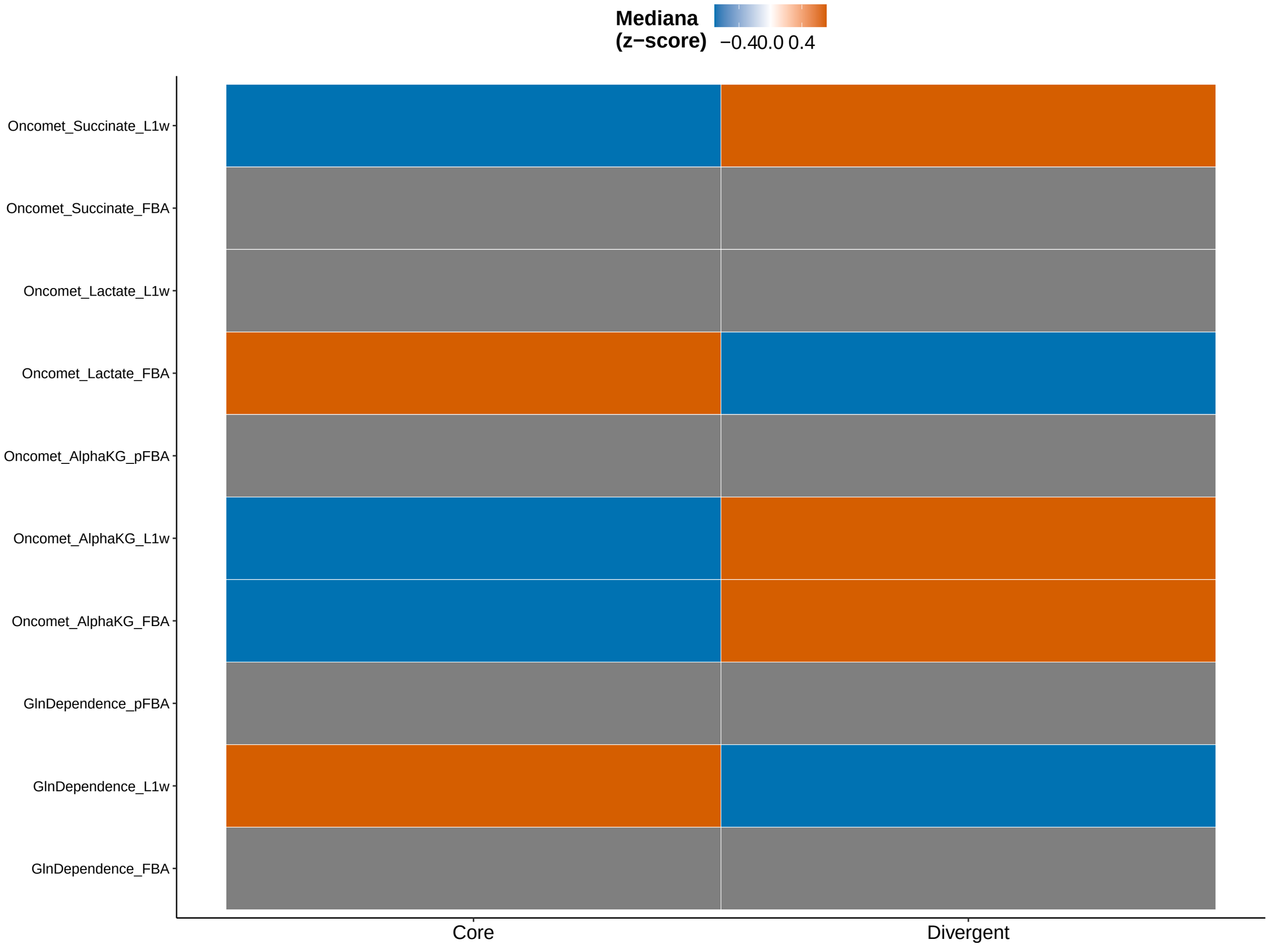
Metabolismo − . 3. Metabolismo Lipídico



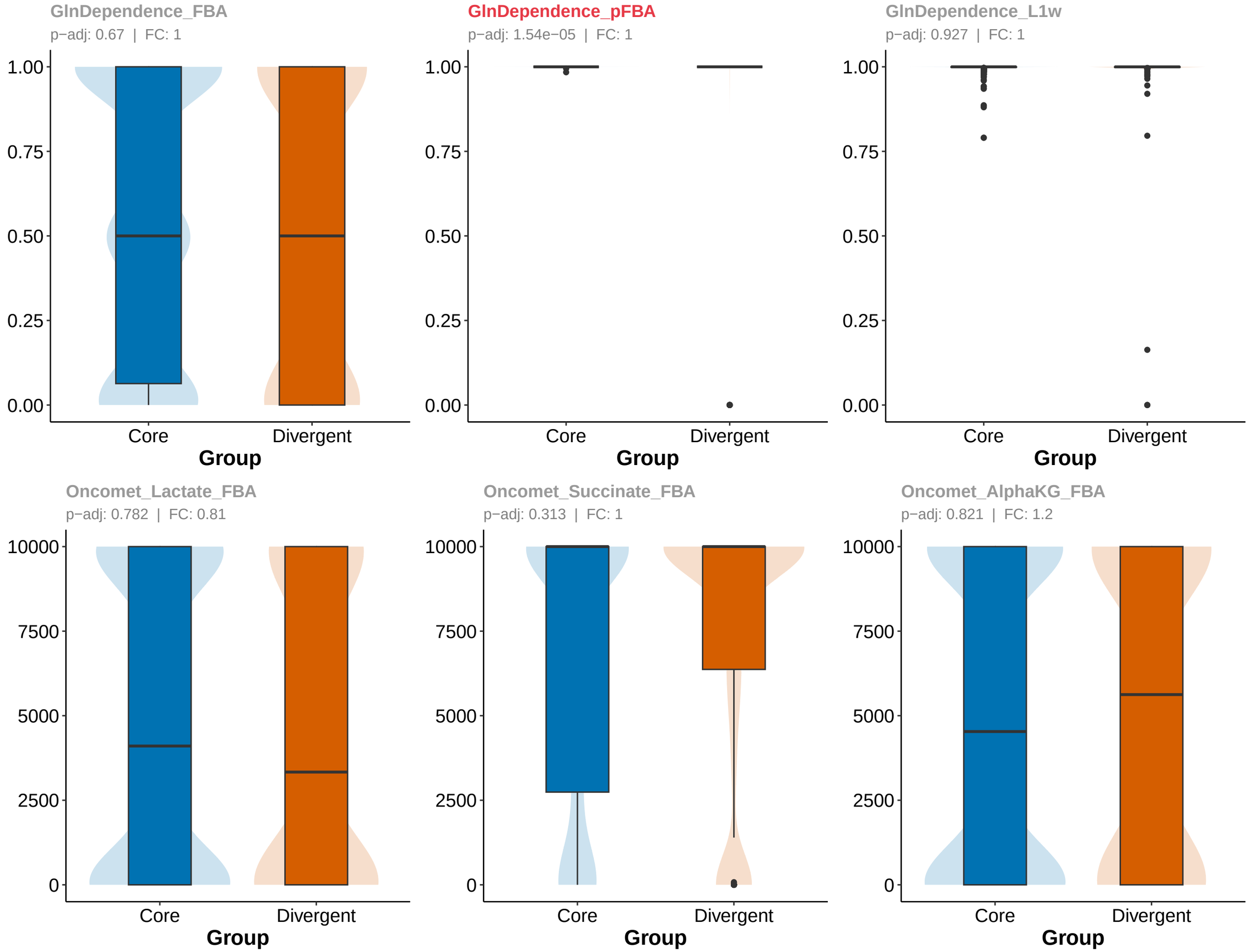
Metabolismo − . 3. Metabolismo Lipídico



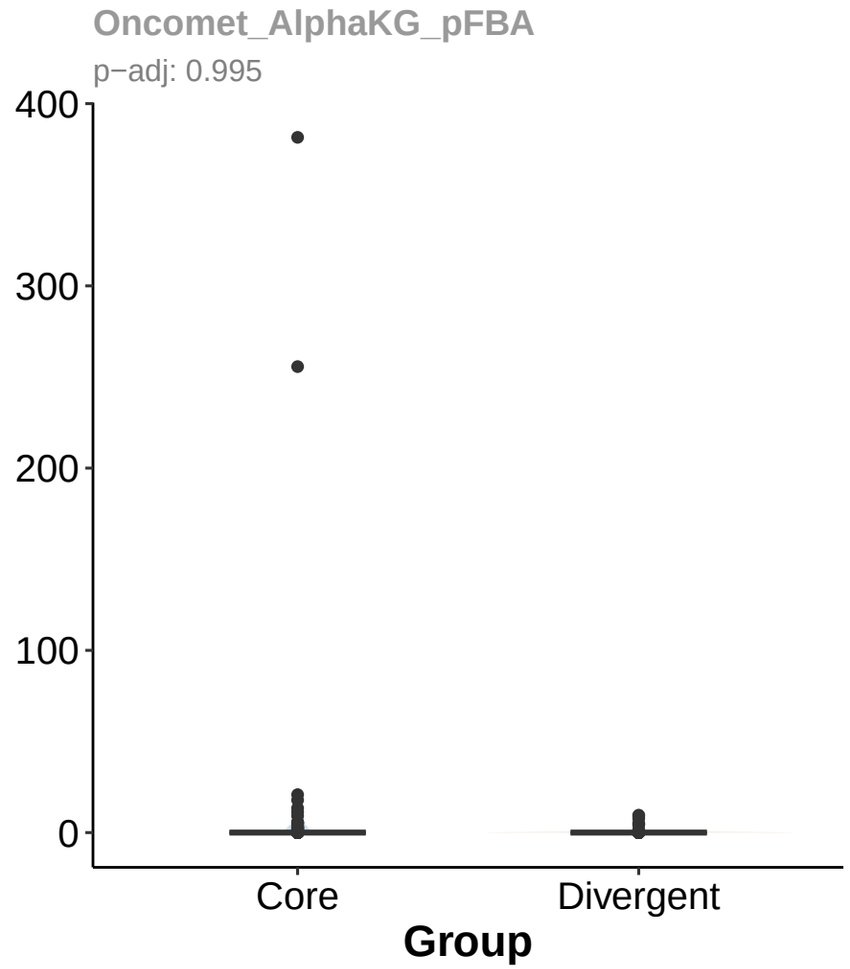
Heatmap: . 4. Dependencias Metabólicas Específicas

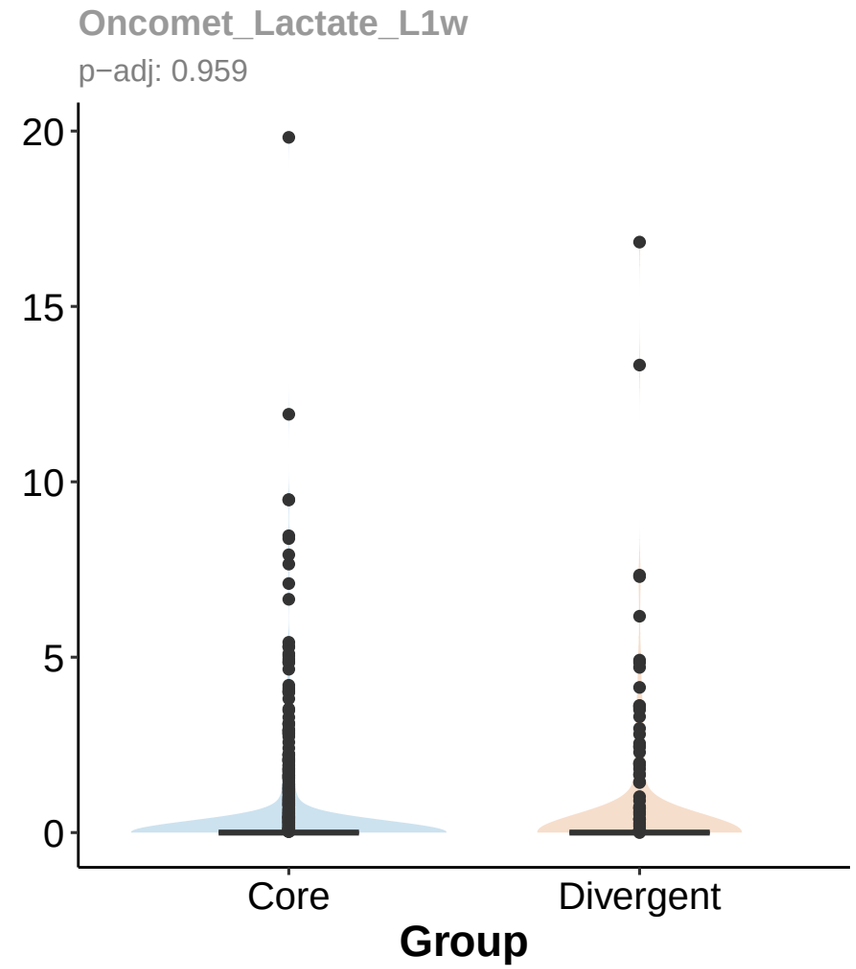


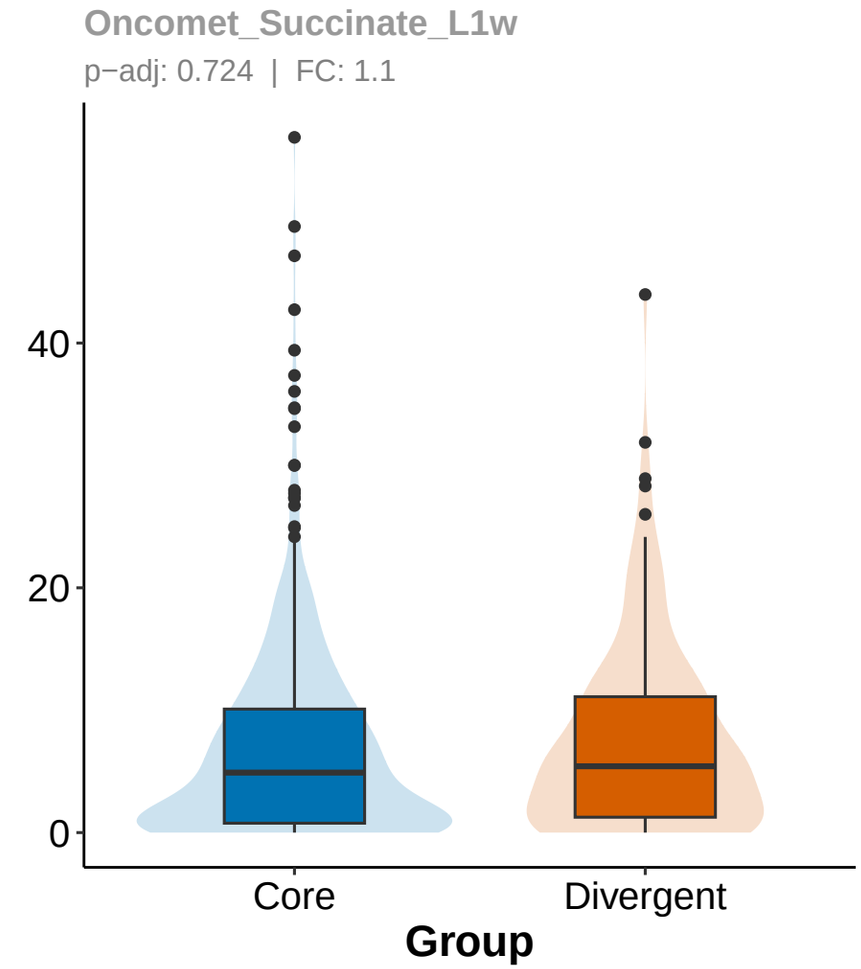
Metabolismo − . 4. Dependencias Metabólicas Específicas

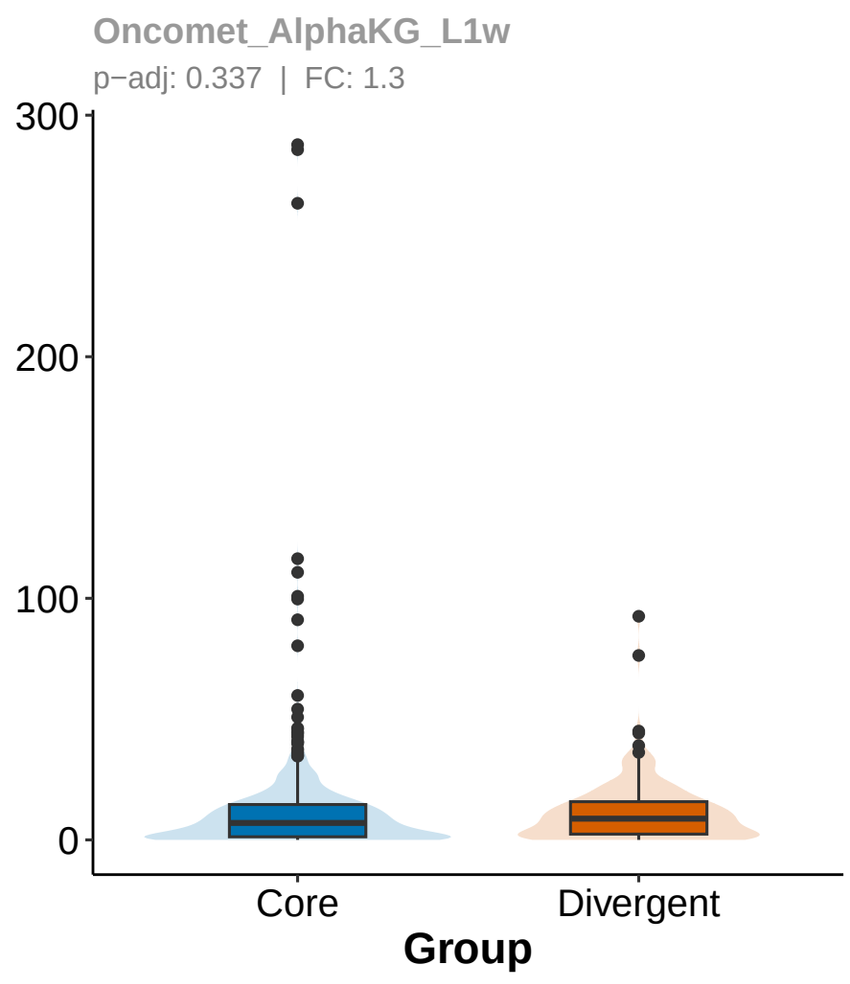


Metabolismo − . 4. Dependencias Metabólicas Específicas

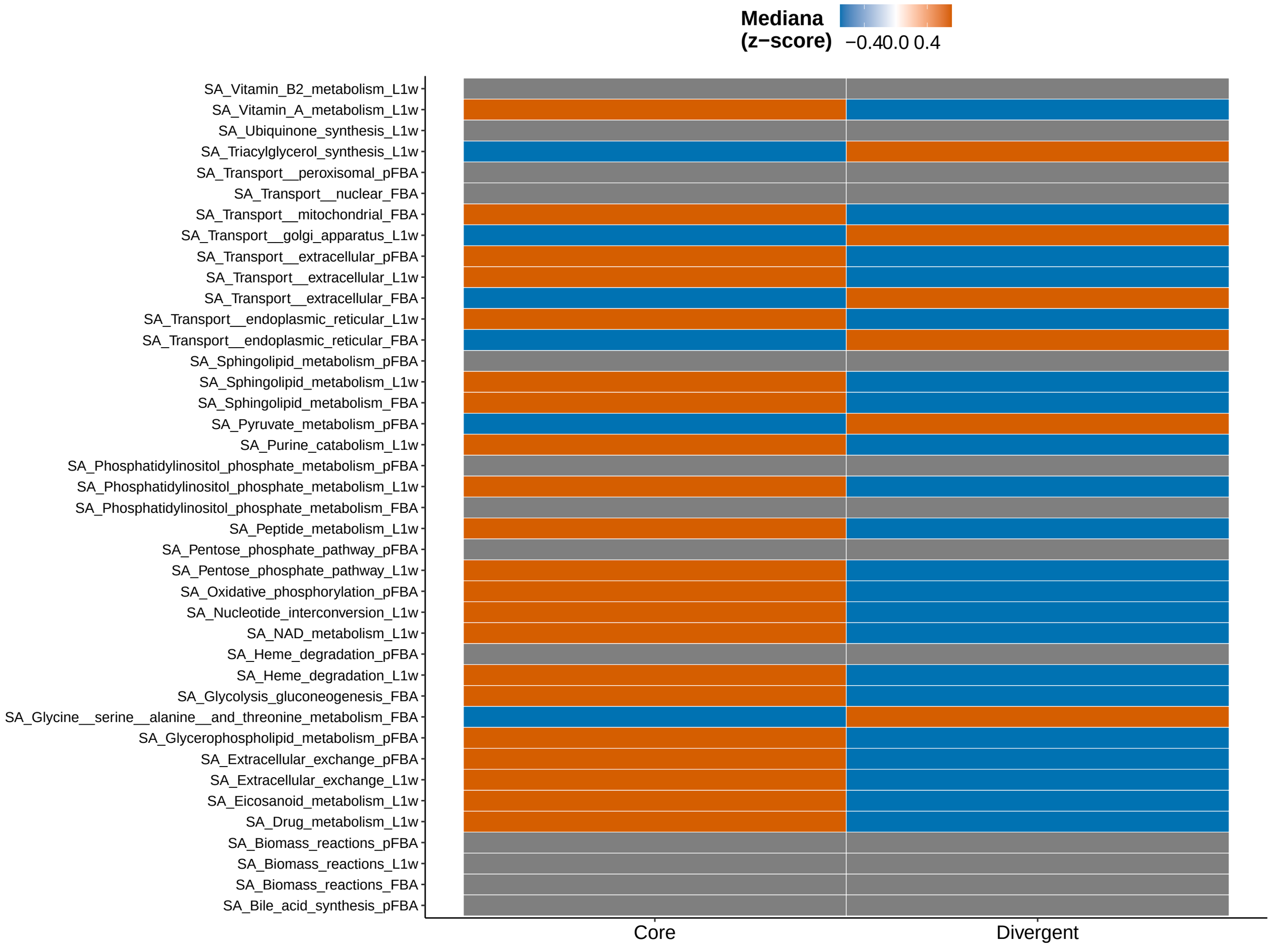








Heatmap: . 5. Reprogramación Sistémica (SA_*)



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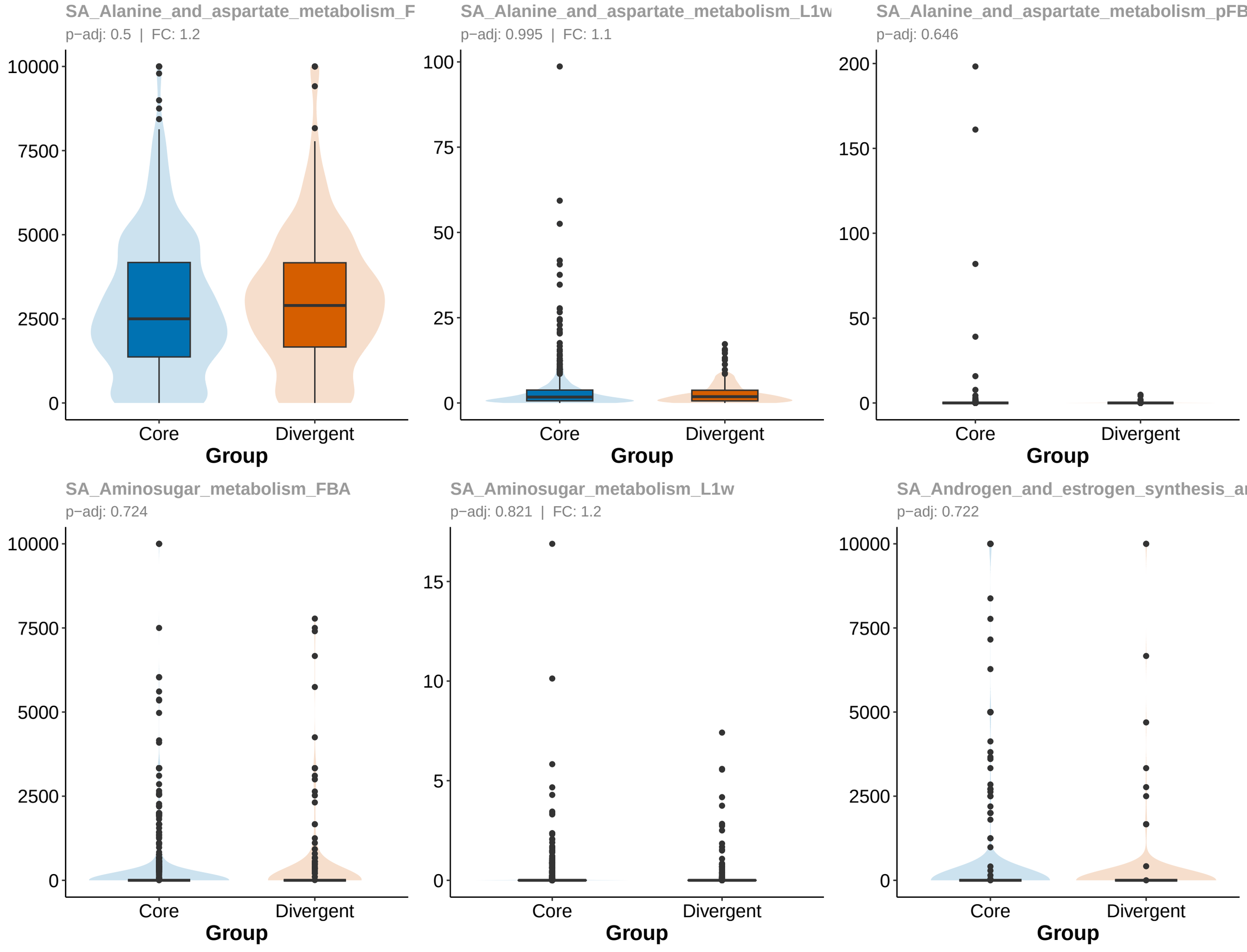

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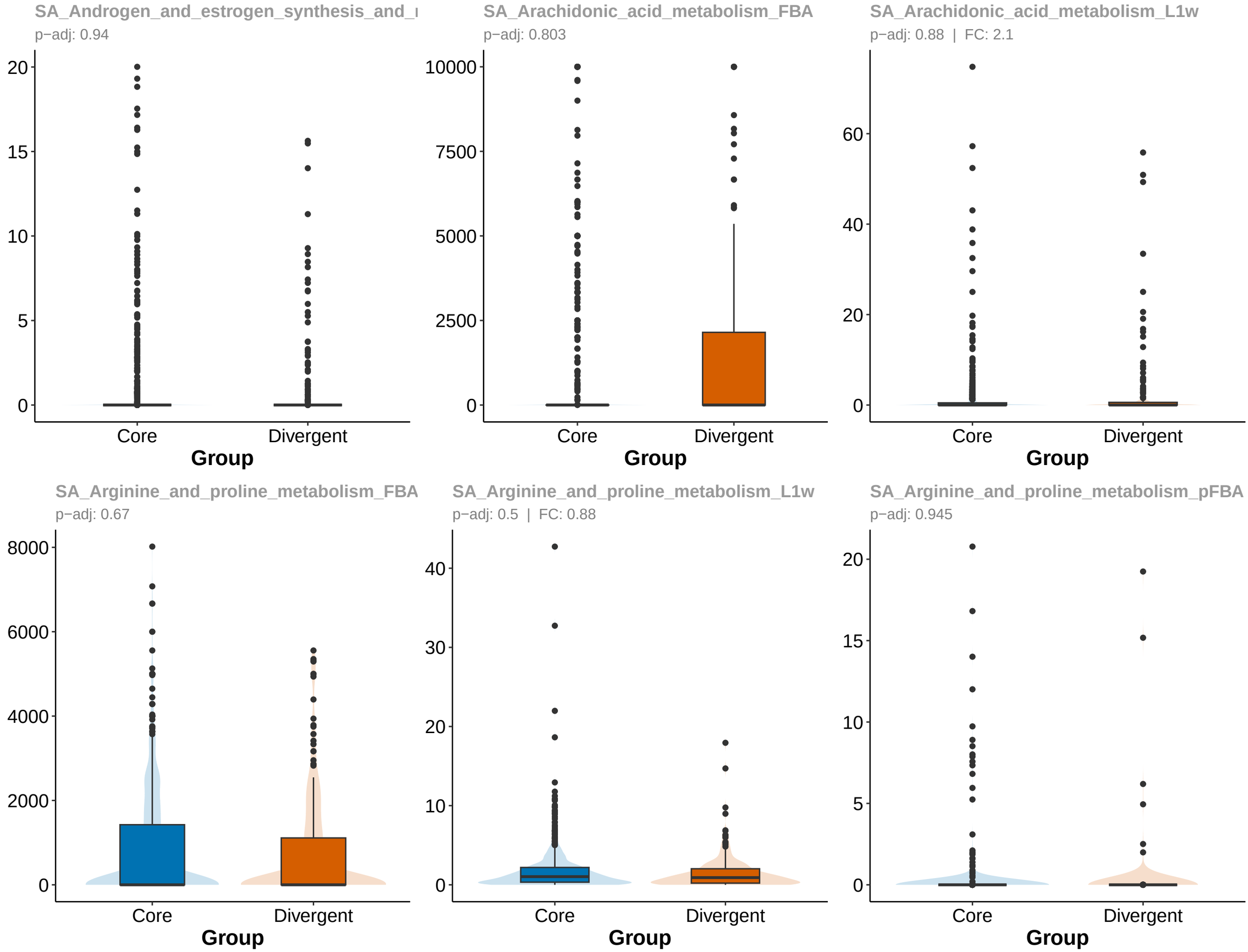

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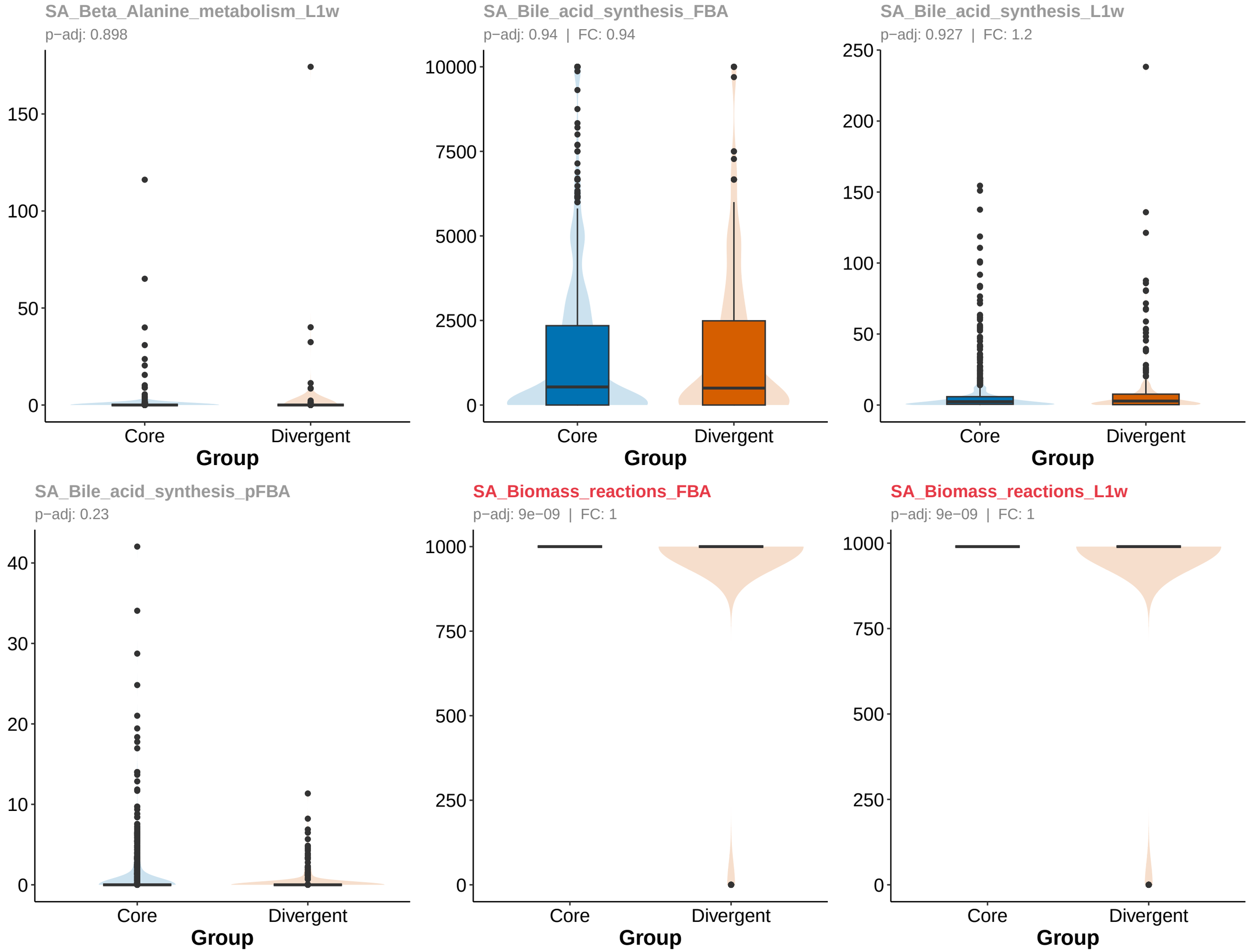

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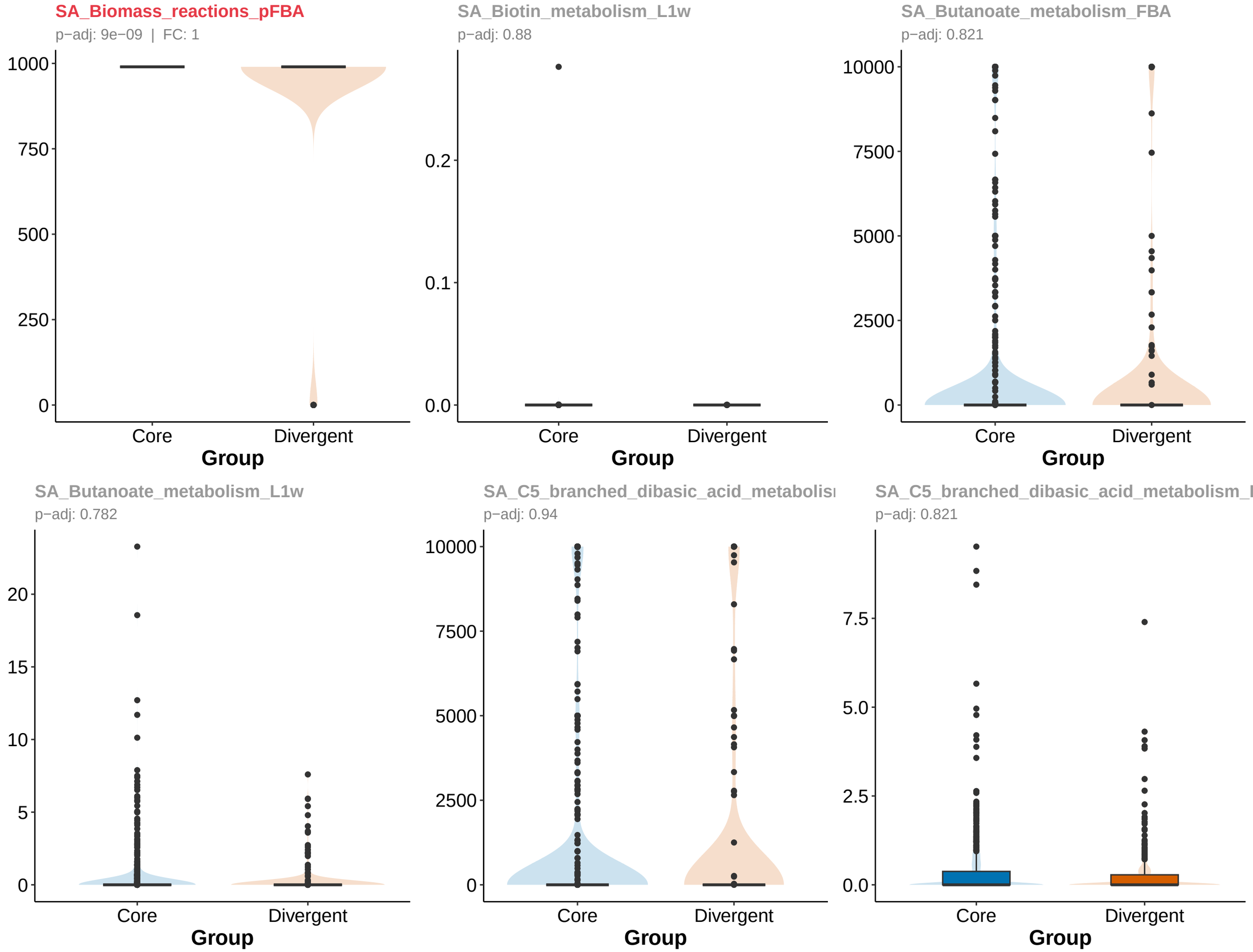

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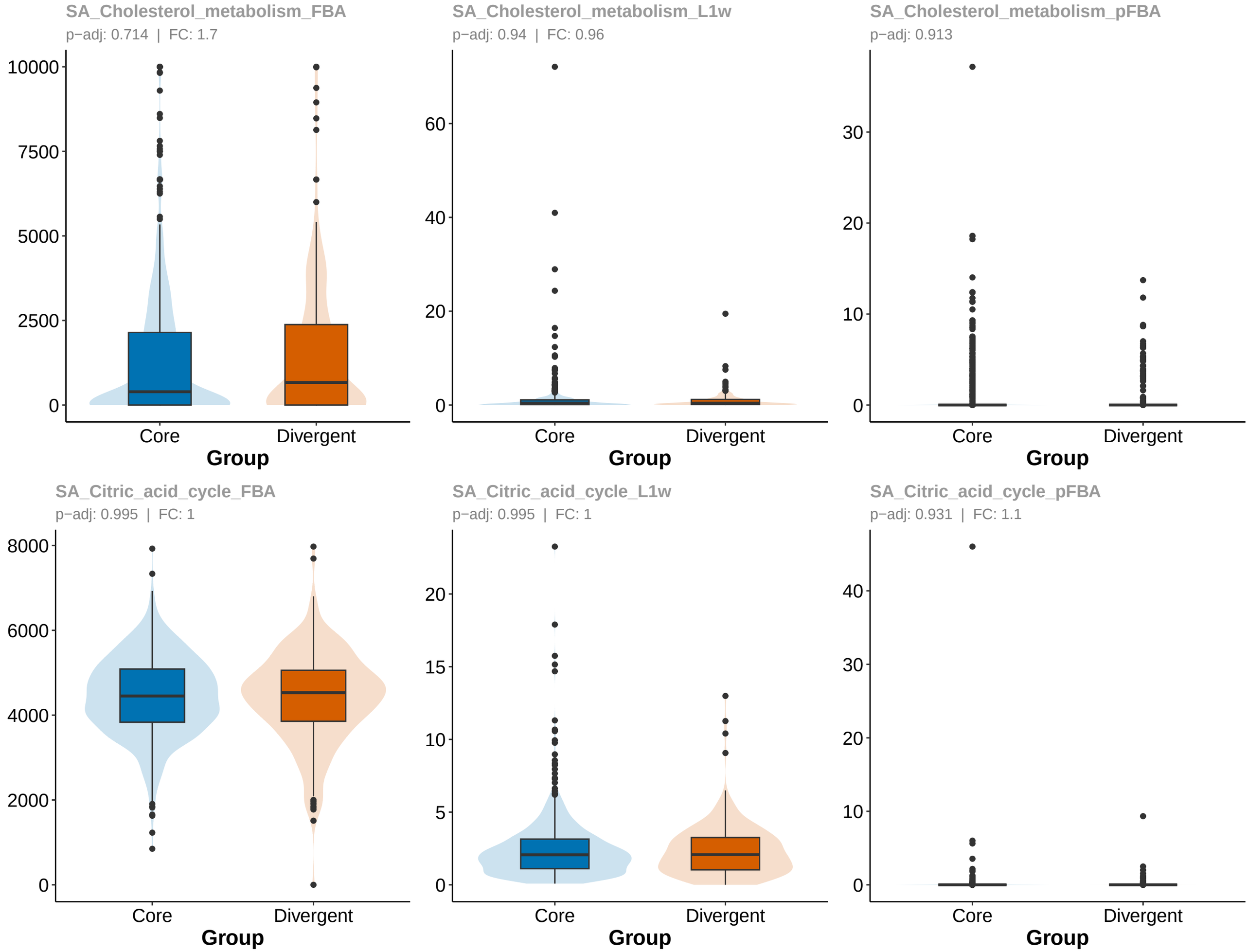

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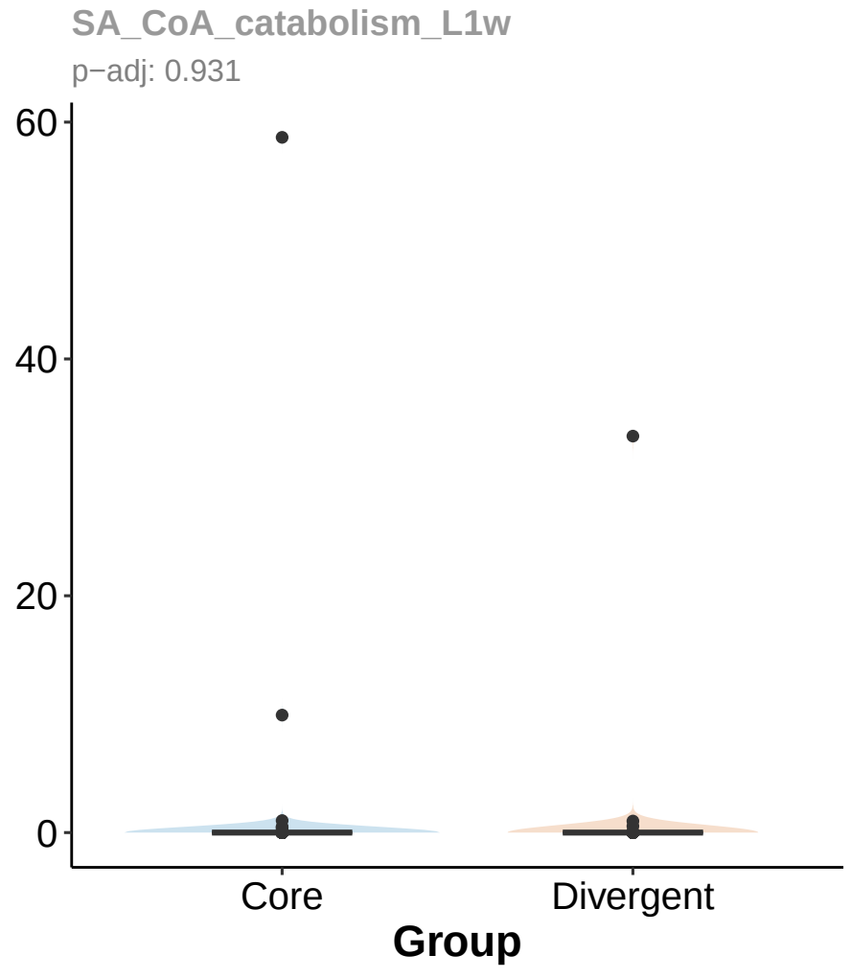

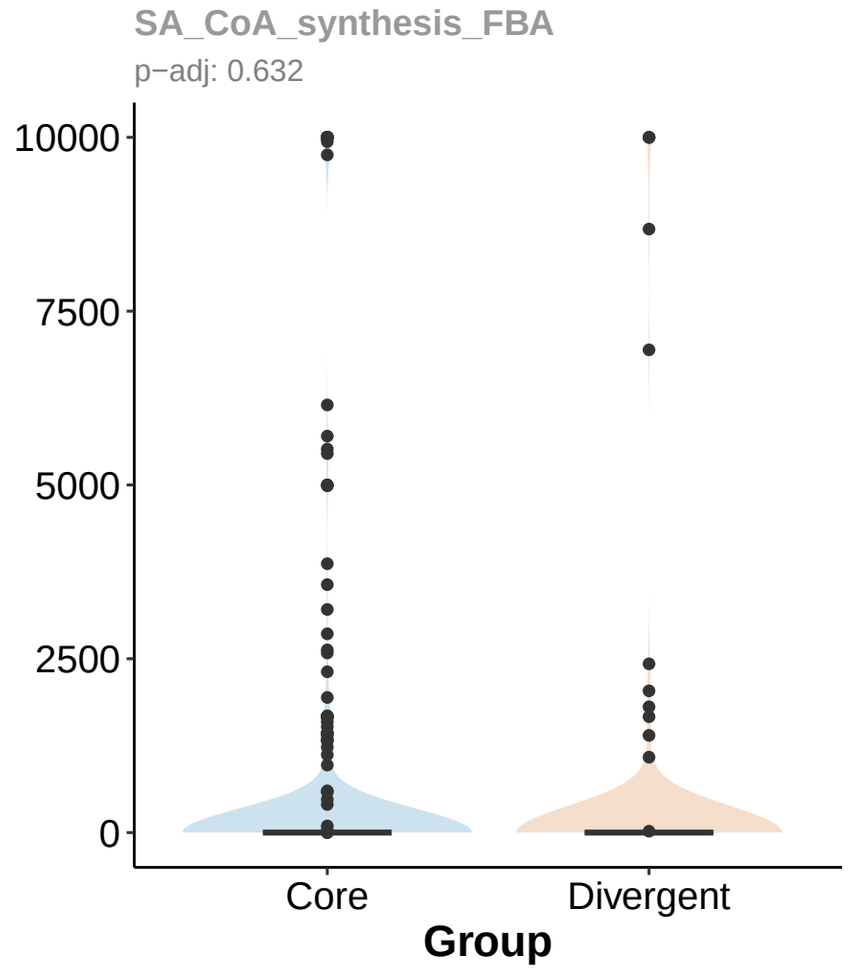

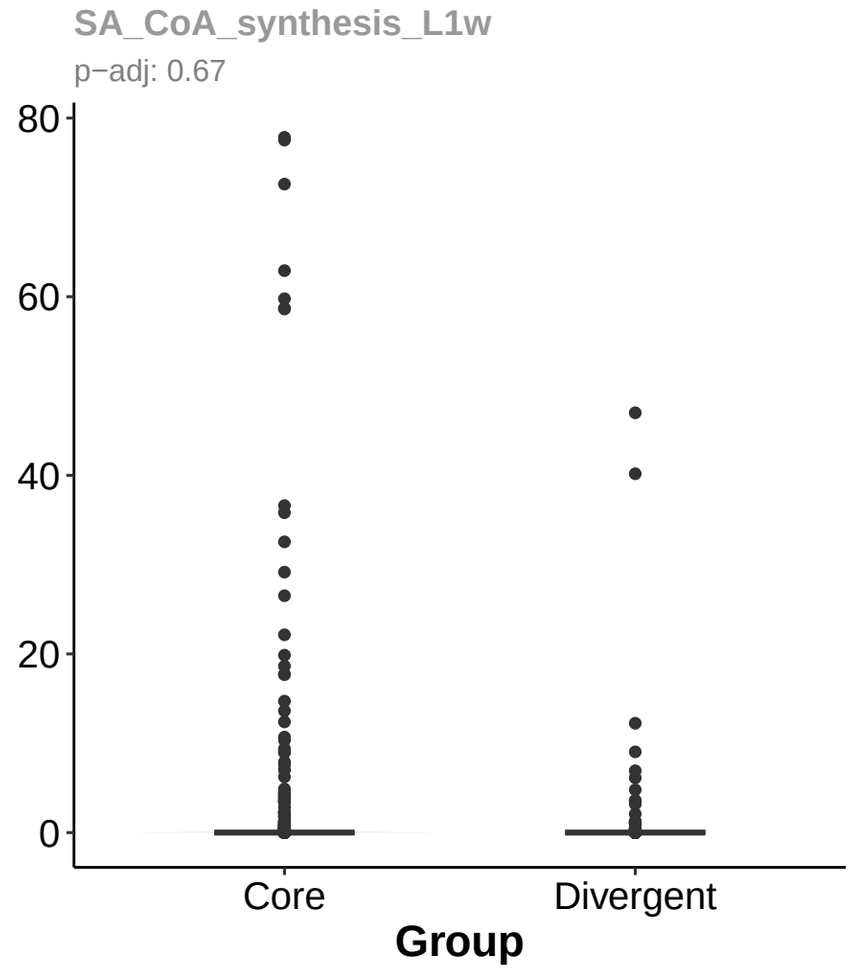

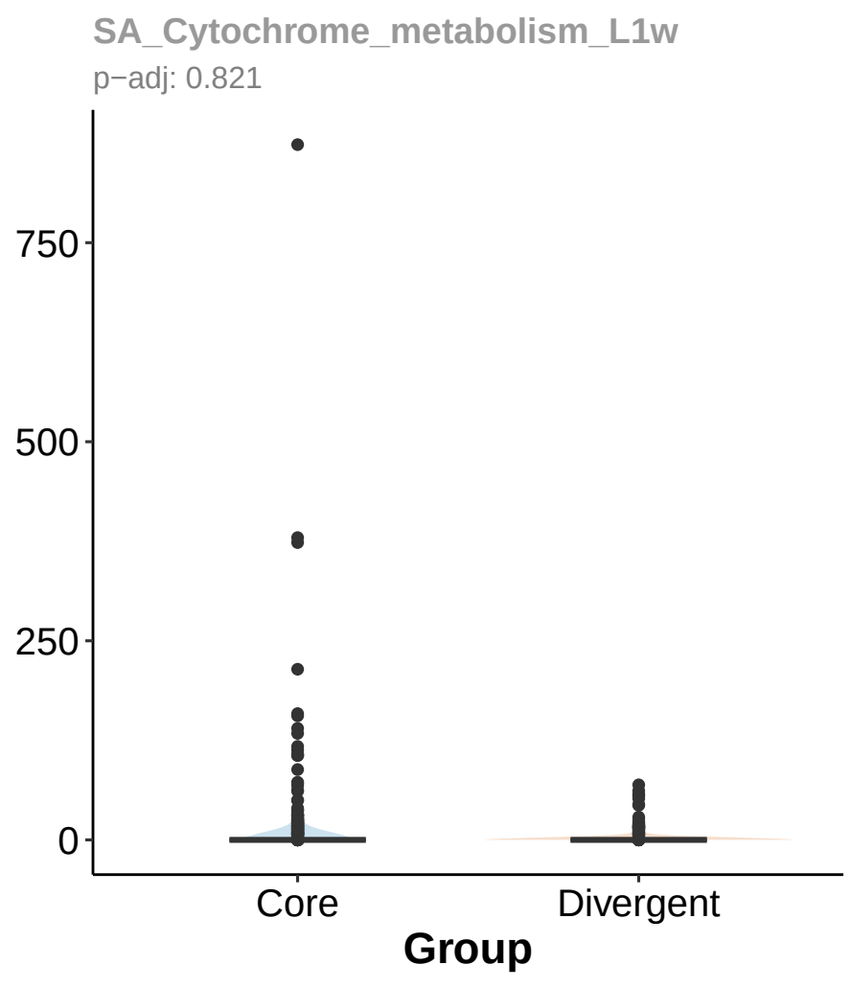

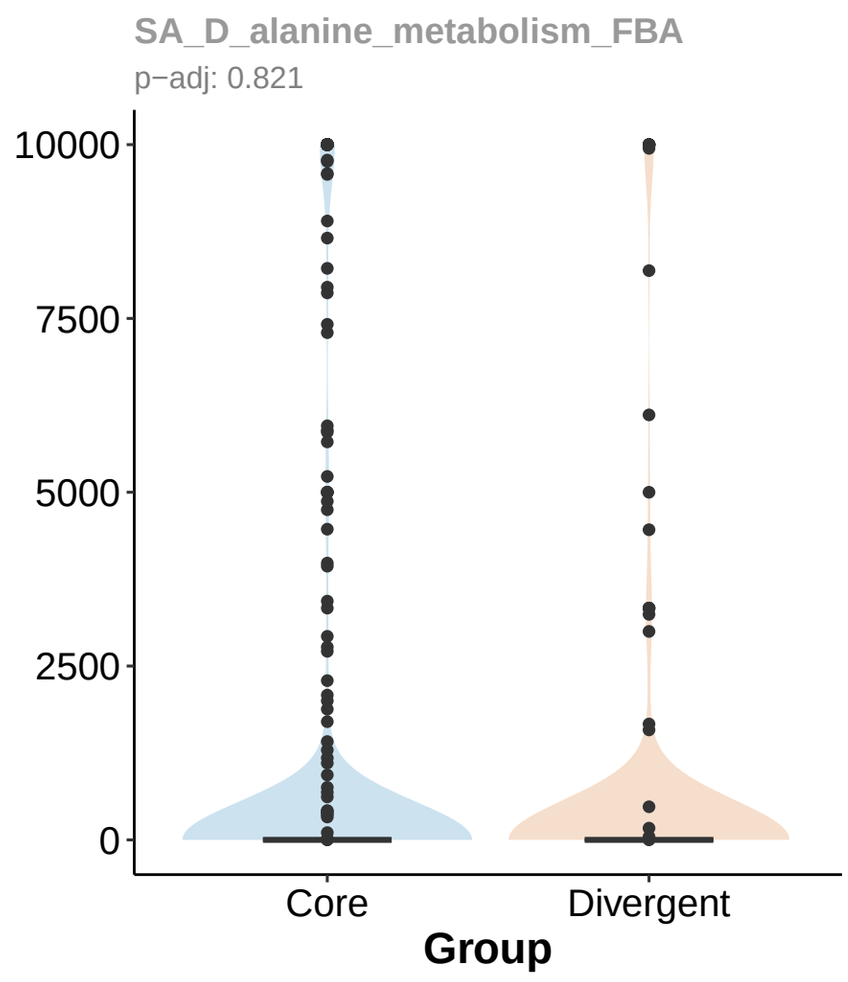

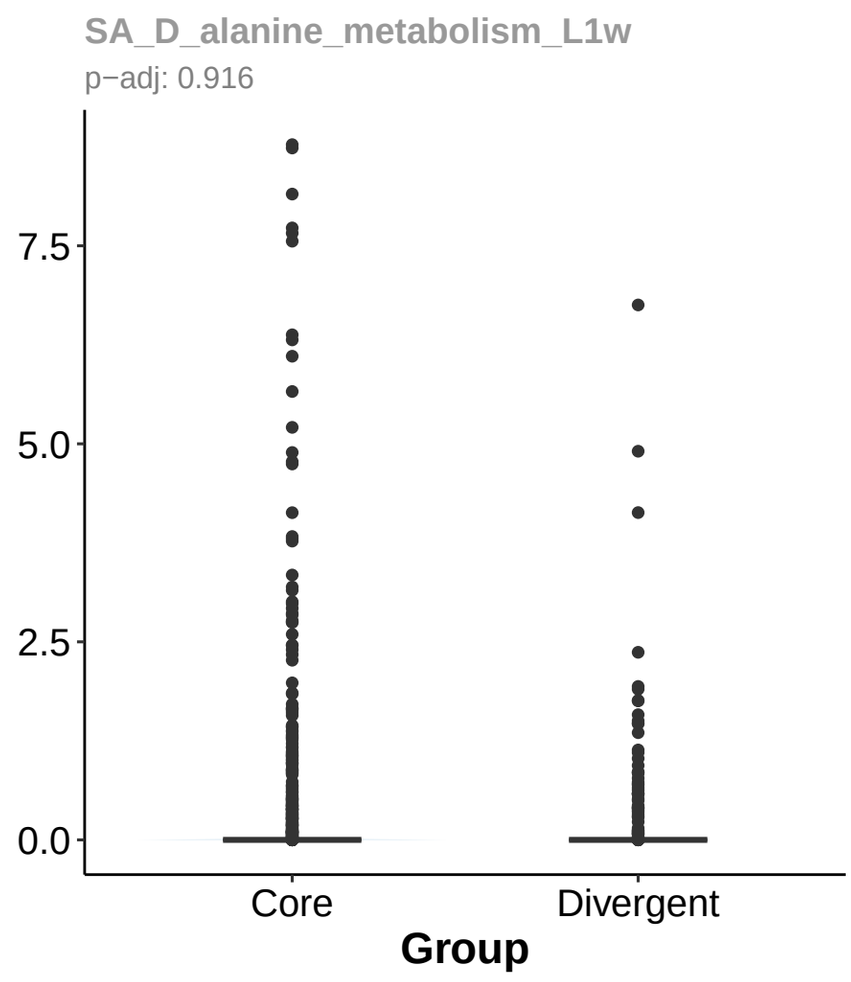

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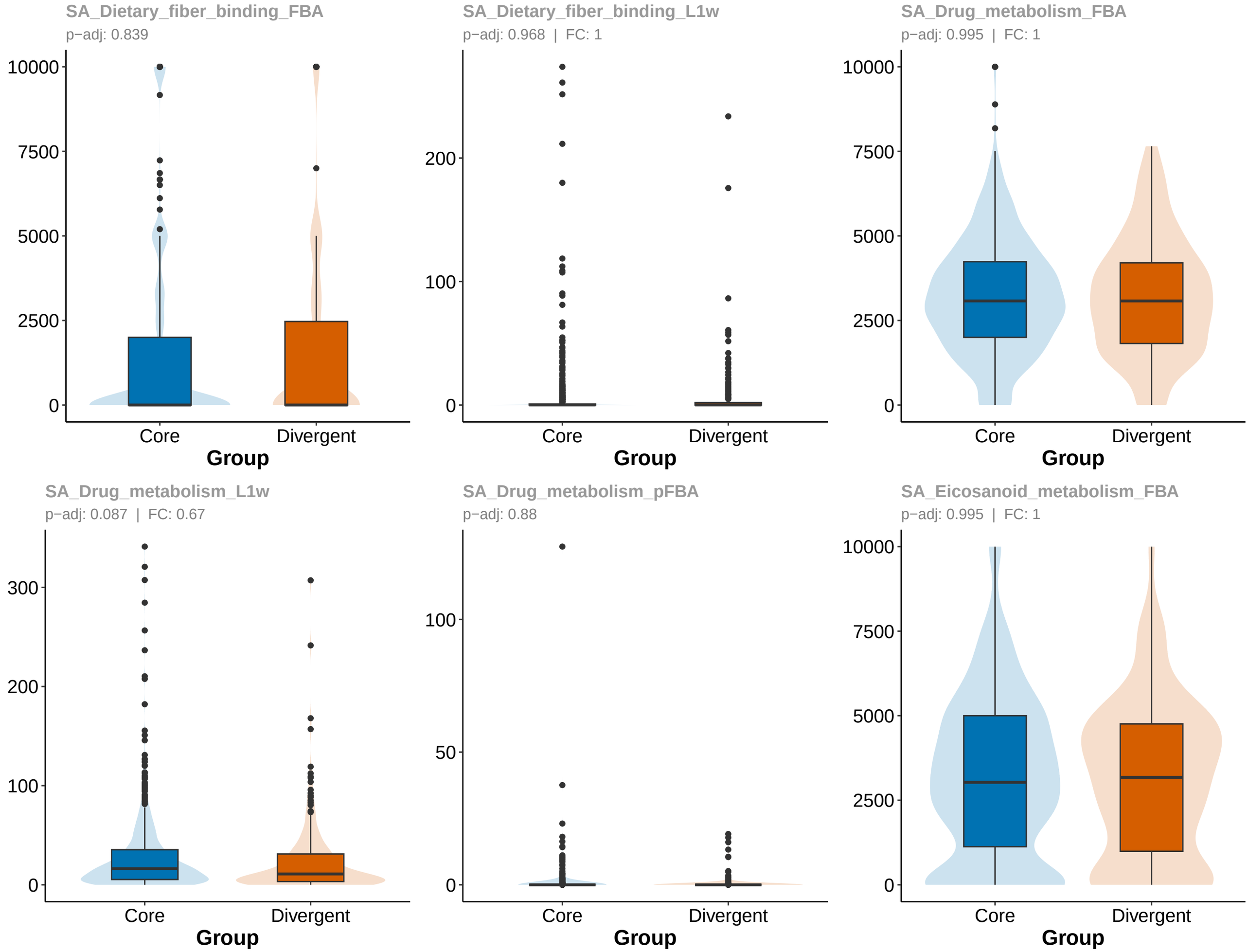
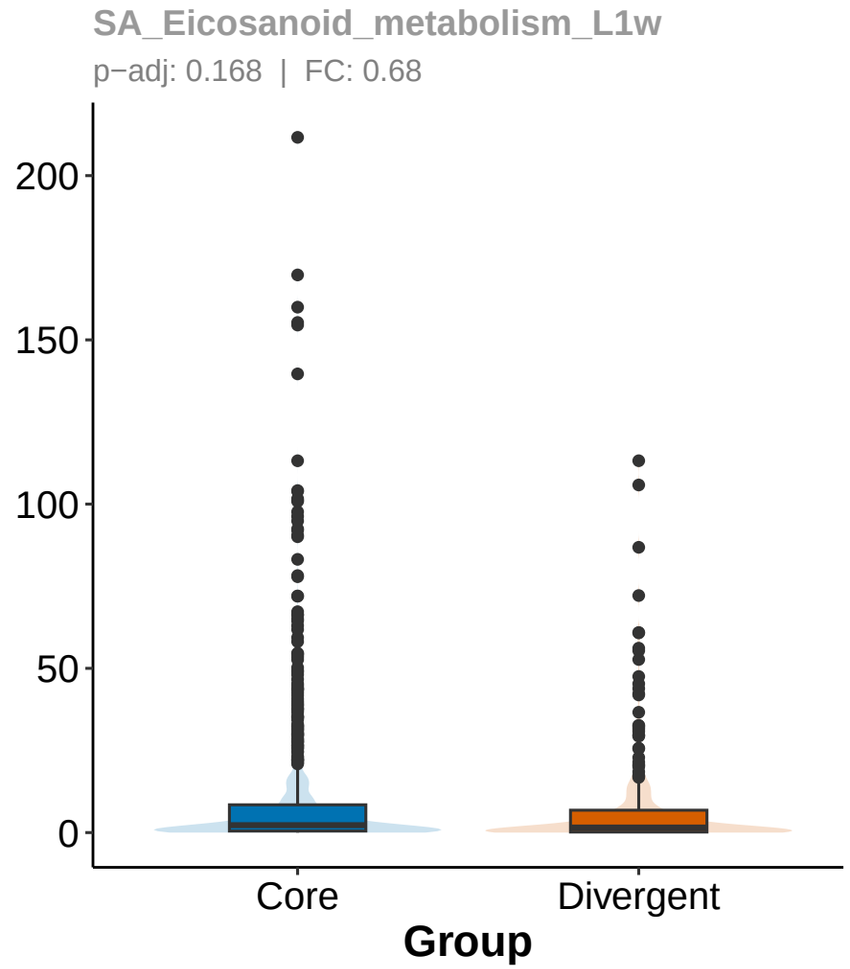
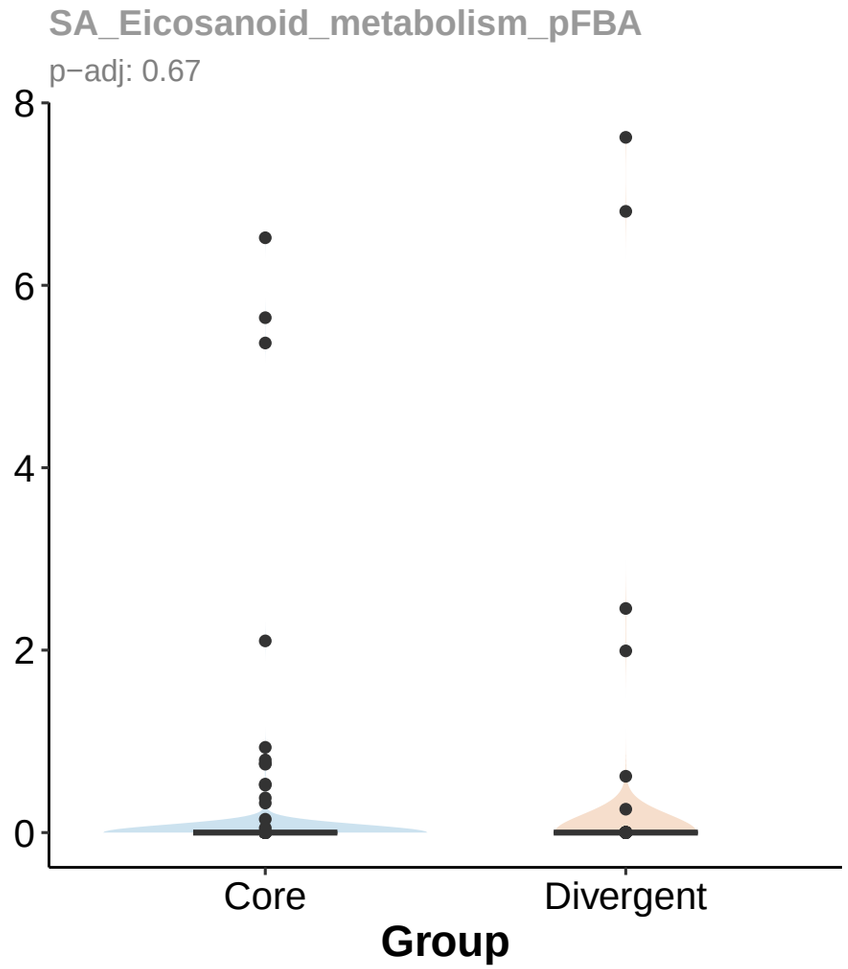
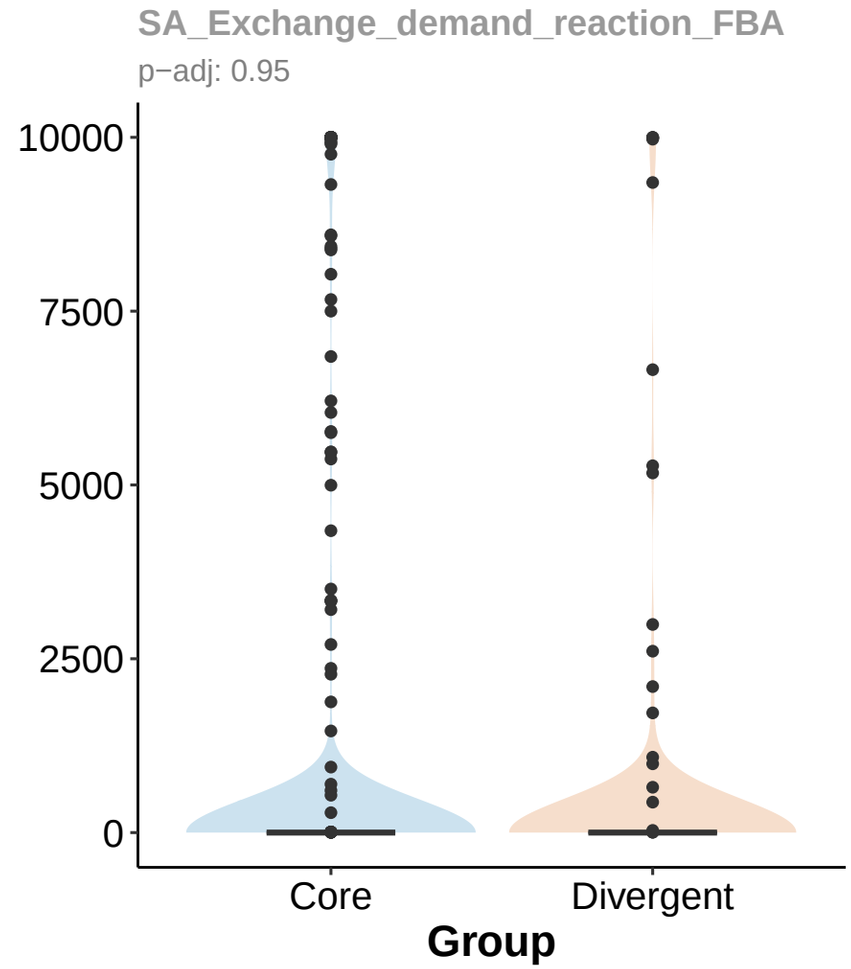
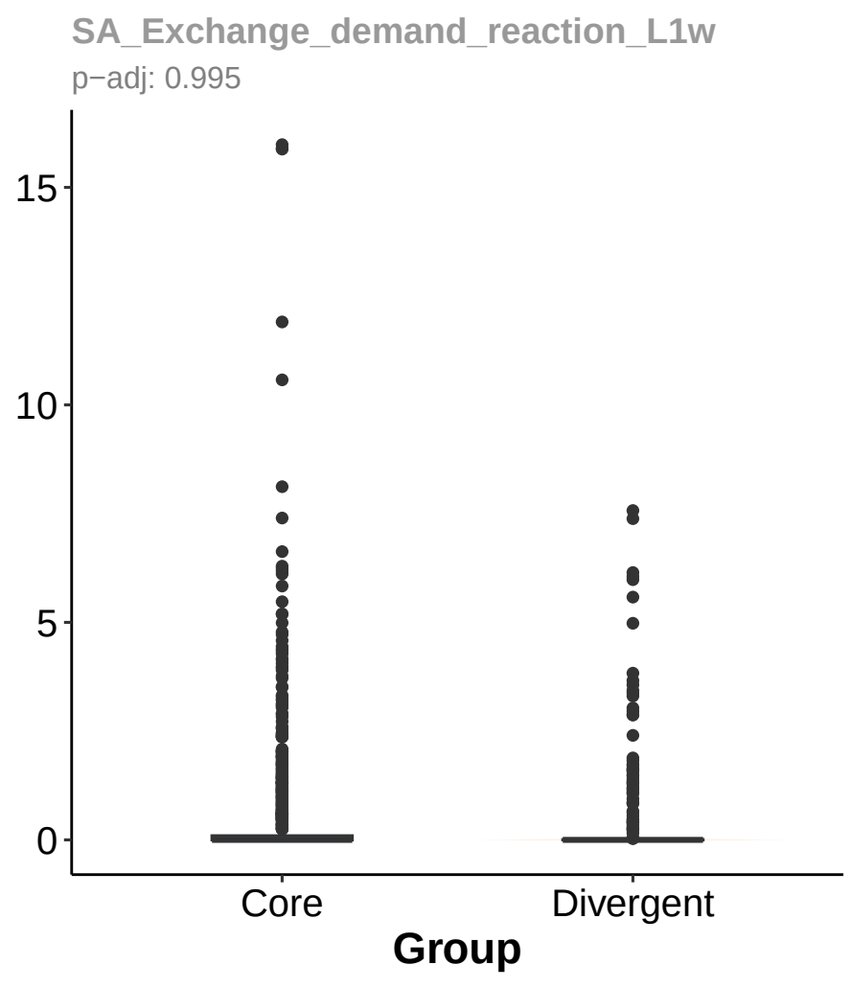


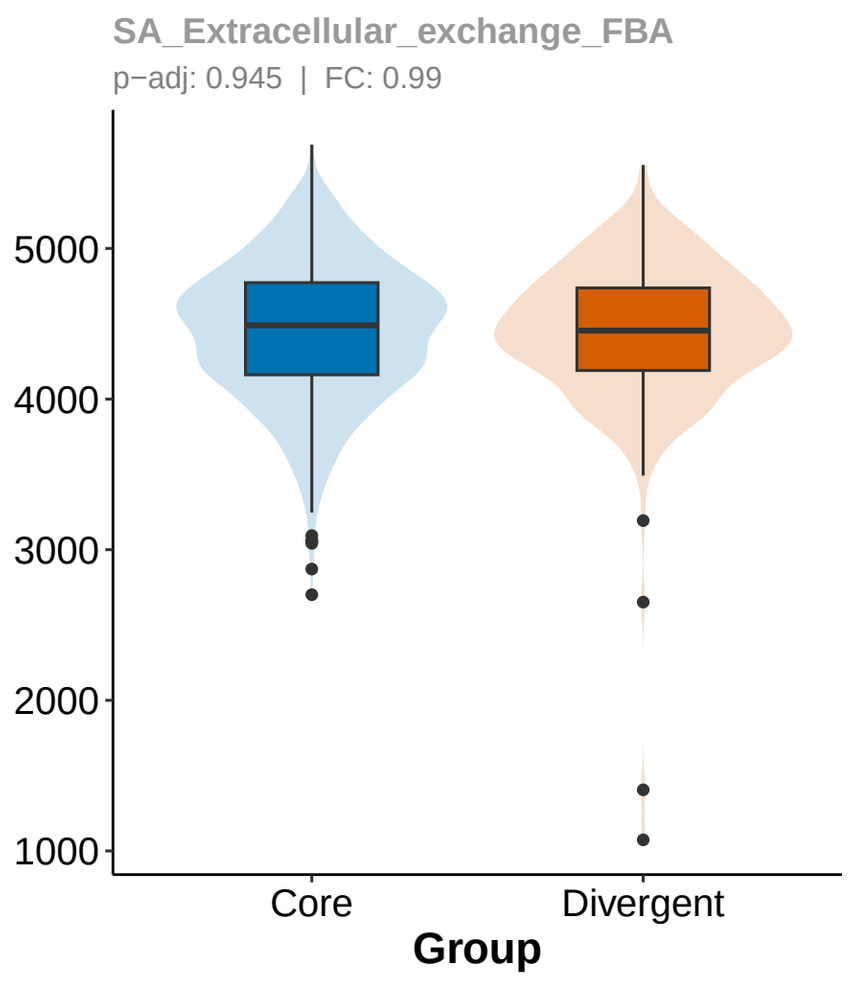
Metabolismo − . 5. Reprogramación Sistémica (SA_*)

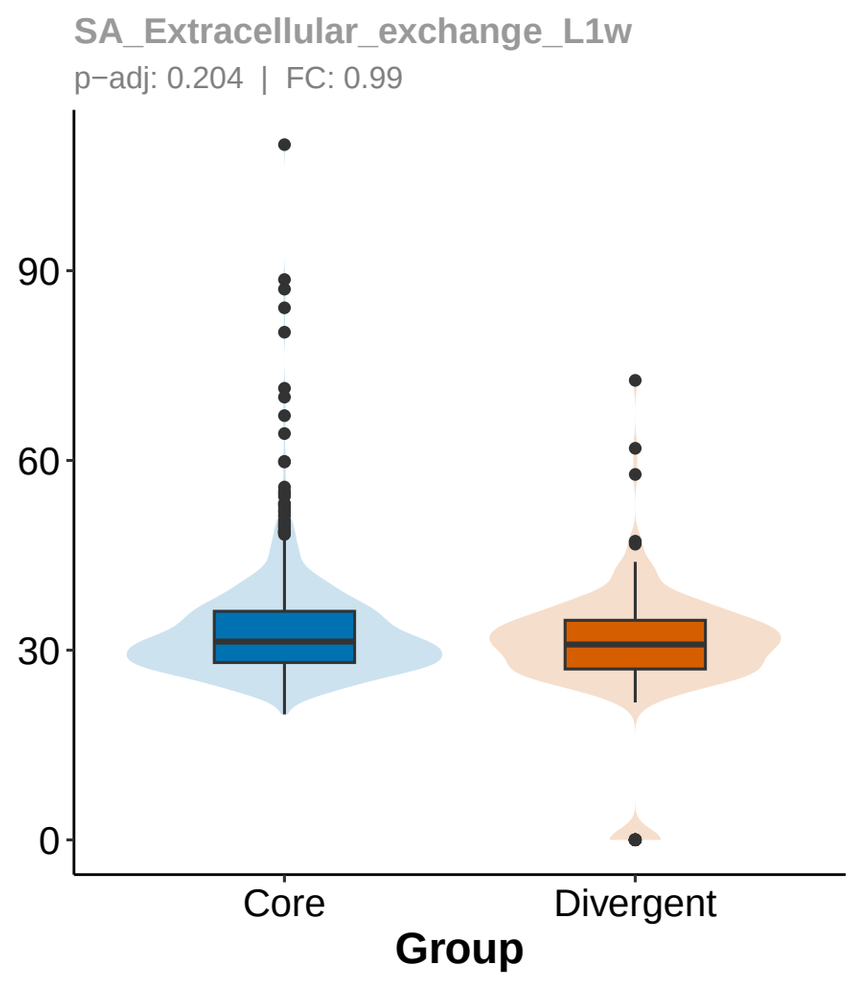












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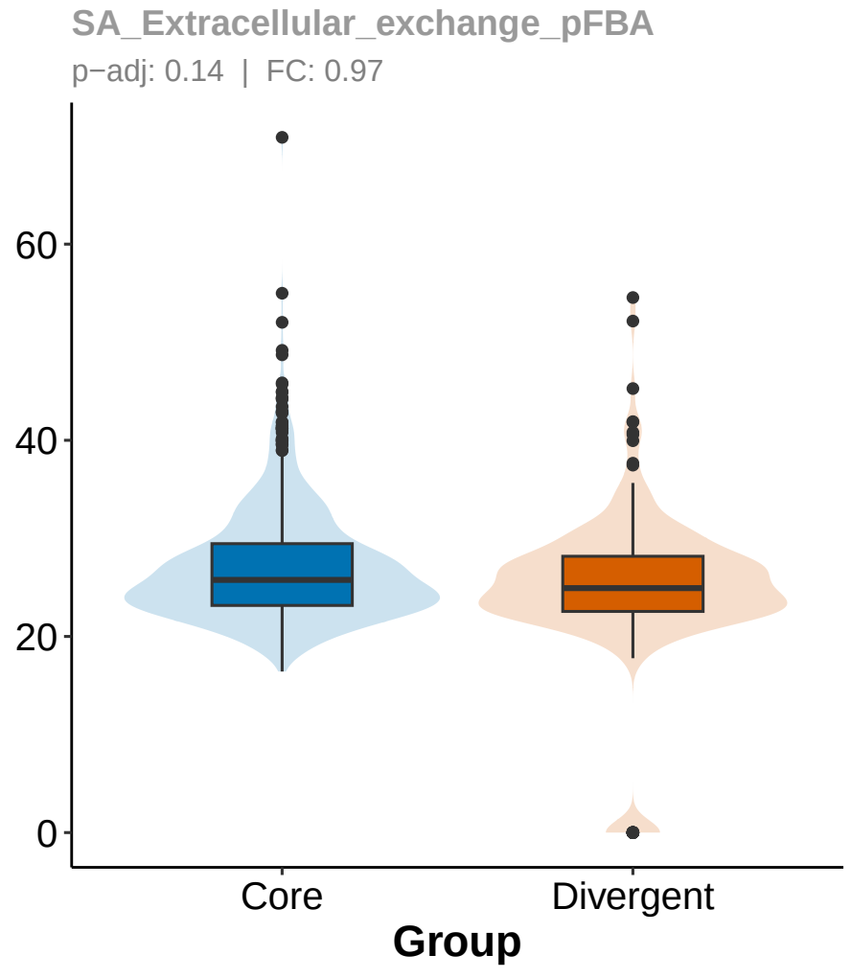

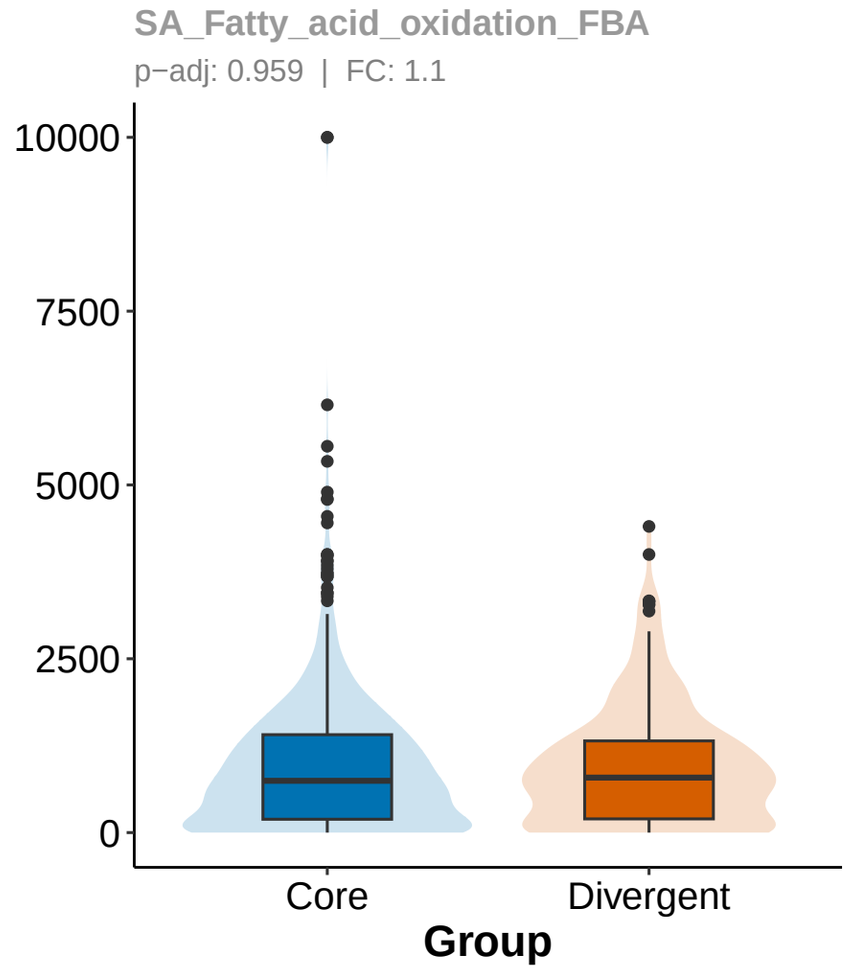

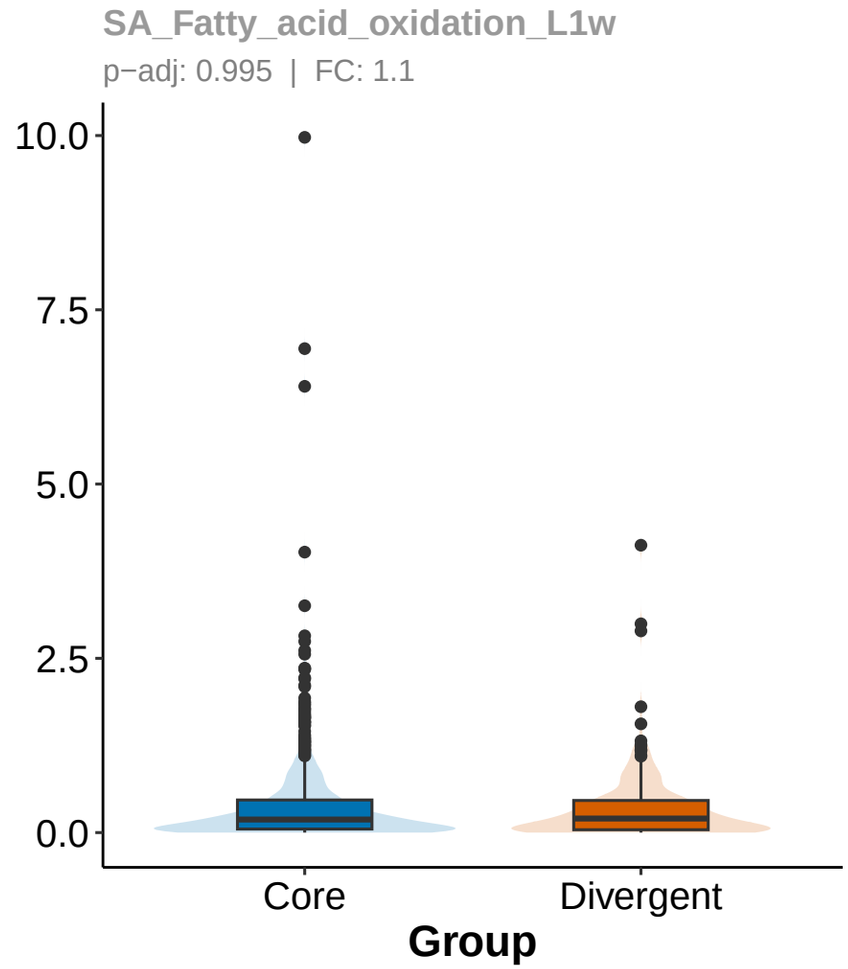

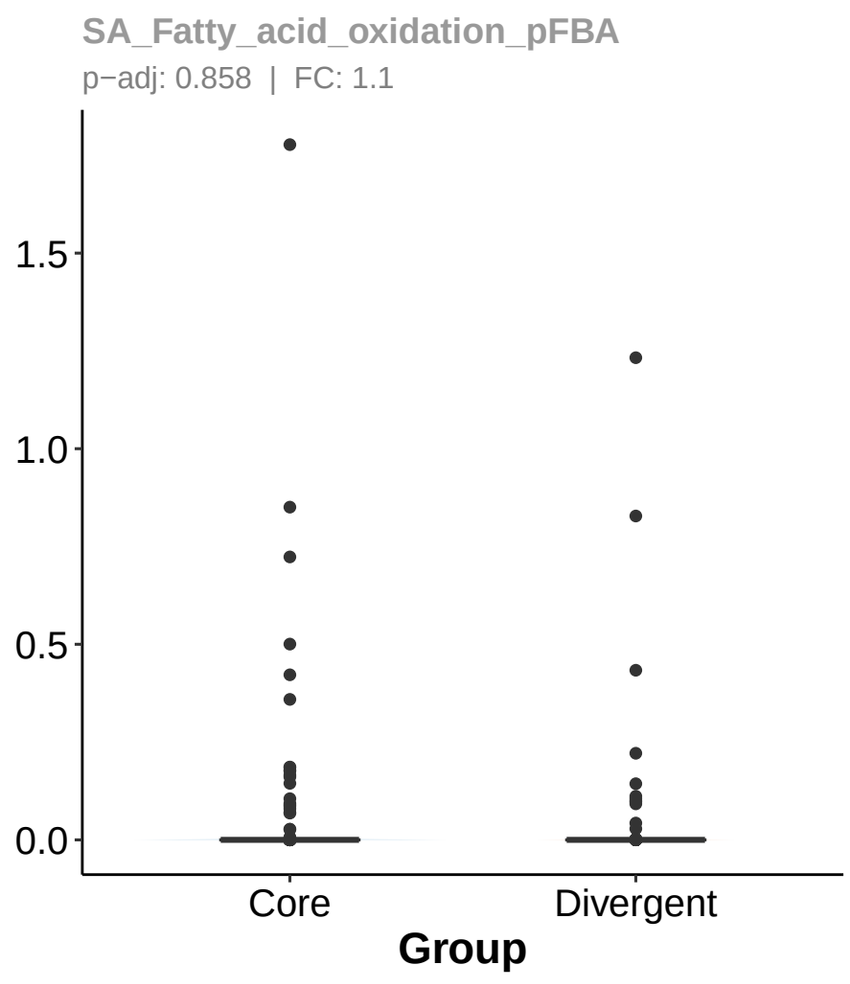

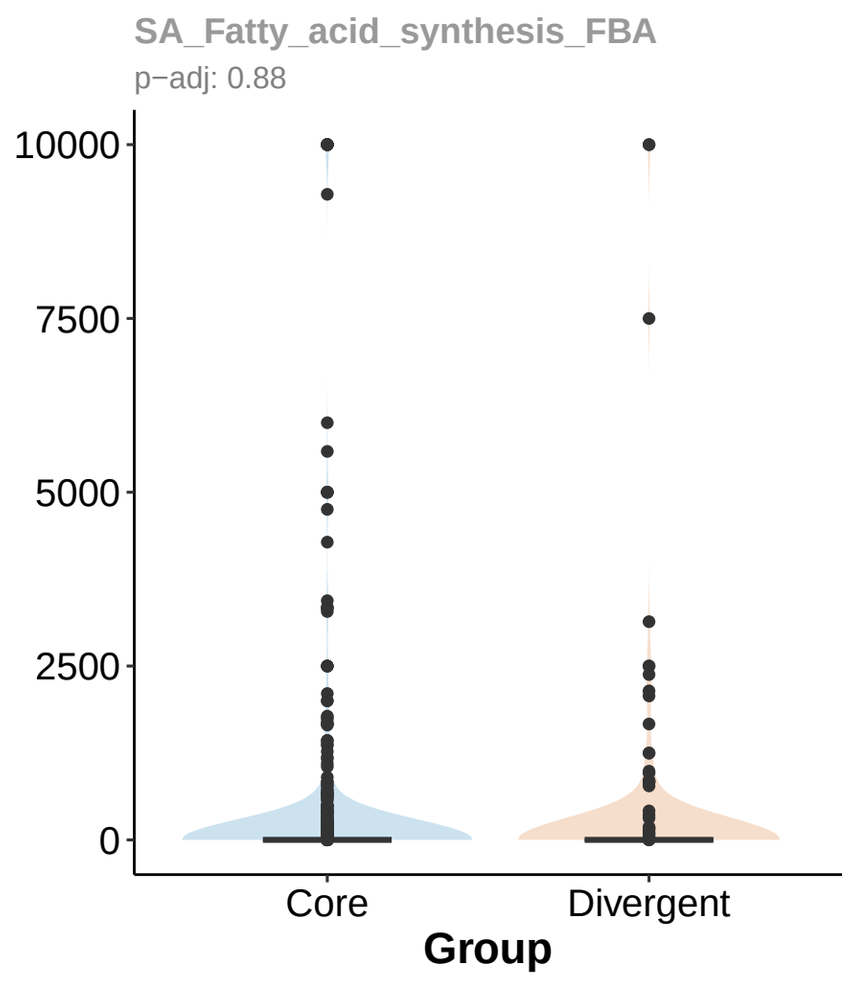

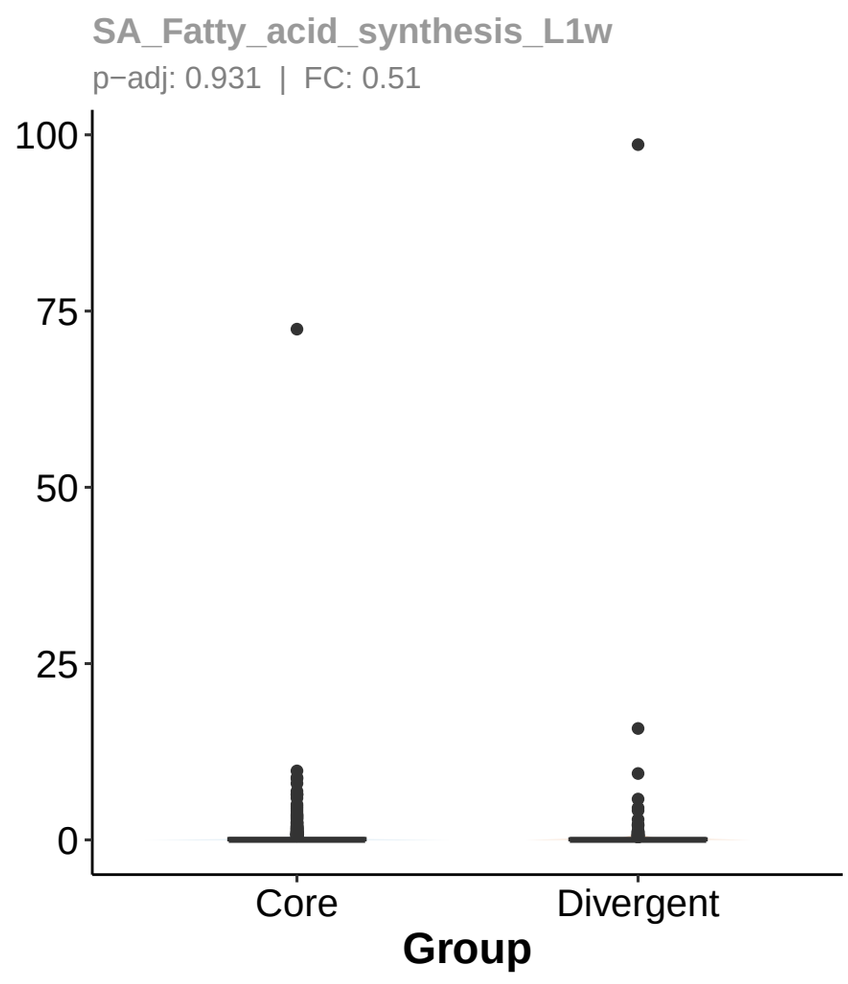

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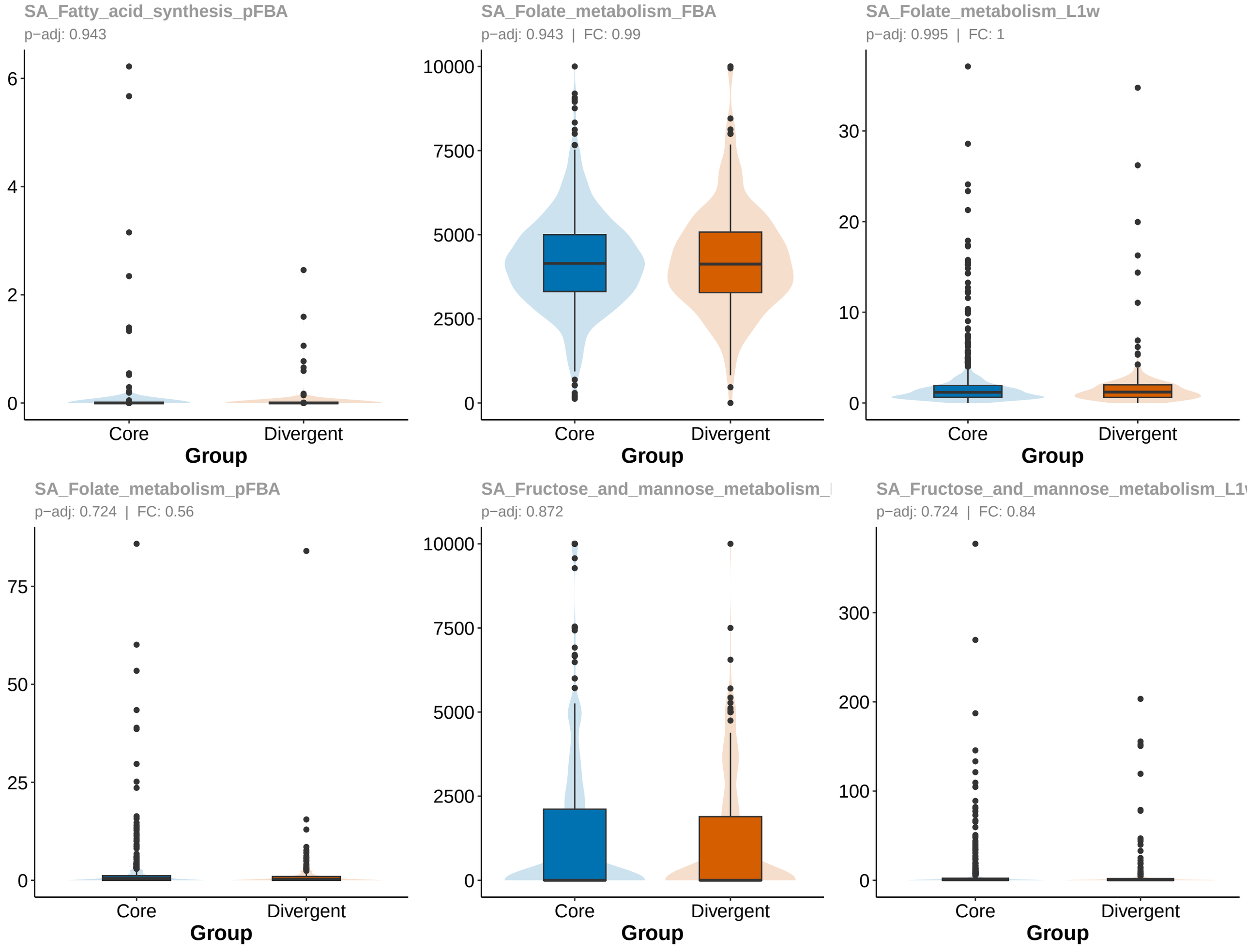

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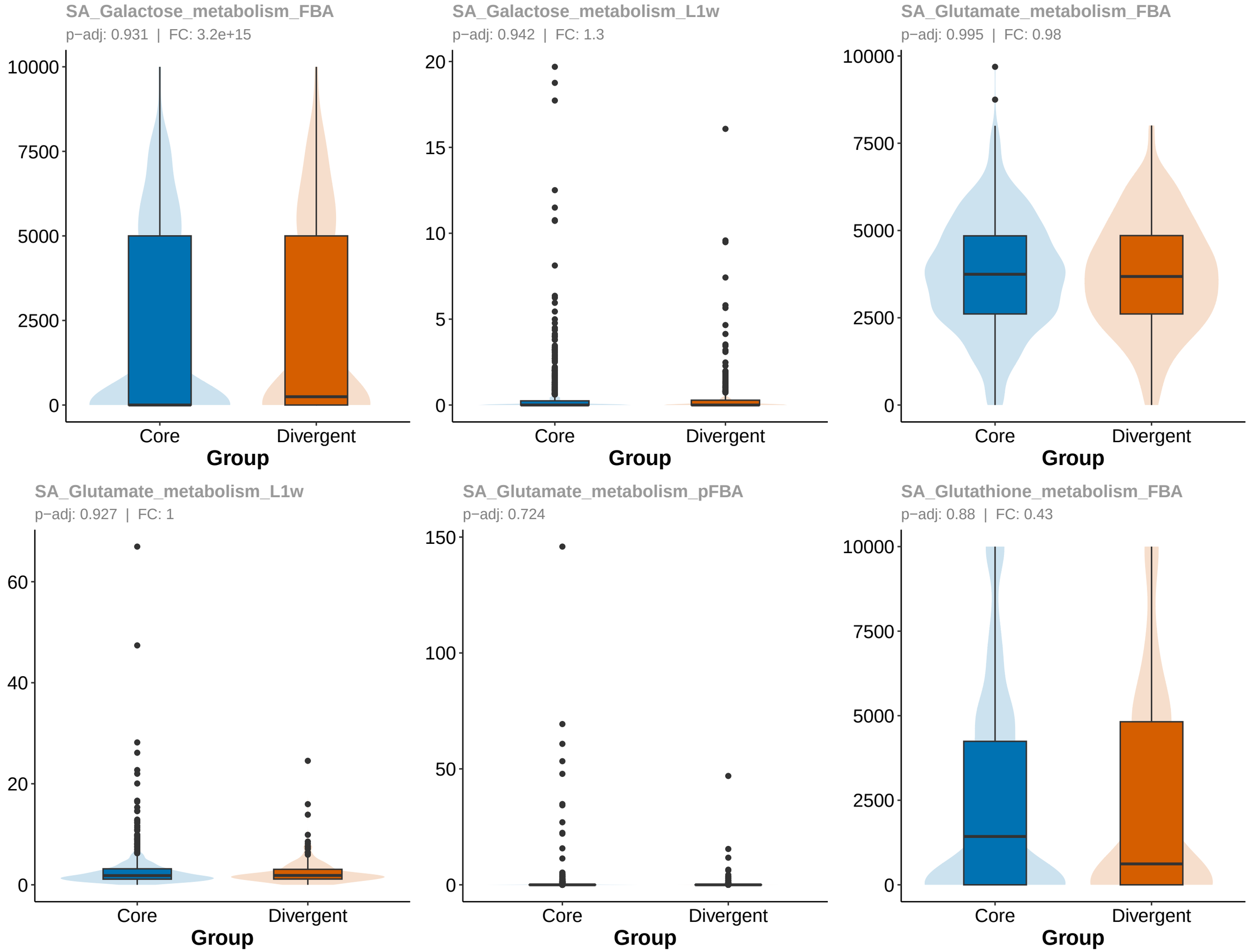

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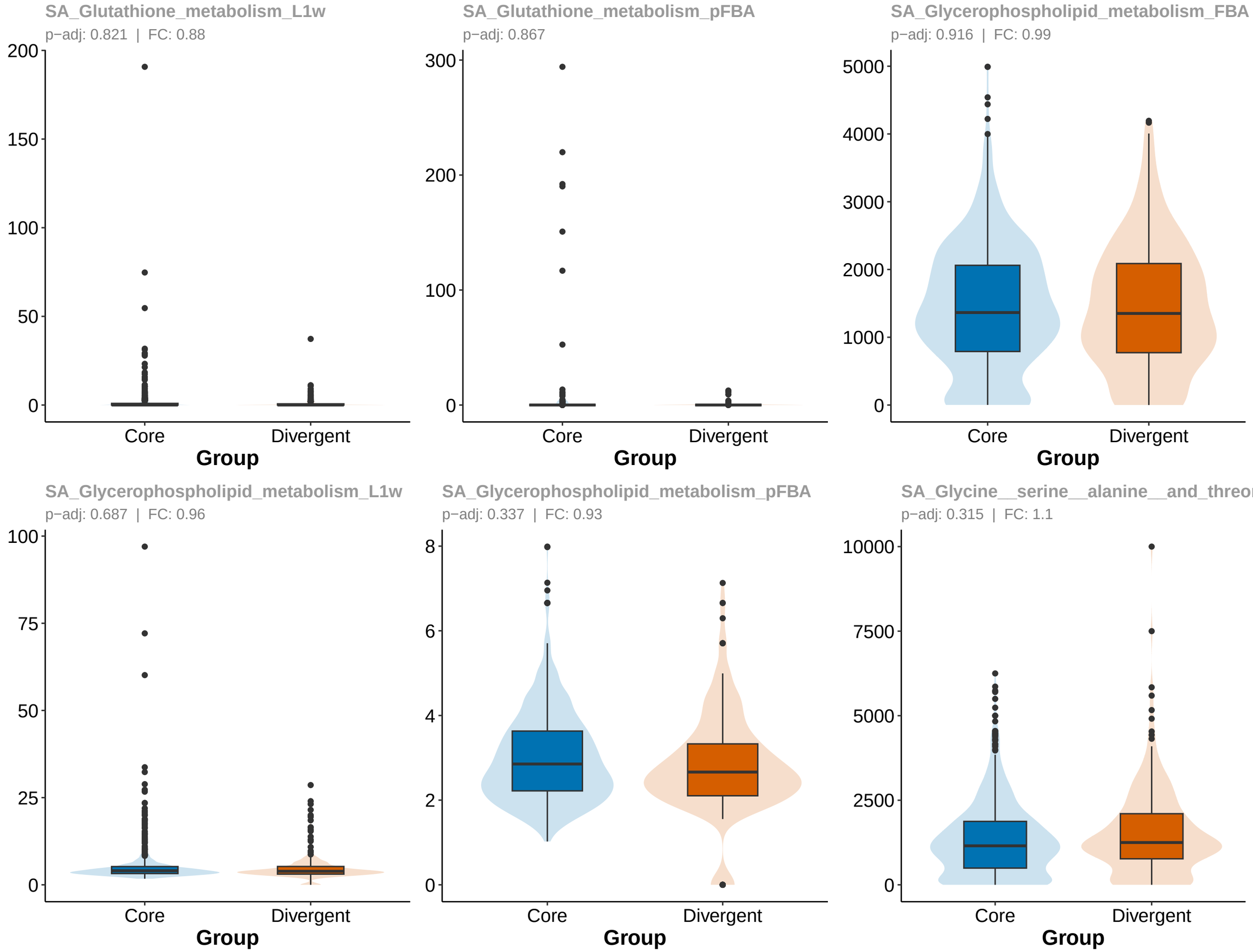

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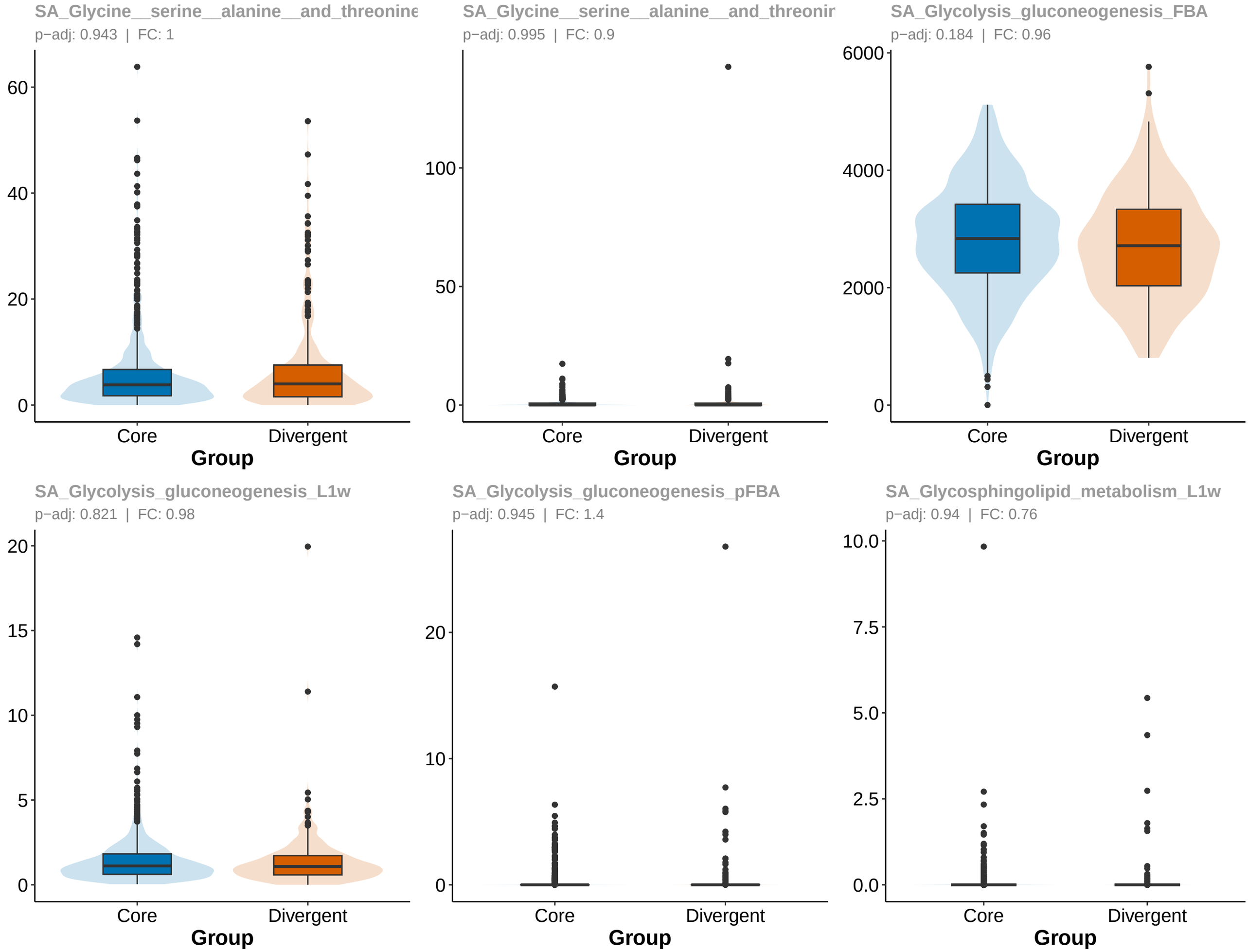

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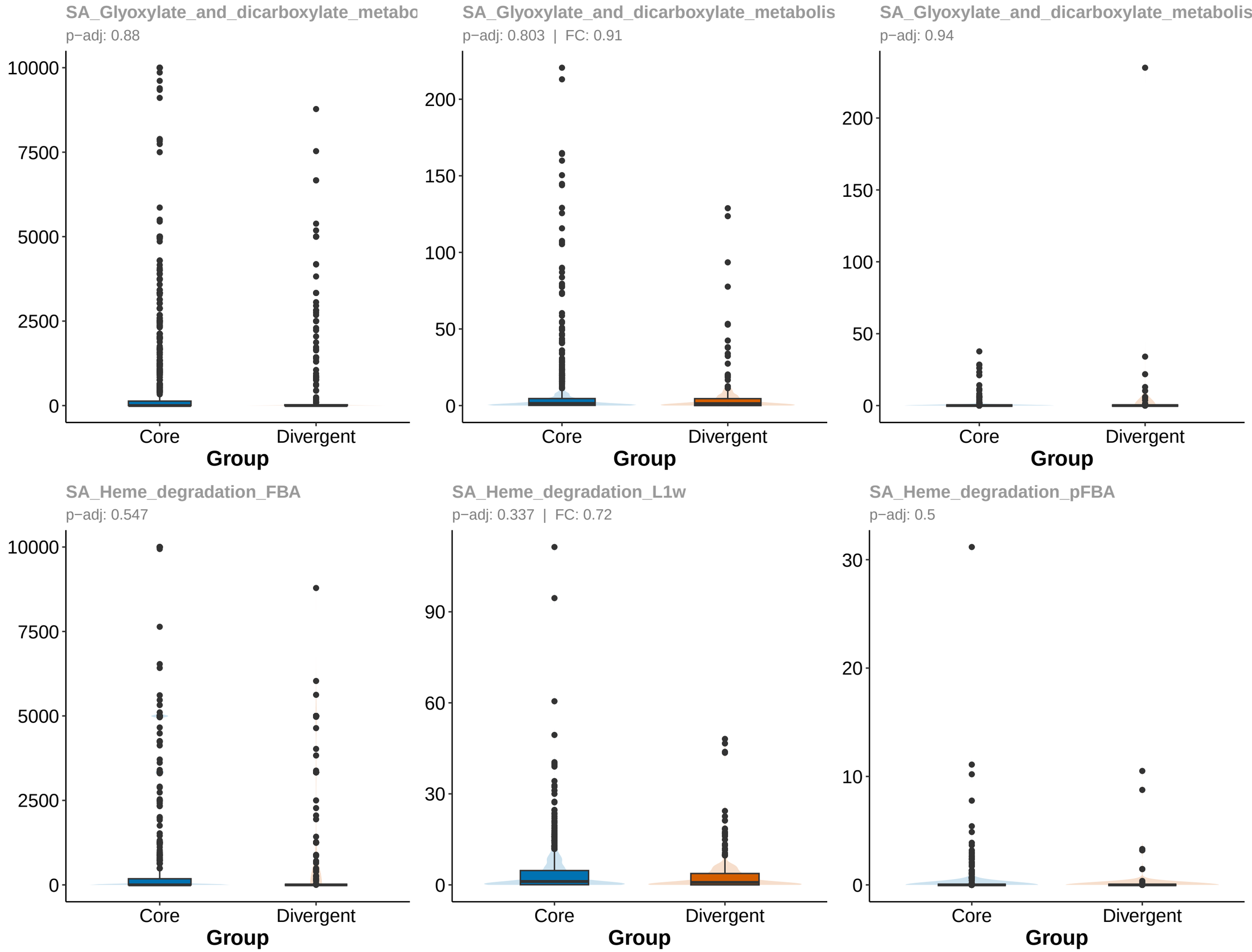

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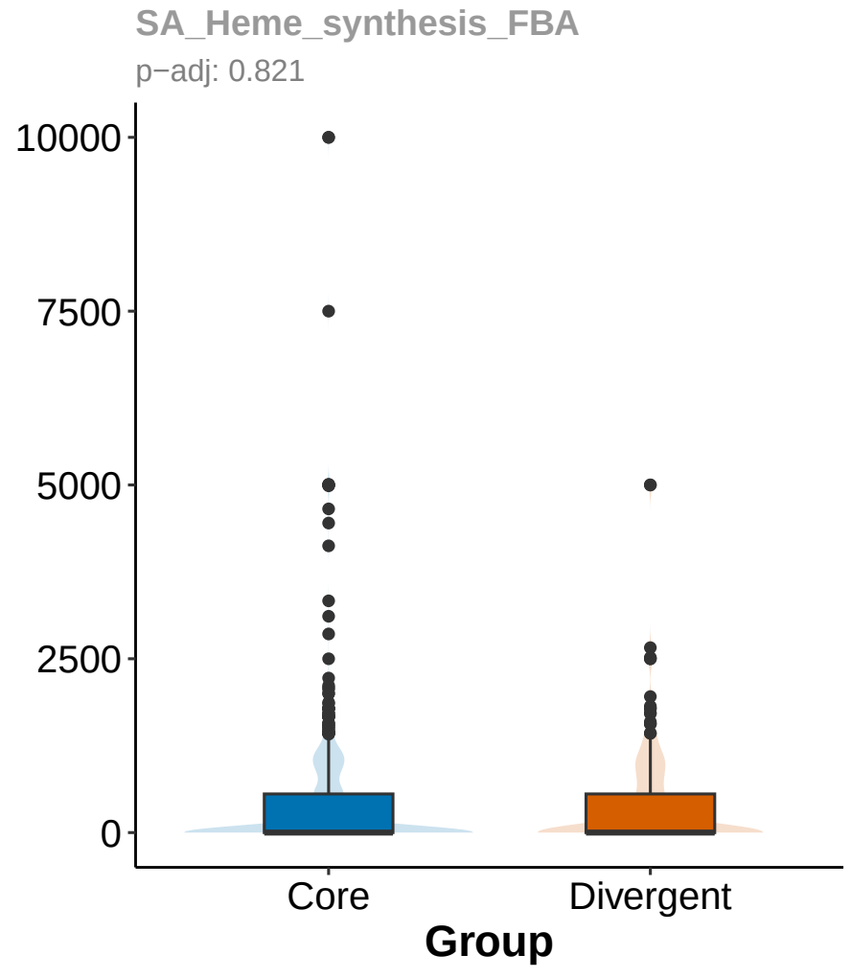

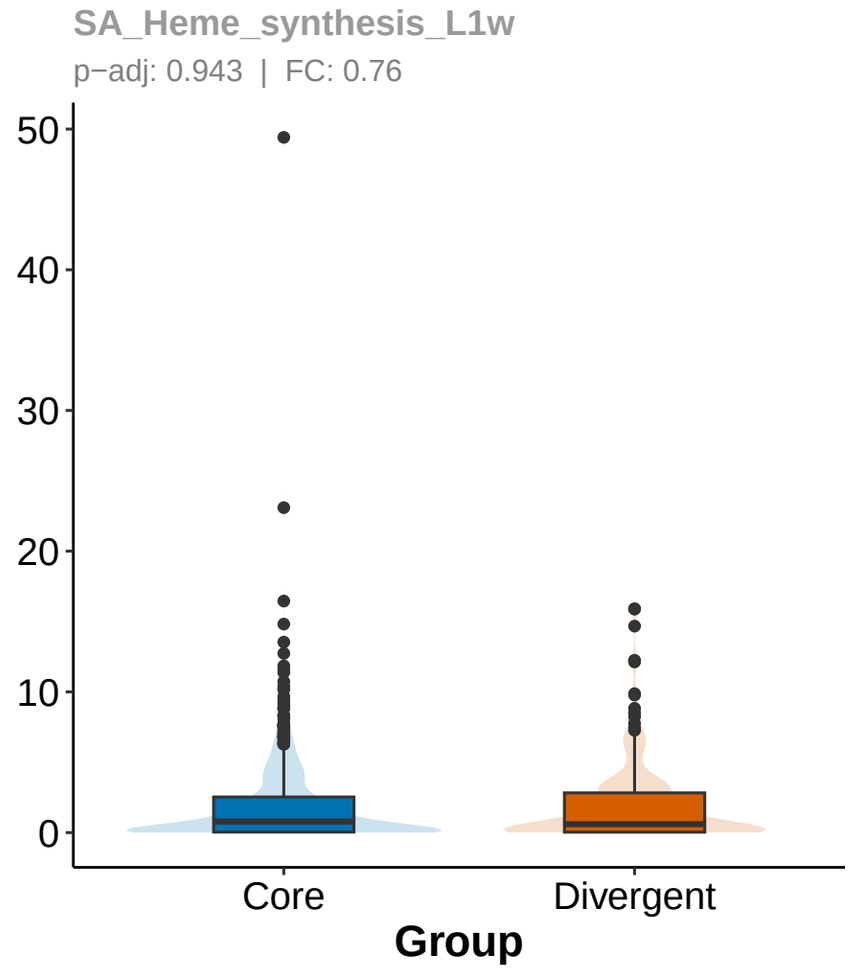

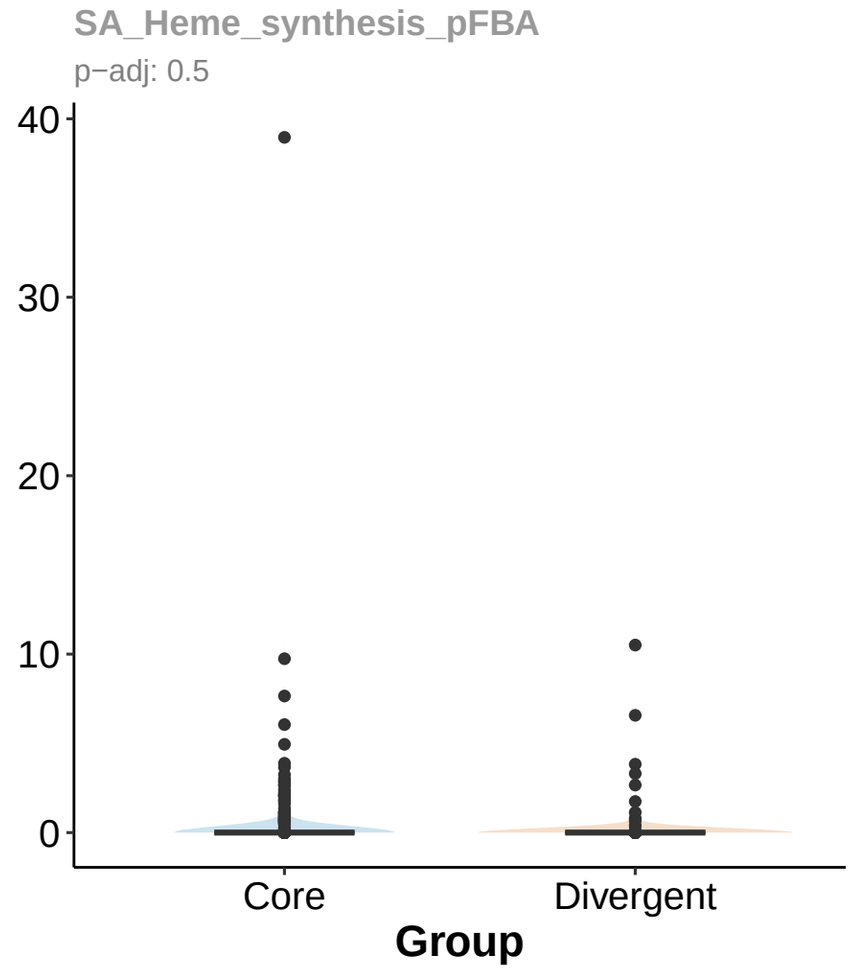

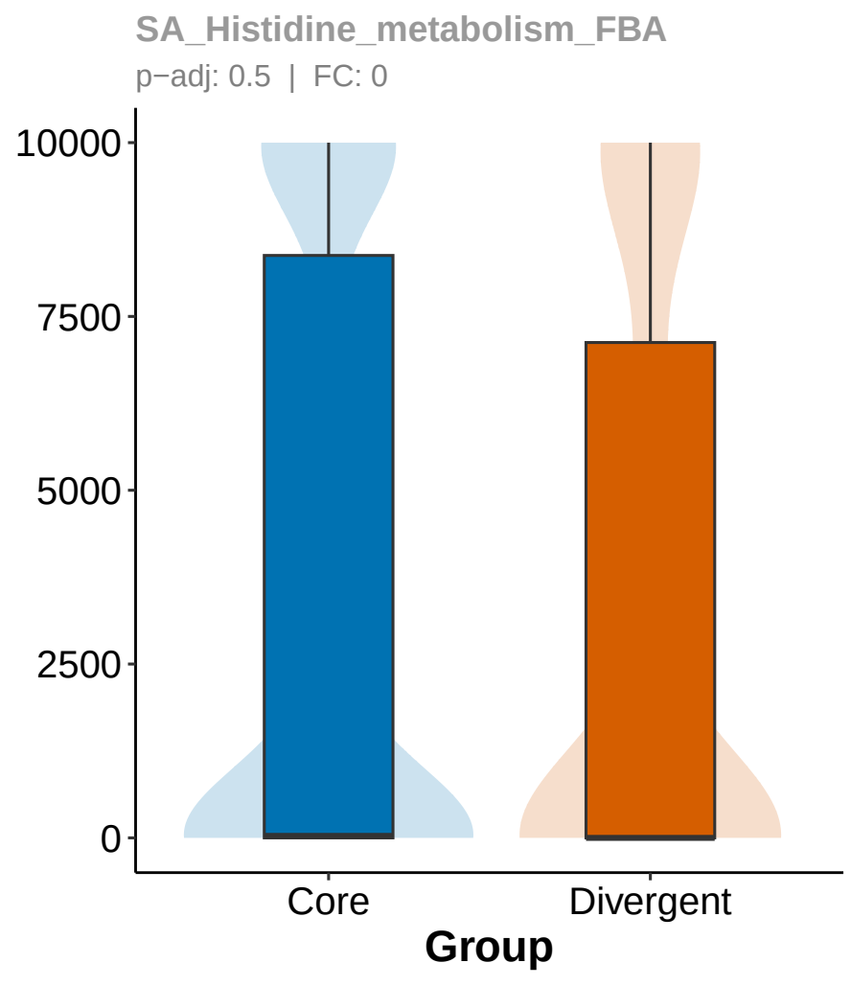

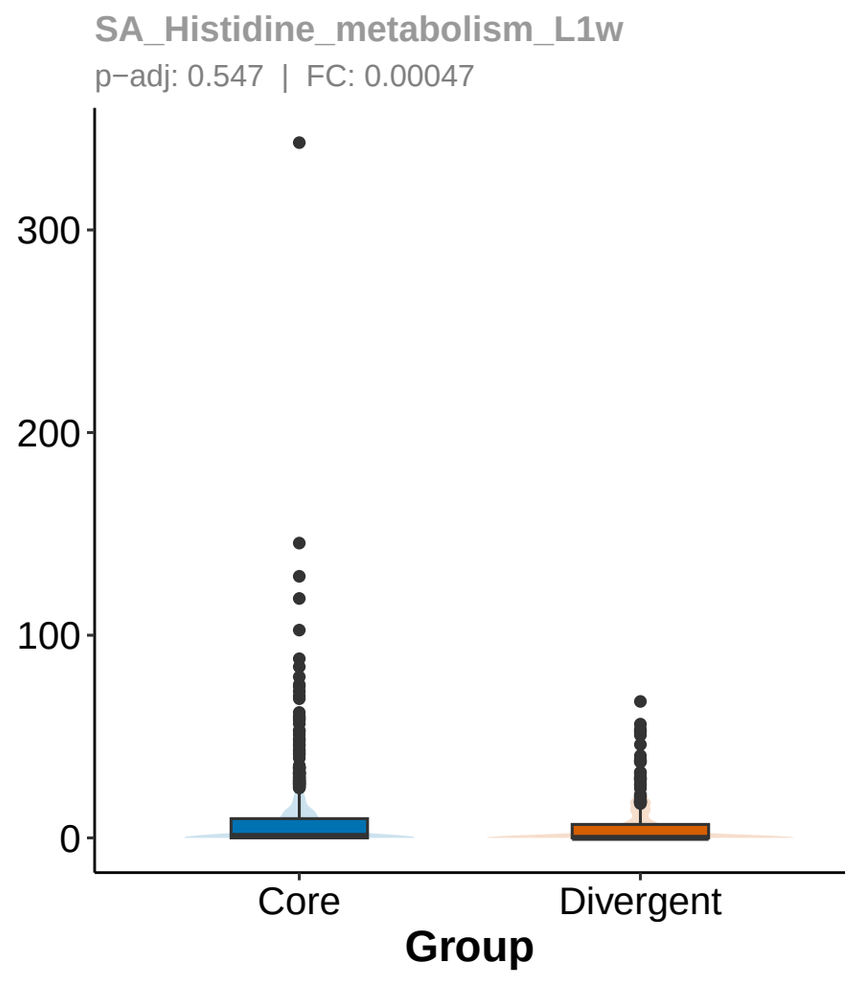

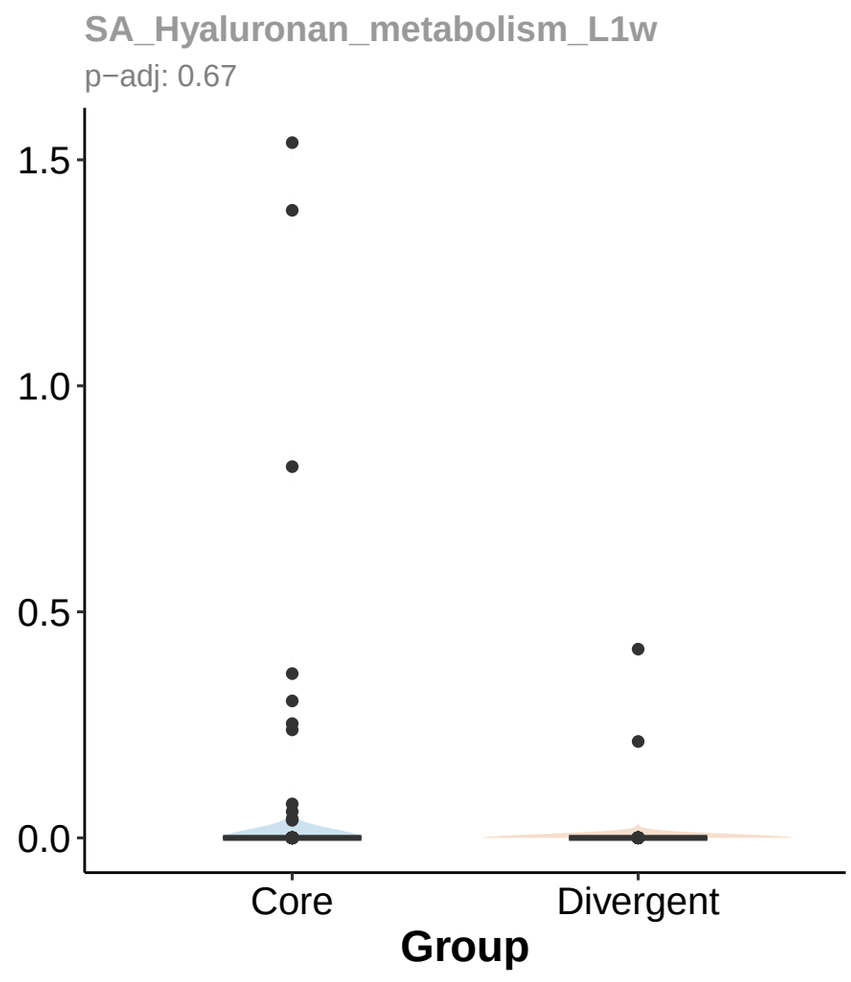

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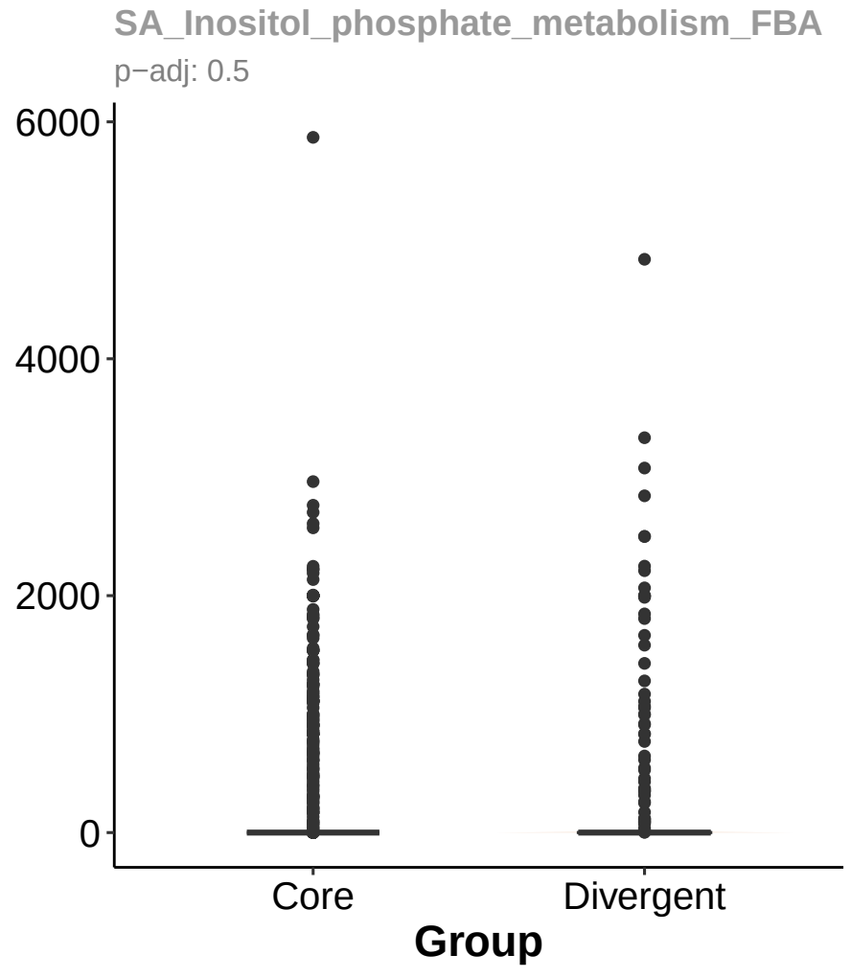
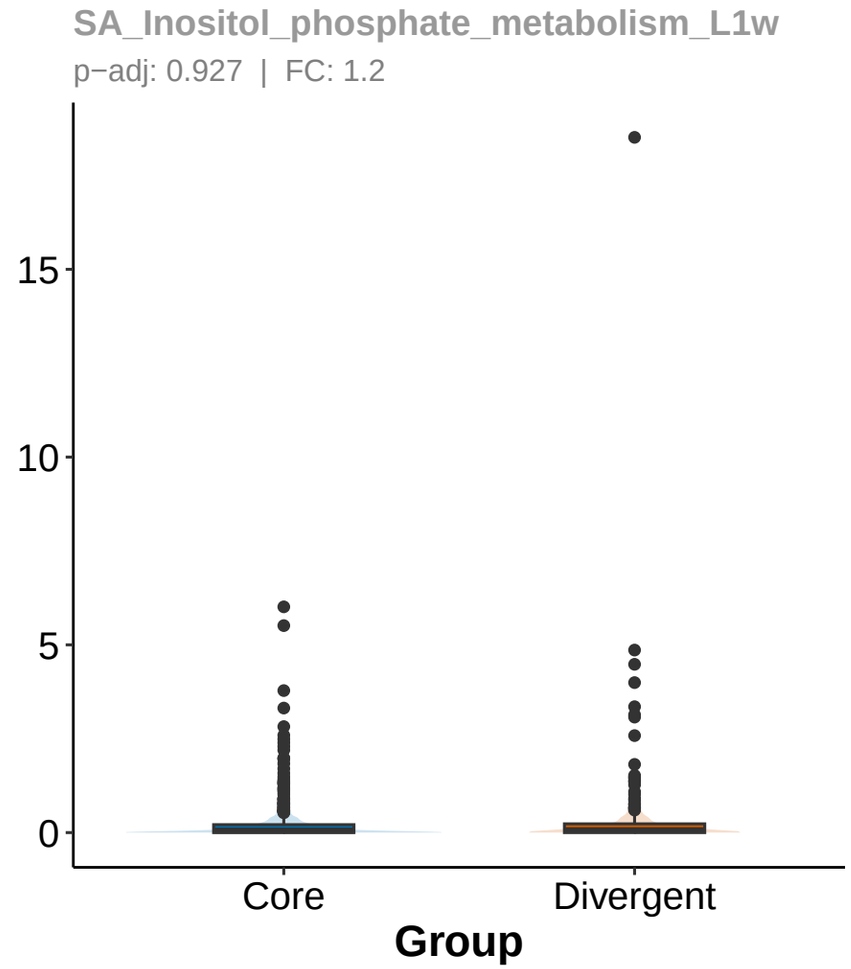
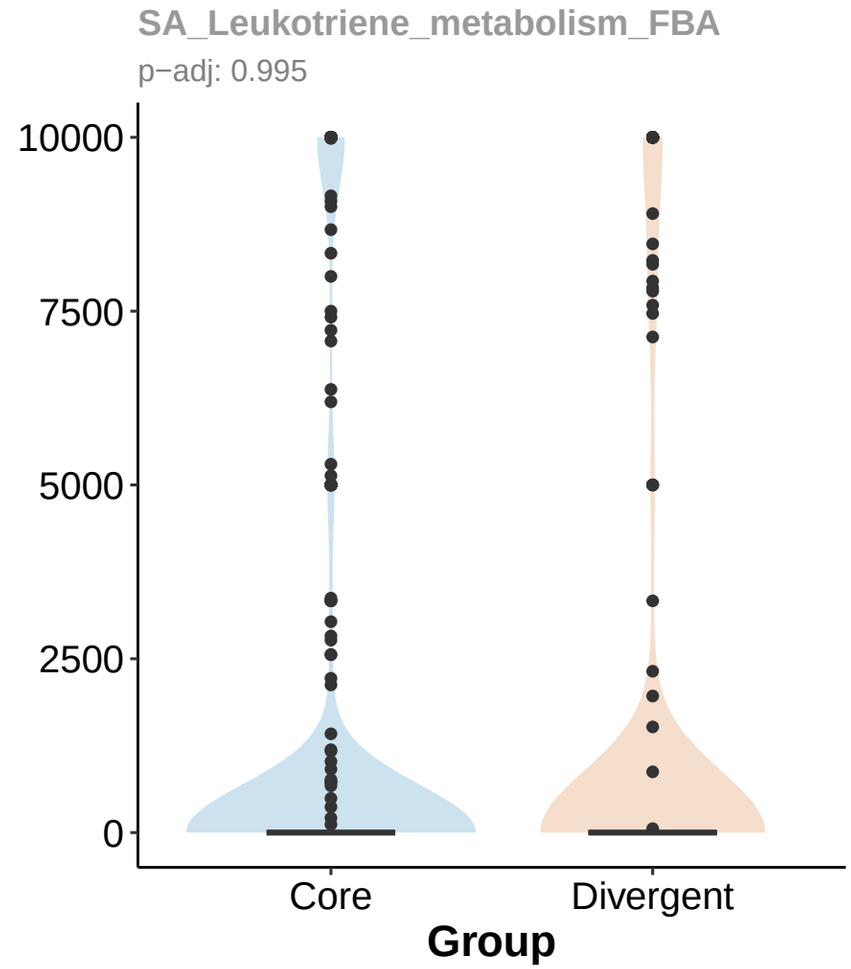
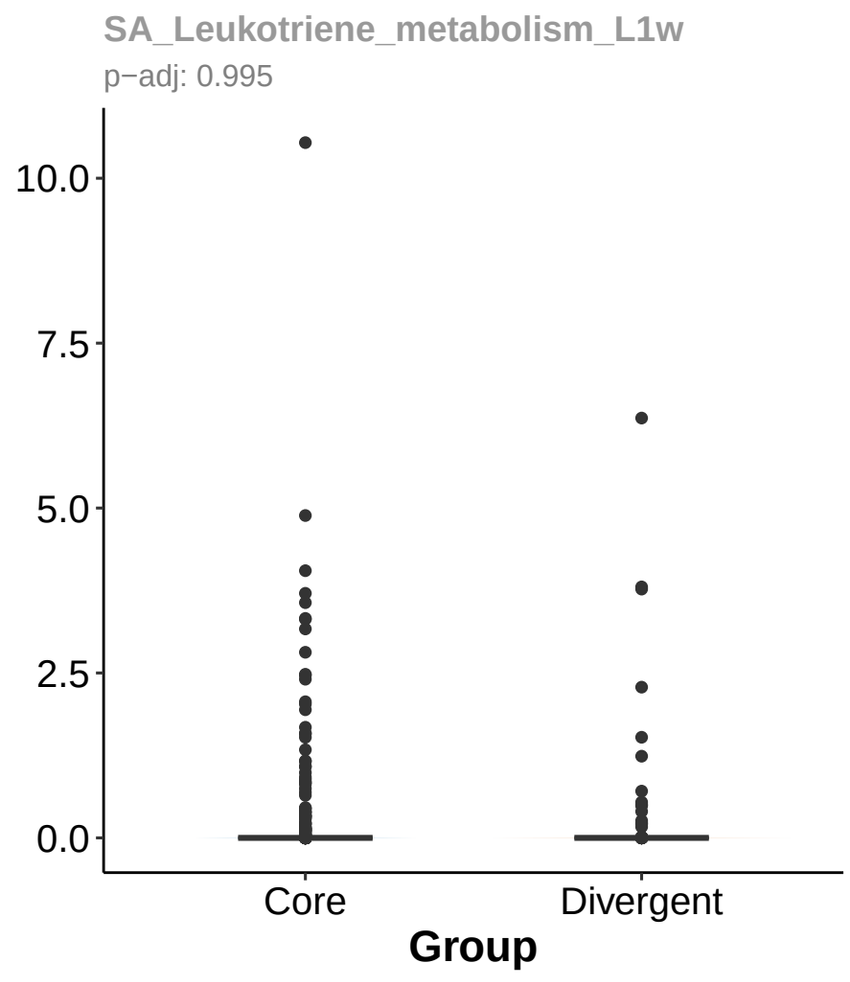
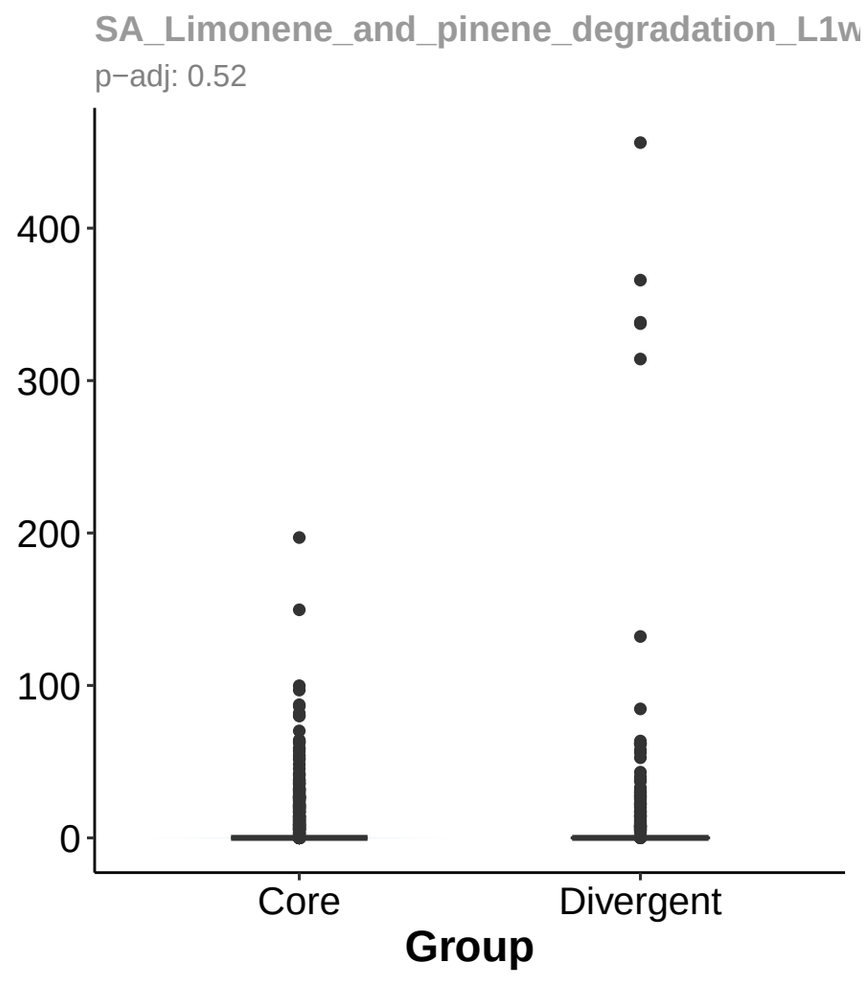
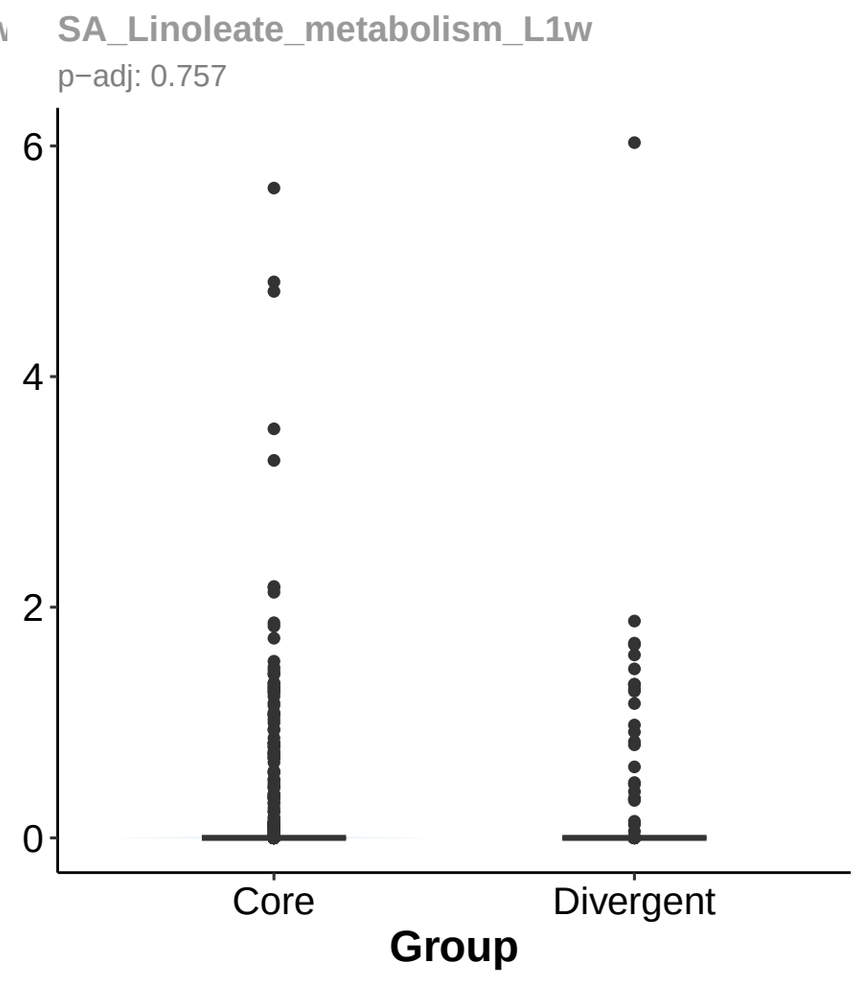




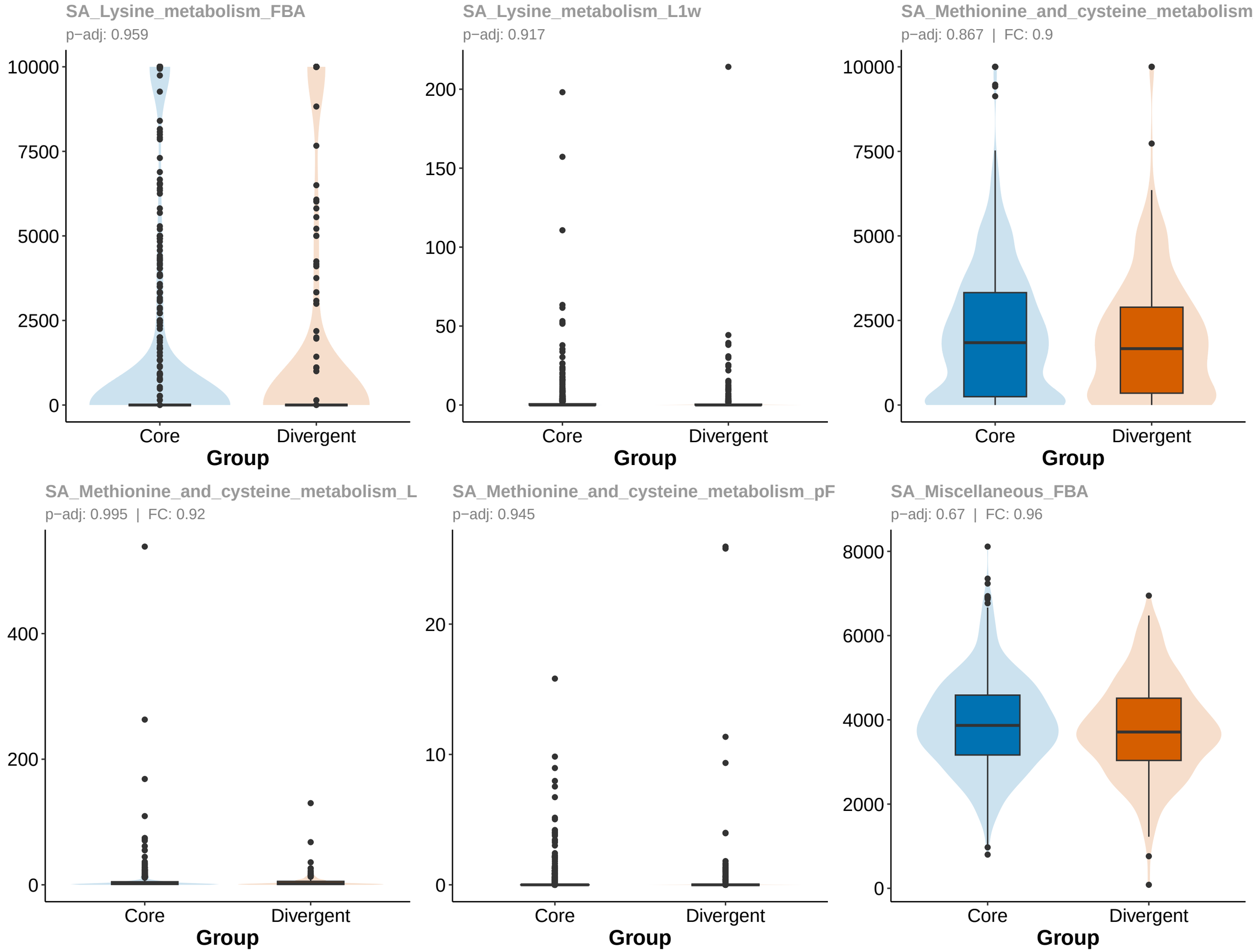








Metabolismo − . 5. Reprogramación Sistémica (SA_*)



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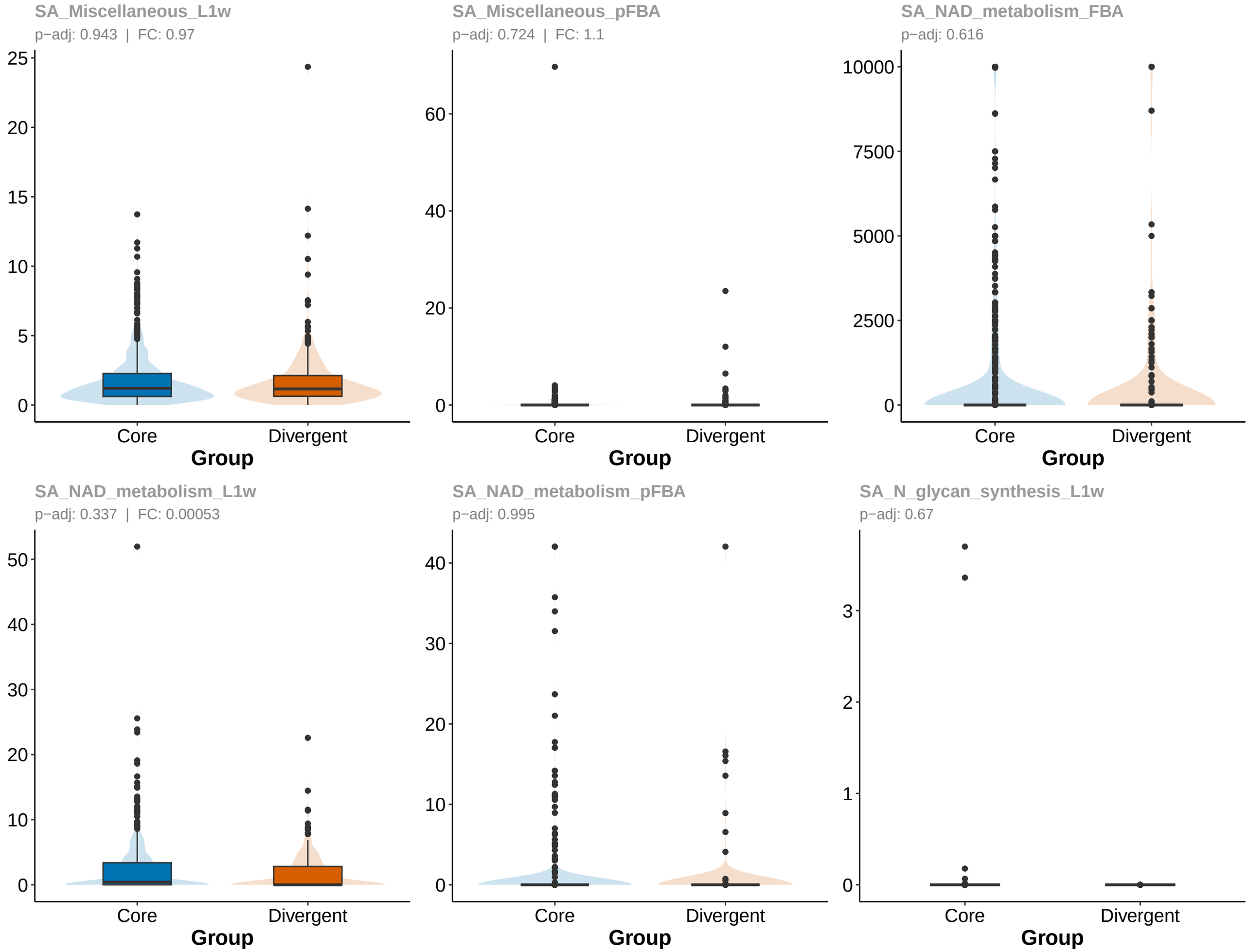

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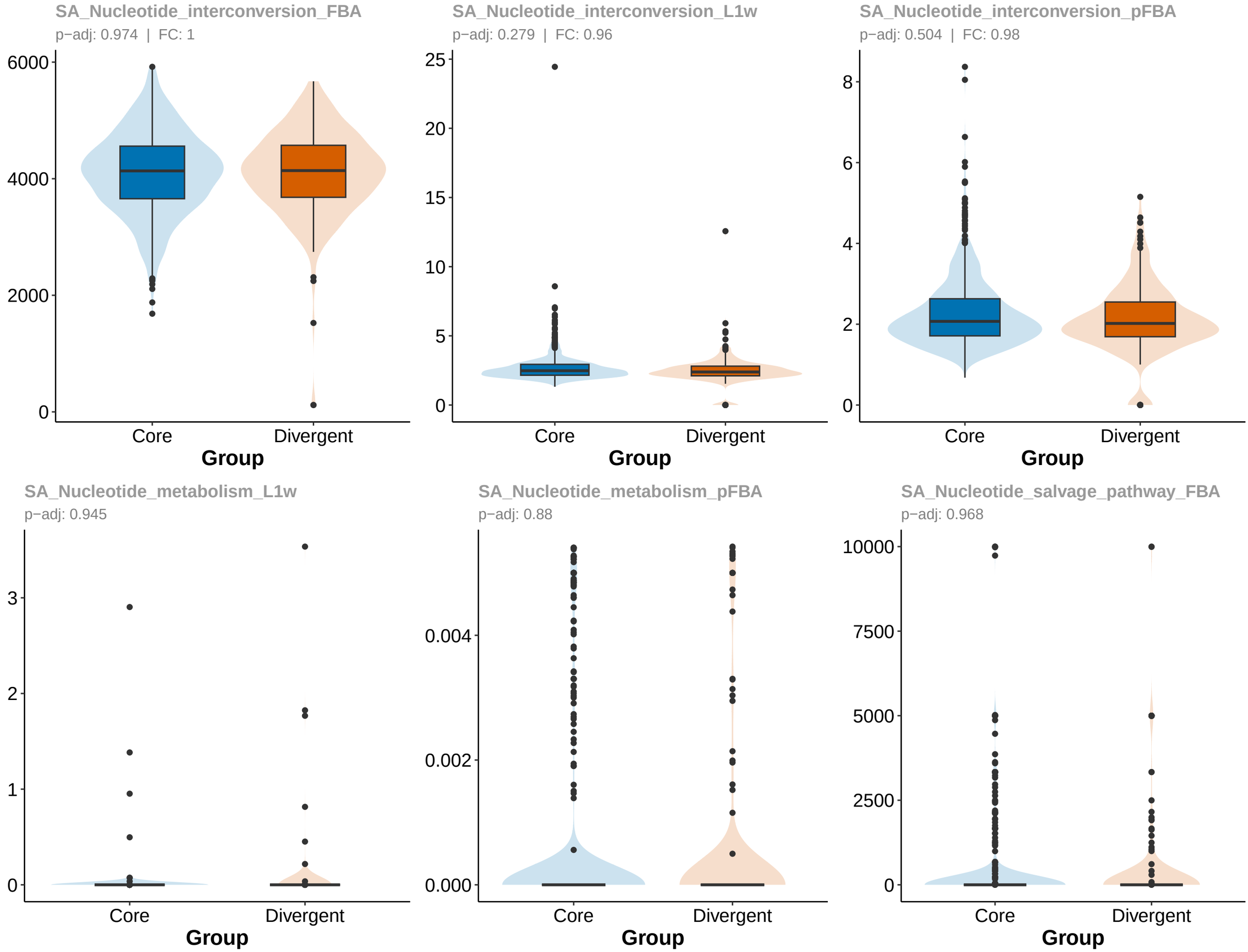

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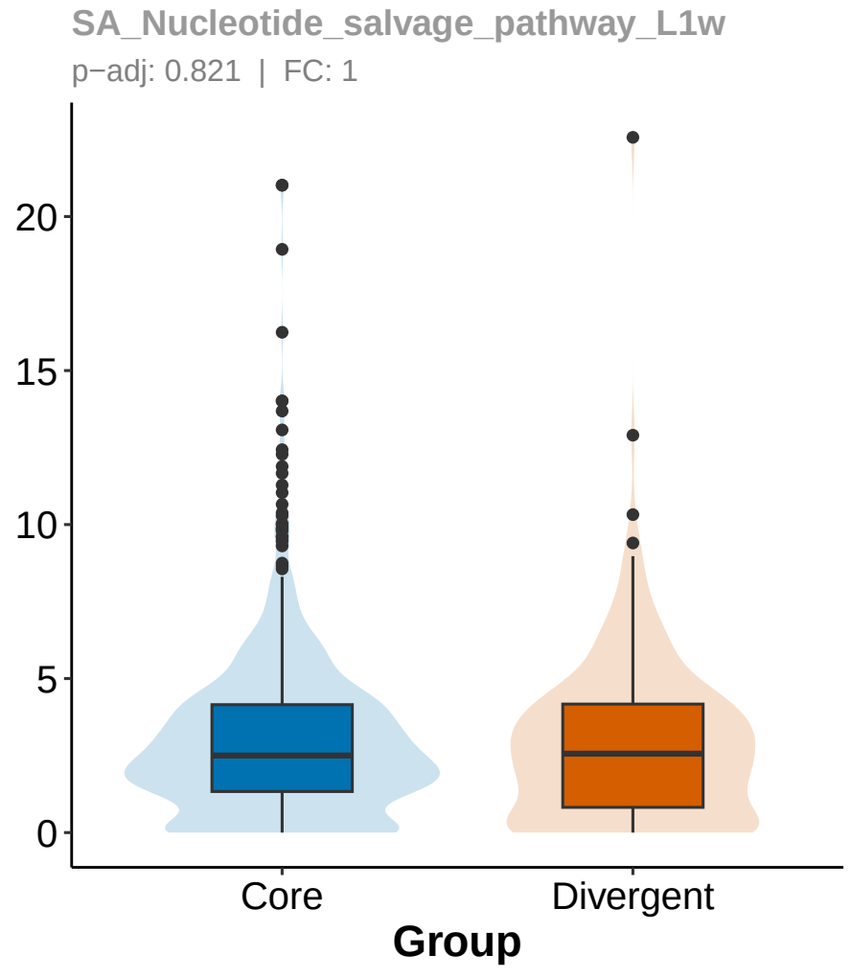
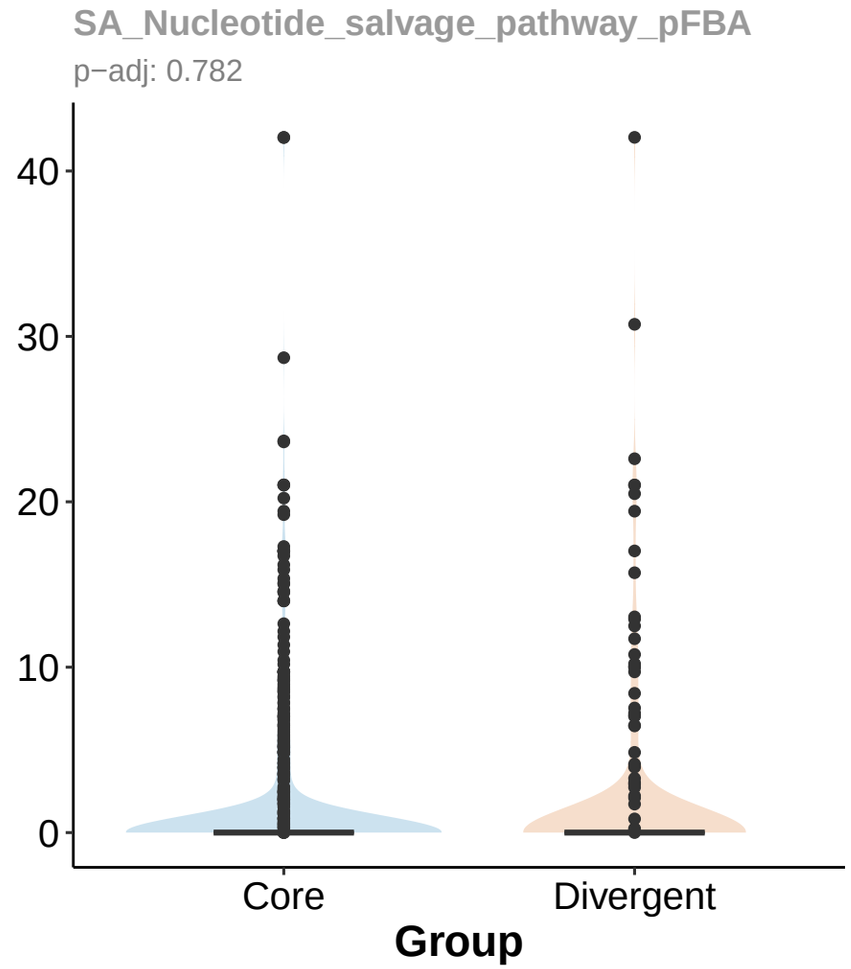
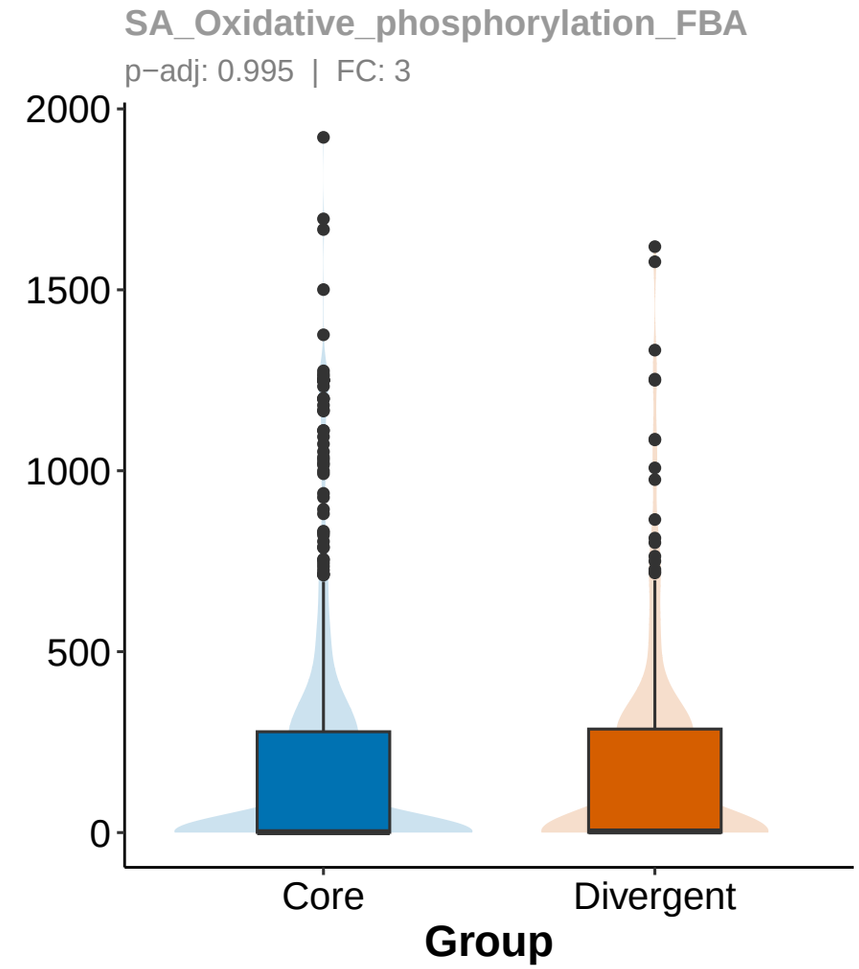
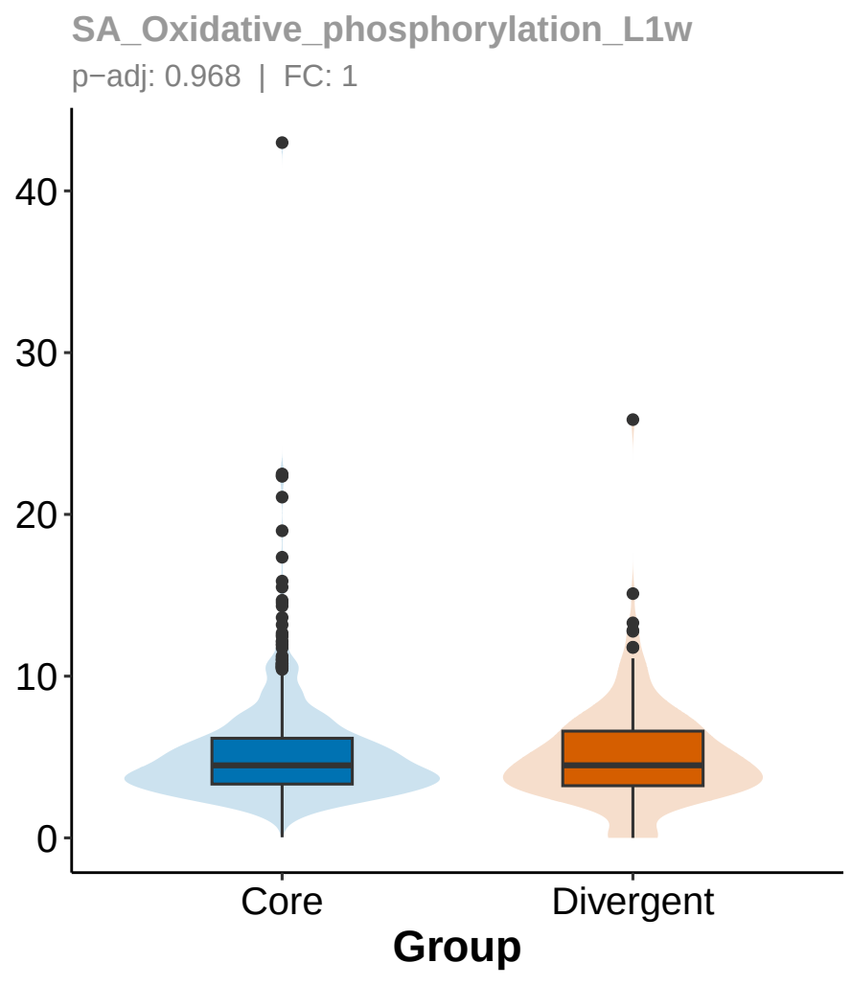
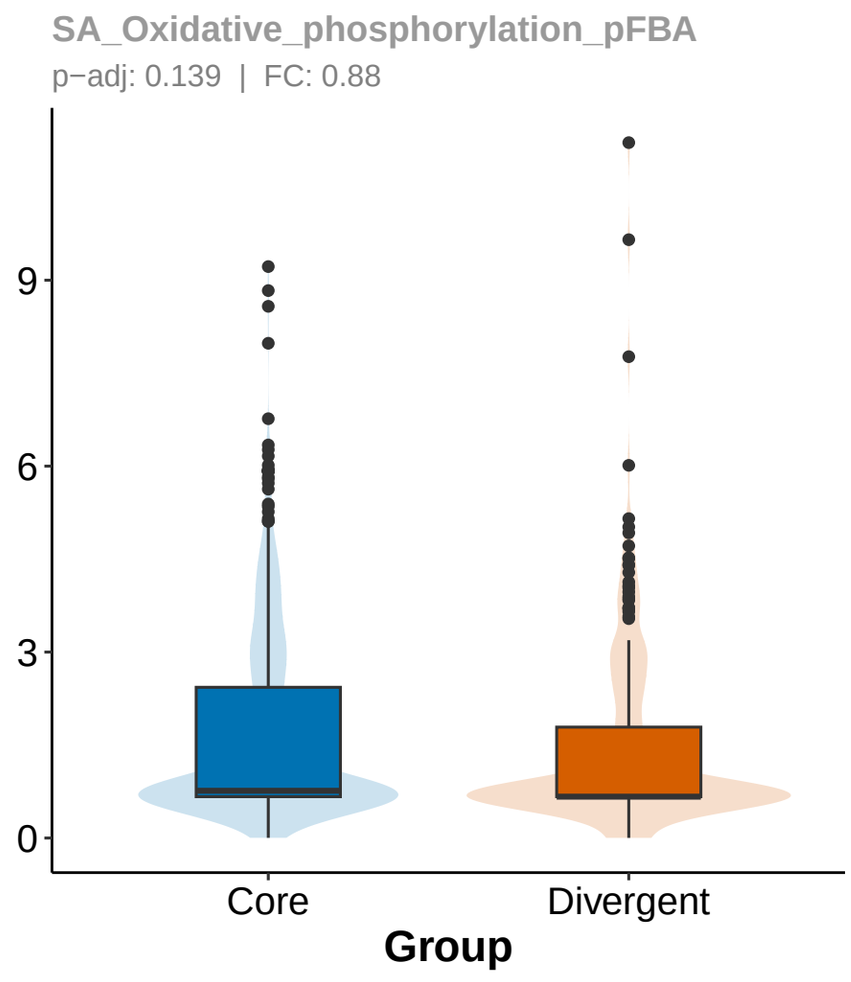
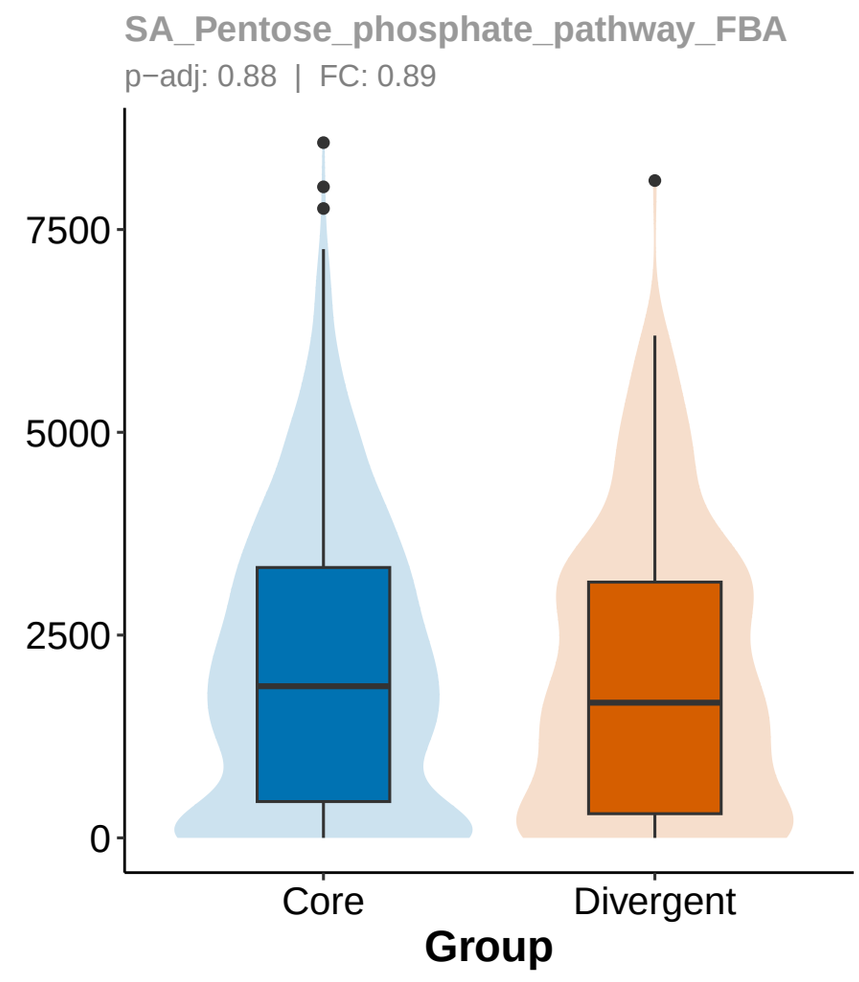




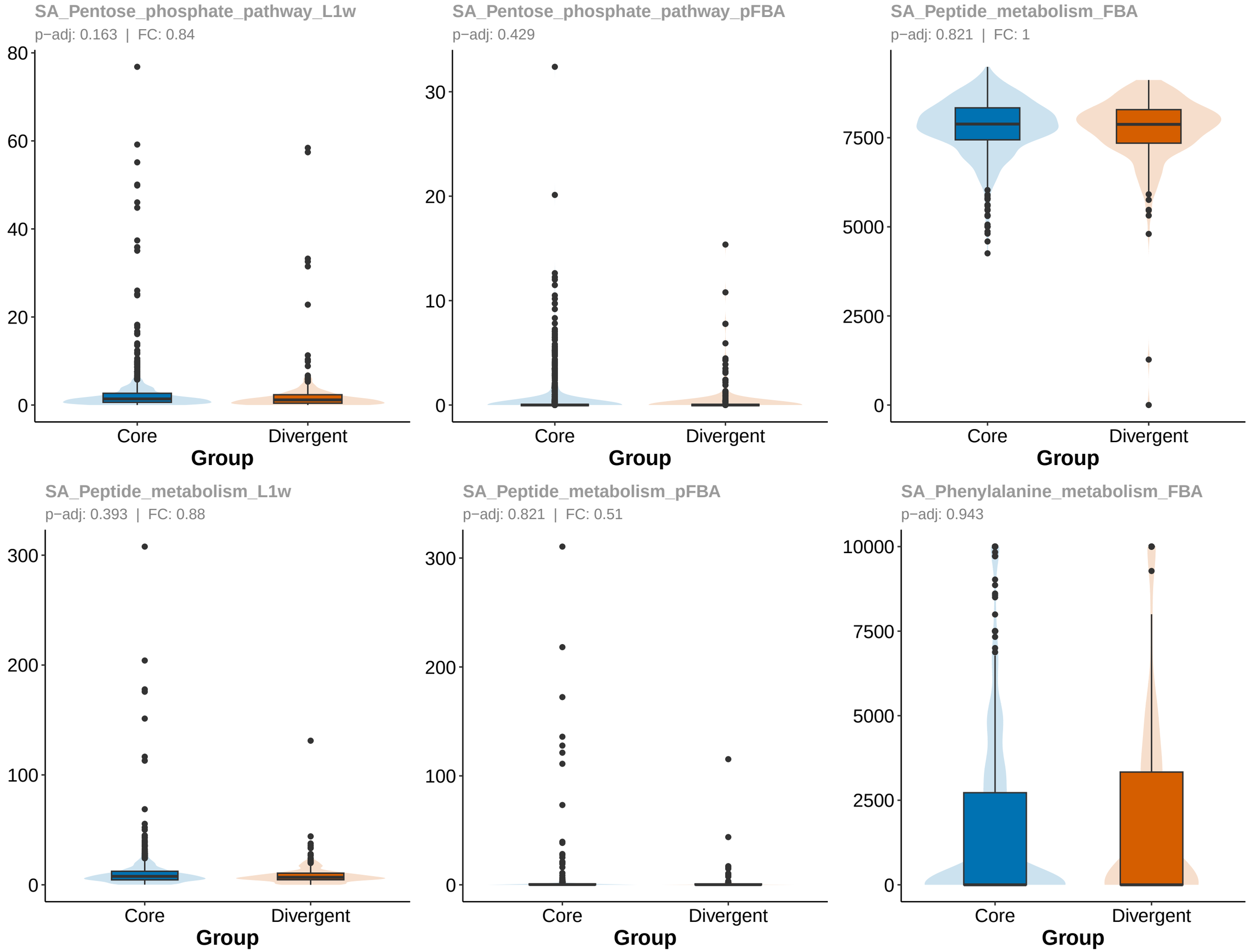








Metabolismo − . 5. Reprogramación Sistémica (SA_*)



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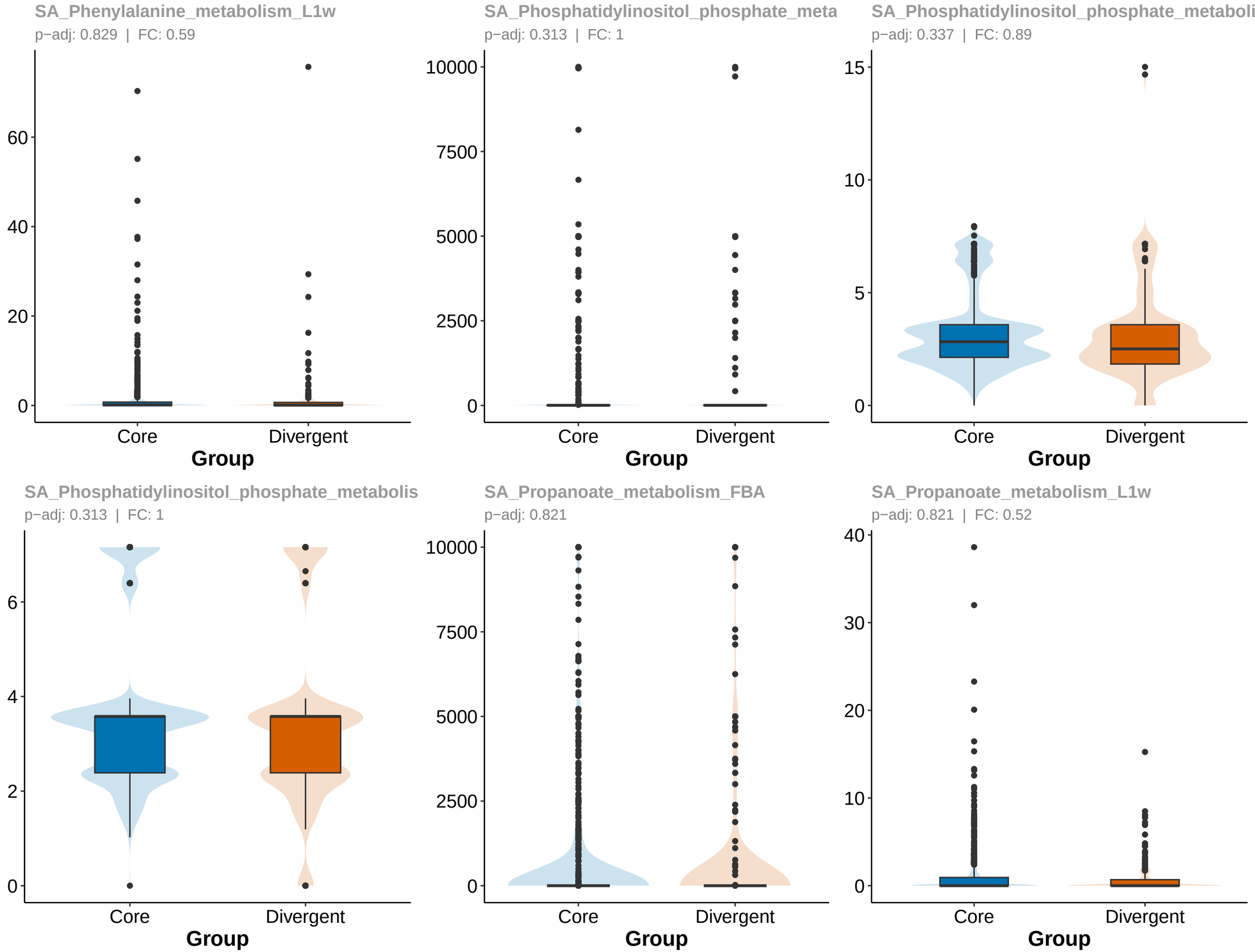

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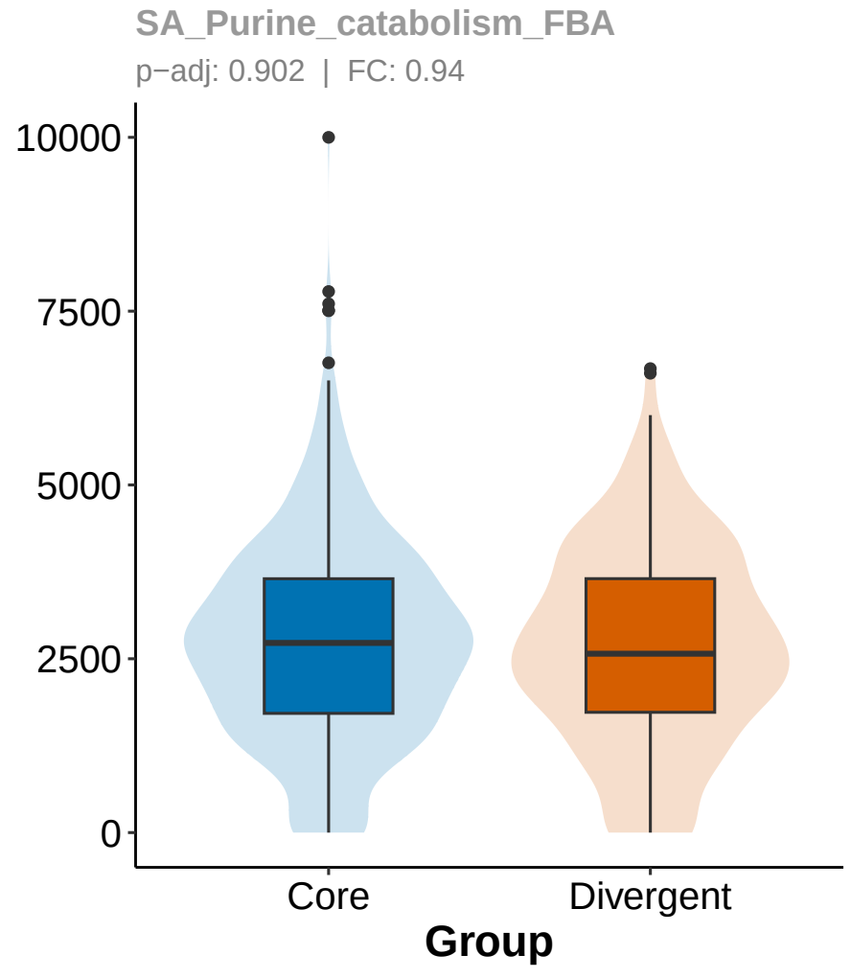

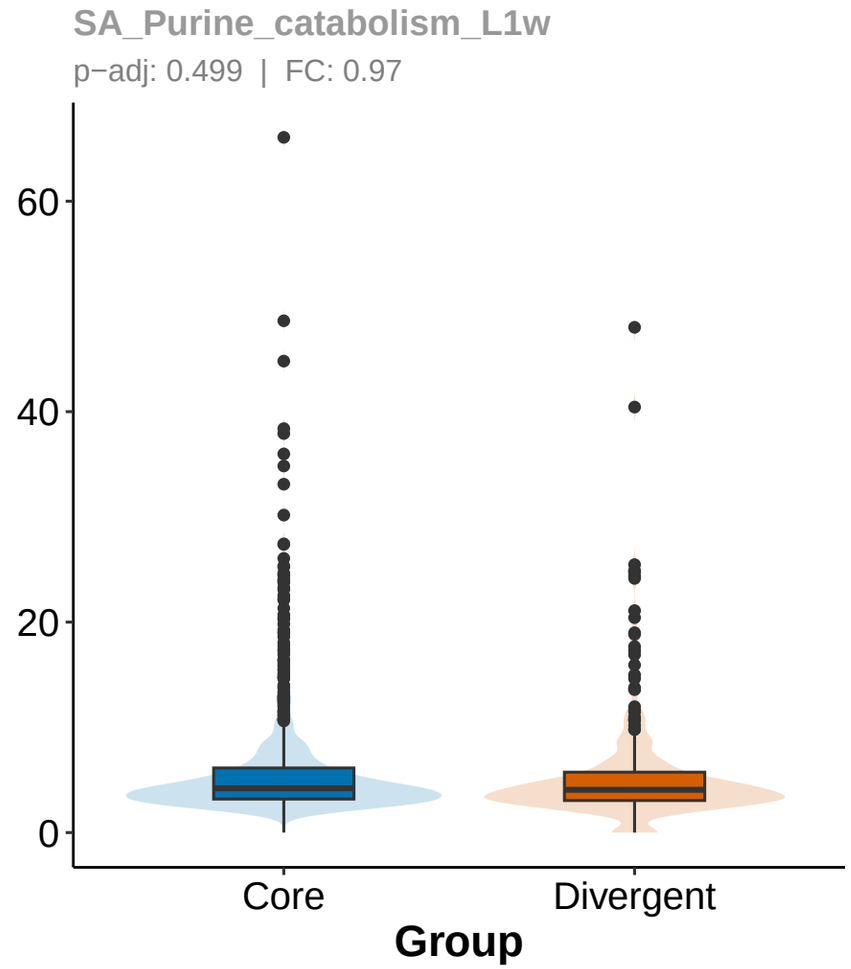

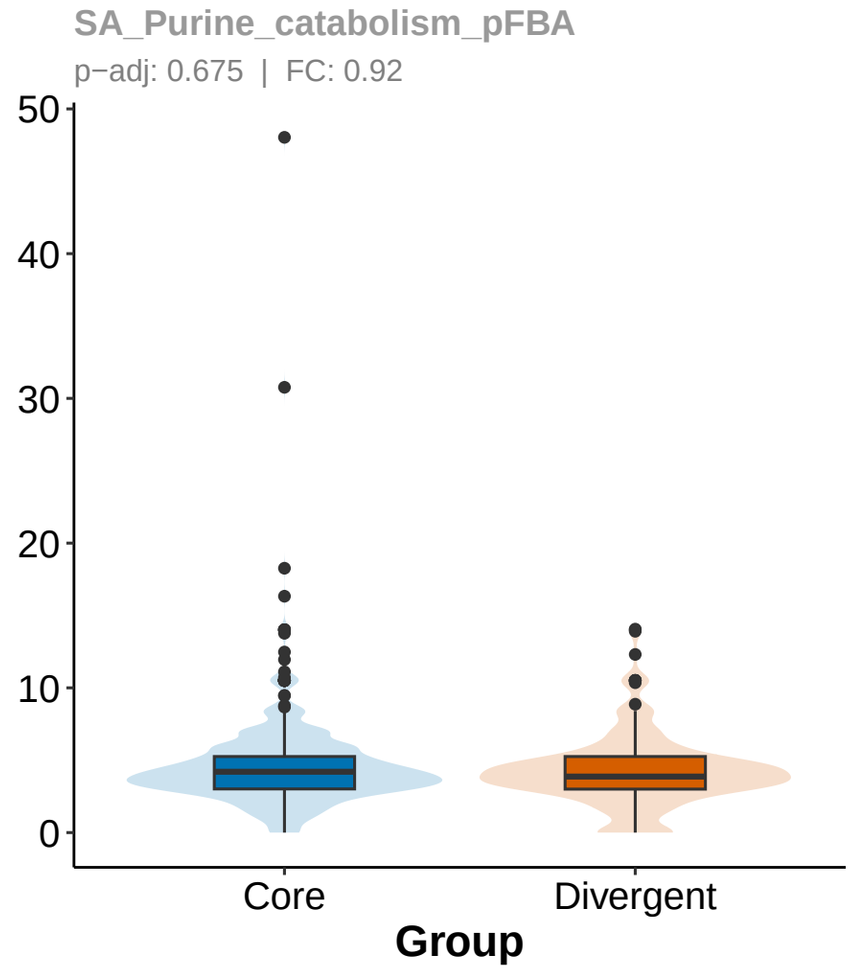

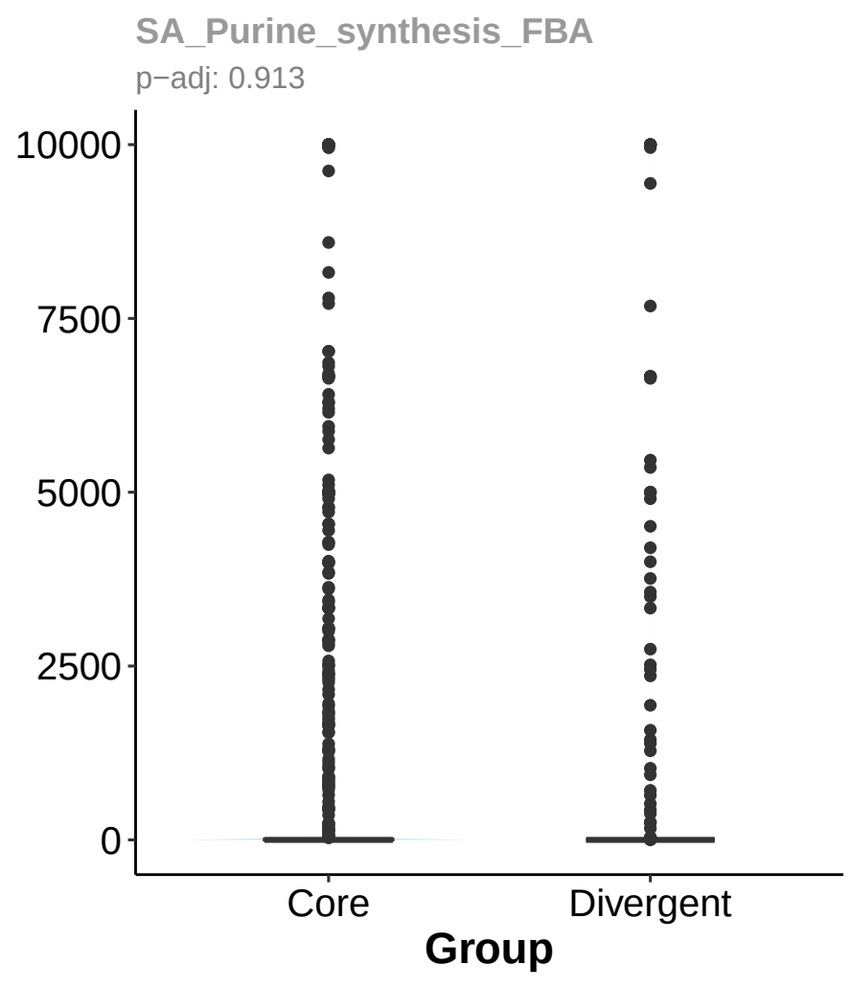

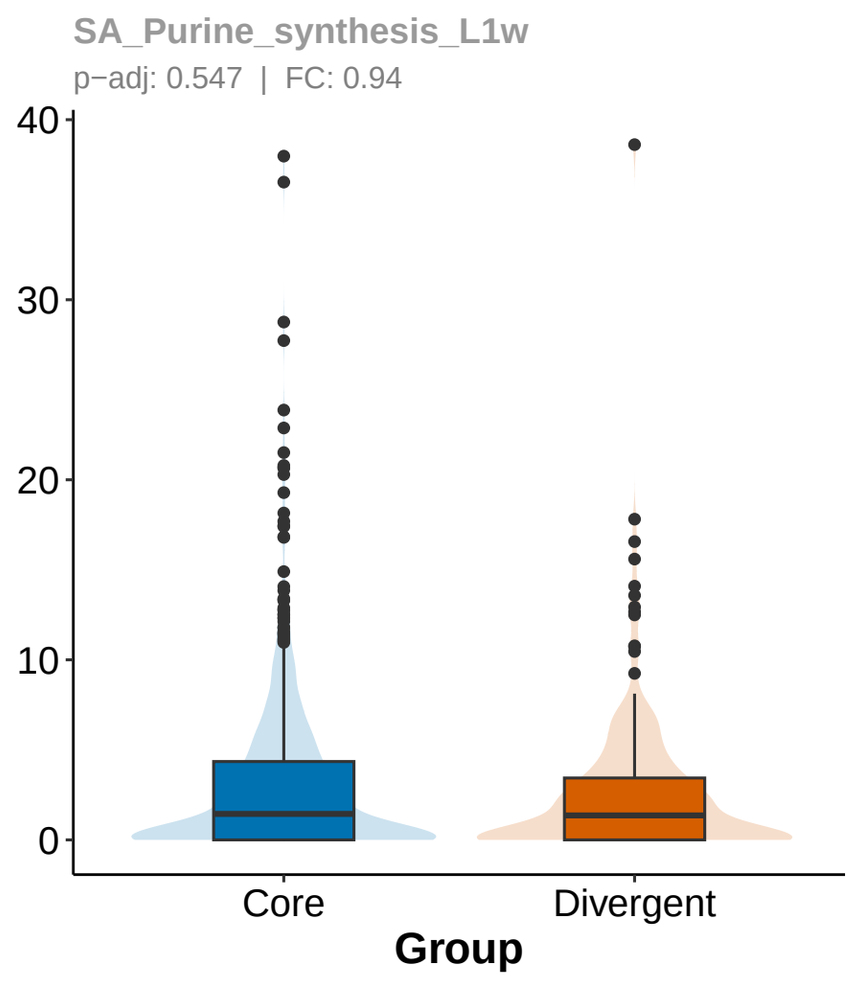

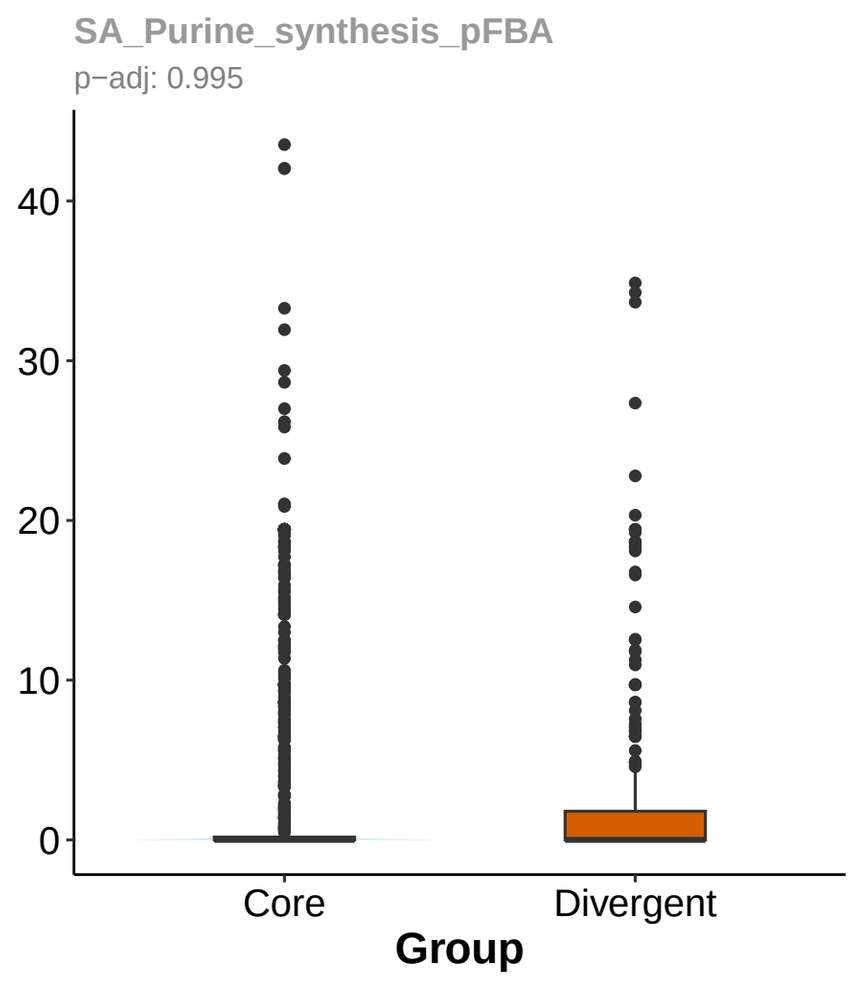

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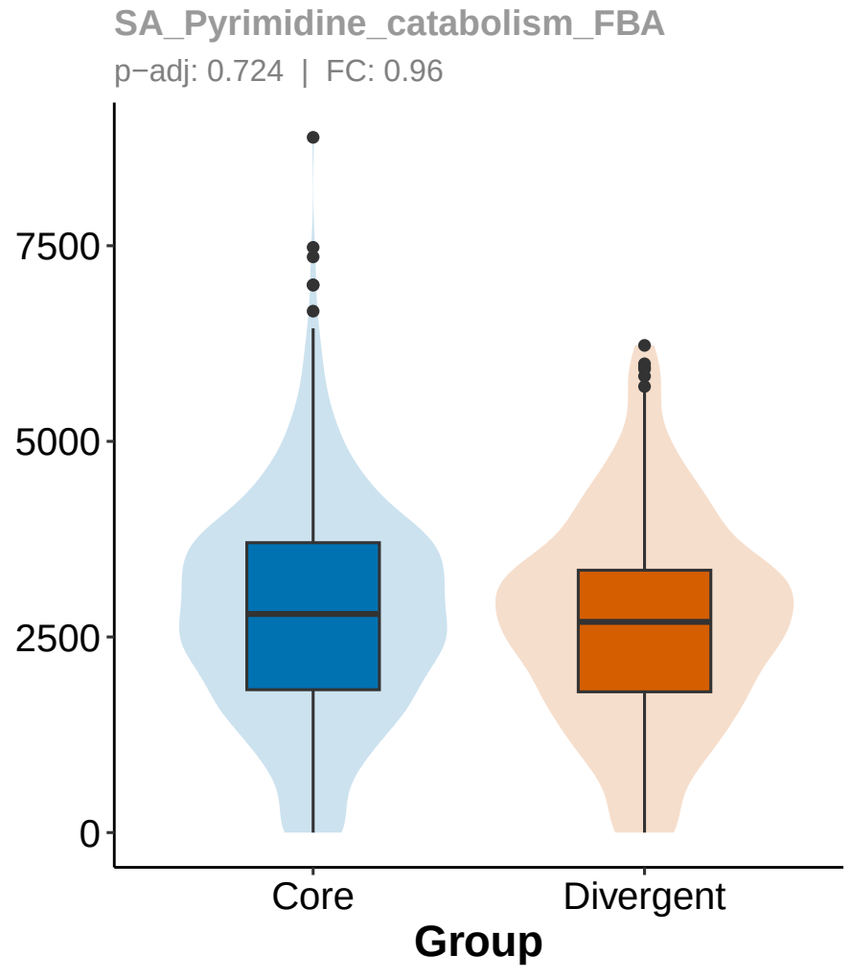

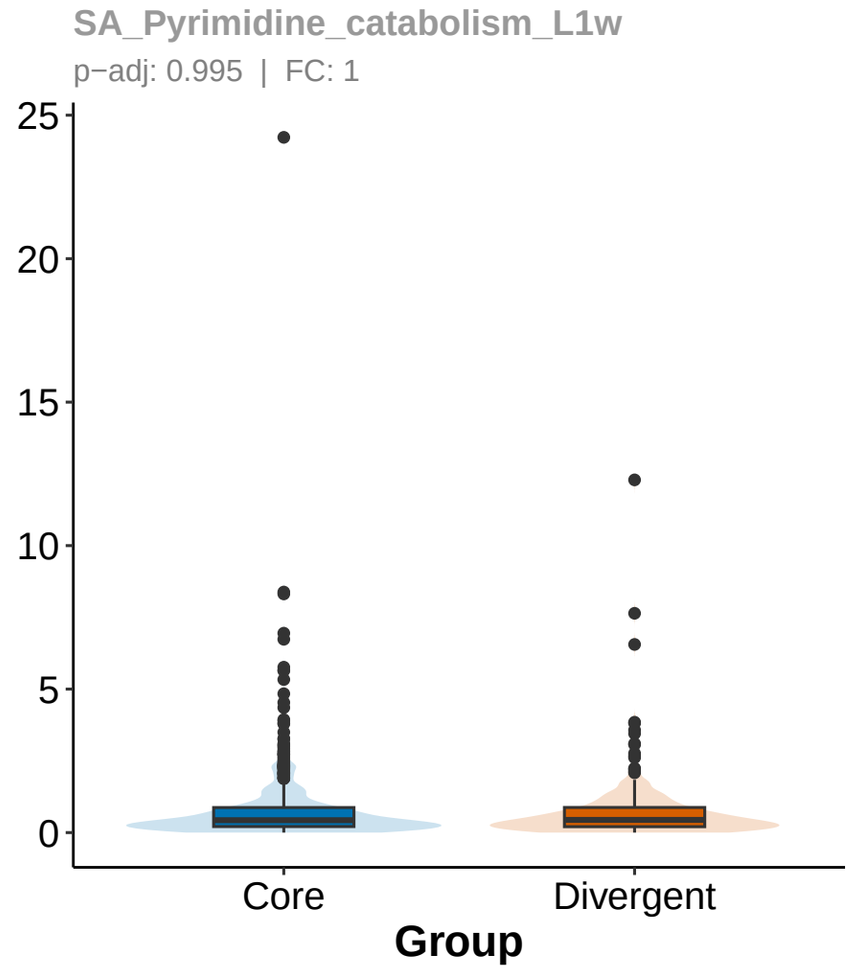

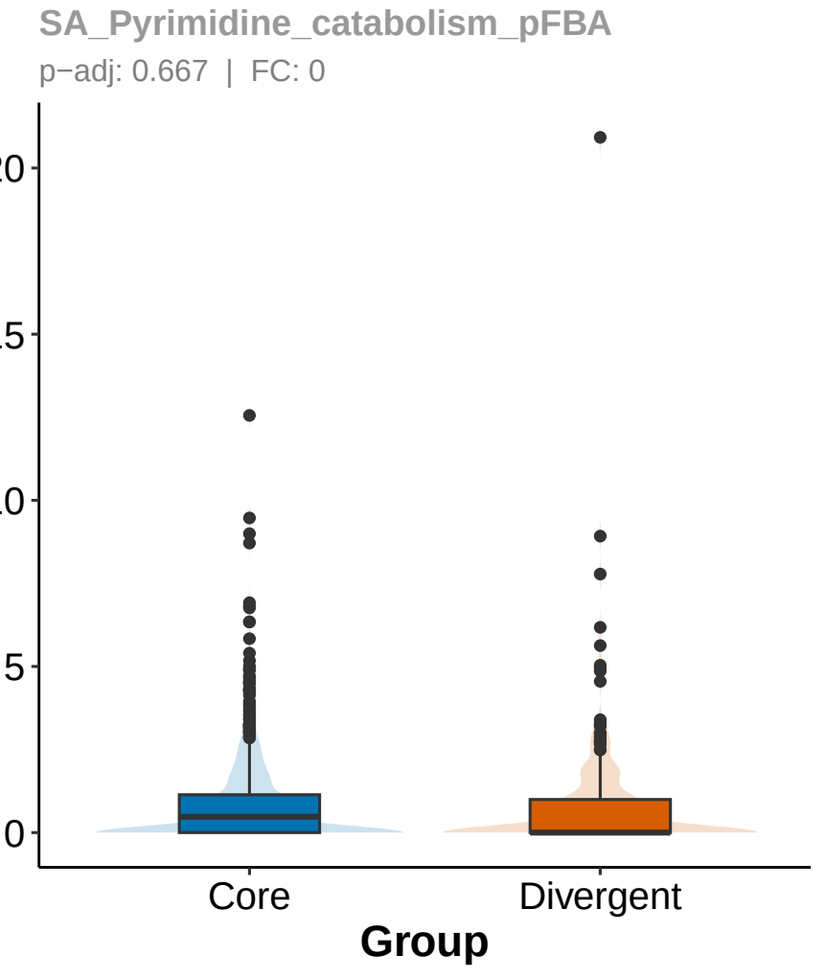

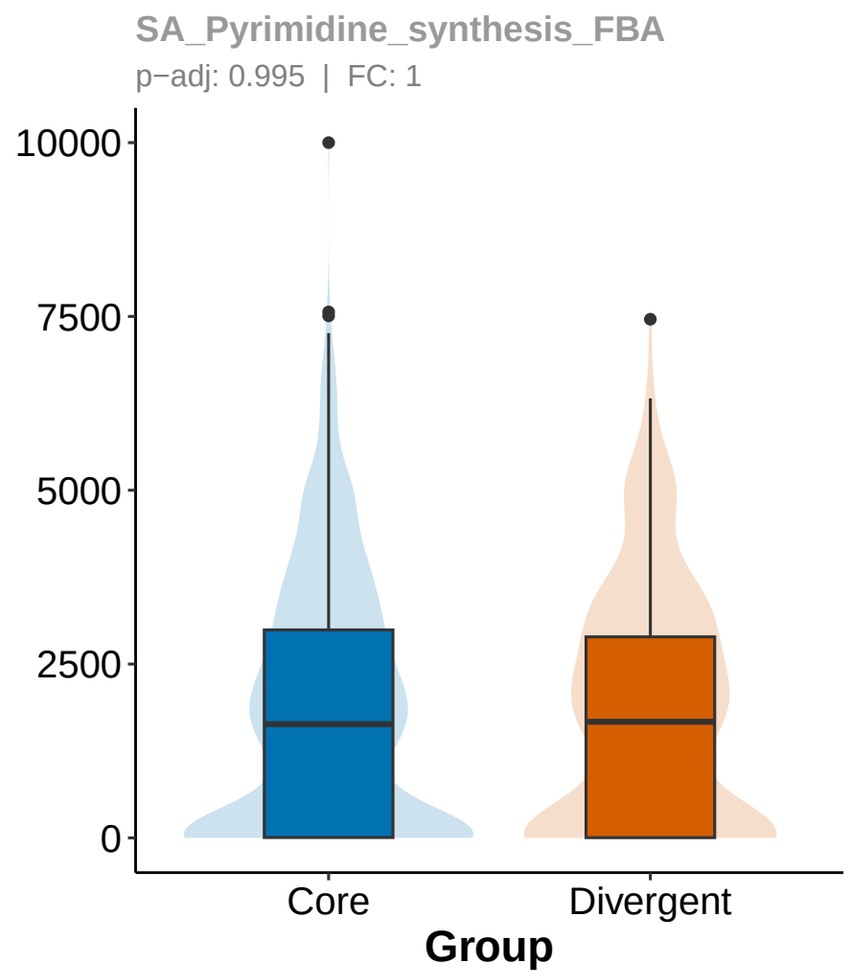

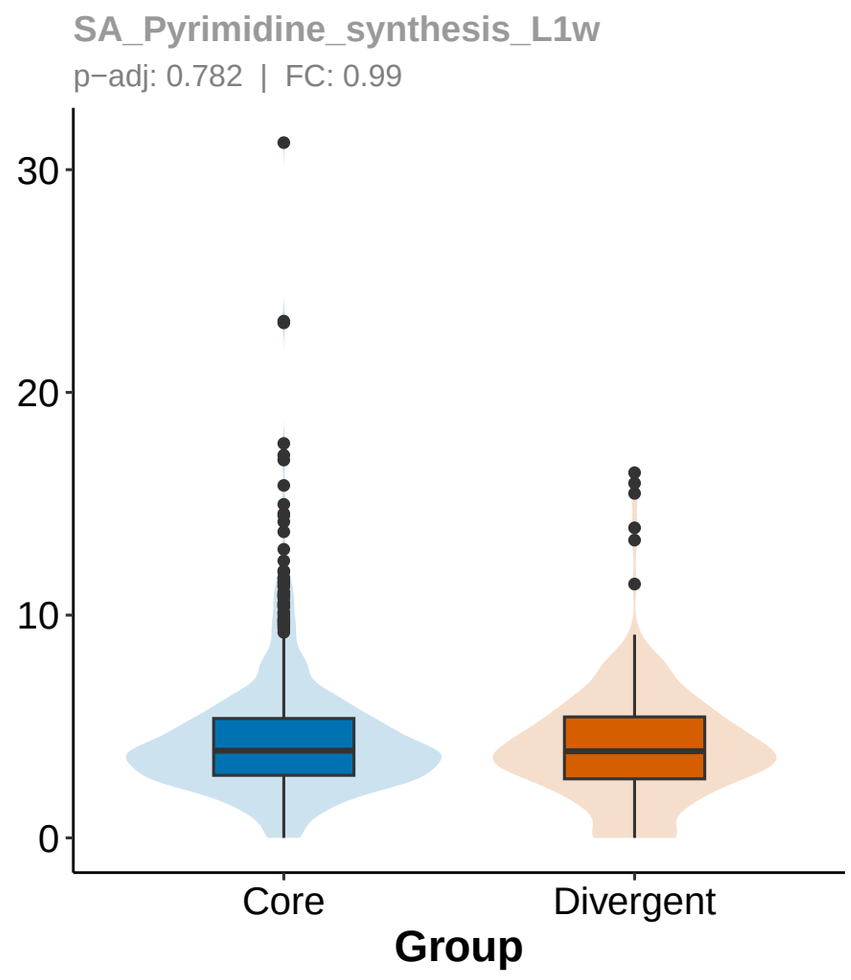

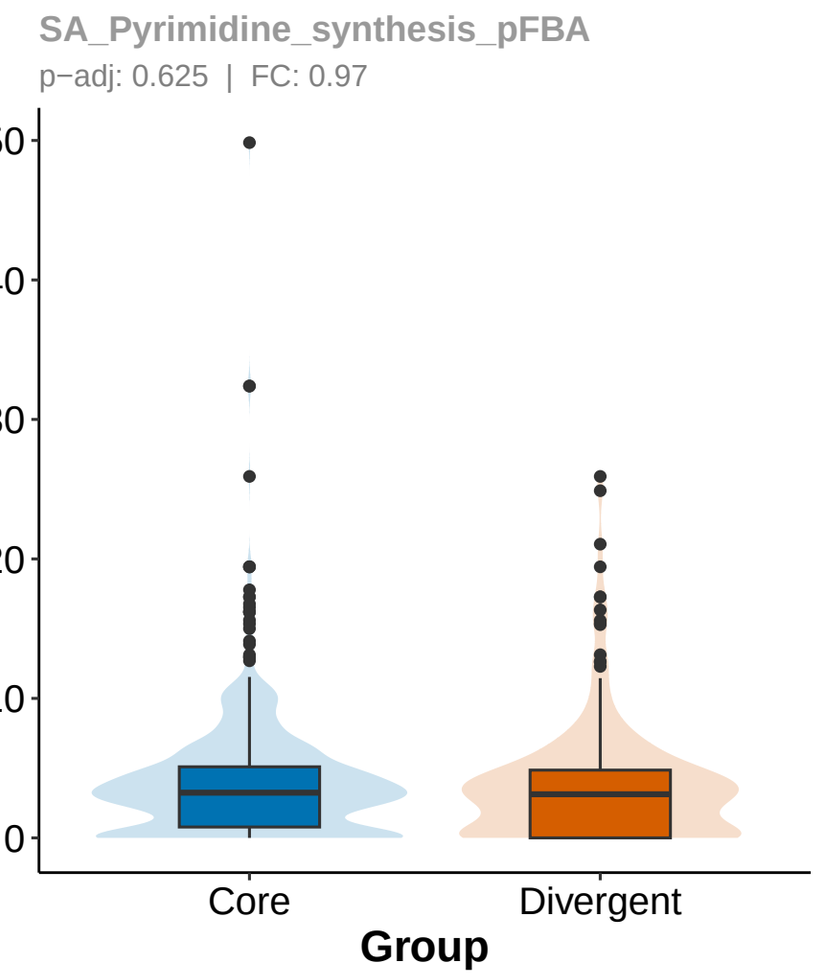

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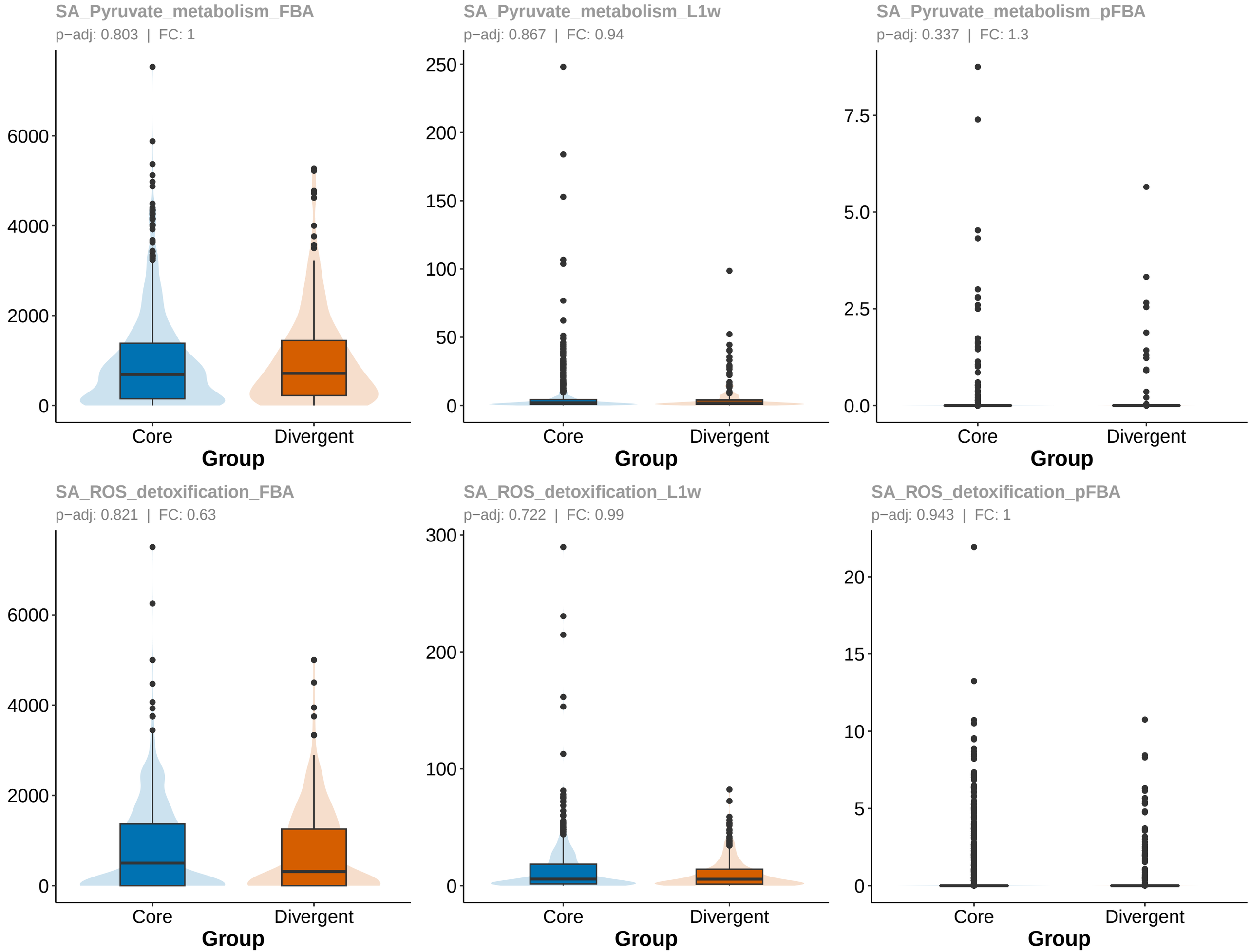

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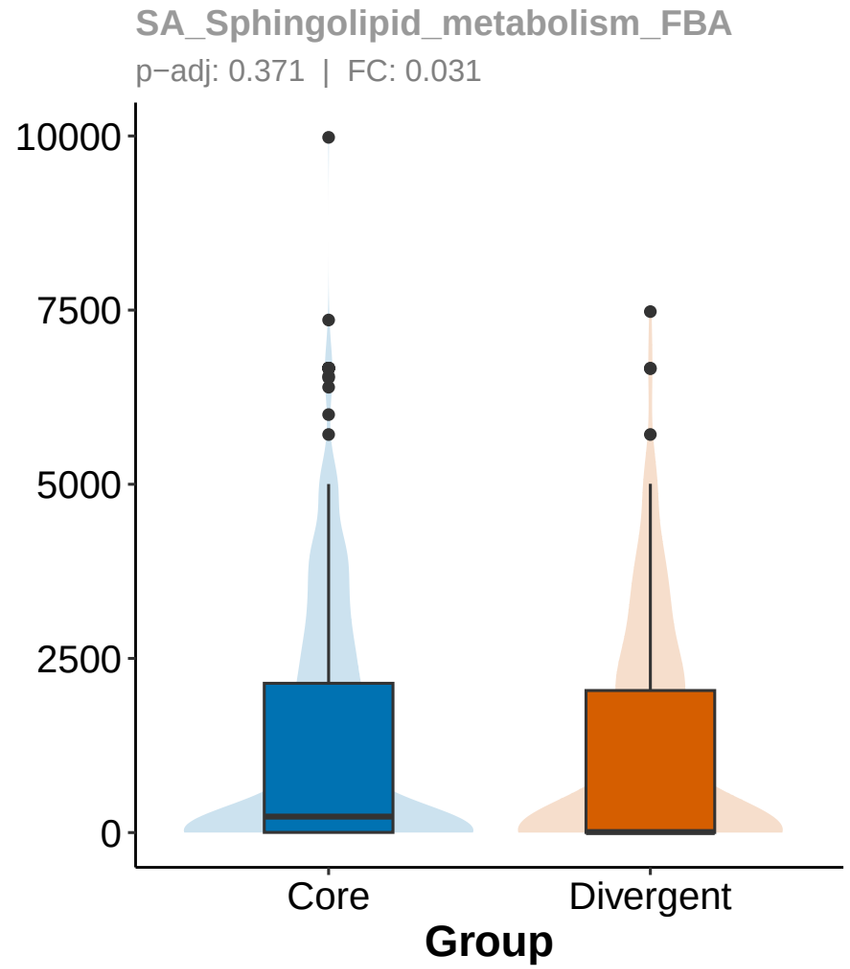
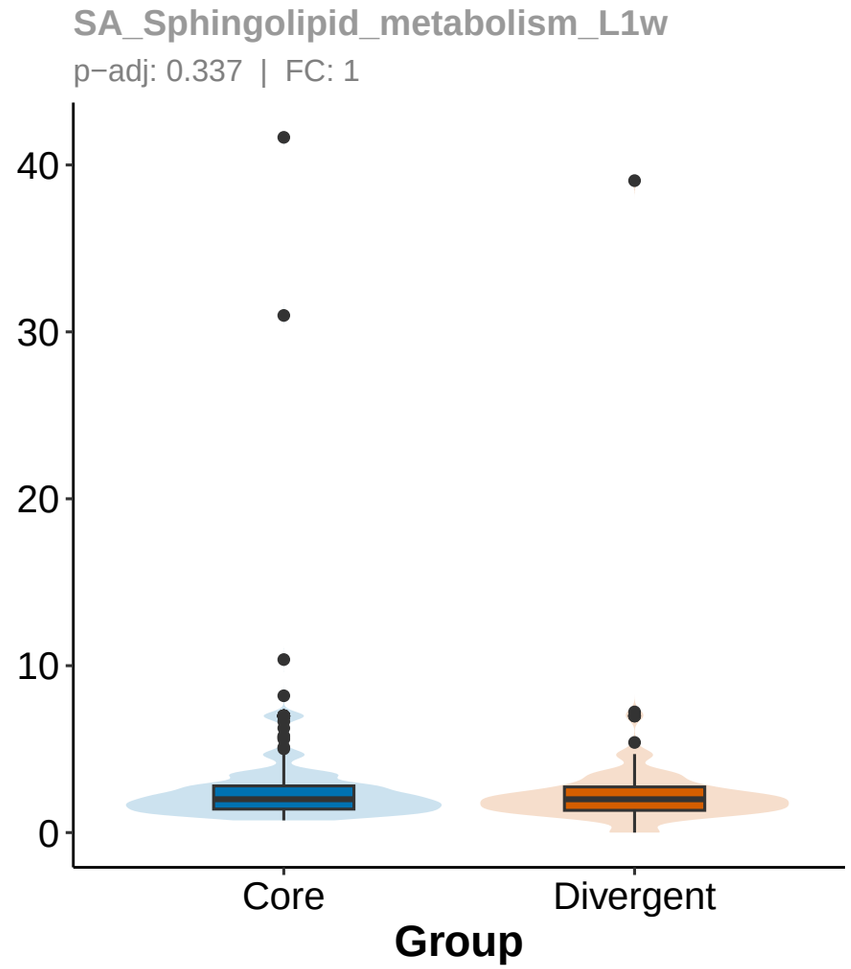
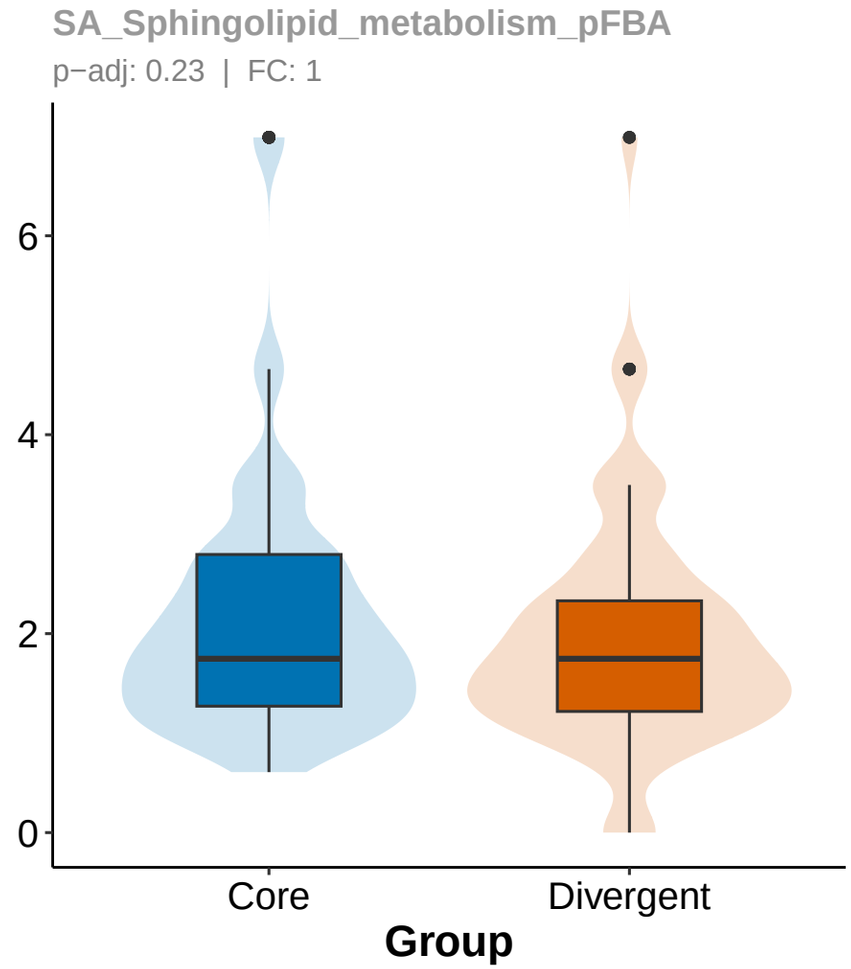
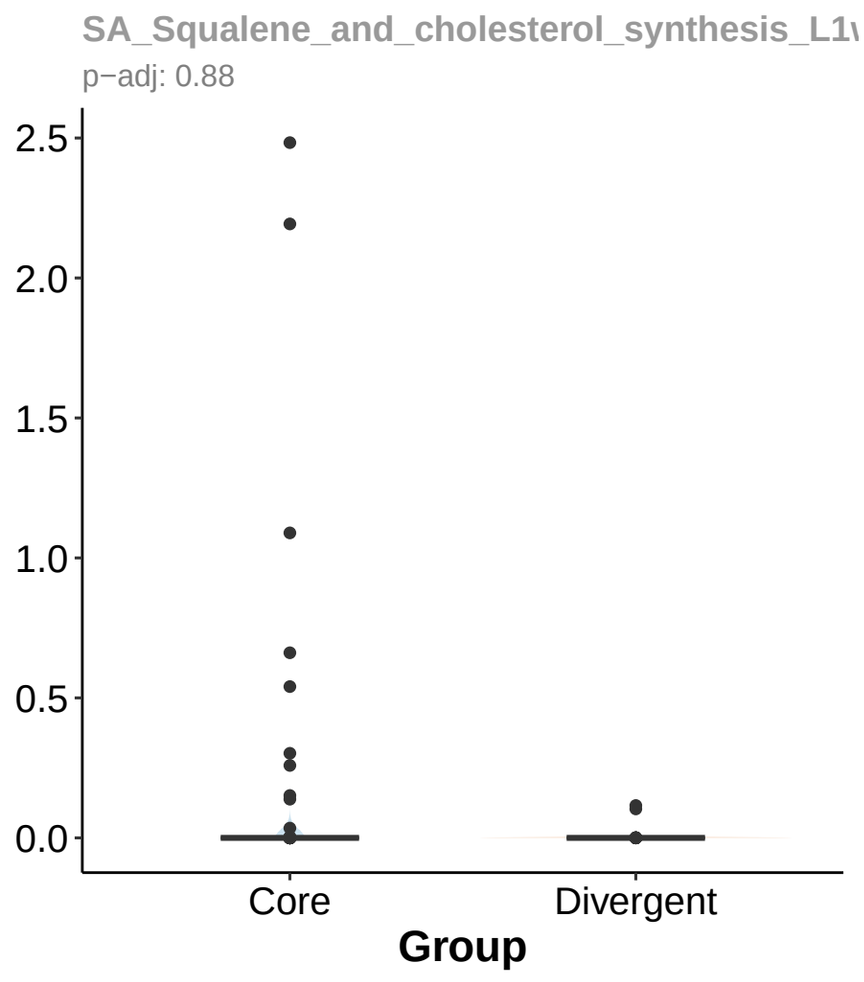
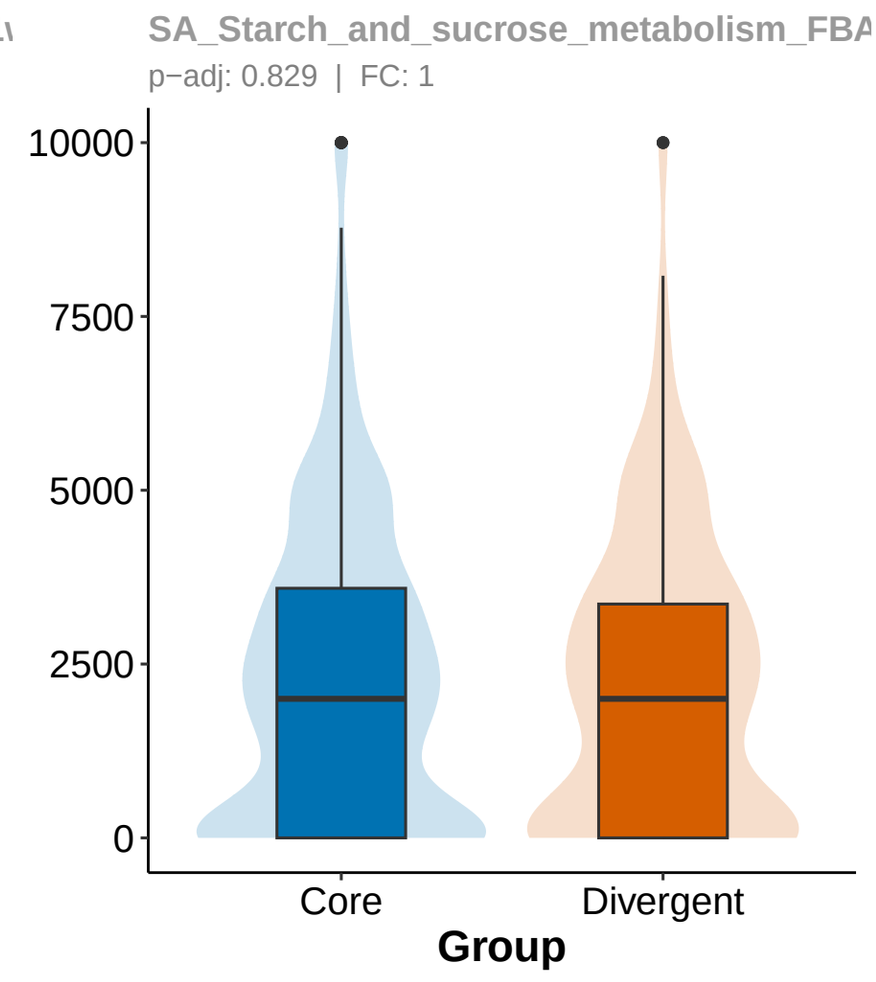
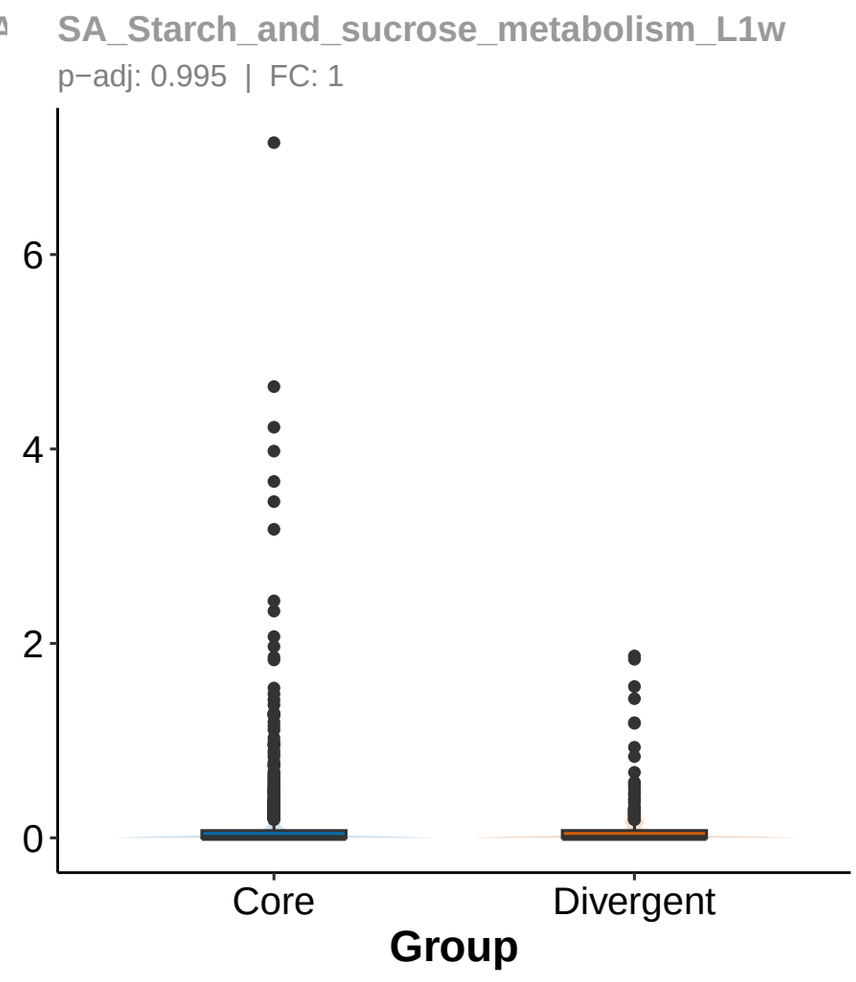




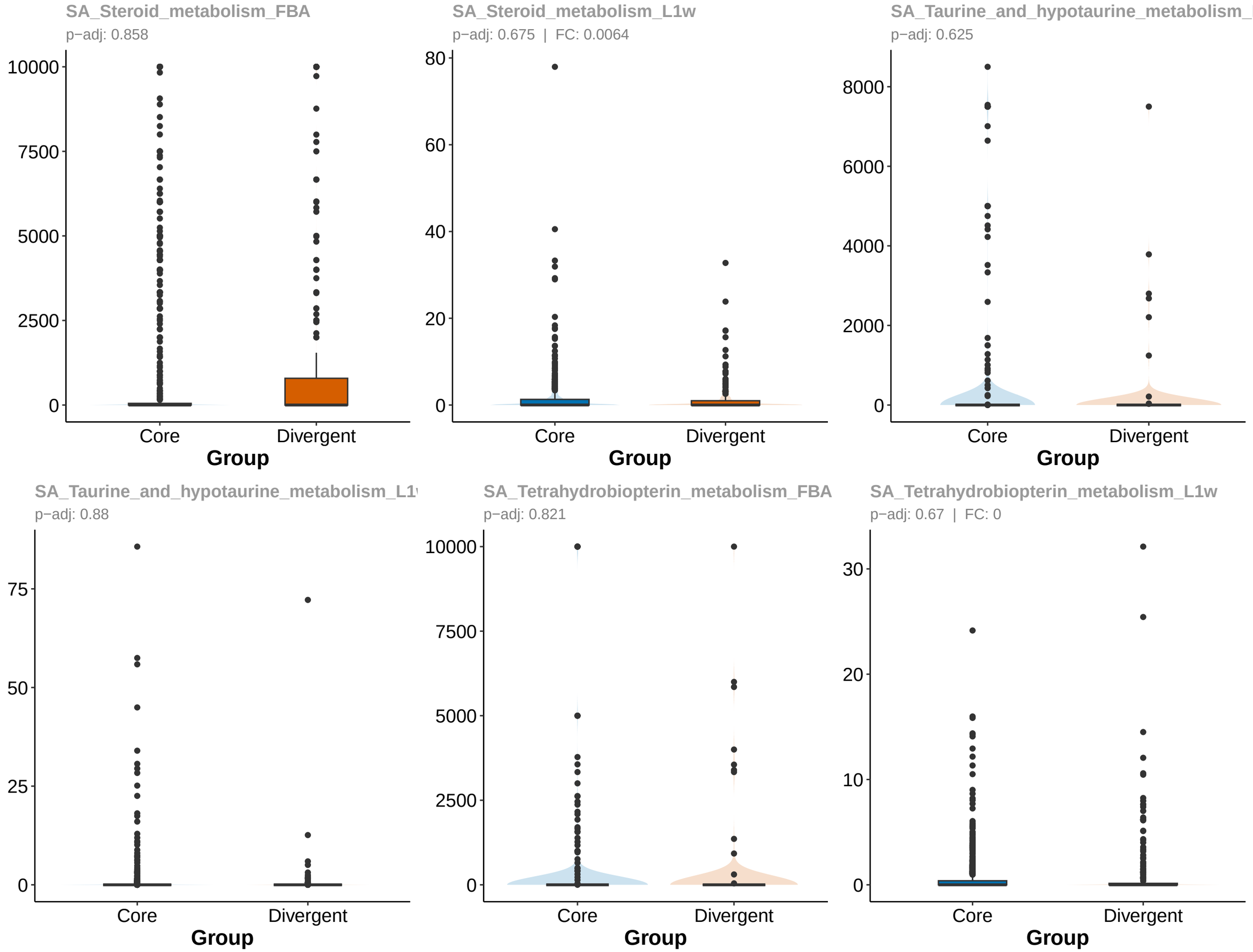








Metabolismo − . 5. Reprogramación Sistémica (SA_*)



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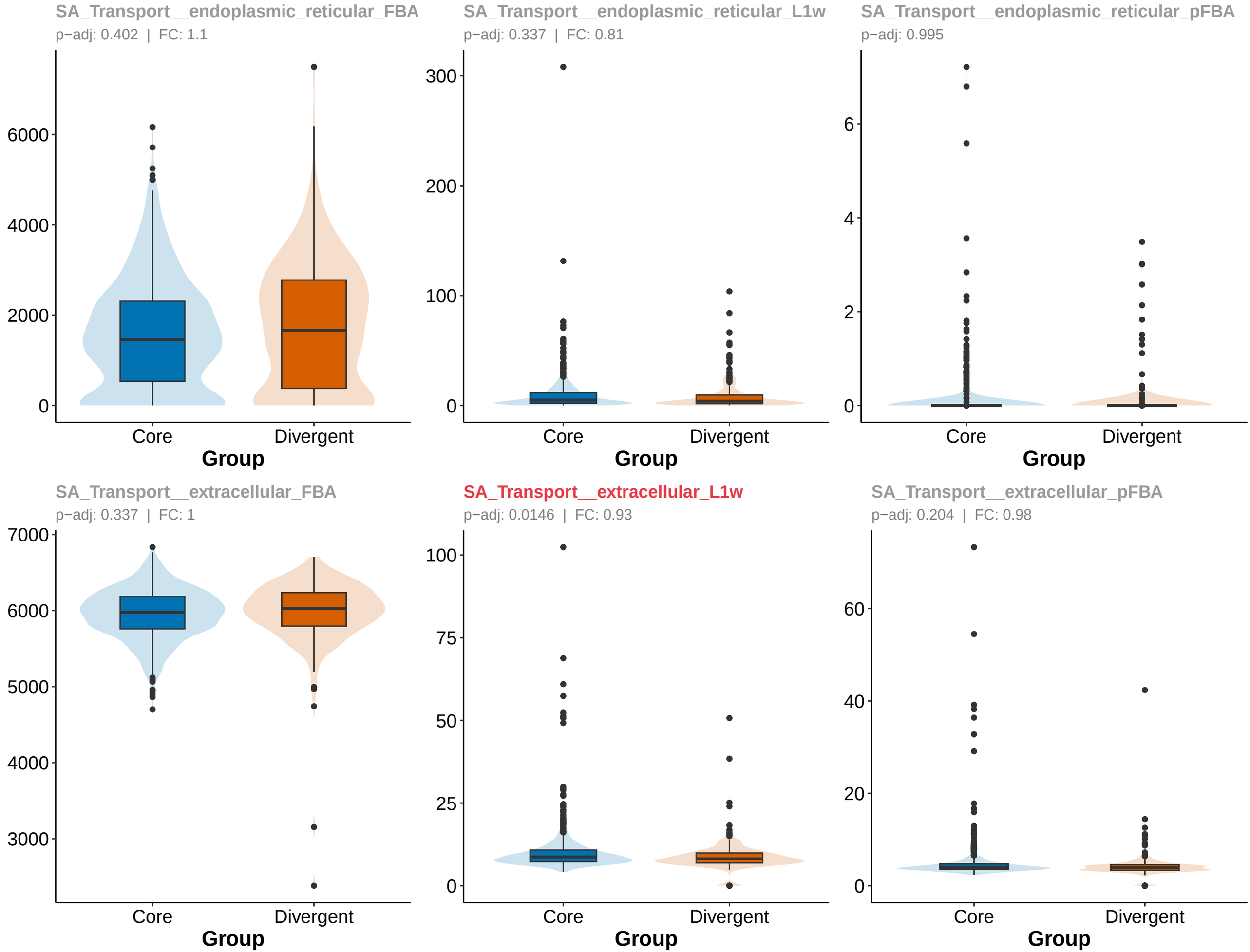

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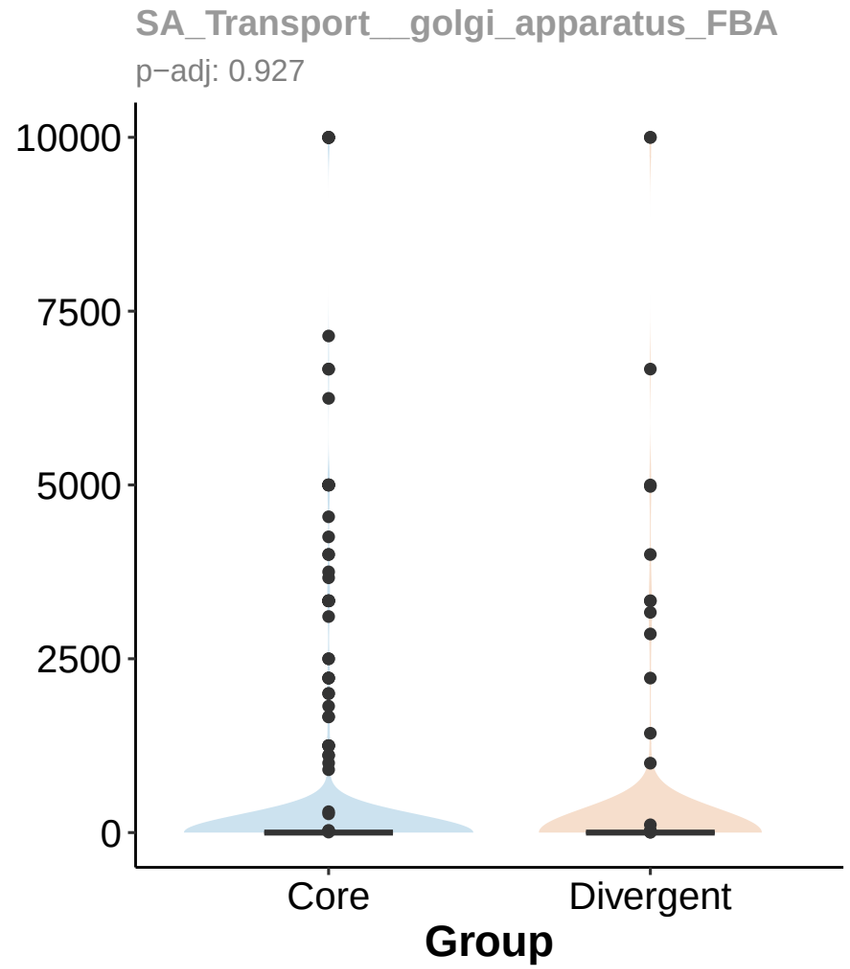
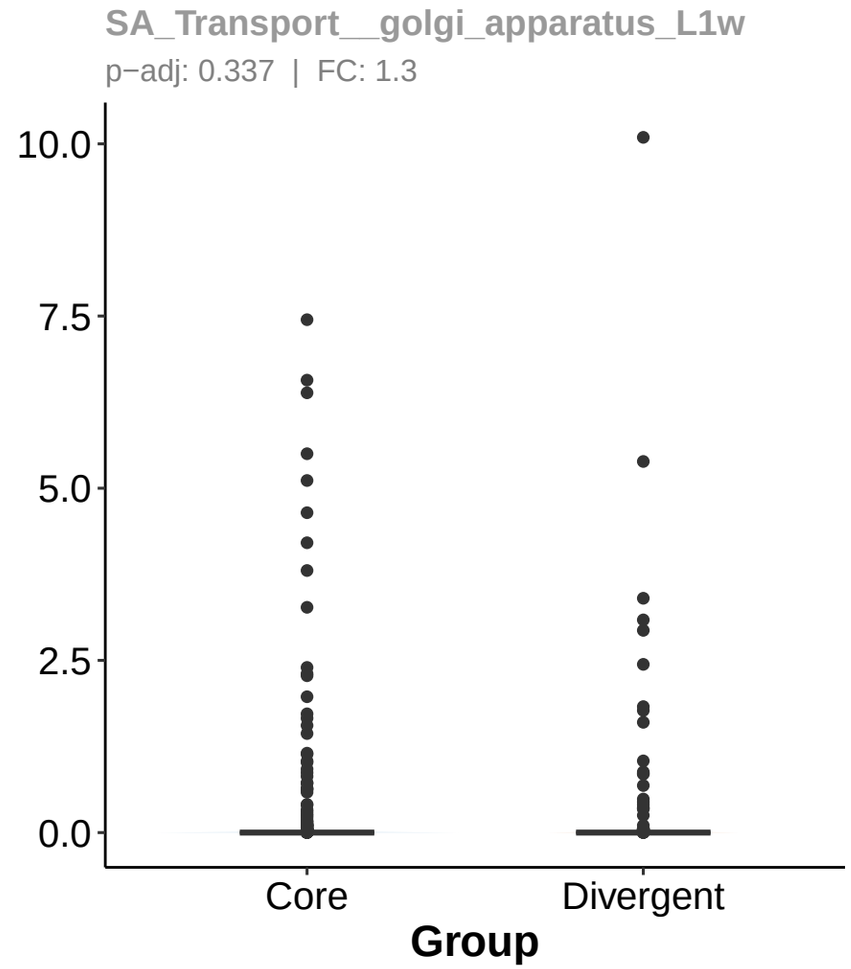
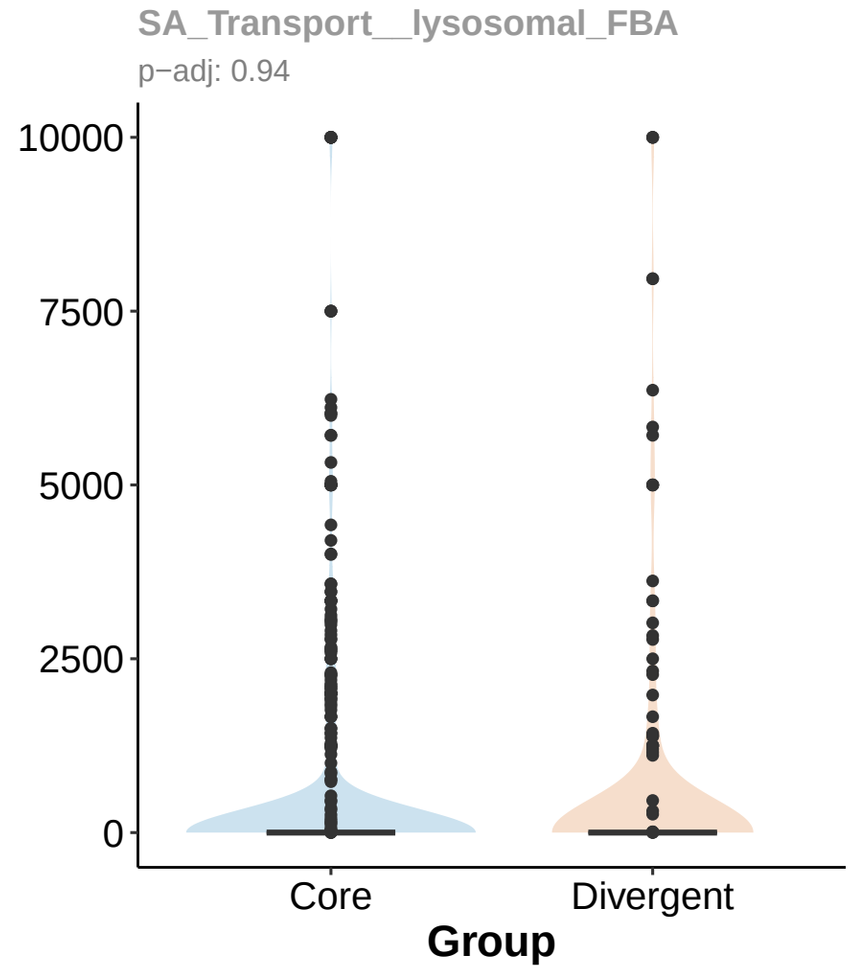
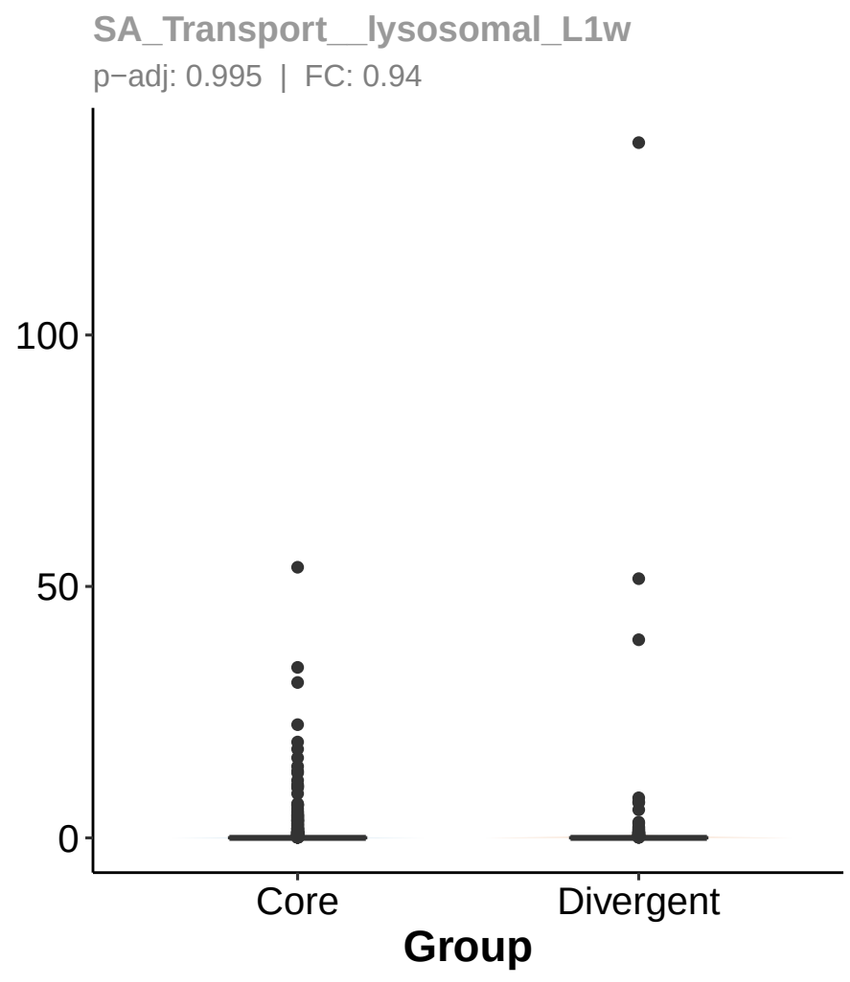
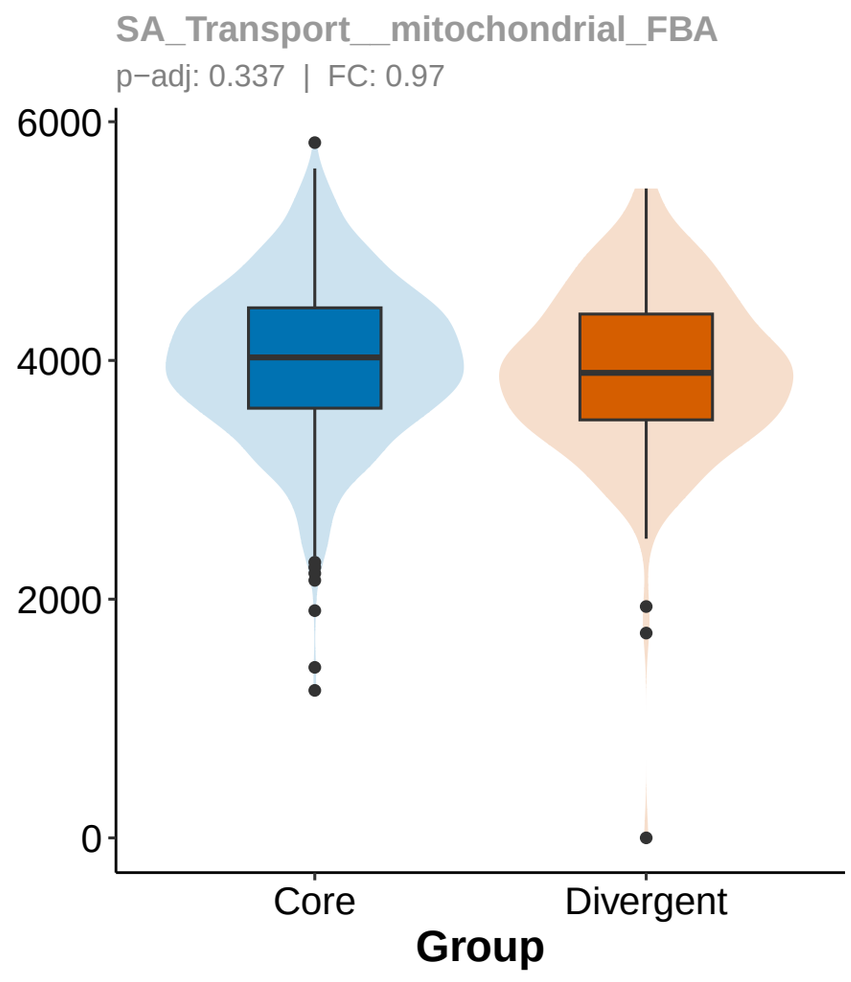
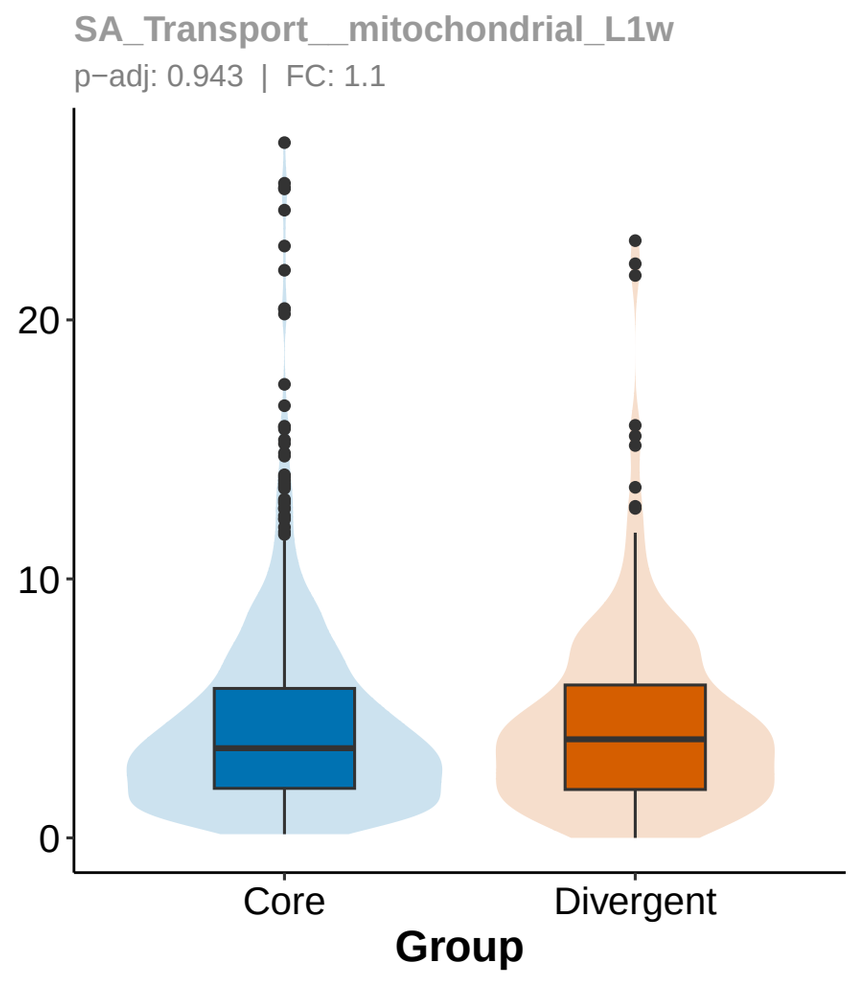




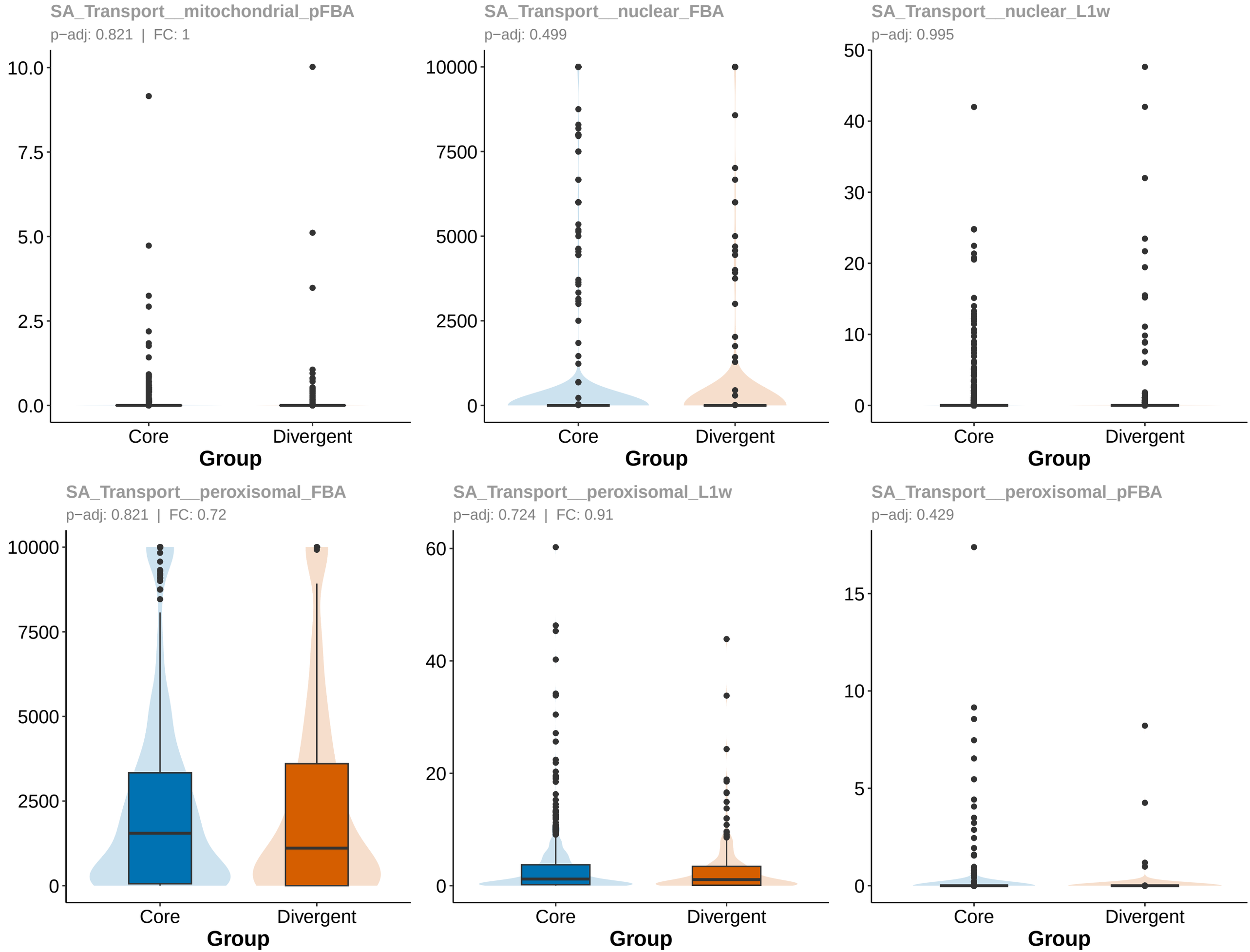




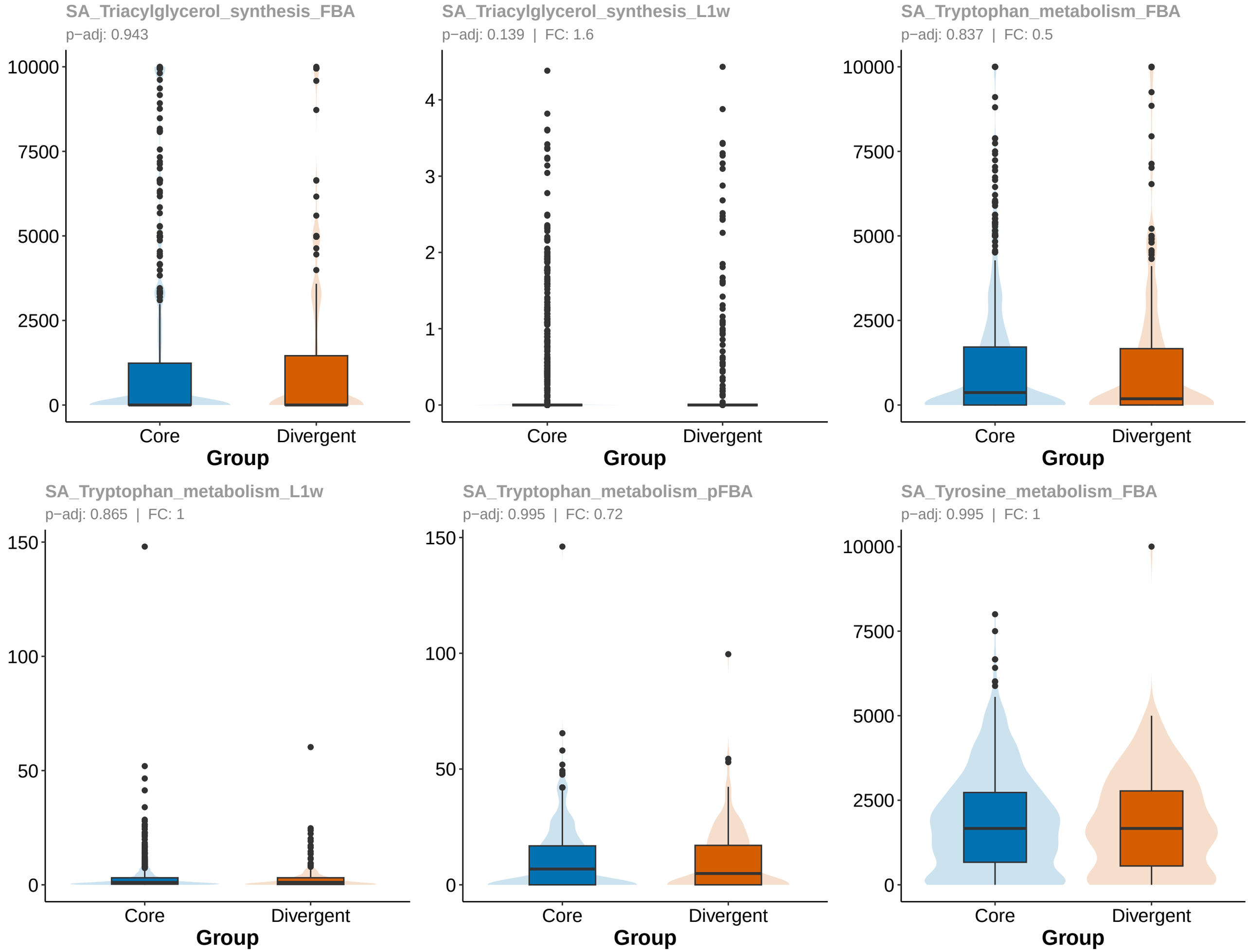




Metabolismo − . 5. Reprogramación Sistémica (SA_*)



Metabolismo − . 5. Reprogramación Sistémica (SA_*)



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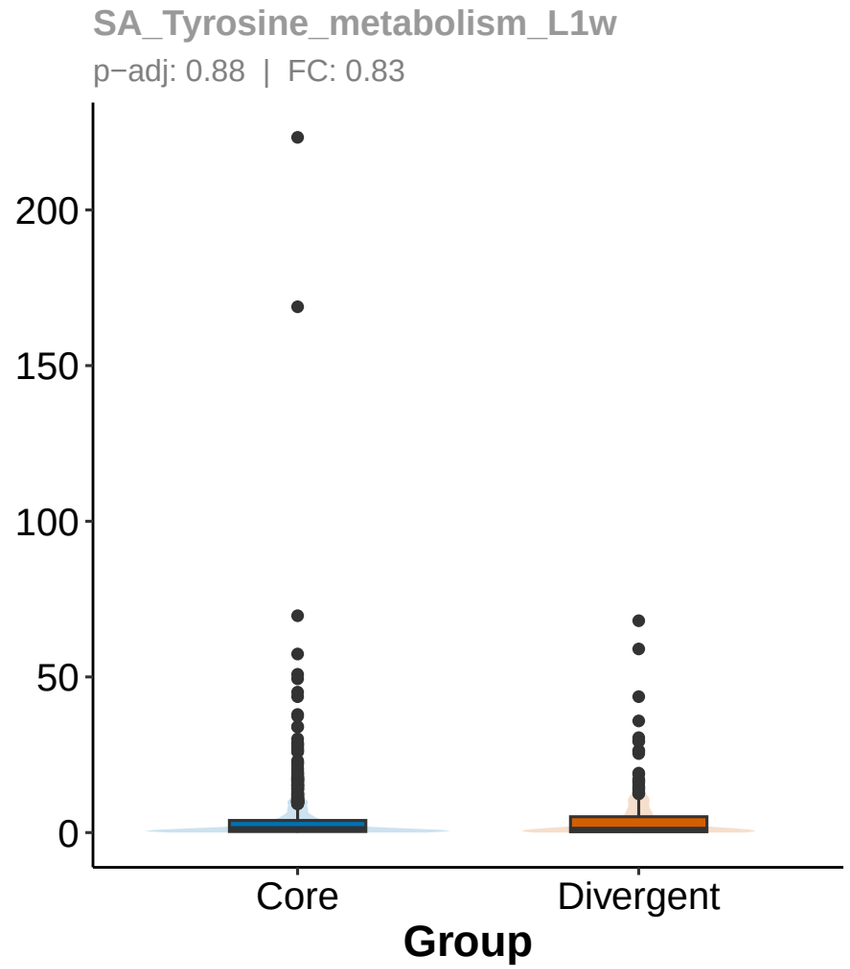

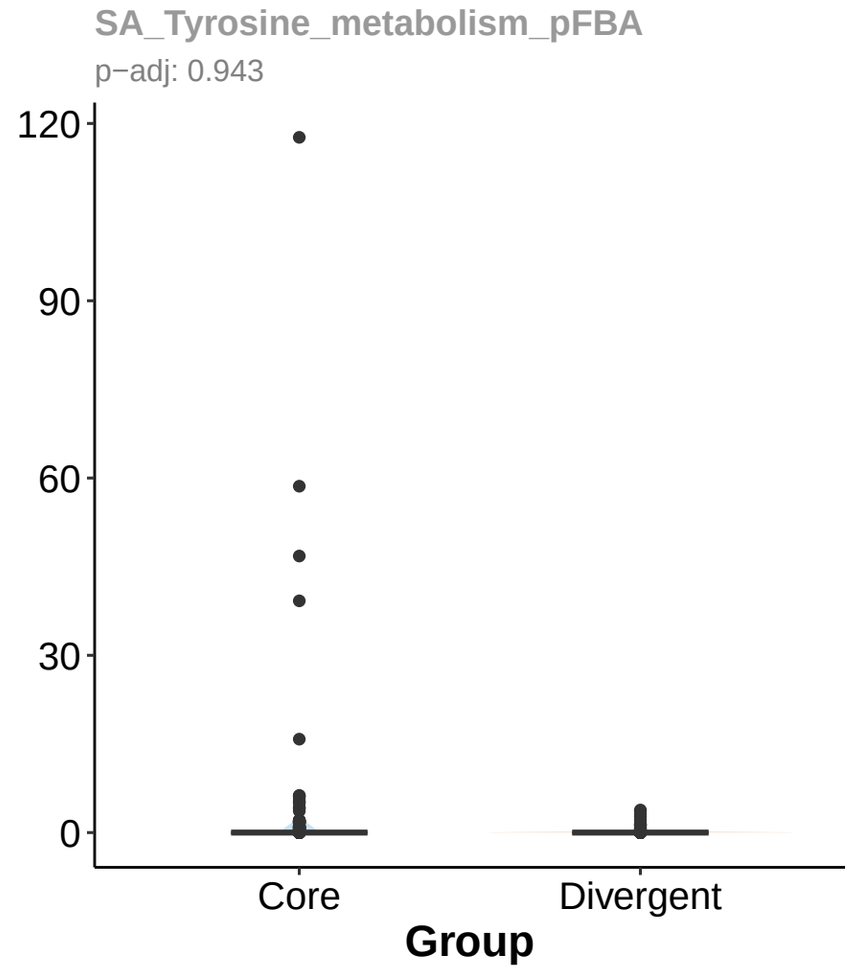

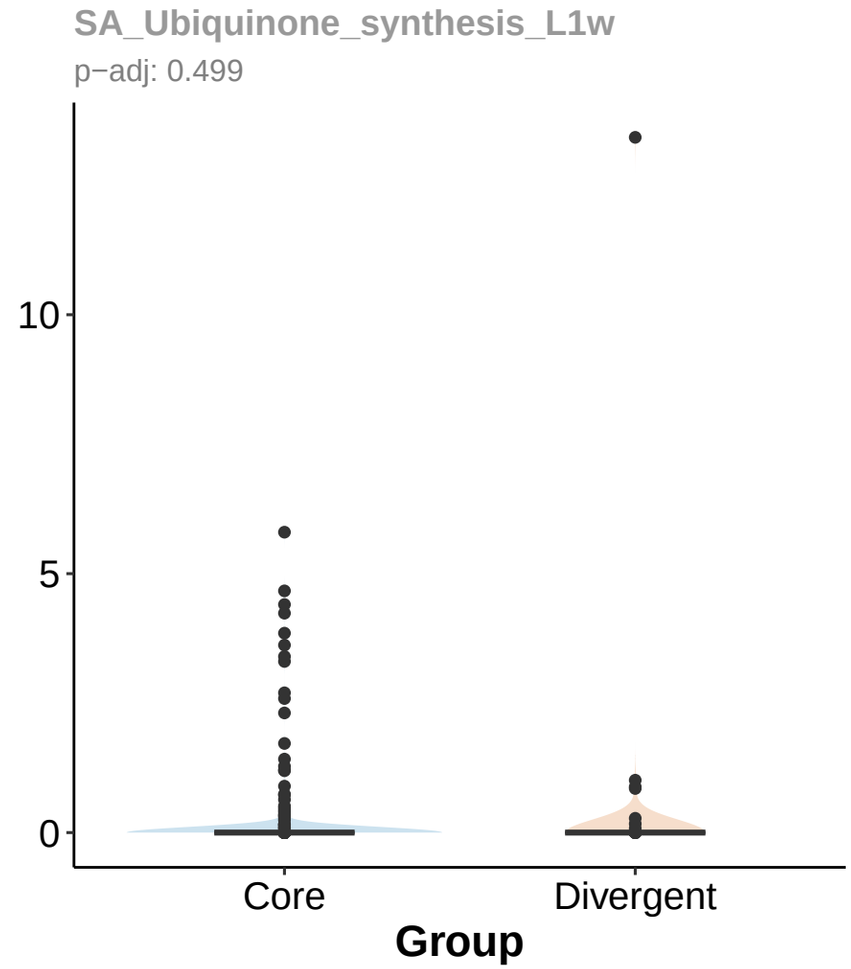

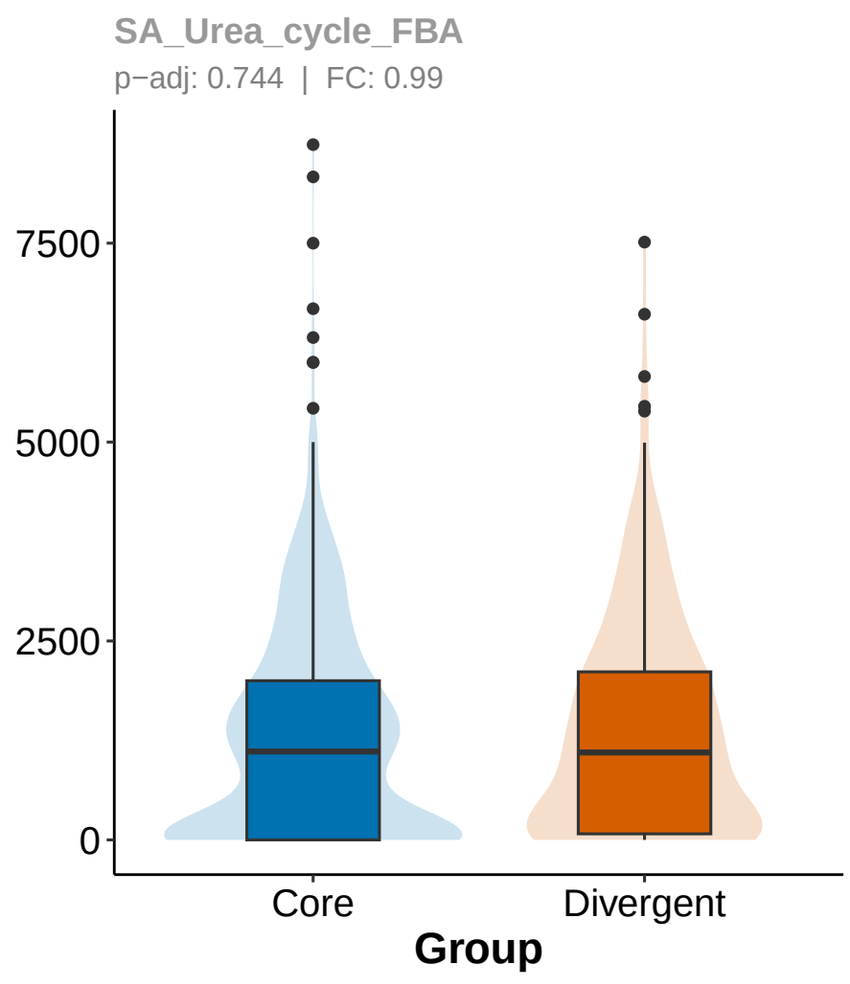

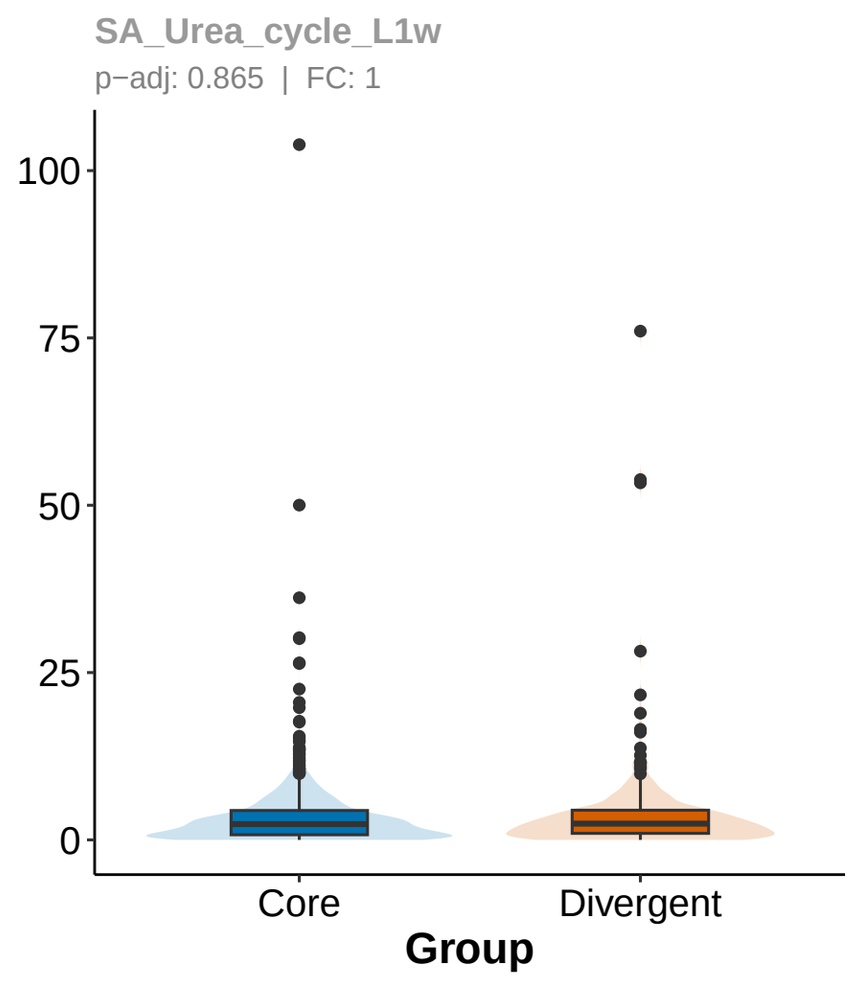

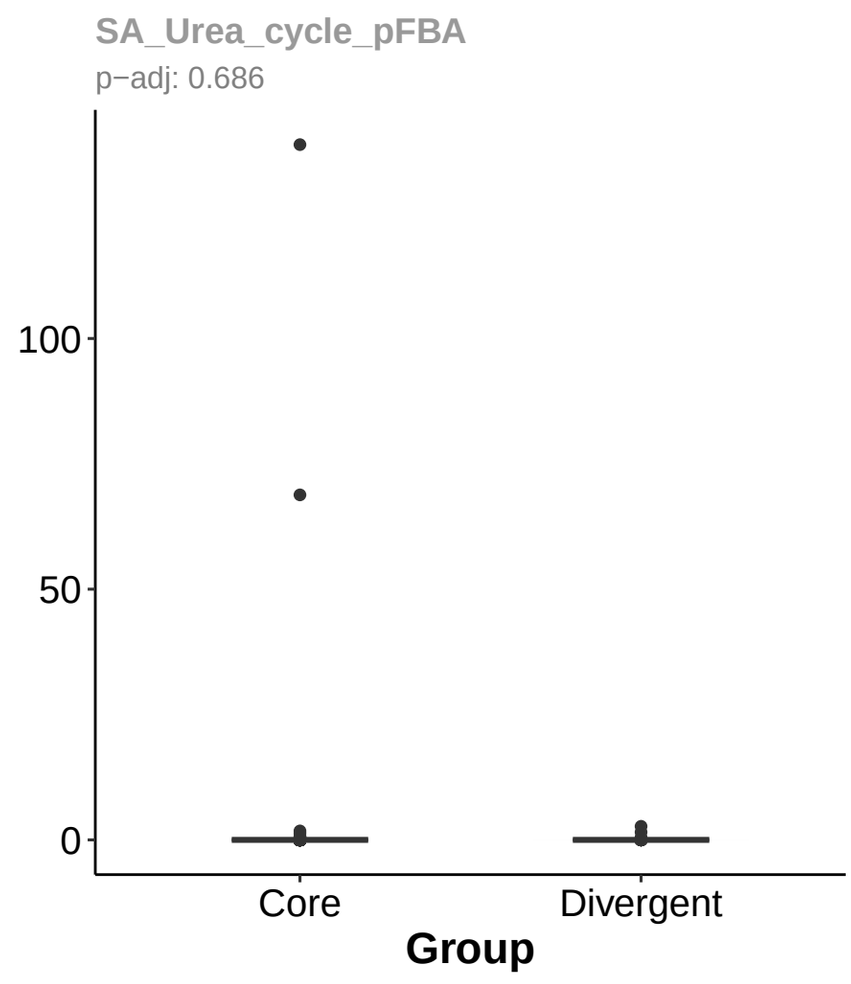

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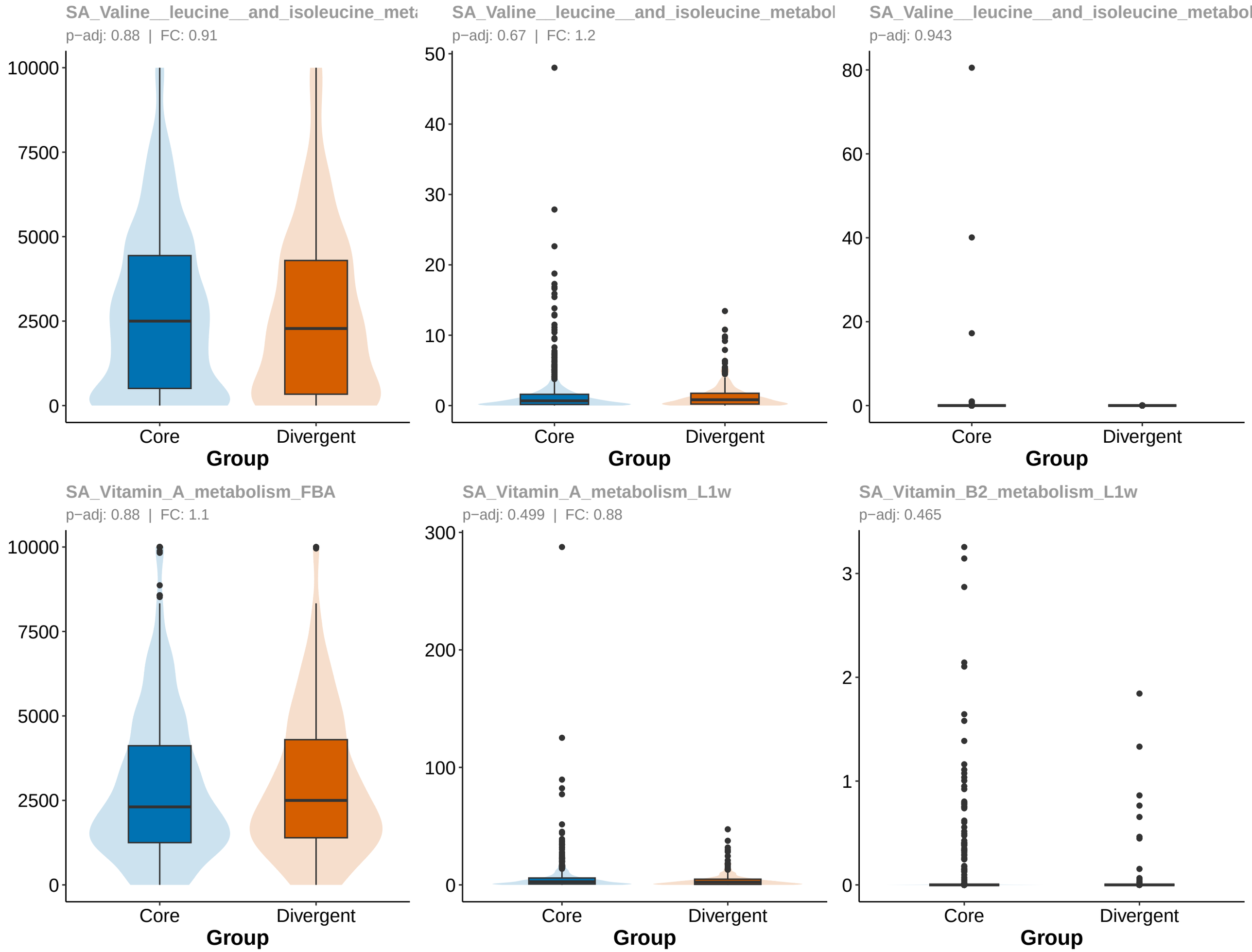
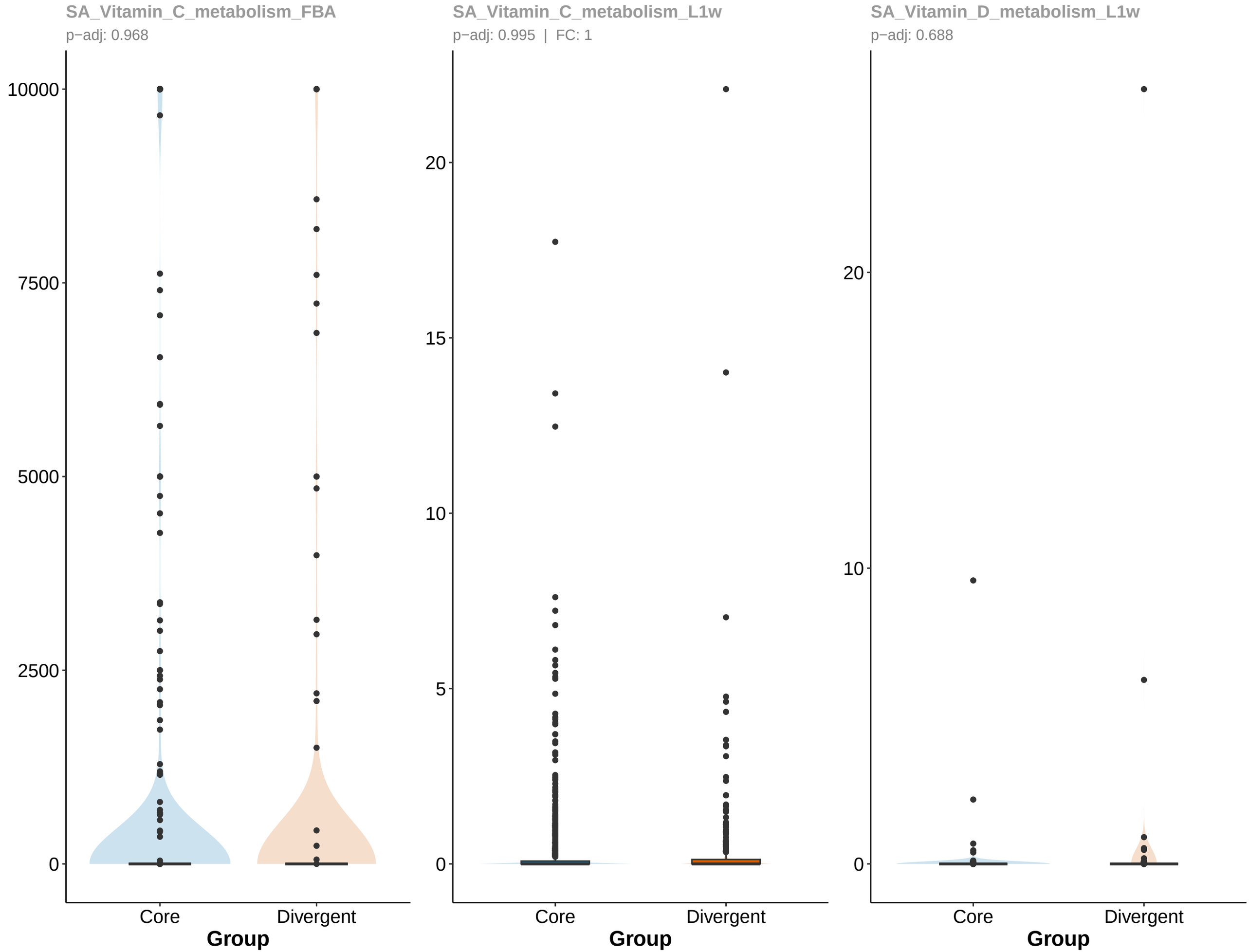
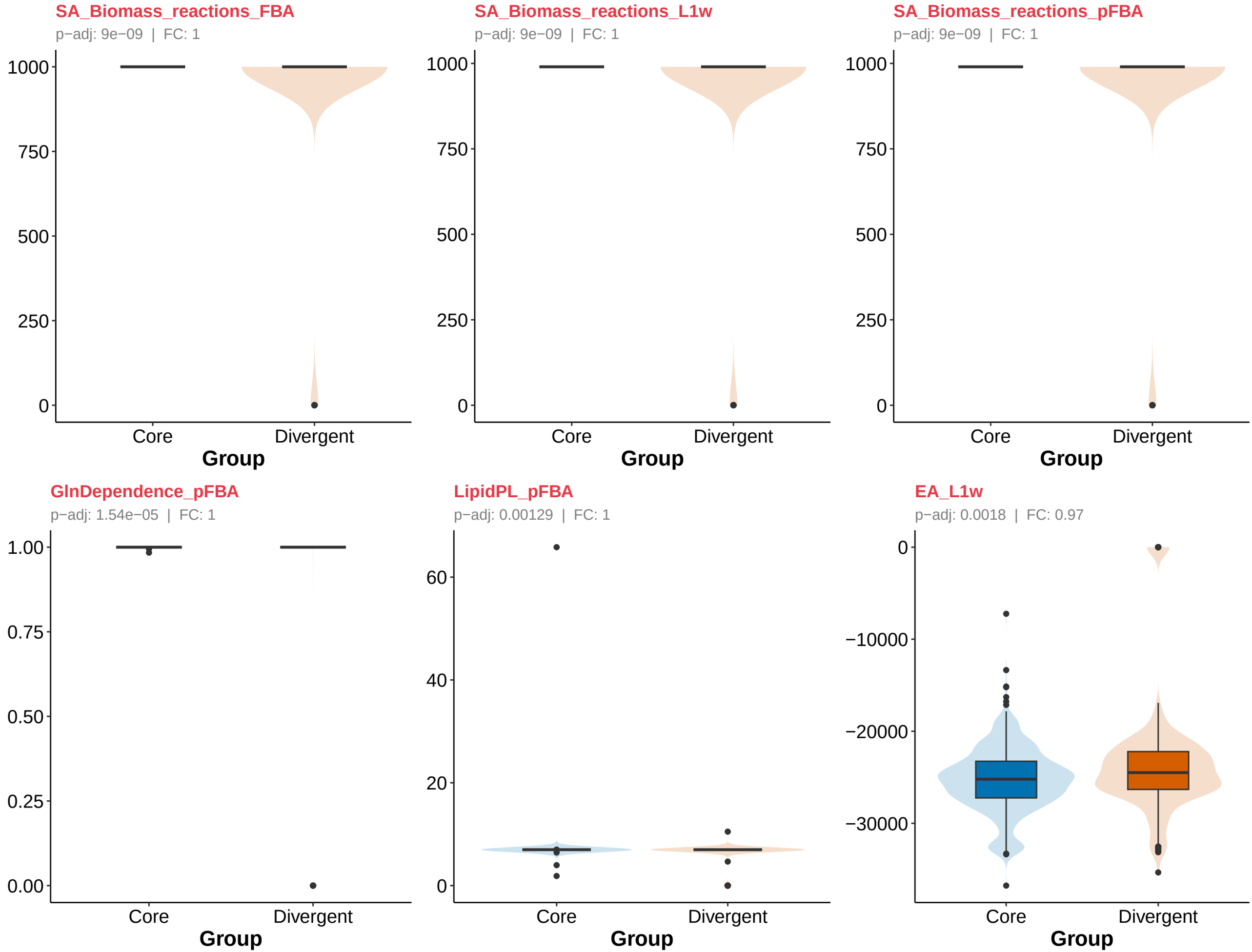


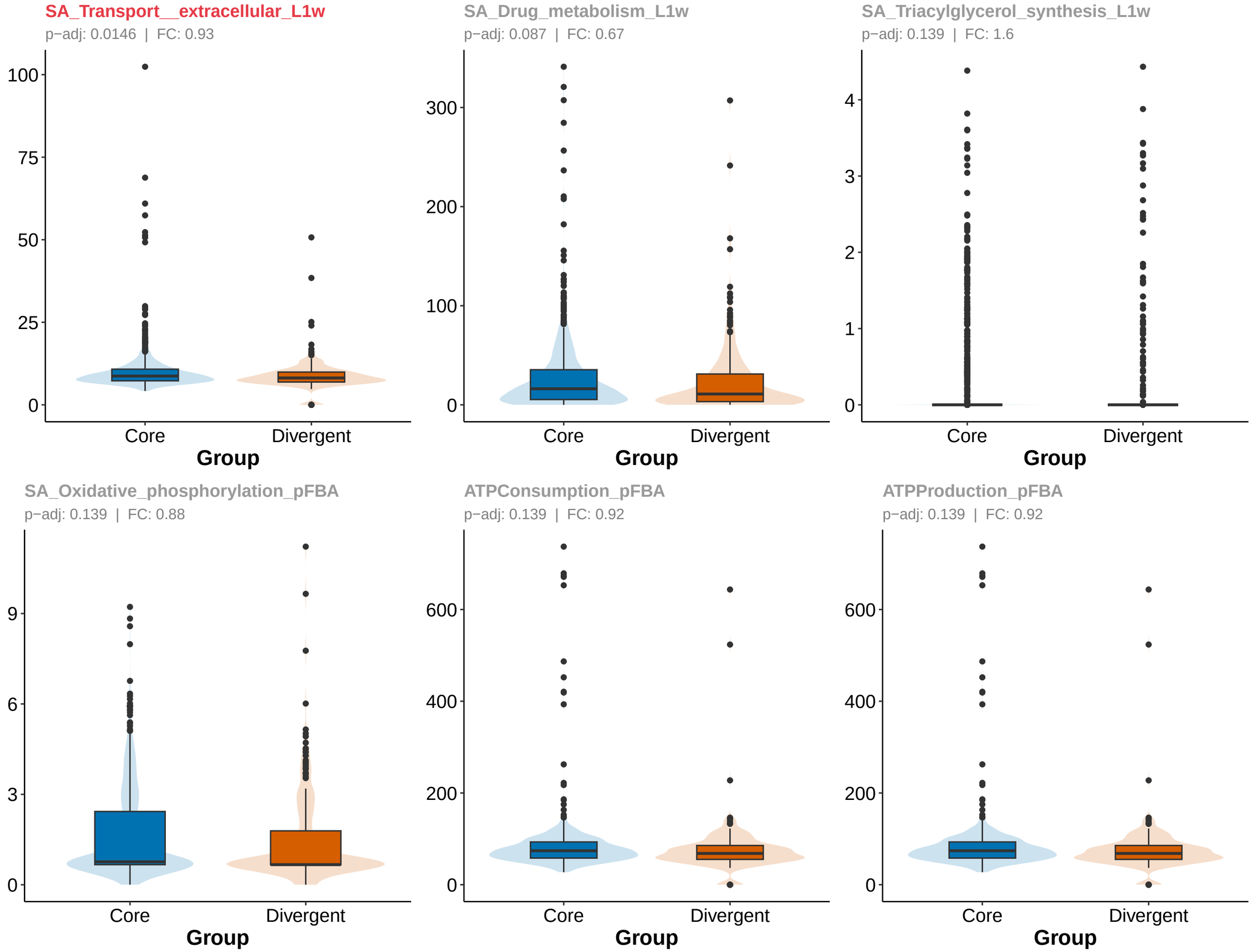
Metabolismo − . 5. Reprogramación Sistémica (SA_*)



. Top 12 Variables Metabólicas (p_adj + efecto)



. Top 12 Variables Metabólicas (p_adj + efecto)

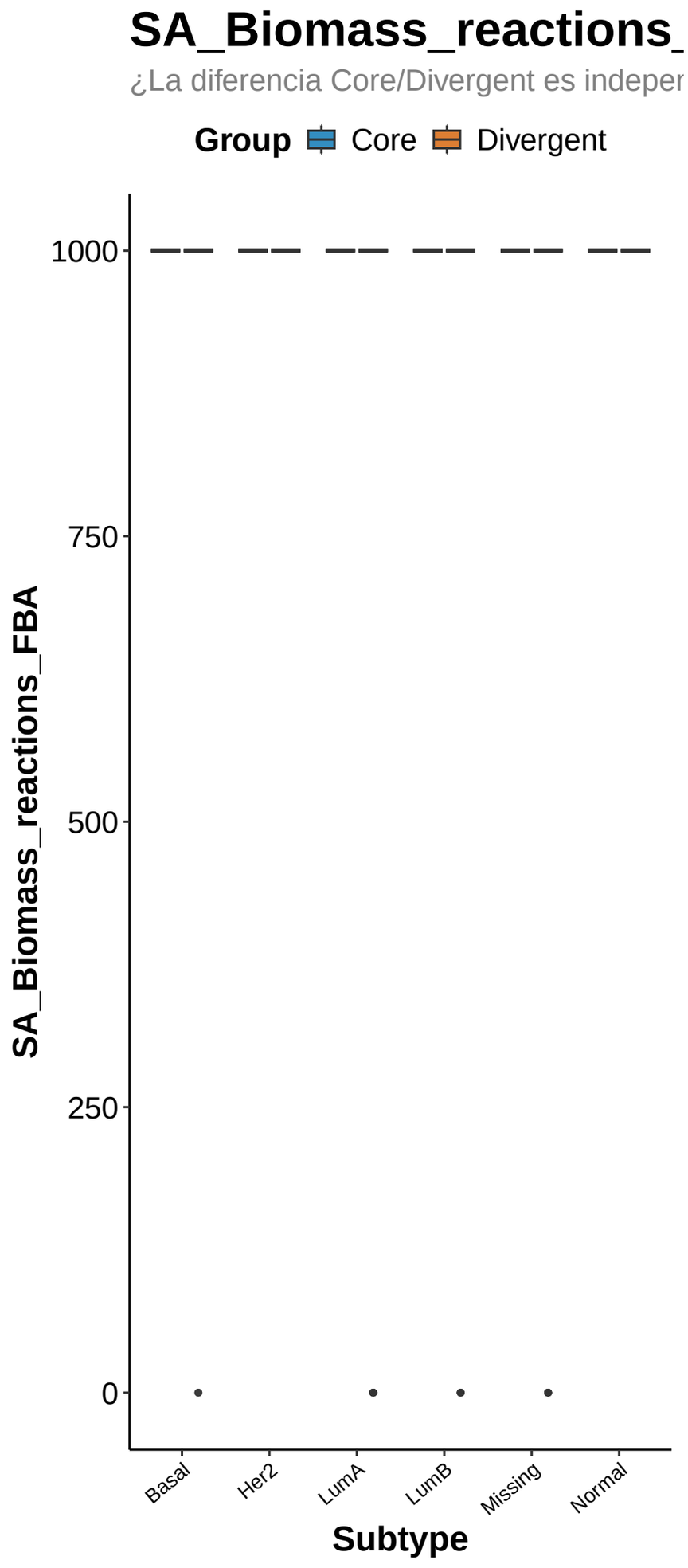


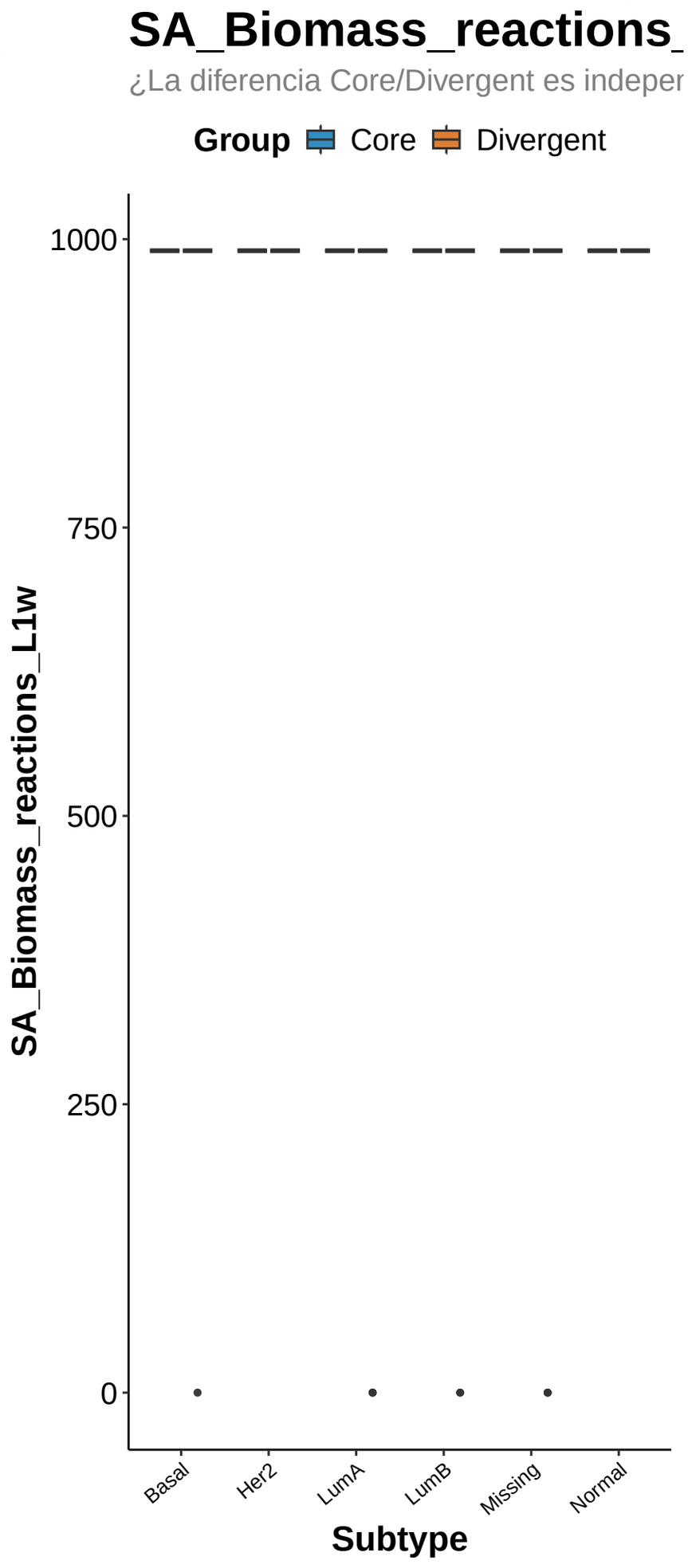
SECCIÓN III

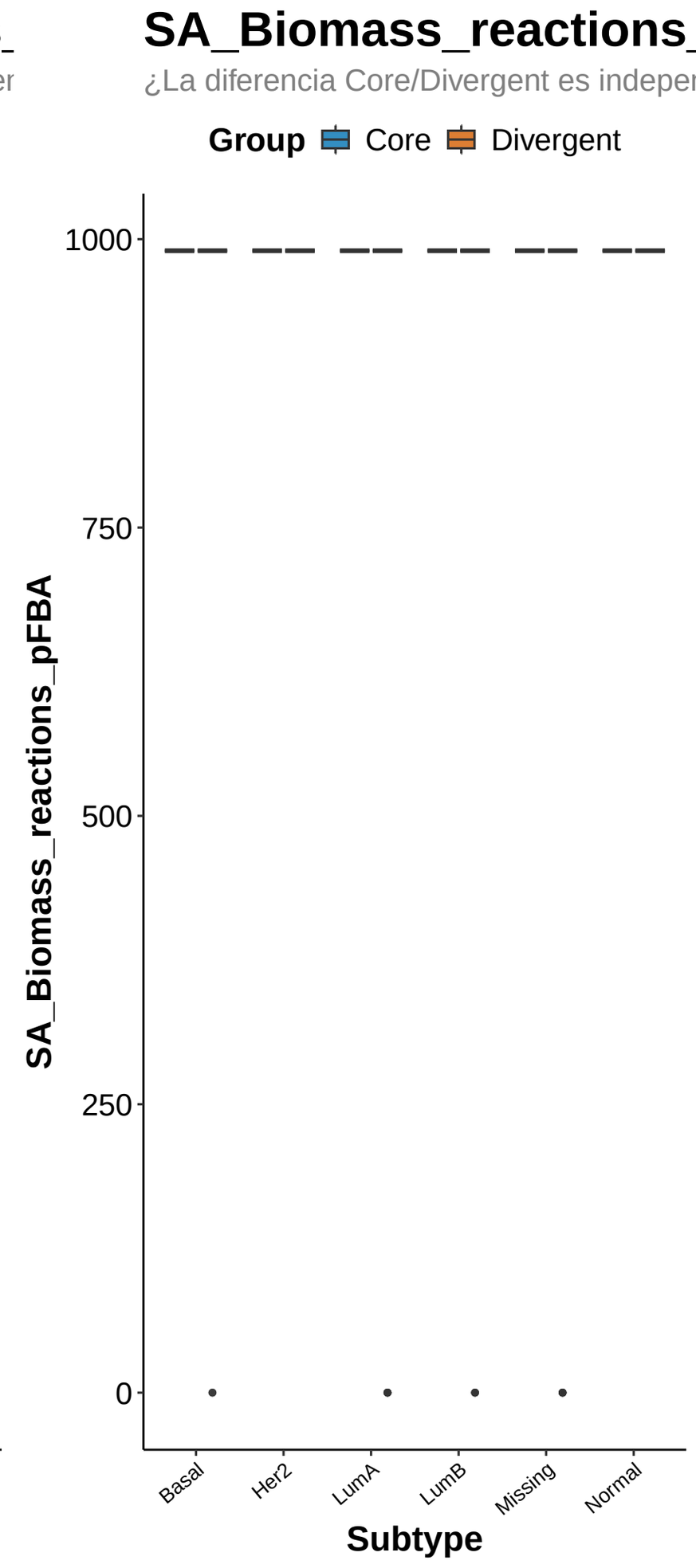
Análisis Integrado: Clínica <–> Metabolismo

¿Core vs Divergent es reflejo del subtipo? ¿O reprogramación metabólica independiente?

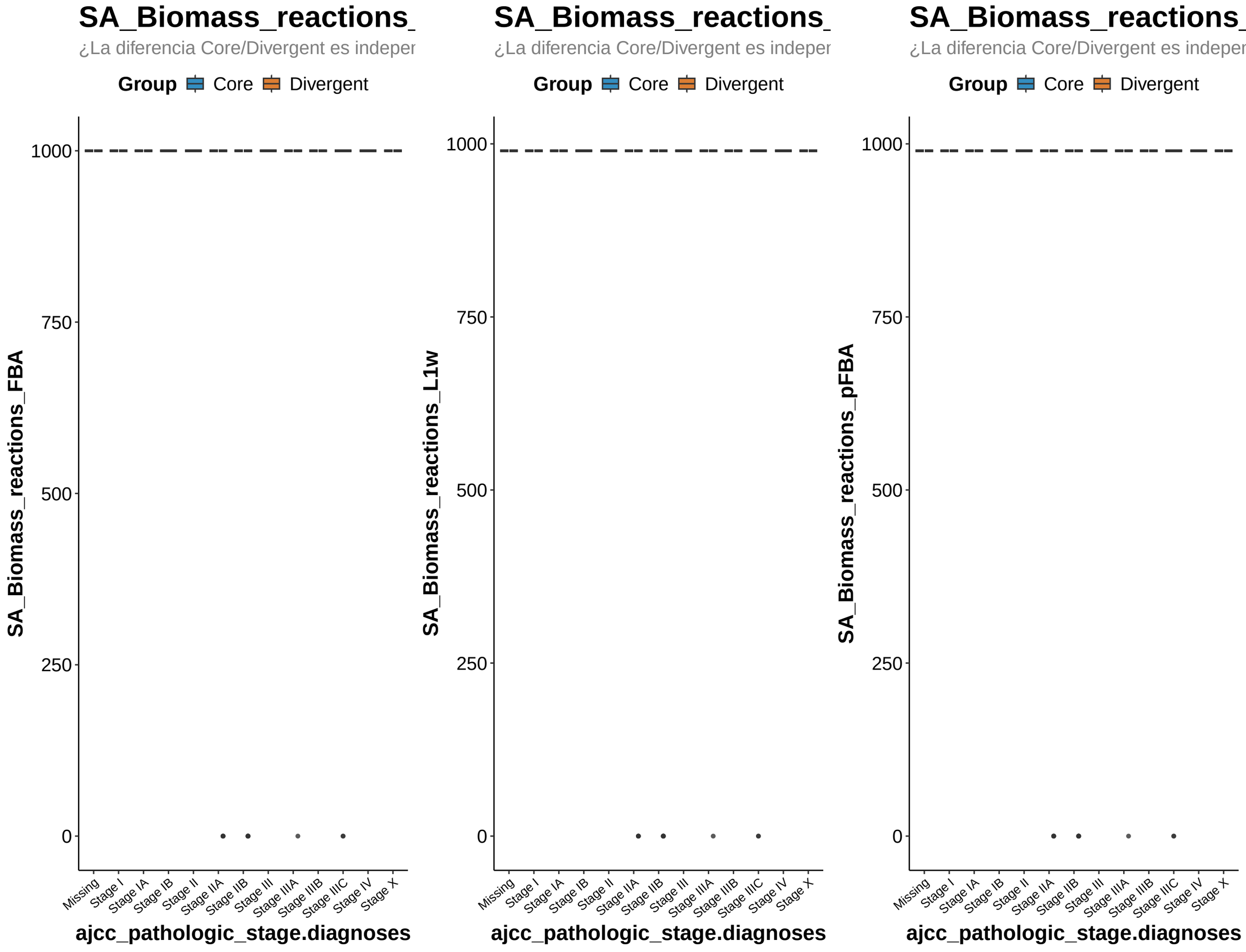
Integración: Top metabólicas × Subtype



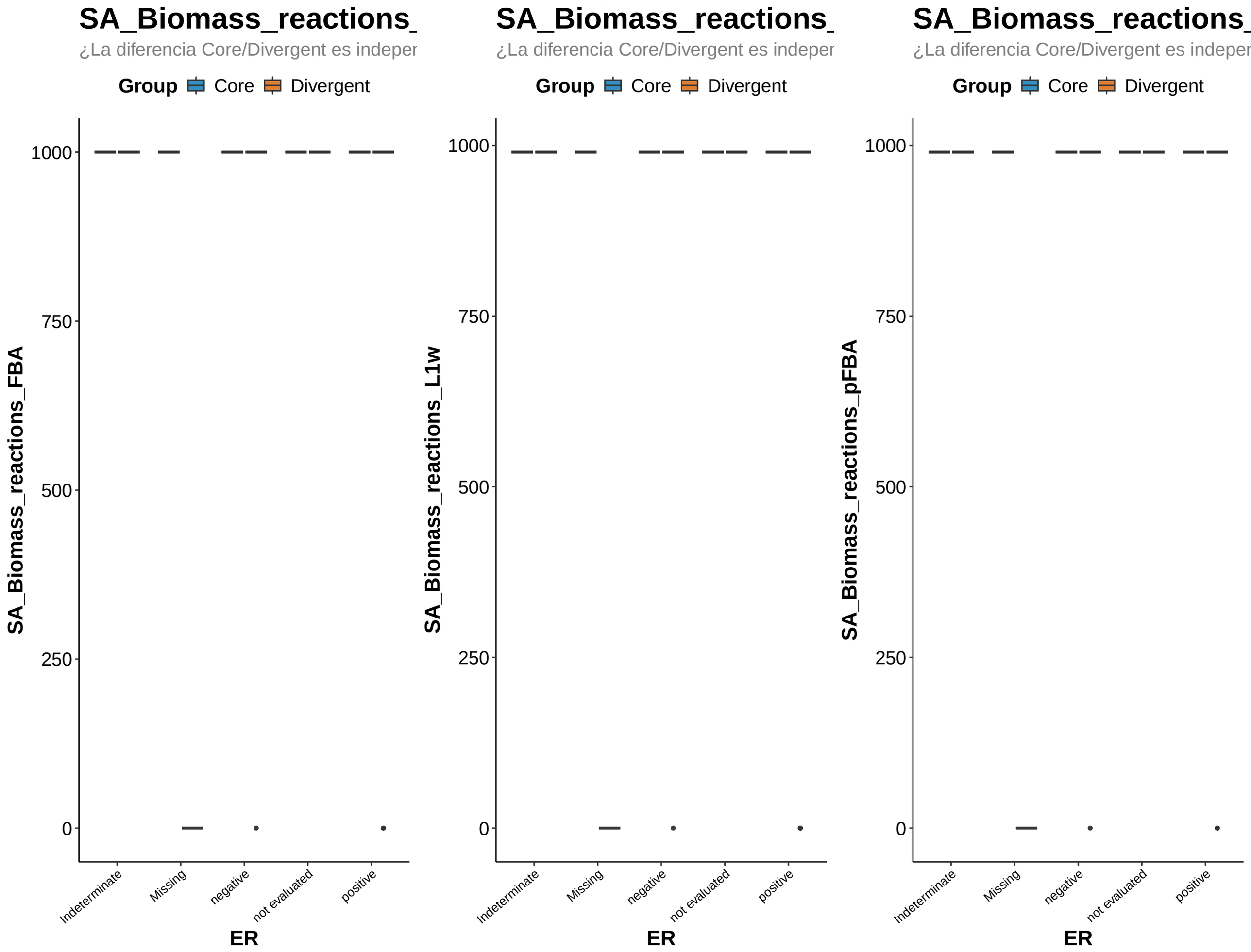




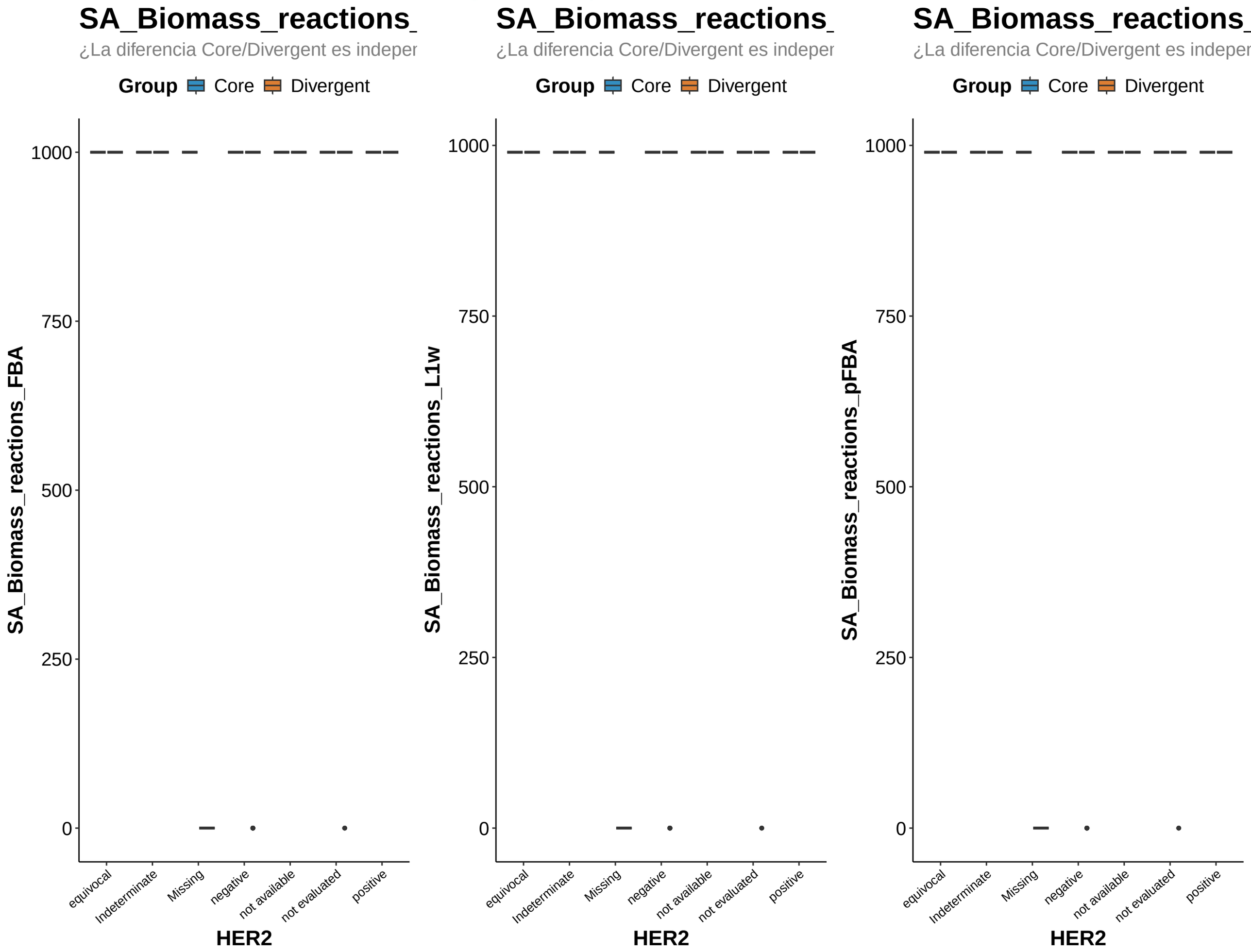
Integración: Top metabólicas × ajcc_pathologic_stage.diagnoses



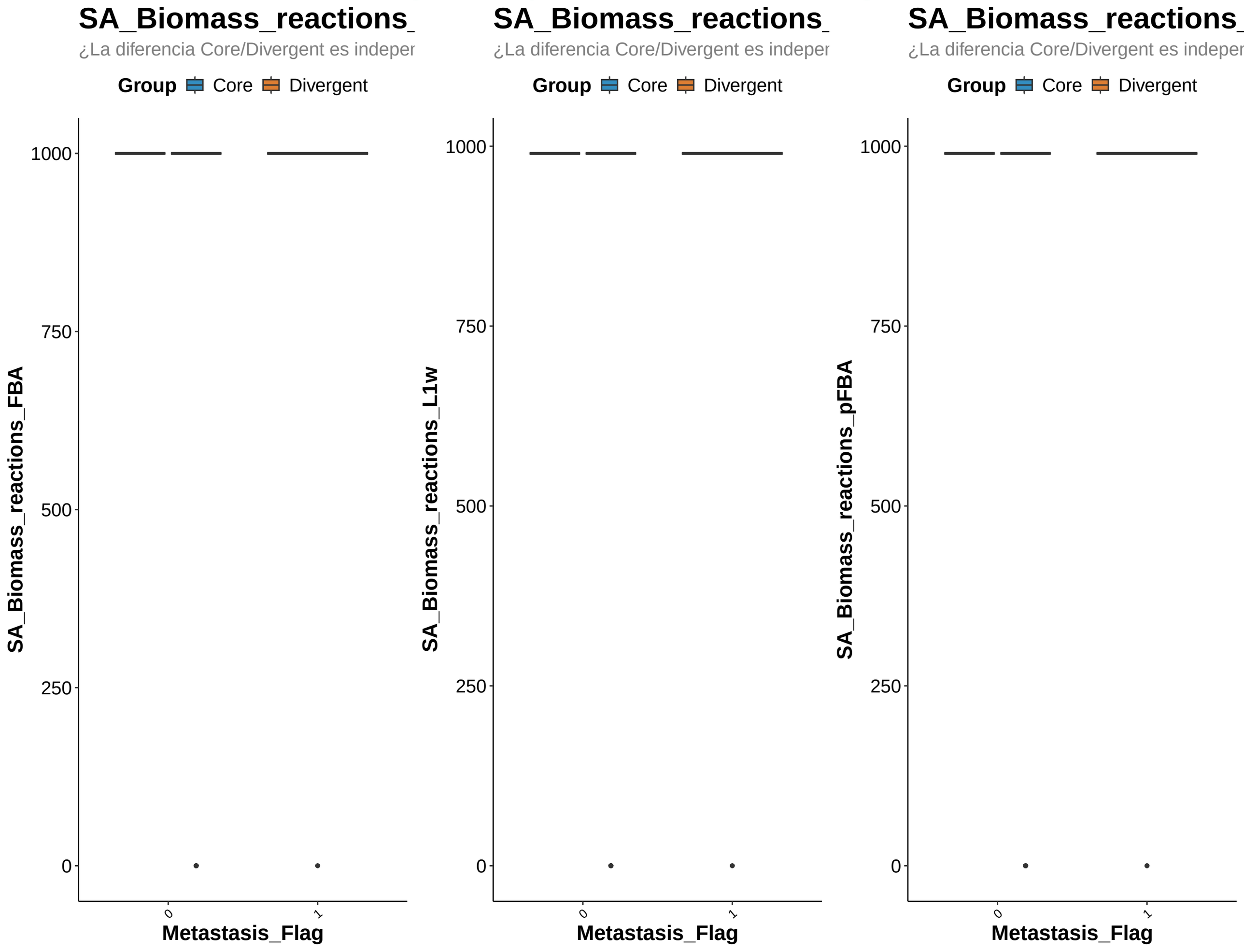
Integración: Top metabólicas × ER



Integración: Top metabólicas × HER2

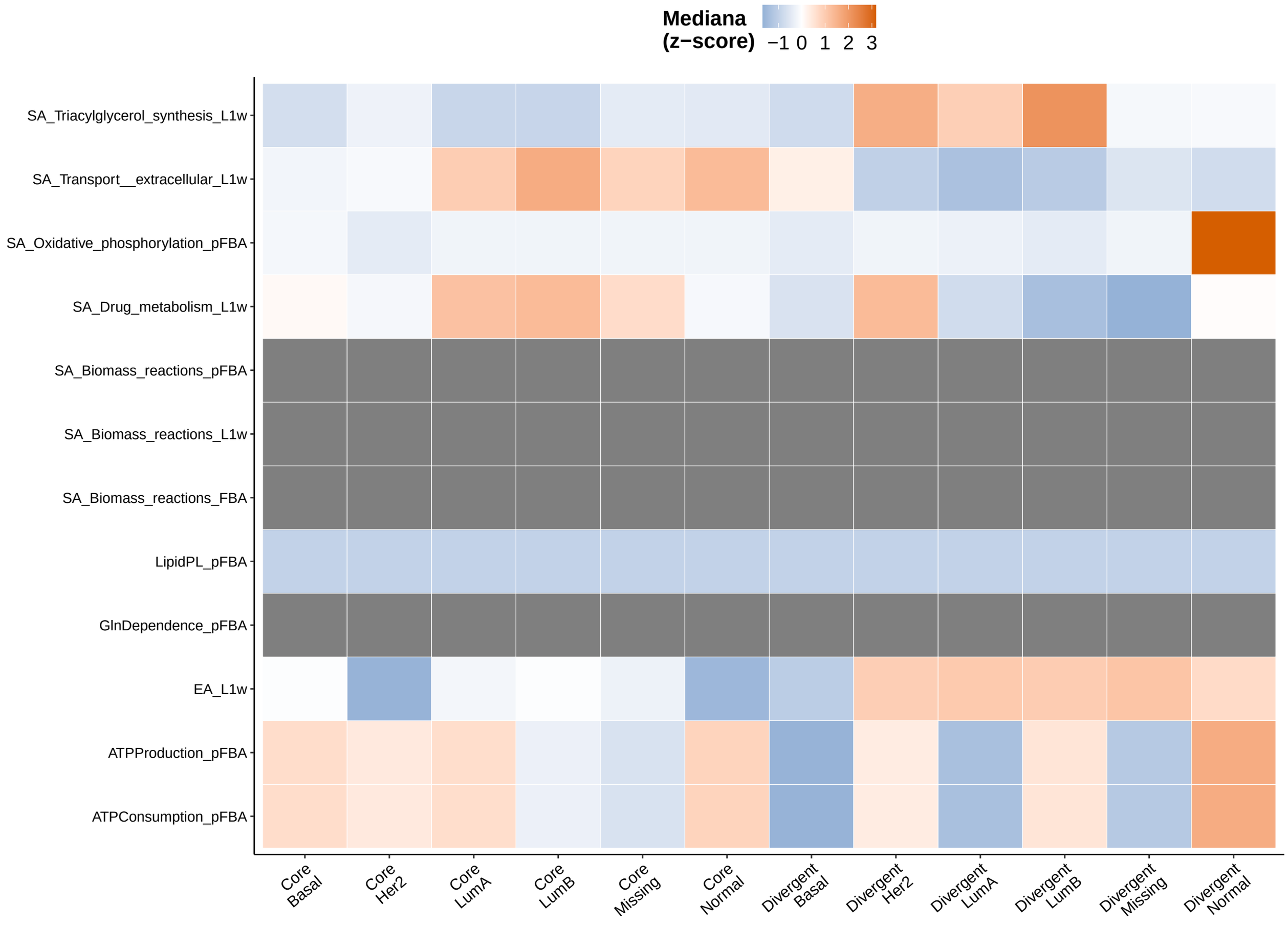


Integración: Top metabólicas × Metastasis_Flag



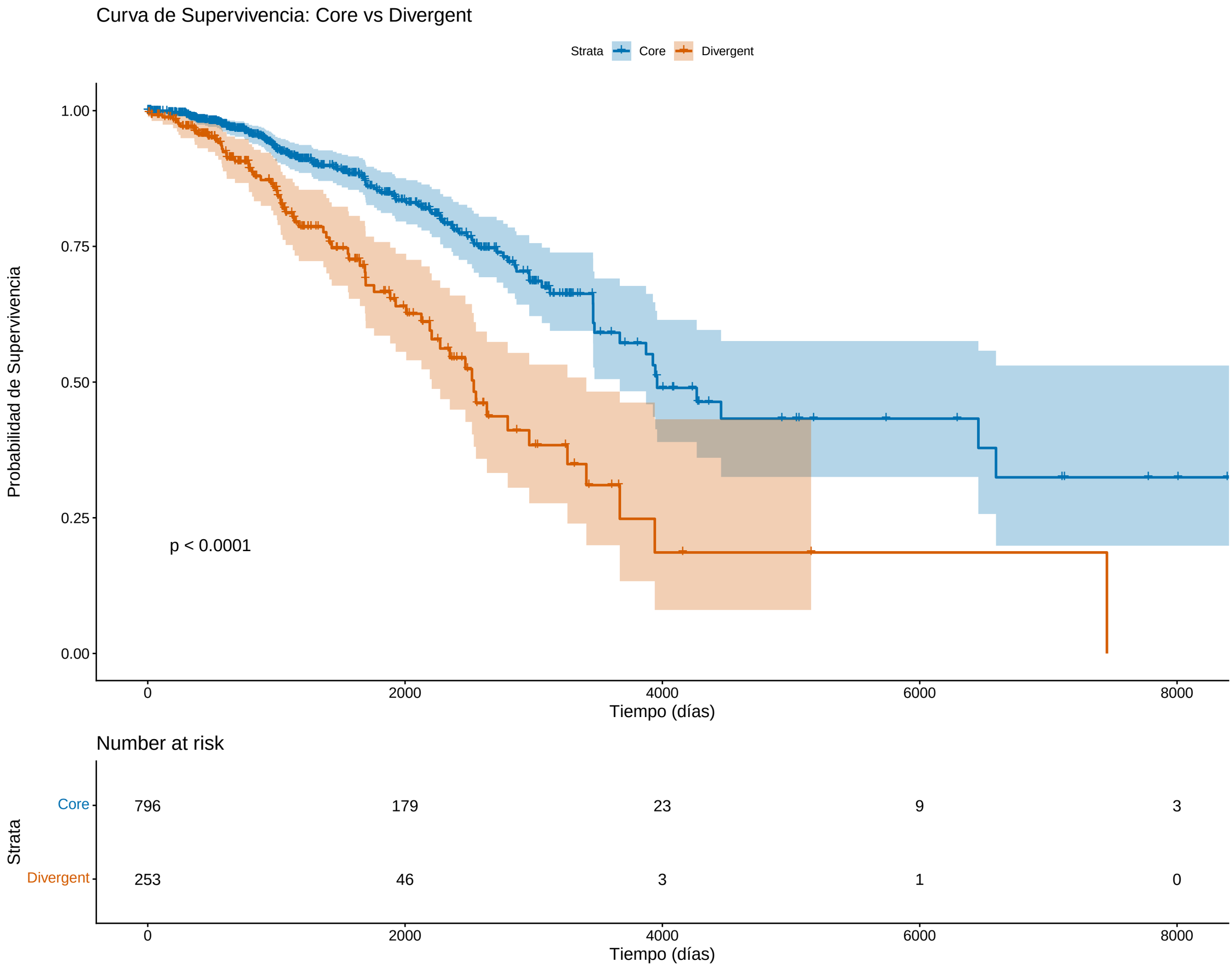
Heatmap Integrado: Top Metabólicas × Subtype × Grupo

Si el patrón Core/Divergent persiste dentro de cada subtipo −> efecto independiente



SECCIÓN IV
Análisis de Supervivencia

¿El fenotipo metabólico Core/Divergent predice pronóstico?


Curva de Supervivencia: Core vs Divergent
Strata
Core
Divergent
Probabilidad de Supervivencia
1.00
0.75
0.50
0.25
0.00
p < 0.0001
0
2000
4000
6000
8000
Tiempo (días)
Number at risk
Strata
Core
796
179
23
9
3
Divergent
253
46
3
1
0
0
2000
4000
6000
8000
Tiempo (días)